

0 Preliminaries

0.1 Who am I?

I am Andrew Rees, more formally written as Dr. D. A. S. Rees. I have been stashed away in 4E in the Department of Mechanical Engineering since 1990 and have been happily whiling away my time doing research and also teaching the ever-increasing numbers of students. In 1991 we had approximately 90 students per year for a three year course, whereas now we have close to 300 on a four year course. In that time the number of academic staff has only doubled.

My office is 4E 2.54 and my email address is ensdasr@bath.ac.uk, which reflects the old naming convention we had of **ensdasr**, which is a shorthand for **engineering**, **staff** and **my initials**. Only a few of us old-stagers still have such an email address. An alias is D.A.S.Rees@bath.ac.uk which I prefer, but the Computer Centre doesn't! In your cases your official email addresses consist of your initials and a random number. I have no idea whether we'll get a fourth convention.....

0.2 My background.

By what authority do I claim to be able to teach you?

I guess that a 1st class degree in Mathematics from Imperial College, London, in 1980 isn't too shabby. I guess too that the fact that Bath University hasn't yet sacked me for being useless is a note in my favour. Somehow I have managed to get away with it for 30 years....

My Ph.D. was in Applied Mathematics from the University of Bristol in 1986 where I concentrated on convective flows and stability in porous materials. Added to this is an ATCL diploma from the Trinity College of Music, London, in 1977 and membership of the National Youth Orchestra of Wales; so I guess that this means that I do have at least one interest outside of maths and that I also get to see people who aren't colleagues.

I did have a short spell from 1980 to 1982 working for what was then called British Aerospace. Based in Filton, they wanted me to do some flutter calculations, and to devise feedback laws which would stop aircraft wings falling off when they flew faster than their original design speed. That's quite important for an aircraft.

After my doctorate I stayed at Bristol University with the same supervisor doing essentially the same things but doing them better and quicker. This was followed by two years' worth of fixed-term lecturing duties in the Maths Department at Exeter University. Then the job in Bath appeared and I just had to say yes to the offer of employment because I had a mortgage to service, and a wife and two young children to feed.

0.3 Provision of 'content'

Everything that I am prepared to give you may be accessed at my dedicated Maths 1 website:

<https://people.bath.ac.uk/ensdasr/ME10304.bho/maths1.html>.

and there is a link to that from the appropriate Moodle page.

You have these notes, and these are also available at my Maths 1 website.

The individual lectures will not be recorded, both those which I recorded in the Autumn 2020, when we were in a fully remote-learning mode, are available at the Maths 1 website. These were done in a conference presentation style, effective but potentially boring — jokes don't work so well when there is no audience! There may be very slight differences in the content, example problems, but the syllabus is exactly the same.

Typically there will be one problem sheet per week which will be given out in the lecture (and also placed at the website). I won't be marking your workings because 270 students times an estimate of one hour per script adds up to more than a week! Nevertheless, I will be posting my solutions a week later at my Maths 1 website.

The two tutorials (really they are problems classes) will see about seven or eight of us (me, a colleague, various postgrads and postdocs) circulating around the two rooms engaging with your mathematical issues.

0.4 Outline of ME10304.

Hopefully we may now describe the situation we are in as being post-covid and therefore this academic year is a return (at least for my unit) to the pre-covid in-person face-to-face lectures and tutorials. Whether this is wise is moot. So there would be two 50-minute lectures each week (timetabled for successive hours) with a two-hour problems class (timetabled as two separate one-hour slots), and I generally present 20 lectures out of the 22 slots available.

The following is a brief breakdown of the content of the unit:

Topic	Number of lectures
Introduction	0.5
Curve sketching	1.5
Complex numbers	2
Differentiation	4
Integration	5
Series	3
Vectors	4

As you can see, many of these topics are familiar. It is my experience that different people will have covered different things and, given that all of these topics are deemed to be important, I will need to cover them all. So, for each person, a subset of the material this semester will be pure revision, but the content of that subset will be different for different people. However, I aim to bring forth some new ideas in various places so that there will be a few nuggets of gold to find within the familiar.

0.5 The detailed syllabus.

The following is how I will split the unit content over the 20 lectures.

	Topic	Content
1.	Curve	Polynomials. Modulus. Exponentials. Hyperbolic functions. Logs.
2.	sketching.	Envelopes . Square roots . Ratios of polynomials .
3.	Complex numbers.	Motivation/need for them. Definition. Arithmetic operations. Geometric interpretation. Polar/exponential form.
4.		de Moivre's theorem . Euler's formula and identity . Roots of complex numbers . Further identities .
5.	Differentiation.	Definition and notations. Use of limits. Higher derivatives. Linearity. Product rule. Product of more than two functions.
6.		Chain rule and proof . Nested functions . Advanced cases . Quotient rule .
7.		Critical points. Primary and secondary criteria. Checklist.
8.		Partial differentiation . Clairaut's theorem . Critical points of surfaces .
9.	Integration.	From sums to integrals. Definite and indefinite. Integration by substitution. f'/f form.
10.		Ratios of polynomials and partial fractions . Top heavy ratios . Repeated factors . Irreducible quadratics .
11.		Integration by parts. Derivation of the Rees method – rules of implementation. Miscellaneous cases.
12.		Applications 1 . Means and RMS . Recurrence relations . Volumes under surfaces in Cartesians and in polar coordinates .
13.		Applications 2 . Length of a line. Volumes and surfaces of revolution. Triple integrals.
14.	Series.	Series and sequences . Binomial theorem and Pascal's triangle . Binomial series .
15.		Taylor's series and Maclaurin. Derivation/justification. Two forms of Taylor's series.
16.		Convergence and d'Alembert's test . Examples of numerical and of power series . Radius of convergence. l'Hôpital's rule. Derivations.
17.	Vectors.	Revision: definition and elementary concepts. Unit vectors. Scalar product. Angle between two vectors.
18.		Vector product . Lexicographical ordering! Area of a triangle using vector products . Coplanarity of three vectors or four points .
19.		Equation of a line. Distance of a line from a point. Closest approach of two lines.
20.		Distance of a point from a plane . Equation of a line - two forms . The direction of the intersection of two planes .

The remaining subsections in this Chapter, Chapter 0, cover a variety of preliminary topics. I am not at all sure that they are emphasized sufficiently in pre-university tuition, but again I would far prefer to play safe by presenting my ideas to you so that you will know what the standards are that I expect.

0.6 Significant figures.

Here's a good question: which is better, 6 decimal places or 6 significant figures?

The answer depends on how large the number is.

The number $\pi/1000 = 0.003142$ is correct to 6 decimal places but only to 4 significant figures, while the number $100\pi = 314.159265$ is also correct to 6 decimal places but is correct to 9 significant figures.

So it is clear that the number of significant figures is the better concept because it is independent of the magnitude of the number being considered.

0.6.1 The number of significant figures.

From the practical point of view of computing numbers, the big question is, **How many significant figures do we need?** To answer this we need to understand what happens when we do arithmetic with limited precision.

In engineering, we frequently need only 3 or perhaps 4 significant figures for comparison with experimental work, for that is often the greatest precision with which we can measure. There are, of course, counter-examples to this such as the mass of the electron, the speed of light, gravitational acceleration and the density of water, for each of these have been measured to a very much greater degree of accuracy.

If one were to do a Google search for the value of g , the acceleration due to gravity, then one website gives **9.8m/s^2** , another gives **9.81m/s^2** while the 'standard value' is generally defined as **9.80665m/s^2** . The first of these has 2SFs, the second has 3SFs, while the third is apparently correct to 6SFs.

So what is the truth of the matter?

If we take the third value of g as being correct, then the first two are clearly approximations, so all is well....or is it? Although the third has been designated the standard value, it gives the misleading impression that it is valid everywhere on the earth's surface. Although we know that g decreases as we rise above the Earth's surface and also as we descend under the sea, the standard value is not correct. In fact it varies between 9.779m/s^2 for Mexico City to 9.819m/s^2 for Oslo; these values represent fairly closely the two extreme cases for the earth's land surface when we confine ourselves to cities. Therefore it is pointless using all six significant figures in the standard value unless you are at one of the precise places where it applies — Zürich is correct to four significant figures! Perhaps we should only use 9.8m/s^2 and simply note that all our calculations will also have only two SFs of accuracy. And even if we know exactly where we are on the globe, g will also vary with the time of day if we are on open water due to the phase of the tide.

What are the consequences of this variation? One trivial one is that one would weigh **0.4%** less in Mexico than in Oslo. Another is that a pendulum in Mexico has a period which is **0.2%** longer than an identical one in Oslo. Admittedly these are small values, but at least we know what the possible error is.

However, despite needing only 3 or 4 figures for experimental work, it is not always good to retain only 3 or 4 figures of accuracy during intermediate theoretical calculations. This is because round-off errors can build up catastrophically. In general, I would recommend using 5 or 6 for all calculations, and if the final result is required to 3 significant figures (SFs), one may then round the final answer to the required accuracy. But even then, one must be aware of how accuracy can get eroded. See **this web site** for further information.

Example 0.1 Calculate $0.301 + 0.478 + 1.42 + 18.4 + 101$ to 3 SFs.

This example illustrates how care must be taken when doing arithmetic. We'll compute the sum using three different 'methods'.

Method (i) Exact arithmetic gives **121.599** which is **122** to 3SFs. This is the way one would do it using a calculator or pencil and paper.

Method (ii) Adding by beginning with the largest number and rounding each computation before adding the next largest and so on.

$$\begin{aligned} 101 + 18.4 &= 119.4 &= 119 \text{ (3SF)} \\ 119 + 1.42 &= 120.42 &= 120 \text{ (3SF)} \\ 120 + 0.478 &= 120.478 &= 120 \text{ (3SF)} \\ 120 + 0.301 &= 120.301 &= 120 \text{ (3SF)} \end{aligned}$$

Method (iii) Adding by beginning with the smallest number and rounding each computation before adding the next smallest and so on.

$$\begin{aligned} 0.301 + 0.478 &= 0.779 \\ 0.779 + 1.42 &= 2.199 &= 2.20 \text{ (3SF)} \\ 2.20 + 18.4 &= 20.60 &= 20.6 \text{ (3SF)} \\ 20.6 + 101 &= 121.6 &= 122 \text{ (3SF)} \end{aligned}$$

Method (i) works well because the precision we are using (namely, exact for pen and paper, 8 to 10 SFs for a calculator, or either 7 or 15 or more for a computer) uses very many more SFs than the precision of the numbers.

Method (ii) is the poorest. By using the largest number first, we are declaring that almost all the information contained after the decimal point in the rest of the data will not contribute to the final result. This is why it provides the least accurate sum. Note also that if we had included another million numbers under **0.5** in magnitude, then this would have had no effect on the final answer. So this is a bad way to proceed.

Method (iii) yields the safest method when working with a highly restricted number of SFs.

A further danger point is that the original data that we have added together isn't consistent in terms of the number of decimal places, i.e. we have no idea whether they were already rounded to 3SFs before we started to sum them. If whomsoever gave us the data had rounded the data to 3 SFs, then the largest value, **101**, could range between **100.500** and **101.500** in real life and this could alter our final answer. We would have more confidence in the answer if either (a) it is stated that each value is exact or (b) all the data were presented to 3DPs. So the moral of the story is that we must enquire about the provenance of the data before we can manipulate or use the data.

The reason I have mentioned the above example is that all calculating devices use a fixed number of SFs, and therefore rounding always happens after each addition, subtraction, exponentiation, multiplication, division, square root, etc... It is rare for this to cause any problems, but it is good to be aware that round-off error can very occasionally yield incorrect results.

However, I often see very severe rounding taking place in computations performed in exams. Even if the method used is correct, the final answer can often be very poor. Always be suspicious of numbers and where they come from.

Example 0.2 Solve the quadratic $x^2 - 2.01x + 1.01 = 0$ giving the answer to 4SFs.

It is straightforward to find the exact answer,

$$\begin{aligned} x &= \frac{2.01 \pm \sqrt{2.01^2 - 4 \times 1.01}}{2} \\ &= \frac{2.01 \pm \sqrt{0.0001}}{2} \\ &= \frac{2.01 \pm 0.01}{2} \\ &= 1.01, 1. \end{aligned}$$

Now let us rerun the analysis rounding to 4SFs after each arithmetic operation. First we note that $2.01^2 = 4.0401$ which is equal to **4.040** to 4SFs. Therefore

$$\begin{aligned} x &= \frac{2.01 \pm \sqrt{2.01^2 - 4 \times 1.01}}{2} \\ &= \frac{2.01 \pm \sqrt{4.040 - 4.04}}{2} \\ &= \frac{2.01 \pm 0}{2} \\ &= 1.005, 1.005 \end{aligned}$$

In this case, we have an error of **0.5%** even with 4SF arithmetic. Although this loss of precision isn't too catastrophic, it has changed the qualitative nature of the solution, which is catastrophic. We had two different roots when using exact arithmetic, and now they are two identical roots. This has arisen because we subtracted two numbers which are almost equal. It is in these situations where the loss of significant figures is felt the most.

An alternative view of the above is to analyse the effect of possible inaccuracies in the coefficients, **2.01** and **1.01**. These values could have been obtained from an experiment, and, given that they have been quoted to 3SFs or 2DPs, the likely maximum error in their values is ± 0.005 . Let us therefore consider the equation

$$x^2 - (2.01 + \epsilon_1)x + (1.01 + \epsilon_2) = 0 \quad \text{where } |\epsilon_1| \leq 0.005 \text{ and } |\epsilon_2| \leq 0.005.$$

Using exact arithmetic, and with the final answers rounded to 3DPs, we get the following possible solutions.

$$\epsilon_1 = 0.005, \quad \epsilon_2 = 0.005 : \quad x = 1.015, 1.000,$$

$$\epsilon_1 = 0.005, \quad \epsilon_2 = -0.005 : \quad x = 1.108, 0.907,$$

$$\epsilon_1 = -0.005, \quad \epsilon_2 = -0.005 : \quad x = 1.005, 1.000,$$

$$\epsilon_1 = -0.005, \quad \epsilon_2 = 0.005 : \quad x = 1.0025 \pm 0.1000i.$$

Note that the last case has complex roots!

So for this example we can get an error of up to **10%** for real answers for only a **0.5%** change in a coefficient. This is a problem which is very sensitive to the degree of accuracy of the basic data — these are dangerous problems! I emphasize that this is an extreme circumstance, but occasionally these circumstances do arise, and it is important to acquire a good intuition of how accurate one's arithmetic truly is.

0.6.2 Too few significant figures?

This is motivated by the fact that I often see students providing answers with too few significant figures in exams. If the exact answer to a maths problem were to be $1/\sqrt{2}$, then how should we present it? First, I have to come clean and say that I am very happy for the answer to be written as $1/\sqrt{2}$, although some people prefer to have the square root in the numerator: $\sqrt{2}/2$. An alternative notation is $2^{-1/2}$. If this were the final answer, then I would prefer one of these three exact solutions to be written. If someone were to prefer to write it out, then five or six SFs is fine. Hence **0.707107** is generally ok.

Suppose now that we are asked to evaluate $\sin 2\pi$. The great majority of us (assuming that 2π is in radians) would say that the value is zero. The following table shows what happens if 2π is first converted to a number with a limited precision before the sine is computed.

#/SFs	2π	$\sin 2\pi$
2	6.3	0.016814
3	6.28	−0.003185
4	6.283	−0.000185
5	6.2832	0.000015
6	6.28319	0.000005
7	6.283185	−0.000000

So the approximation to 2π using 3 or 4 SFs isn't particularly good given that the maximum value of the sine function is **1** and that the exact value of $\sin 2\pi$ should be zero. It is much better to retain as much accuracy as possible.

0.6.3 Too many significant figures?

The above went into detail about what could happen if we take too few significant figures. But how many is sufficient?

One may quote π to 36 decimal places,

$$\pi \sim \mathbf{3.141\ 592\ 653\ 589\ 793\ 238\ 462\ 643\ 383\ 279\ 502\ 884},$$

as a clever party trick, but the last significant digit shown, which is in the 36th decimal place, represents a very incredibly small quantity. Just how small is it? One way of attempting a visualization is to consider 10^{36} sugar crystals where I will assume that each crystal is a cube of side **0.5mm**. It doesn't take long to find out that 10^{36} crystals is equivalent to a cube of side **5,000,000km**, which is roughly 12 times the distance of the moon from Earth. If we compare one sugar crystal with this cube, then that is 10^{-36} . Therefore I conclude that 36 significant figures is a waste of effort!

I tend to use roughly 6 SFs for my calculations and I will only increase that if it turns out to be necessary to do so. So keep your error-checking radar on at all times.

0.6.4 How accurate is the usual approximation to the value of π ?

Well, we know from school days that $\pi \simeq 22/7$, but how good is this? So we need to compare $22/7 \simeq 3.142857142857\dots$ with π as given above. To do this we may define the relative error as being,

$$\text{R.E.} = \frac{22/7 - \pi}{\pi},$$

and this comes to **0.00040**, i.e. four parts in 10,000, or **0.04** percent, which isn't at all bad.

Is there a better one that isn't too complicated? How about $355/113$? This is **3.1415929204** to 10 decimal places. Here's the comparison:

$$\begin{array}{r} 3.1415929204\dots \\ 3.14159265358979\dots \end{array}$$

The relative error is now about 8.5×10^{-8} which is considerably better than for $22/7$.

No doubt you can think of a nice easy way of remembering $355/113$.

Question: How good are the following as approximations to $\sqrt{2}$: $99/70$ and $8119/5741$? I will leave this question to you.

0.6.5 To conclude....

I have just read that lot through again and I reckon that it's somewhat heavy, especially as it is the very first technical material on a Maths unit at university. So perhaps I ought to summarize it in a few recommendations.

1. Check your data. What is its provenance? How accurate is it?
2. Do not do any rounding off until you get to the very final answer.
3. Do not use too few (it's dangerous) or too many (it's pointless) significant figures.
4. Always be on the lookout for potential accuracy issues, e.g. the subtraction of almost equal numbers.

0.7 How to structure a mathematical argument.

The aim of this section is to cover briefly the frequently-neglected subject of the structuring of a mathematical argument and how to write it down.

When one undertakes a piece of mathematics, be it algebra, trigonometry or calculus or a combination of these, then the mathematics should proceed by sequence of logical steps. It is important to be able to write these steps out in a manner which reflects that logic and which allows someone else to be able to follow that logic. A poor example is the following.

Example 0.3 (bad) Evaluate the polynomial $(x + 5)^2 - x + 10$ when $x = 2$.

I have seen the following sequence of steps:

$$(2 + 5)^2 = 49 - 2 = 47 + 10 = 57. \tag{0.1}$$

The answer is correct, but what do the various equals signs mean in this context? Only the last one is used correctly. This way of doing things is a mathematical version of a 'stream of consciousness' writing.

Apart from the obvious simple substitution, $(2 + 5)^2 - 2 + 10 = 49 - 2 + 10 = 57$, which is fine because all of the equals signs have been used correctly, one could be very pedantic and write the problem out this way:

When $x = 2$, then $(x + 5)^2 = 49$, and hence

$$\begin{aligned}(x + 5)^2 - x + 10 &= 49 - 2 + 10 \\ &= 57.\end{aligned}\tag{0.2}$$

Example 0.4 (good) Solve the quadratic equation, $x^2 + 2x - 8 = 0$.

We'll do this by completing the square.

$$\begin{aligned}&x^2 + 2x - 8 = 0 && \text{Need to add 9 to both sides...} \\ \Rightarrow &x^2 + 2x + 1 = 9 && \text{...since } x^2 + 2x + 1 \text{ is square} \\ \Rightarrow &(x + 1)^2 = 9 \\ \Rightarrow &x + 1 = \pm 3 && \text{taking square roots} \\ \Rightarrow &x = -1 \pm 3 \\ &= -4, 2.\end{aligned}\tag{0.3}$$

The 'implies' symbol, $\Rightarrow$, means that the current line follows logically from the preceding line. It must not be used to replace the equals sign. Note that I did not need to use the implies sign on the last line leading to the final solution in Eq. (0.3), because that was the result of a simple calculation (the same is true for Eq. (0.2)). Note also the hints on the right, which might aid the reader understand what the logical step was.

So the aim is to write out a mathematical argument the logic of which I (as the marker of your exams) should be able to follow quite easily. This would also mean that you would be able to follow your own work months from now when revising for exams.

One could think of the mathematical argument leading to (0.3) as being a condensed version of a verbal statement: *Given that $x^2 + 2x - 8 = 0$, this implies that it may be rearranged in the form, $x^2 + 2x + 1 = 9$, because the left hand side is now a perfect square and we have a fleeting suspicion that this might be useful to us. Now we take square roots of both sides etc. etc....*

0.8 Notations and functions and the avoidance of ambiguity.

0.8.1 Inverse functions.

It is a real pain that inverse functions such as the inverse sine are denoted almost universally by $\sin^{-1}$ these days and in the UK. It happens for other functions too, such as $\cos^{-1}$, $\tan^{-1}$ and $\cosh^{-1}$. There is even a general notation: if $y = f(x)$ then $x = f^{-1}(y)$. The pain is that it looks like a reciprocal.

In the olden days, whenever they were, the inverse sine was written as **arcsin** which isn't at all ambiguous.

If you really wished to write down the reciprocal of $\sin x$, then there are various ways of doing so:

$$\frac{1}{\sin x}, \quad (\sin x)^{-1} \quad \text{and} \quad \operatorname{cosec} x.$$

What about the following four functions; are they the same as one another:

$$(\sin x)^{-2}, \quad (\sin^{-1} x)^2, \quad \arcsin^2(x) \quad \text{and} \quad \sin^{-2} x?$$

The first is the -2 power of $\sin x$. The second is the square of the inverse sine of x , as is the third. The fourth one is ambiguous. Some would say that it is the same as the first one because the power isn't -1 , which might be said to be reserved to denote the inverse function. Others would say that it is the second, although I

would hope that almost no-one would fall into that trap. As for me, my view is that, because it is ambiguous, it should never appear; one's mathematical writing *must* never have even the smallest taint of ambiguity.

As for the reciprocal of the inverse sine — I have never seen anyone use this — then the following four will work:

$$(\sin^{-1} x)^{-1}, \quad \frac{1}{\sin^{-1} x}, \quad (\arcsin x)^{-1}, \quad \frac{1}{\arcsin x}.$$

All of these are correct although the first is an absolute horror.

0.8.2 Double negatives.

Ah, now these have become very popular in the last ten years or so. By this I mean constructions such as,

$$3 - -2 = 5.$$

In one sense this is ok. However, the first minus sign is formally a subtraction operator while the second indicates the negative of **2**, so it isn't really a double negative. Just as we saw that -1 as a superscript could mean two different things and that these usages may be confused, here the difficulty/danger is due to where the two minus signs are when written down. I will illustrate that using the following, which is something that I have often witnessed on exam scripts:

$$3 - -2 = 3 - -2 = 3 -- 2 = 3 - 2 = 3 - 2 = 1.$$

The above generally appears on successive lines of an exam script as the more complicated parts of an equation are simplified, but the minus signs just keep getting closer and closer! So although this isn't really an ambiguity, my dislike of the double negative is primarily because this often happens. So I would recommend writing, $3 - (-2) = 5$, as a safe way to play the mathematical game.

Being a bit of a purist (and it's also a Great British sport to complain about irrelevant matters!), I would far prefer the following usage: $3 \times (-2) = -6$, the brackets still signifying the negative number, as opposed to 3×-2 — uuurgh.

0.9 Logarithms.

There are three types of logarithm which are used in Engineering. The first is the base-10 logarithm, which is the first one that is taught and which is known as the common logarithm. It is usually written as $y = \log_{10} x$, so if y is increased by 1 (i.e. the addition of 1), then x is multiplied by a factor of 10. Scales such as the Richter Scale behave in the same way: an earthquake which measures 7 on the Richter scale has 10 times the energy of one which measures 6. The amplitude of sound also operates in the same way. I guess that the unit is the *bel* where 10 decibels is equivalent to 1 bel. So a sound of 80 decibels has 100 times the power of one which is rated at 60 decibels.

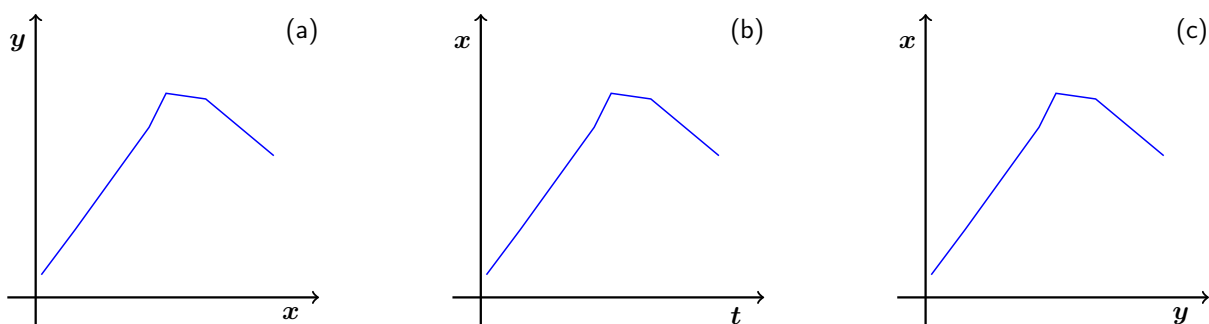
The natural logarithm, invented by Napier, uses $e = 2.7182818284\dots$ as its base. This rather strange number arises naturally in very many places and it has many very interesting properties such as the fact that the derivative of the function, e^x , is also e^x . The natural is generally written as $y = \ln x$, and we simply say that "y is equal to log x" when talking about it, and certainly wehn discussing anything involving calculus. There is no need to say, "lun x", because the base-10 version is hardly ever used in Mechanical Engineering. So $\ln x = \log_e x$.

The base-2 or binary logarithm is written as $y = \log_2 x$. It too is not used often in Mechanical Engineering, but has great use in Computer Science, Information Theory and Music Theory. Given that many of you will also be musicians it is worth dwelling on this for a moment. Violins in an orchestra generally tune their instruments to a note called A which has a frequency of 440Hz. Perhaps this isn't too well-known, but the highest note that the solo violin plays in Dvřák's Violin Concerto is also an A, but its frequency is 3520Hz. The ratio of

these frequencies is precisely 8 which is also 2^3 . Given that $\log_2 8 = 3$, this tells us that this frequency ratio is precisely 3 octaves. Thus the binary logarithm of a frequency ratio tells us the equivalent number of octaves.

[As an aside for the more geeky musicians, a perfect fifth on a stringed instrument corresponds to a frequency ratio of precisely 1.5, whereas the equivalent on an equally-tempered piano, i.e. seven semitones, corresponds to a frequency ratio of $2^{7/12}$ which is 1.498307. Using $\log_2$, the interval for a violin is 0.584963 of an octave, while for the piano it is $7/12 = 0.583333$ (6SFs). This slight difference between these two intervals is why violinists find it difficult to tune to a piano when both the D and the A are sounded. If the violin A is tuned to the piano A then the two D's do not agree, and vice versa. If the violinist tunes the strings perfectly to the piano, then the simultaneous playing of the D and A strings sounds out of tune. Such matters are unresolvable....]

0.10 Naming the axes.



These three graphs look the same. If the first one, labelled (a), were to be written in the form, $y = f(x)$, then the other two are $x = f(t)$ and $x = f(y)$, in turn.

The primary reason for introducing these is the very common but very often inaccurate naming of the vertical axis as the y -axis and the horizontal one as the x -axis. Formally this is correct only for the first one. I have noticed that this can even occur in, say, official government graphs of, to choose an example, the rate of inflation against time, where the rate of inflation is then referred to by the TV reporter as the y -axis and the month by the x -axis. Oops....

In the figure labelled, (b), it seems odd to me that one should say that the y -axis corresponds to the value of x while the x -axis corresponds to the value of t . Even worse, should we say that the y -axis gives the value of x while the x -axis gives the value of y in the figure labelled (c)? Ambiguity is certain.

One simple resolution is to call them the **vertical and horizontal axes** — no-one will misunderstand you. One could also refer to them as the **ordinate** and the **abscissa**, respectively, which are now rather old-fashioned words that are rarely seen. My general recommendation is to use 'vertical' and 'horizontal' because again there is no potential for ambiguity.

The final word. Sometimes one needs 3D graphs. They could take various forms: (i) a projection of the full 3D curve onto say the (x, y) -plane, (ii) a stereoscopic pair where one needs to go slightly cross-eyed to be able to view its 3D nature, or (iii) an anaglyph form which requires red/blue glasses. So such graphs could be formed of lines (e.g. the trajectory of a particle in 3D space) or of surfaces (e.g. the depiction of the height above sea level of an area of land). The three axes could be (x, y, z) or else (x, y, t) or many others. In all these cases it is best to refer to a chosen axis by the name of the coordinate along that axis. Many examples of stereopairs and anaglyphs may be found using Google.

1 Curve sketching

1.1 Preamble

The chief aim of this short section of the Maths 1 unit is to develop one's understanding of what shapes the various standard functions have so that more complicated functions may be sketched. We will attempt to sketch functions by looking at the detailed formula for that function. It is hoped that this will also allow one to be able to give a reasonable guess for what function represents a given graph, the reverse process. We'll begin with polynomials and then complicate issues with the modulus function, exponentials, envelopes, square roots, and finally ratios of polynomials.

We will not employ any methods from calculus in this section. This is simply because we wish to determine the rough outline of what the graph looks like, but not precisely where maxima or minima are located. We will often need to ask questions such as, where are the zeroes, where are the asymptotes, what is the large- x behaviour?

Graphs have been sketched by hand and scanned into this chapter — this is not because I am lazy, but rather it indicates how I require you to do it, particularly in the exam! The sketches will show all the salient features of the graph.

1.2 Polynomials

By polynomials I mean seemingly straightforward functions of x , say, such as those given by $y = x^3 - x$ and $y = (x - 5)^2(x - 1)x$. We will start from the utterly trivial and make our way to slightly more complicated shapes.

The straight line, $y = x$, should be quite straightforward! The bullet indicates the single zero or root where $y = 0$.

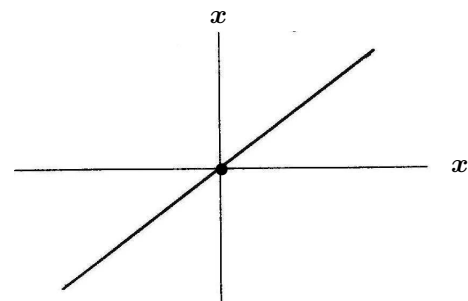


Figure 1.1.

This is a parabola, of course. It has a zero slope at $x = 0$. From the point of view of its factorization, $y = x \times x$, we may say that it also has a 'double zero' at $x = 0$. This will be important below. The bottom of the parabola (double bullet) shows what a **double zero** looks like.

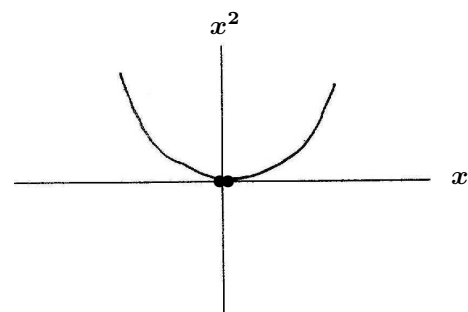


Figure 1.2.

This is a downward-facing parabola. All parabolae, $y = ax^2 + bx + c$, will look like this when $a < 0$, although the maximum will not be at the origin in general.

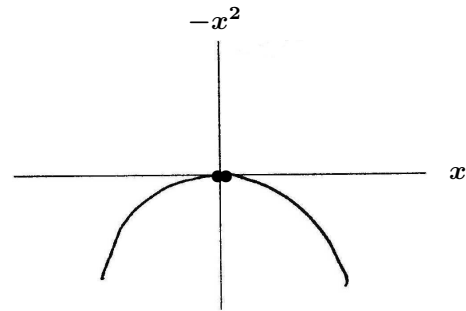


Figure 1.3.

This parabola is identical to the one in Figure 1.2, but has been shifted by 1 in the x -direction. We see that there is a double root at $x = 1$ — note the repeated $(x - 1)$ factor.

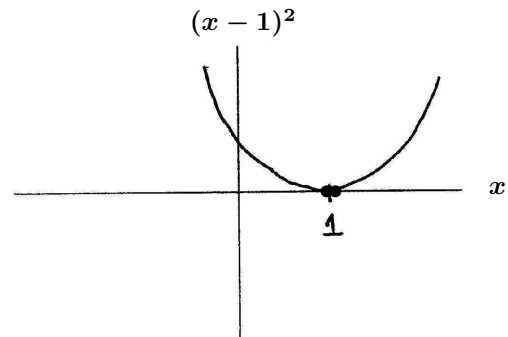


Figure 1.4.

This parabola is identical to the one in Figure 1.4, but it has been shifted downwards by 1. The resulting expression for y may be factorised into $y = x(x - 2)$ and hence there are two zeros: $x = 0$ and $x = 2$.

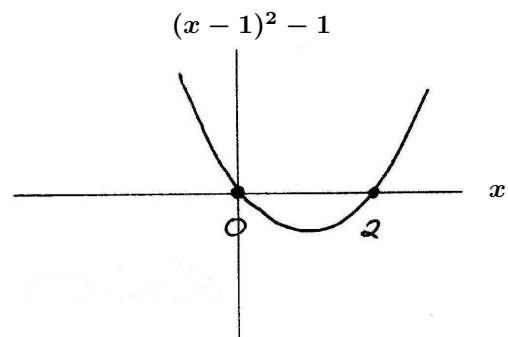


Figure 1.5.

Cubic functions

We now move on to cubic functions. These are of the general form, $y = ax^3 + bx^2 + cx + d$ where a must be nonzero, otherwise the function is a quadratic. The general shape that a cubic displays comes in three varieties which will be illustrated in the next three Figures. They are: (i) with a local maximum and a local minimum; (ii) with a point of inflexion; (iii) without a maximum, minimum or point of inflexion.

This cubic has a local maximum and a local minimum. Given that it may be factorised into the form, $y = x(x - 1)(x + 1)$, it has the three roots, $x = -1, 0, 1$. Note that cubics of this type don't always have three roots: if this cubic were moved upwards by 100 to give, $y = x^3 - x + 100$, then there would only be one root.

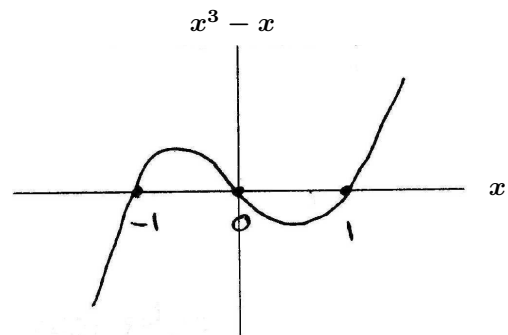


Figure 1.6.

This is the standard cubic function. It has a point of inflexion at $x = 0$, which means that both the slope and the second derivative are zero there. Given that we may write this as $y = x \times x \times x$, we see that a point of inflexion sitting on the horizontal axis corresponds to a **triple zero**.

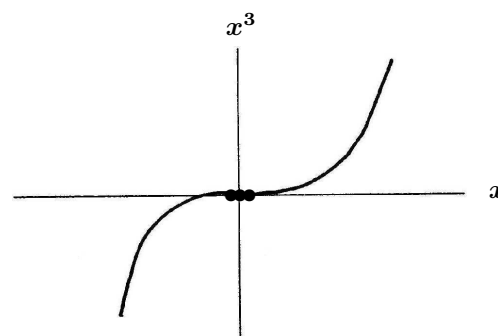


Figure 1.7.

Cubics such as this have no critical points, by which I mean maxima, minima or points of inflexion. They always have one zero, though. In this case it is at $x = 0$.

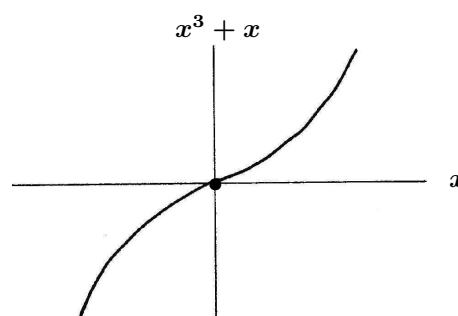


Figure 1.8.

Quartic functions

Last of all we will deal with quartic functions — note the difference between the words, *quadratic* and *quartic*. These have the general form, $y = ax^4 + bx^3 + cx^2 + dx + e$ where a is again nonzero. Given the number of constants, we could also write,

$$y = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 = \sum_{n=0}^4 a_nx^n.$$

As the largest exponent increases, we obtain more options for the types of curve, and therefore we will stop at quartics. Quartics come in the following varieties: (i) two maxima and one minimum (or vice versa); (ii) a point of inflexion and either a maximum or a minimum; (iii) a quartic minimum (or maximum); (iv) a standard parabolic minimum or maximum.

This quartic curve has three **extrema**: two minima and one maximum. It also has four zeros, namely $x = -2, -1, 1$ and 2 , which may be found directly from the function itself. Note that if we were to add exactly the right constant (it turns out to be $9/4$) to this function in order to raise the minima so that they both lie on the x -axis, then we would now have a pair of double zeros.

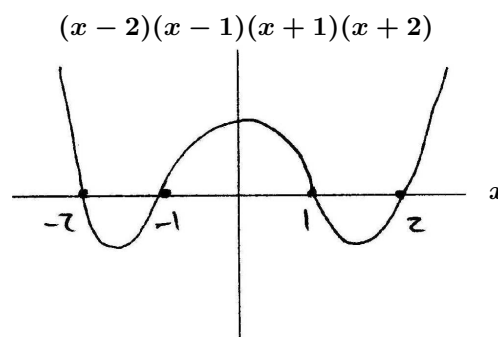


Figure 1.9.

This curve has four zeros: $x = 0, 0, 0$ and 2 . Therefore there is a point of inflexion at $x = 0$ (i.e. a triple zero) and a simple zero at $x = 2$.

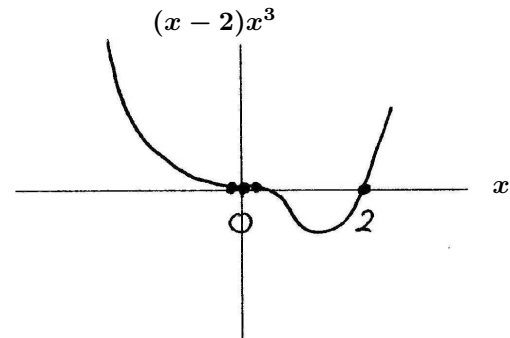


Figure 1.10.

The pure quartic function: $y = x^4$. The first three derivatives are zero at $x = 0$ and therefore this curve has a much flatter base than the parabola does, and this must be shown clearly in the sketch. We also have a four-times repeated root, and therefore $x = 0$ is a **quadruple zero**.

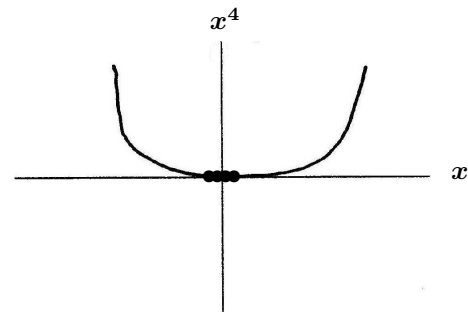


Figure 1.11.

This looks like a parabola because the x^2 term is much larger than x^4 is when x is small. The value, $x = 0$, corresponds to a double zero because $x^4 + x^2 = x^2(x^2 + 1)$. However, the function grows much faster as x increases than a parabola does because of the x^4 term, although this is quite difficult to show on a simple sketch.

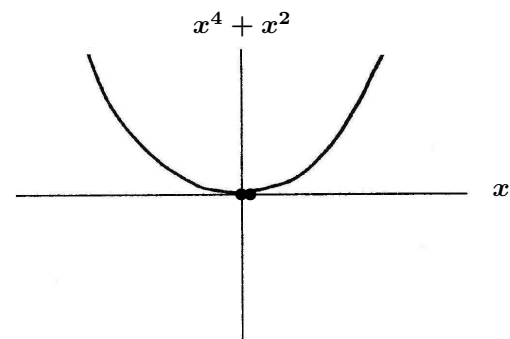


Figure 1.12.

This quartic curve is very similar to that displayed in Figure 1.9. While the general shape is the same, this one has single zeros at $x = \pm 1$ and a double zero at $x = 0$ because $x^4 - x^2 = x^2(x + 1)(x - 1)$.

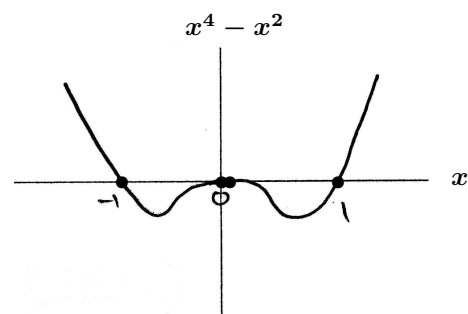


Figure 1.13.

This final quartic curve has been inferred from the given factorisation. There is a double zero at $x = -2$ and single zeros at $x = -1$ and $x = 1$. Given that the coefficient of x^4 is positive, the function becomes large and positive when $x \rightarrow \infty$. The sketch of $-(x^2 - 1)(x + 2)^2$ is the present one turned upside-down.

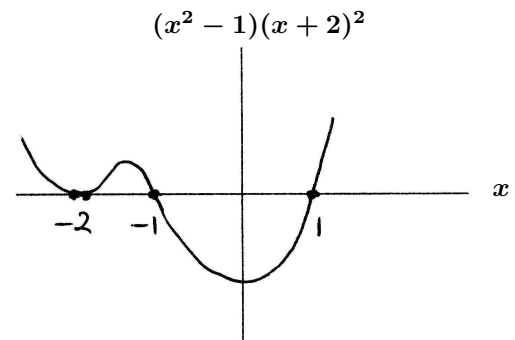


Figure 1.14.

1.3 Moduli

The modulus of a function is its absolute value. We may write this mathematically as follows:

$$|f(x)| = f(x) \quad \text{when} \quad f(x) \geq 0, \quad |f(x)| = -f(x) \quad \text{when} \quad f(x) \leq 0.$$

We could also define it as the positive square root of the square of the function:

$$|f(x)| = +\sqrt{[f(x)]^2}.$$

Therefore the modulus function is always either positive or zero. Colloquially we speak of $|f(x)|$ as '**mod** f ' or '**mod** f of x '. A few examples follow.

We see that $|x|$ is positive everywhere except at the origin where it is zero.

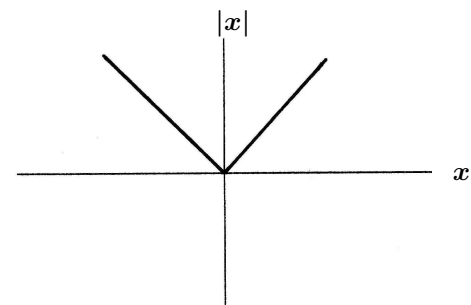


Figure 1.15.

This is 'modulus' version of Figure 1.9. I have constructed this by drawing the curve given in Figure 1.9 using short dashes. Those values which are positive have been overdrawn with a continuous line, while those parts which are negative have been multiplied by -1 and then they are drawn in with a continuous line.

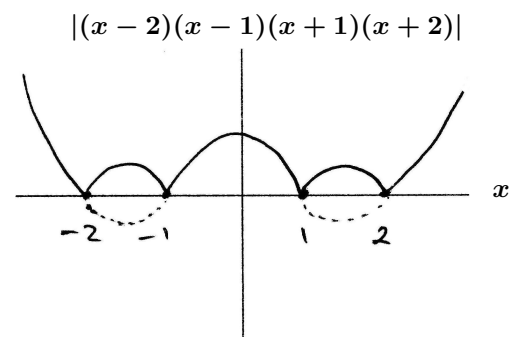


Figure 1.16.

This has been constructed in the same way as for the previous Figure. For this particular function, we may also refer to it as a '**rectified sine wave**' — this is often used in Electrical Engineering.

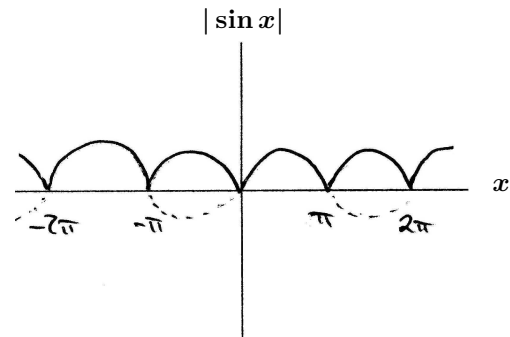


Figure 1.17.

1.4 Exponential and hyperbolic functions.

The exponential function is typified by e^x where $e = 2.7182818284590452$ to 16 decimal places. This strange number arises in many places. One of the most common is the following:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

a result which we will prove later in the unit, but if we set $x = 1$ we obtain a series from which e may be evaluated:

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots.$$

The second is linked to compound interest. If one is faced with the question, do you wish to have bank interest added at the rate of **100%** once a year, the rate of **50%** twice a year, **10%** ten times a year, or whatever you fancy based on this type of formula, then what is the best option? Well, if one has interest added n times in the year, the amount of money you will have at the end of the year will have increased by a factor of,

$$F = \left(1 + \frac{1}{n}\right)^n.$$

It turns out that F increases as n increases, and the limiting case of microscopic rates being added infinitely often yields $F = e$. We will prove this later in the unit using l'Hôpital's rule.

The value, e , is also the base for the natural logarithm: **$y = \ln x$** means that **$x = e^y$** .

The exponential function. As x increases, it rises faster than any power of x . As x becomes increasingly negative, it decreases faster than any inverse power of x .

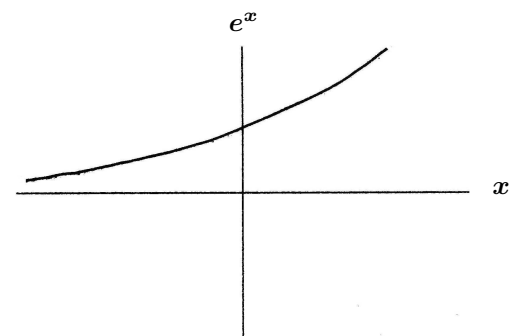


Figure 1.18.

The decreasing exponential function. This is simply the reciprocal of e^x .

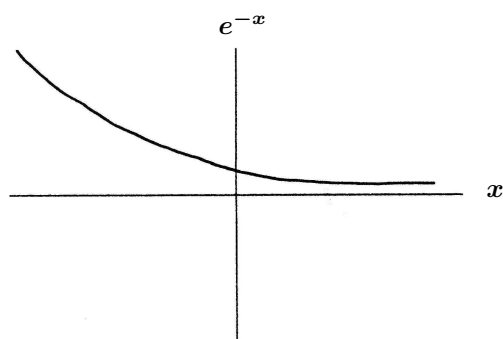


Figure 1.19.

Welcome to your first hyperbolic function. It is defined as $\cosh x = (e^x + e^{-x})/2$ and so it is an 'average' of the previous two Figures. It is the hyperbolic counterpart to $\cos x$ in that it is sometimes referred to as the hyperbolic cosine. We usually say 'cosh x ' — cosh, as in the weapon used by 19th century miscreants.

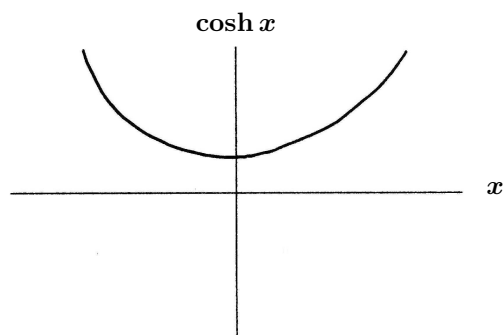


Figure 1.20.

Welcome to your second hyperbolic function. It is defined as $\sinh x = (e^x - e^{-x})/2$ and so it is half the difference between the curves in Figures 1.18 and 1.19. It is the hyperbolic counterpart to $\sin x$ in that it is sometimes referred to as the hyperbolic sine. For this one we usually say 'shine x '.

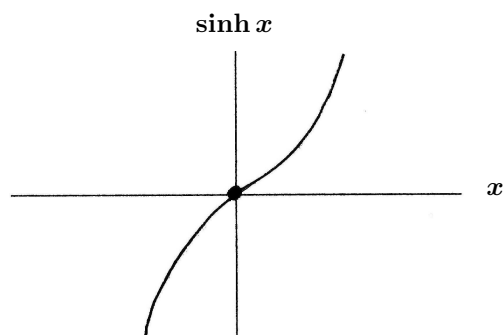


Figure 1.21.

This is the quotient of $\sinh x$ and $\cosh x$. Although written as $\tanh x$ we say, 'tanch x '. It is, of course, the hyperbolic tangent function. Given that

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}},$$

it is clear that $\tanh x \rightarrow 1$ when $x \rightarrow \infty$. A similar argument leads to $\tanh x \rightarrow -1$ when $x \rightarrow -\infty$. It's a much nicer function than $\tan x$ because there are no infinities!

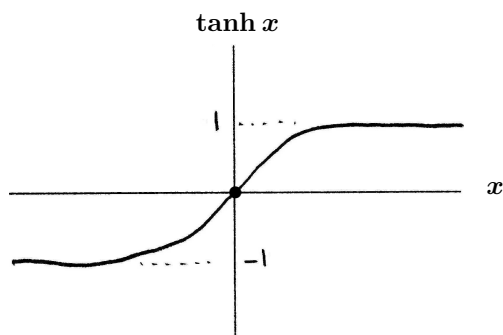


Figure 1.22.

Another nice-looking function. This one arises in Normal Probability Distributions. Hopefully it is clear to see why this decays when $x \rightarrow \pm\infty$. Strictly speaking, this function doesn't decay exponentially, but it does so super-exponentially, which is considerably faster.

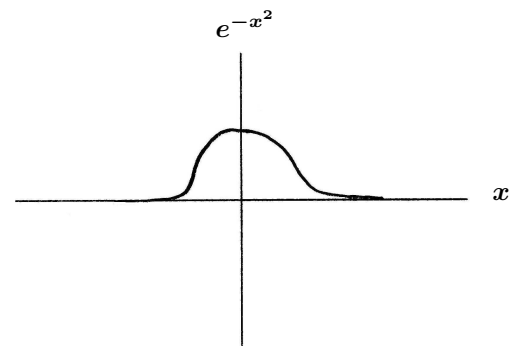


Figure 1.23.

We are now entering the realms of the crazy. First, we must consider the exponent — how does $1/x^2$ behave? Well, when $x \rightarrow \infty$, then $1/x^2 \rightarrow 0$ and hence $\exp(1/x^2) \rightarrow 1$. The same goes for when $x \rightarrow -\infty$. When $x \rightarrow 0$, then $1/x^2 \rightarrow \infty$ and so $\exp(1/x^2) \rightarrow \infty$. All of this may be seen in the Figure.

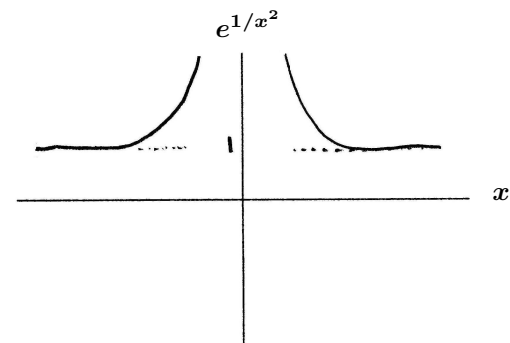


Figure 1.24.

The reciprocal of the function in Figure 1.24, we may mentally find that reciprocal and then draw it. Alternatively, we could follow the same type of argument as above. You may be interested to know that not only is this function equal to zero when $x = 0$, but all of its derivatives are too. This is exceptionally flat.

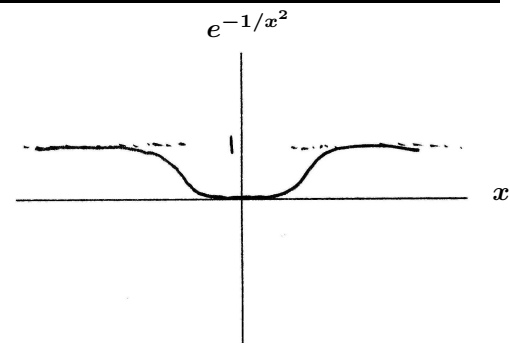


Figure 1.25.

1.5 Comments on symmetries.

We'll take a breather at this point and consider symmetries. While geometrical shapes may have symmetries (e.g. a square has four axes of reflective symmetry, and three different rotational symmetries), mathematical functions may also be said to have symmetries. These come in three main kinds, namely (i) symmetric (i.e. even), (ii) antisymmetric (i.e. odd) and (iii) asymmetric (i.e. no symmetry — the prefix, *a*, means 'without', as in the words, atheist, amoral, amuse, apnea).

An even function is one which has mirror symmetry in the vertical axis. Figures 1.2, 1.9, 1.15 and 1.20 are examples of even functions, although there are more above. Mathematically we may say that an even function satisfies $f(-x) = f(x)$. Examples of this include, $\cos(-x) = \cos(x)$, and $(-x)^2 = x^2$.

The easiest pictorial way of describing an odd function is to say that it stays the same when one rotates it through 180° about the origin. Figures 1.1, 1.6, 1.7, 1.8, 1.21 and 1.22 are all odd. Mathematically we have $f(-x) = -f(x)$ for odd functions. Let us try it out: $\sin(-x) = -\sin(x)$ and $(-x)^3 = -x^3$, therefore we conclude that both $\sin x$ and x^3 are odd. Occasionally one must be careful, though: $|\sin(-x)| =$

$|\sin(x)| = |\sin(x)|$, shows us that $|\sin x|$ is even, as seen in Figure 1.17. The same sort of argument tells us that both $\sin^2 x$ and $\sin(x^2)$ are even.

The third type is asymmetric, and so the function is neither even nor odd. Figures 1.4 and 1.14 are examples of this. The simple addition of an even function and an odd function guarantees the generation of an asymmetric function: $x + x^2$ is an easy example.

One aspect which will be used very soon is to consider what happens when one multiplies two functions together. We know that both x and x^3 are odd, but their product, x^4 , is even. This is indeed a general property: **odd $\times$ odd = even**. By taking other simple examples, we may see that **even $\times$ even = even** and that **even $\times$ odd = odd** are also true in general.

Finally, it is important to note that our definitions have been made with regard to the properties of the function relative to $x = 0$. We could say that Figures 1.4 and 1.5 are both even about $x = 1$. It is also possible to state that, while $\sin x$ is an odd function, it is nevertheless even about $x = \pi/2$, and asymmetric about $x = \pi/4$.

1.6 Envelopes.

This class of function generally involves the product of two functions, one of which oscillates. We will look at the functions, $x \cos x$, $x \sin x$, $(\cos x)/x$ and $e^{-|x|} \sin 100x$.

As a preamble to presenting examples, let us consider the general case of $f(x) \sin x$. We can imagine $f(x)$ to be a fairly simple function, although it is not advisable at first to think of it as a sinusoidal function. Given that $\sin x$ lies between -1 and 1 , the function, $f(x) \sin x$, must lie between $\pm|f(x)|$ everywhere. Therefore a good strategy for sketching envelopes would be to draw both $f(x)$ and $-f(x)$ first, then mark in where $\sin x = 0$, check any possible symmetries that might apply (see below for specific examples), and finally attempt to sketch the curve.

We first mark in the functions x and $-x$ as the dashed lines, and therefore we know that $x \cos x$ must lie between these two extremes. The zeros of $\cos x$ are at $x = \pm\pi/2, \pm3\pi/2$ and so on, so these are marked by black disks on the x -axis. Another zero arises at $x = 0$ due to the factor, x , in $x \cos x$. Finally, we note that the function is odd (odd $\times$ even).

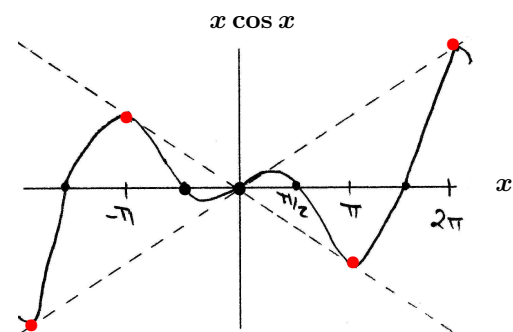


Figure 1.26.

This example is very similar to that shown in Figure 1.26. However, the function is even because $x \sin x$ is odd $\times$ odd. While $\sin x$ has single zeros at $\pm\pi, \pm2\pi, \pm3\pi$ and so on, there is a double zero at $x = 0$ because both the factors in $x \sin x$ have simple zeros there. Therefore the curve will look like a parabola near $x = 0$.

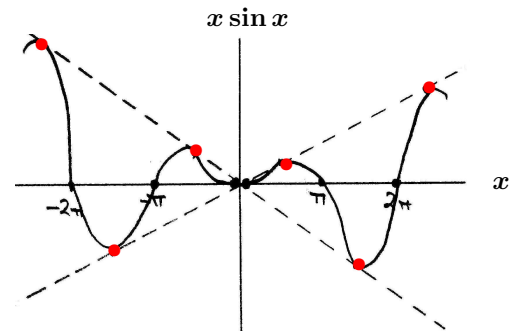


Figure 1.27.

We first mark in the functions $1/x$ and $-1/x$. These tend to plus or minus infinity as $x \rightarrow 0$. This is an odd function and the sketch follows after finding where the zeros of $\cos x$ are, and knowing that both $\cos x$ and x are positive when x is slightly greater than zero.

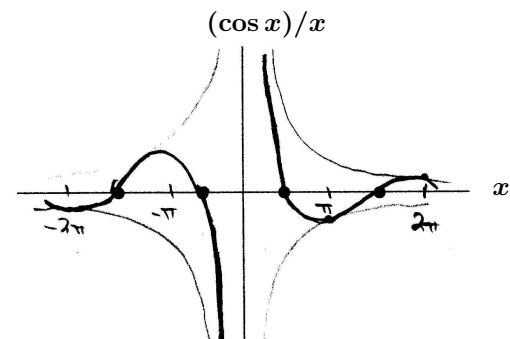


Figure 1.28.

The function $e^{-|x|}$ decays exponentially as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. It is even. We have plotted $\pm e^{-|x|}$ to act as the guide lines, i.e. the envelope. The odd function $\sin 100x$ oscillates very quickly (the period is $\pi/50$). Hence $e^{-|x|} \sin 100x$ is an odd function because it is the product of one even and one odd function. This is an example of an AM radio signal: the carrier wave ($\sin 100x$) has its amplitude modulated by the signal, $e^{-|x|}$.

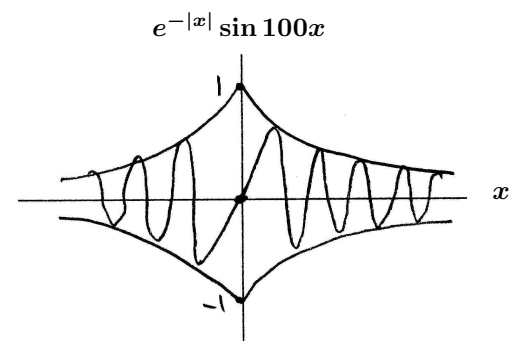


Figure 1.29.

1.7 Square roots of functions.

A little bit of care needs to be taken whenever one needs to sketch the square root of a function. The question is always whether one needs to take both square roots, although that question is often resolved when considering what the application is. Of more importance is what happens when one takes the square root of a function which has a single zero. It is best to illustrate this by plotting $y = \sqrt{x}$.

We have taken the positive root only in this sketch. If we square both sides of $y = \sqrt{x}$, then we get $y^2 = x$, which is a parabola, but one which is rotated through 90° compared with the one in Figure 1.2. Thus the slope of the present graph at the origin is infinite, i.e. the tangent to the curve is a vertical line. This will always be true when encountering the square root of a linear factor at the value of x where the factor is zero.

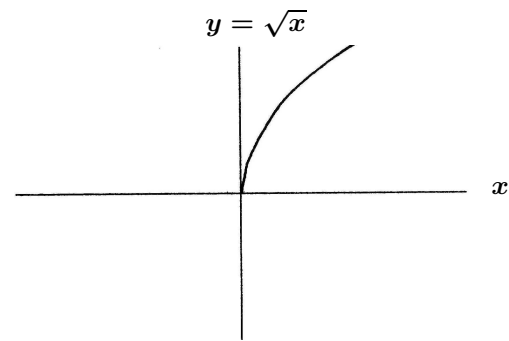


Figure 1.30.

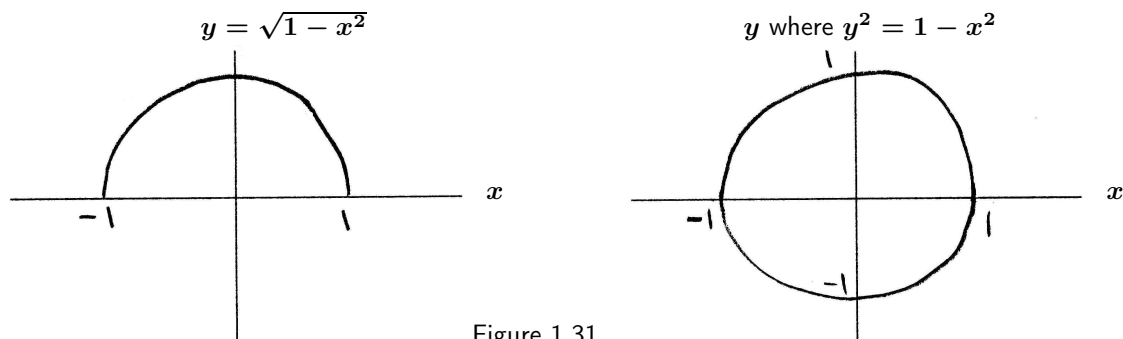


Figure 1.31.

Here I am simply making the distinction between writing $y = \sqrt{1 - x^2}$ explicitly, and writing $y^2 = 1 - x^2$ in terms of what one plots. In the right hand graph the negative values of y are allowed, whereas they are not allowed on the left.

That this is a circle follows from rearranging $y^2 = 1 - x^2$ into the form $x^2 + y^2 = 1$. Obviously the circle is a well-known shape, and therefore we are not surprised that there is a vertical tangent (or infinite slope) at $x = 1$. But this also follows from the short discussion on Figure 1.30, as will now be explained. For the present equation, namely, $y = \sqrt{1 - x^2}$, we may factorise it into the form, $y = \sqrt{(1 + x)(1 - x)}$. When x is close to 1 but slightly below it, then $(1 + x) \simeq 2$ and therefore the equation becomes $y \simeq \sqrt{2(1 - x)}$. This is the square root of a linear factor, just as we have displayed in Figure 1.30.

This is the square root of the function which was plotted in Figure 1.9. That function has two intervals within which it is negative and therefore the square roots doesn't exist, at least for the purposes of sketching here. That function has four simple zeros, and therefore we expect the square root of that function to look like sideways parabolae at those points. We also note that $y \simeq x^2$ when $x \rightarrow \pm\infty$. The dotted lines yield the negative part of the sketch that I would have produced should I have asked for

$$y^2 = (x - 2)(x - 1)(x + 1)(x + 2).$$

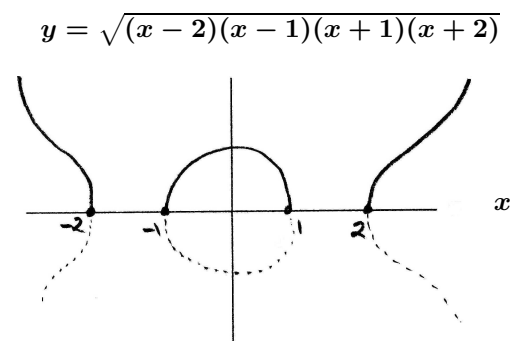


Figure 1.32.

1.8 Ratios of polynomials.

This is the final topic. It sounds scary, but this is very systematic. An example might be

$$y = \frac{(x-1)^2(x+2)}{x(x-3)^2}.$$

The general set of information that one must collect at the outset is: (i) the locations of all zeros and their multiplicity (i.e. single, double and so on); (ii) the locations of all poles (i.e. infinities) which are the zeros of the denominator, and we also need their multiplicity; (iii) the large- x behaviour. For the above example we have the following information:

Zeros at $x = 1, 1, -2$ i.e. $x = 1$ twice and $x = -2$ once,

Infinities at $x = 0, 3, 3$ i.e. $x = 3$ twice and $x = 0$ once,

$$y \rightarrow 1 \text{ as } x \rightarrow \pm\infty.$$

This last piece of information comes from the observation that, when x is large (either positively so or negatively) then the constants are negligible, and therefore $y \simeq x^3/x^3 = 1$.

We won't sketch this particular one, but it is important to know that the 'infinities' are also known as **singularities**. More precisely we would say that $x = 0$ is called either a **simple pole** or a **first order pole**, and that $x = 3$ is called either a **double pole** or a **second order pole**.

A straightforward case of a pole at $x = 0$. As x increases past $x = 0$, the sign of y changes, but y will have descended down to $-\infty$ before re-emerging at $+\infty$. This infinite discontinuity always happens at a single pole.

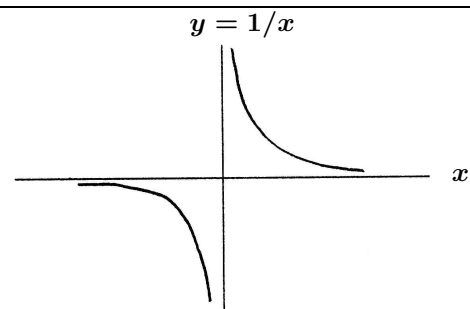


Figure 1.33.

As Figure 1.33 but the pole is now at $x = 1$.

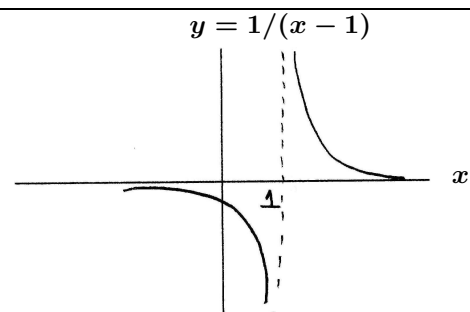


Figure 1.34.

This is a **double pole** at $x = 1$. I have indicated the location of the double pole by a pair of dashed lines. The curve's shape is qualitatively different from that of the single pole shown in Figure 1.34. The presence of the square means that the function is always positive, but more importantly we need to note that as x increases past $x = 1$, then the sign does not change. Hence the curve ascends to $+\infty$ and then returns from $+\infty$.

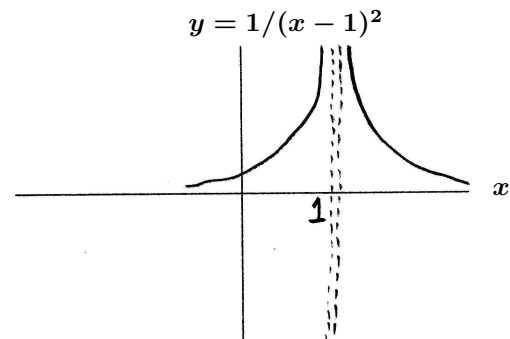


Figure 1.35.

This is a **triple pole** at $x = 1$, the presence of which is indicated by three dashed lines. The fact that it is a cube means that there will be a sign change as x increases past $x = 1$. So this looks very similar to the single pole case.

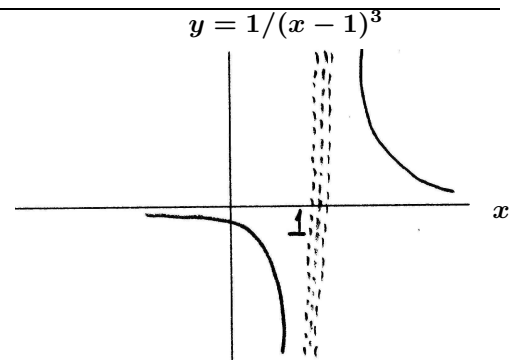


Figure 1.36.

$$y = \frac{1}{x^2 - 1}.$$

There are no zeros.

Poles at $x = -1$ and $x = 1$

because $x^2 - 1 = (x - 1)(x + 1)$.

When $x \rightarrow \pm\infty$ then $y \rightarrow 0$.

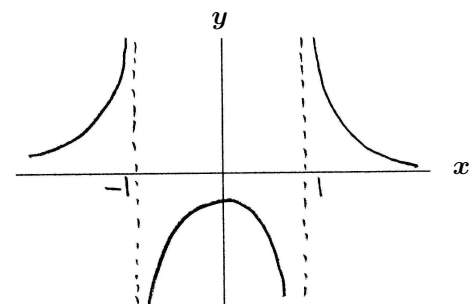


Figure 1.37.

Beginning at a large positive value of x we note that $y > 0$. Therefore the curve must rise from the x -axis as x decreases. As x approaches $x = 1$ from above, y must rise all the way to infinity and re-emerge from $-\infty$ because of the single pole at $x = 1$. As x decreases further towards the second pole at $x = -1$, the curve must turn back downwards towards $-\infty$; it cannot travel through the x -axis because there is no zero between $x = -1$ and $x = 1$ — this is the only option for the curve. On the negative side of $x = -1$ the curve re-emerges from $+\infty$ and descends back down towards $y = 0$, which is the large- $|x|$ value.

All of our examples will follow a similar logic. The behaviour of the curve will then depend on the multiplicity of the zeros and poles.

$$y = \frac{x}{x^2 - 1}.$$

Zero at $x = 0$.

Poles at $x = -1$ and $x = 1$.

When $x \rightarrow \pm\infty$ then $y \rightarrow 0$.

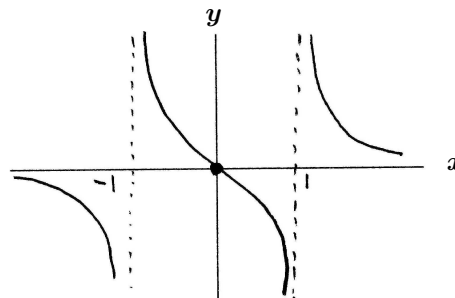


Figure 1.38.

The logic begins in the same way as for Figure 1.37 except that there is now a zero in-between $x = 1$ and $x = -1$. Therefore the curve must pass through that zero and continue to ascend to $+\infty$ at $x = -1$. It is worth comparing Figures 1.37 and 1.38 to see the difference caused by the addition of a zero.

$$y = \frac{x(x-1)}{(x+1)(x-2)}.$$

Zeros at $x = 0, 1$.

Poles at $x = -1, 2$.

When $x \rightarrow \pm\infty$ then $y \rightarrow 1$ since $y \sim \frac{x^2}{x^2}$.

Similar logic to the previous example.

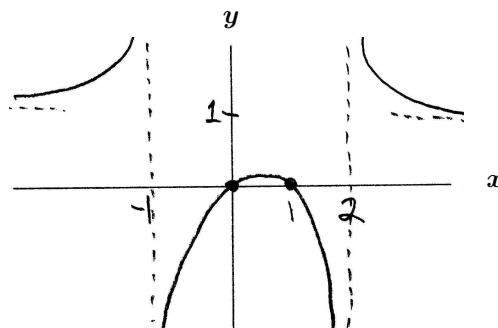


Figure 1.39.

$$y = \frac{x(x-1)^2}{(x+1)^2(x-2)}.$$

Zeros at $x = 0, 1, 1$.

Poles at $x = -1, -1, 2$.

When $x \rightarrow \pm\infty$ then $y \rightarrow 1$.

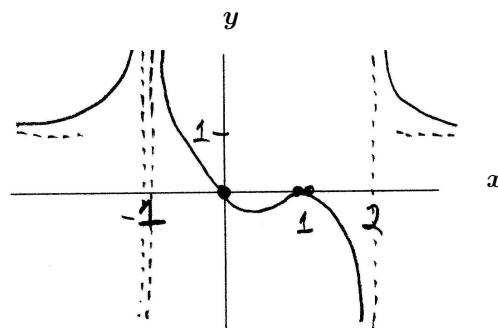


Figure 1.40.

This example has both a double zero (at $x = 1$) and a double pole (at $x = -1$). Therefore the curve resembles a parabola near to $x = 1$, a downward-facing one. The double pole has the same feature as the one that was discussed for Figure 1.35, namely that the sign doesn't change.

$$y = \frac{x(x-1)^2}{(x+1)(x-2)}.$$

Zeros at $x = 0, 1, 1$.

Poles at $x = -1, 2$.

When $x \rightarrow \pm\infty$ then $y \sim x$.

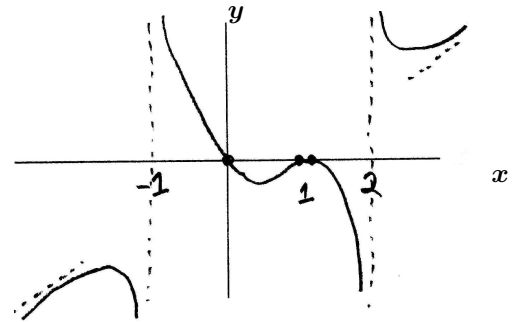


Figure 1.41.

A slightly modified version of both Figures 1.39 and 1.40. Note the different large- x behaviour here.

Final note.

There are some further examples of the sketching of ratios of polynomials available from the Handouts section of the unit webpage. This includes a detailed account of quite a complicated example. The direct link, should you wish to type it, is

<http://staff.bath.ac.uk/ensdasr/ME10304.bho/extra.curves.pdf>.

There is some extra information in the Miscellaneous section of the unit webpage on hyperbolic functions. There's a little bit of calculus, some hyperbolic versions of the multiple angle formulae, some graphs as opposed to sketches, and an explanation by way of differential equations why there is a similarity between the hyperbolic and the trigonometric functions. The direct link is,

<http://staff.bath.ac.uk/ensdasr/ME10304.bho/hyperbolics.pdf>.

2 Complex Numbers

2.1 Historical Note [For interest only]

As mathematics has developed from early times it has suffered many revolutions in terms of the ideas which are taken as acceptable. Before and up to the time of Pythagoras (approximately 569 BC) it was believed that all quantities could be expressed in terms of whole numbers. What we now call rational fractions were allowed, since they are ratios of whole numbers (e.g. $\frac{3}{4}$), but concepts such as π and $\sqrt{2}$ were unacceptable. Pythagoras himself believed all of this, but this belief sat uncomfortably beside the fact that the diagonal of a square cannot be written in terms of a rational fraction times the side of the square. The technical term for this is that the diagonal is **incommensurate** with its side. Note that the hypotenuse of the right-angled triangle with sides, **3**, **4** and **5**, is commensurate with the other two sides because the ratios of those sides to the hypotenuse are $\frac{3}{5}$ and $\frac{4}{5}$, i.e. ratios of whole numbers. Neither π nor $\sqrt{2}$ can be written like this.

Returning to the diagonal of a square, Pythagoras's theorem applied to two sides of length **1** which subtend a right angle shows that the length of the hypotenuse is $\sqrt{2}$ — this is well-known. If we wish to prove that this number cannot be expressed as a ratio of whole numbers we may first assume the opposite, namely that $\sqrt{2} = n/m$ where n and m are whole numbers that do not have any common factors. If we square both sides and rearrange slightly we get

$$n^2 = 2m^2. \quad (2.1)$$

As the right hand side is even, this implies that n^2 is even, and, given that only even numbers have squares which are even, this implies that n must be even and so we can write $n = 2p$ where p is a whole number. On substitution into (2.1) we get

$$4p^2 = 2m^2 \quad \Rightarrow \quad m^2 = 2p^2. \quad (2.2)$$

In turn this latest expression implies that m is also even. But we have now shown that both n and m have a common factor, **2**, which is in contradiction to the original assumption. Therefore $\sqrt{2}$ cannot be expressed as a rational fraction. Consequently it is an example of an **irrational** number.

The scientific community, Pythagoras notwithstanding, eventually accepted irrational numbers since they often arise when solving polynomial equations such as quadratic equations. It has also been shown that numbers like π and e , the base of the natural logarithm, are also irrational, but since they do not arise from solving polynomial equations they are also termed **transcendental**.

Although equations such as

$$x^2 - 6x + 8 = 0 \quad \text{and} \quad y^2 - 6y + 7 = 0$$

may be written in the forms

$$(x - 3)^2 = 1 \quad \text{and} \quad (y - 3)^2 = 2$$

and hence have the solutions,

$$x = 2, 4 \quad \text{and} \quad y = 3 \pm \sqrt{2},$$

equations such as,

$$z^2 - 6z + 10 = 0, \quad \text{or} \quad (z - 3)^2 = -1, \quad (2.3)$$

were deemed for a long time not to have a solution. The reason simply was that $z = 3 \pm \sqrt{-1}$ was considered to be nonsensical because there is no number whose square could possibly be equal to -1 .

Such a state of affairs lasted for quite some time, and it was only around the time when Cardano (1501-1576) discovered how to solve a cubic equation of the form, $ax^3 + bx^2 + cx + d = 0$, that the resistance to $\sqrt{-1}$ finally faded. The reason for this is that, while a cubic equation may have either one real root or three real

roots, Cardan's formula giving the three roots, when they are all real, involves using the square roots of negative numbers; the detailed formula is given in the next section.

The following is a quotation from *Significant Figures* by Ian Stewart which I think is of interest in this context.

Ars Magna [by Cardano] is significant for one other reason. Cardano applied his algebraic methods to find two numbers whose sum is 10 and product is 40, and got the answer $5 + \sqrt{-15}$ and $5 - \sqrt{-15}$. Since negative numbers have no square roots, he declared this result to be 'as subtle as it is useless'. The formula for cubics also leads to such quantities when all three solutions are real, and in 1572 Rafael Bombelli observed that if you ignore what such expressions mean and just do the sums, you get the correct real solutions. Eventually this line of thinking led to the creation of the system of complex numbers in which -1 has a square root. This extension of the real number system is essential to today's mathematics, physics and engineering.

Thus it was that the symbol i (or, when within the engineering fraternity, j) came to signify the "value" of $\sqrt{-1}$, since it could at least be interpreted as a mathematical trick to obtain results which are nevertheless correct. The symbols i and j denote imaginary numbers, and, together with the real numbers, are termed complex numbers. The solution of Equation (2.3) for z may therefore be written in the form $z = 3 \pm j$.

It is interesting to note throughout this very brief historical sketch the type of words that have been used to denote the new kinds of numbers: irrational, transcendental and imaginary! Perhaps these mark a certain amount of distrust or perhaps awe when compared with nice safe "rational" numbers. Much more recently the term "surreal numbers" has been invented (see wikipedia) and so the old tradition of curious names continues.

2.2 Cardan's formula for the cubic equation [For interest only]

The solution for the **quadratic equation** is well-known. If z satisfies

$$az^2 + bz + c = 0, \quad (2.4)$$

then

$$z_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad z_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \quad (2.5)$$

where, if we set $q = b^2 - 4ac$, we have the three cases

$$\begin{aligned} q > 0 &\Rightarrow \text{two real roots} \\ q = 0 &\Rightarrow \text{two equal real roots} \\ q < 0 &\Rightarrow \text{two complex roots.} \end{aligned} \quad (2.6)$$

The value, q , is called the discriminant: its value allows us to distinguish between the three potential cases.

The above categorisation holds only when a , b and c are real. But the formulae for z_1 and z_2 remain valid for complex values of a , b and c .

For the **cubic equation**, $z^3 + a_2z^2 + a_1z + a_0 = 0$, we let

$$q = \frac{1}{3}a_1 - \frac{1}{9}a_2^2 \quad r = \frac{1}{6}(a_1a_2 - 3a_0) - \frac{1}{27}a_2^3. \quad (2.7)$$

Again we have three cases:

$$\begin{aligned} q^3 + r^2 > 0 &\Rightarrow \text{one real root and a pair of complex roots} \\ q^3 + r^2 = 0 &\Rightarrow \text{all roots are real with at least two equal} \\ q^3 + r^2 < 0 &\Rightarrow \text{all roots are real.} \end{aligned} \quad (2.8)$$

To complete the solution we set

$$s_1 = \left[r + \sqrt{(q^3 + r^2)} \right]^{1/3} \quad s_2 = \left[r - \sqrt{(q^3 + r^2)} \right]^{1/3} \quad (2.9)$$

and then

$$\begin{aligned} z_1 &= (s_1 + s_2) - \frac{1}{3}a_2 \\ z_2 &= -\frac{1}{2}(s_1 + s_2) - \frac{1}{3}a_2 + \frac{1}{2}\sqrt{3}(s_1 - s_2)j \\ z_3 &= -\frac{1}{2}(s_1 + s_2) - \frac{1}{3}a_2 - \frac{1}{2}\sqrt{3}(s_1 - s_2)j. \end{aligned} \quad (2.10)$$

There is a much longer formula for the solution of quartic equations (i.e. fourth order polynomials) but there are no general formulae for the solution of quintics or of higher order polynomials. This was proved in 1839 by Abel.

That completes the historical stuff! From now on we shall look at the properties of complex numbers, determine how they may be used and manipulated, and then finally we'll consider a general method for finding the roots of complex numbers (by which we mean the fractional powers of complex numbers such as square roots).

2.3 Manipulation of Complex Numbers

The basic rules for adding, subtracting, multiplying and dividing complex numbers are not too far removed from those involving solely real numbers, except that we will often need to use the result that $j^2 = -1$, and that real and imaginary numbers are usually treated separately.

Addition and Subtraction:

Here we add and subtract in the way one would expect by collecting like terms. For example

$$(1 + 5j) + (2 - 3j) = (1 + 2) + (5 - 3)j = 3 + 2j, \quad (2.11)$$

and

$$(1 + 5j) - (2 - 3j) = (1 - 2) + (5 + 3)j = -1 + 8j. \quad (2.12)$$

So these work in the same way as $(x + 5y) + (2x - 3y) = 3x + 2y$.

Multiplication:

Here we expand the product in the same way as we do for the product,

$$(a + b)(c + d) = ac + bd + ad + bc,$$

except that a little tidying up takes place afterwards using $j^2 = -1$. In general, we have,

$$(a + bj)(c + dj) = ac + bd \mathbf{j^2} + adj + bcj = (ac - bd) + (ad + bc)j. \quad (2.13)$$

A numerical example:

$$\begin{aligned} (1 + 5j)(2 + 3j) &= (1 \times 2) + (5j \times 3j) + (1 \times 3j) + (5j \times 2) \\ &= 2 + 15 \mathbf{j^2} + 3j + 10j \\ &= 2 - 15 + 13j \\ &= -13 + 13j. \end{aligned} \quad (2.14)$$

Note: Integer powers of complex numbers (like squares and cubes and 10th powers etc.) may be evaluated this way but there is a better way which will follow later.

Division:

This is the most complicated operation of the four and it is described best using the concept of the complex conjugate. If $z = x + yj$, then the **complex conjugate** of z is written as $\bar{z}$ and defined as

$$\bar{z} = x - yj. \quad (2.15)$$

Clearly $\overline{3 + 4j} = 3 - 4j$, $\bar{6} = 6$ and $\overline{5j} = -5j$. One very useful consequence of the complex conjugate is that the product, $z\bar{z}$, is real:

$$\begin{aligned} z\bar{z} &= (x + yj)(x - yj) && \text{by definition} \\ &= x^2 - y^2 j^2 + xyj - xyj && \text{multiplying out} \\ &= x^2 + y^2 && \text{using } j^2 = -1. \end{aligned} \quad (2.16)$$

This provides us with a sneaky way for evaluating the quotient of two complex numbers: we multiply the denominator and the numerator by the complex conjugate of the denominator, and this guarantees that the new denominator is real. Here is an example:

$$\begin{aligned} \frac{1}{3 + 4j} &= \frac{1}{3 + 4j} \times \frac{3 - 4j}{3 - 4j} \\ &= \frac{3 - 4j}{(3 + 4j)(3 - 4j)} \\ &= \frac{3 - 4j}{25} \\ &= \frac{3}{25} - \frac{4}{25}j. \end{aligned} \quad (2.17)$$

In general we have

$$\frac{1}{x + yj} = \frac{x - yj}{(x + yj)(x - yj)} = \frac{x - yj}{x^2 + y^2} = \frac{x}{x^2 + y^2} - \frac{y}{x^2 + y^2}j. \quad (2.18)$$

Another numerical example is:

$$\frac{2 + j}{1 + j} = \frac{(2 + j)(1 - j)}{(1 + j)(1 - j)} = \frac{3 - j}{2} = \frac{3}{2} - \frac{1}{2}j.$$

Note: This idea of using a complex conjugate may also help us in an entirely different context, one which involves only real numbers. Here's an example:

$$\begin{aligned} \frac{8}{1 + \sqrt{5}} &= \frac{8}{(1 + \sqrt{5})} \times \frac{(1 - \sqrt{5})}{(1 - \sqrt{5})} \\ &= \frac{8(1 - \sqrt{5})}{1 - 5} = 2(\sqrt{5} - 1). \end{aligned} \quad (2.19)$$

2.4 Geometrical Interpretation

This will give us some further physical understanding of complex numbers, but will also allow us to do things like finding square roots, tenth roots and $2/5$ th powers of complex numbers.

The complex number $z = a + bj$ may be represented as the coordinates of a point in what is called the **complex plane**:

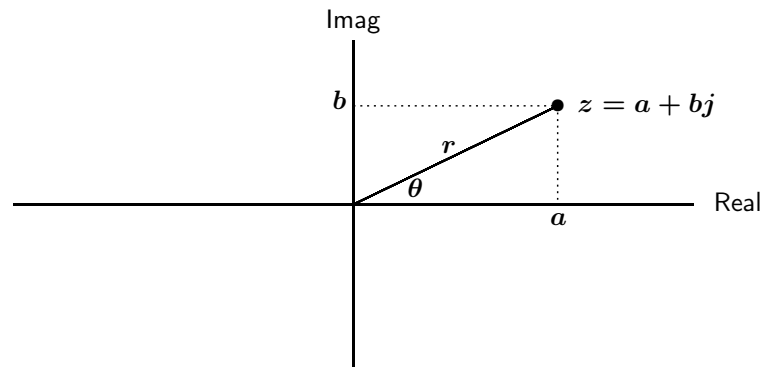


Figure 2.1. Showing the location of the complex number, $z = a + bj$, in the Argand Diagram (or Complex Plane). Also shown are $r = |z| = \sqrt{a^2 + b^2}$ and $\theta = \arg(z) = \tan^{-1}(b/a)$.

The horizontal axis (abscissa) is called the **real axis**, while the vertical axis (ordinate) is called the **imaginary axis**. The whole Figure is also referred to as the **Argand diagram**.

The length of the line joining the origin to z is $\sqrt{a^2 + b^2}$ which is precisely $\sqrt{z\bar{z}}$; this is called the **modulus** of z and is denoted by $|z|$.

The angle that the line joining the origin to z makes with the real axis (measured anti-clockwise) is called the **argument** of z , and is denoted as $\arg z$. If we let $\theta = \arg(z)$ then $\tan \theta = b/a$. Conversely we have $\arg(z) = \tan^{-1}(b/a)$, although it is necessary to specify which of the two possible angles is correct. For example, if $b = a = 1$, then the diagram indicates that $\arg z = \pi/4$, but the inverse tan formula simply says that $\theta = \tan^{-1}(1)$ for which we may have either $\theta = \pi/4$ or $\theta = 5\pi/4$; therefore we either need to state in addition that we require θ to lie in the first quadrant (i.e. $0 \leq \theta \leq \pi/2$), or else quote the correct value in radians. **We note that θ is always measured in the anticlockwise direction.**

The conjugate of z may be located by taking the mirror image of z in the real axis; see Figure 2.2. From this diagram we see that $|z| = |\bar{z}|$ and that $\arg(z) = -\arg(\bar{z})$.

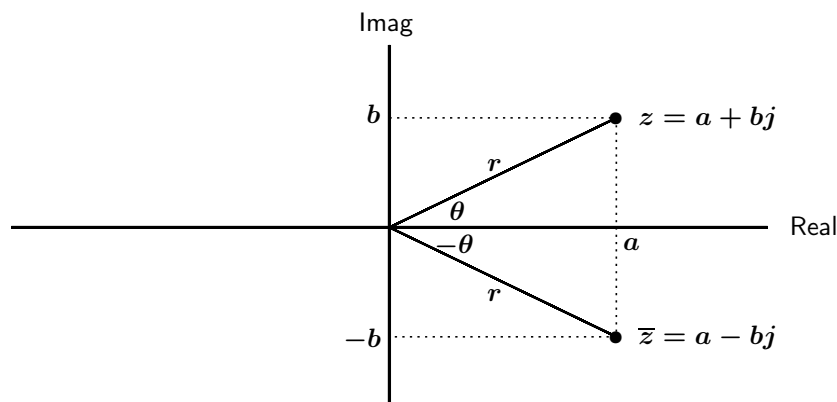


Figure 2.2. Showing the locations of the complex number, $z = a + bj$ and its conjugate, $\bar{z} = a - bj$, in the Complex Plane.

Addition: In the Argand diagram this appears to follow the same rule as the addition of two vectors in that the sum is obtained by completing the rhombus; see Figure 2.3.

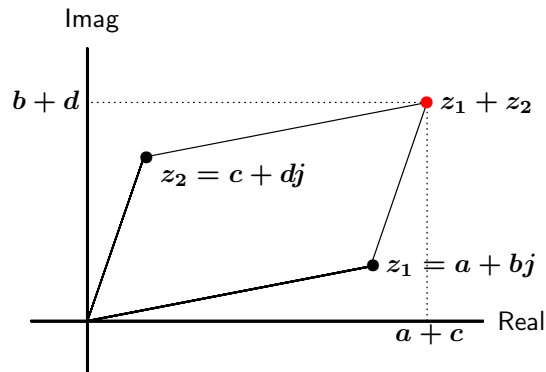


Figure 2.3. Showing how two complex numbers, $z_1 = a + bj$ and $z_2 = c + dj$, may be added by completing the rhombus. Equivalently, we add the real and imaginary parts separately.

Multiplication: For example, we have $(2 + 4j) \times (1 + j) = -2 + 6j$. On the Complex Plane we have:

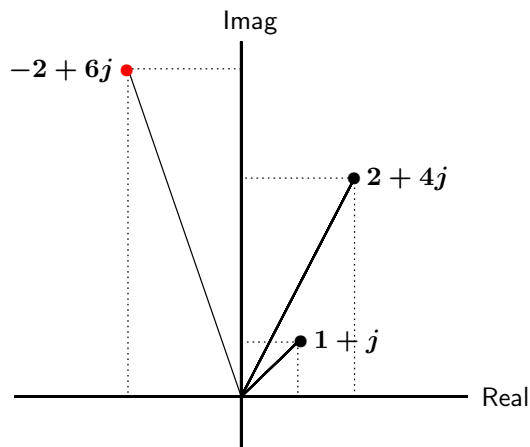


Figure 2.4. Showing the multiplication of the two given complex numbers.

From this Figure it is not immediately obvious what is happening geometrically when we multiply complex numbers. However, if we find the moduli of each of the three numbers:

$$|2 + 4j| = \sqrt{20}, \quad |1 + j| = \sqrt{2}, \quad |-2 + 6j| = \sqrt{40} = \sqrt{2} \times \sqrt{20},$$

then we see that the modulus of the product is equal to the product of the moduli. It is also clear from the Argand diagram that the argument of the product is greater than the (positive) argument of individual numbers forming the product.

We may investigate this in more detail by first considering a partial translation of a complex number into a polar coordinate form. In Fig. 2.1 we can see that

$$a = r \cos \theta \quad \text{and} \quad b = r \sin \theta,$$

or that,

$$a + bj = r(\cos \theta + j \sin \theta). \quad (2.20)$$

Therefore we may write down similar general forms for the following two complex numbers:

$$z_1 = r_1(\cos \theta_1 + j \sin \theta_1) \quad \text{and} \quad z_2 = r_2(\cos \theta_2 + j \sin \theta_2). \quad (2.21)$$

Clearly, the moduli of these numbers are r_1 and r_2 , respectively, and their arguments are θ_1 and θ_2 . Hence

$$\begin{aligned} z_1 z_2 &= r_1 r_2 (\cos \theta_1 + j \sin \theta_1)(\cos \theta_2 + j \sin \theta_2), \\ &= r_1 r_2 \left[(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + j(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \right], \\ &= r_1 r_2 \left[\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2) \right]. \end{aligned} \quad (2.22)$$

where the last line was formed by appealing to some multiple-angle formulae.

This result shows that the **modulus of the product is equal to the product of the moduli**, but also that the **argument of the product is equal to the sum of the arguments**. This leads us into what is called the polar (or exponential or complex exponential) representation of complex numbers.

2.5 Complex Exponential form

Recalling the expression given in Eq. (2.20), we may write down the following:

$$\begin{aligned} z &= a + bj && \text{by definition} \\ &= (r \cos \theta) + j(r \sin \theta) && \text{using Eq. (2.20)} \\ &= r(\cos \theta + j \sin \theta) \\ &= r e^{j\theta}. \end{aligned} \quad (2.23)$$

This last step of rewriting the coefficient of r as a complex exponential needs a little explanation, but this cannot be done satisfactorily without resorting to either Taylor Series, which is a device for writing functions as an infinitely long polynomial, or to the theory of Ordinary Differential Equations; these will be covered later in ME10304 this semester and ME10305 (Mathematics 2) next semester, respectively. You will need to take Eq. (2.23) on trust for now...

Thus complex numbers may be written in either Cartesian form $z = a + bj$ or Polar/Exponential form $z = r e^{j\theta}$. [The word, Cartesian, is always spelt with a capital because it is named for Descartes, otherwise known as Cartesius in its Latinized form.] An alternative notation is $z = r \angle \theta$; this is often used in Electrical Engineering contexts, where θ can be allowed to be in degrees. Another occasional notation is $r \operatorname{cis} \theta$, where cis is a shorthand for $\cos \theta + j \sin \theta$ — this was invented by Hamilton in 1866, but is rarely used now. So the complex exponential form is the generally accepted way.

Warning: When writing $r e^{j\theta}$, the angle, θ , must always be measured in radians. The same is true for sine and cosine. The reason is that derivatives are incorrect when using degrees.

Addition and subtraction are most easily carried out in Cartesian form, while multiplication and division are most easily carried out in Polar form:

$$\begin{aligned} z_1 z_2 &= [r_1 e^{j\theta_1}] [r_2 e^{j\theta_2}] = (r_1 r_2) e^{j(\theta_1 + \theta_2)}, \\ z_1 / z_2 &= [r_1 e^{j\theta_1}] / [r_2 e^{j\theta_2}] = (r_1 / r_2) e^{j(\theta_1 - \theta_2)}. \end{aligned}$$

That said, multiplication and division while in Cartesian form are also pretty straightforward.

The conversion of a complex number from polar/exponential form to Cartesian form is straightforward if one remembers the geometry of the Argand diagram:

$$2e^{j\pi/4} = 2 \left[\cos(\pi/4) + j \sin(\pi/4) \right] = 2 \left[\frac{1}{\sqrt{2}} + j \frac{1}{\sqrt{2}} \right] = \sqrt{2} + \sqrt{2}j.$$

But the conversion from Cartesian to polar/exponential form requires a little more care due to the fact that the inverse tangent function has more than one solution. For example, $\tan^{-1} 1 = \frac{1}{4}\pi$ or $\frac{5}{4}\pi$. Thus conversion from Cartesian to polar form also requires one to choose the correct inverse tangent.

Example 2.1: Express $-1 + 2j$ in complex exponential form.

We have $-1 + 2j = re^{j\theta}$. Here $r^2 = (-1)^2 + (2)^2 = 5 \Rightarrow r = \sqrt{5}$. The argument satisfies $\tan \theta = (2)/(-1) = -2$, and therefore θ is either -1.107149 (directly from my calculator) or 2.034444 (which is π plus the first value). Given where the complex number lies in the complex plane, i.e. the second quadrant, then $\theta = 2.034444$ or 116.565° .

Note: It is usual to quote angles to lie in the range, $-\pi < \theta \leq \pi$ although there may very occasionally be the need to use $0 \leq \theta < 2\pi$.

2.6 de Moivre's Theorem

This is a very useful result and is known as a theorem despite the fact its proof takes only two lines (i.e. just one step). It is based on the following well-known property of the powers of exponentials,

$$(a^b)^c = a^{bc}. \quad (2.24)$$

We apply this result to $e^{\theta j}$:

$$\begin{aligned} (e^{\theta j})^n &= e^{n\theta j} \\ \Rightarrow [\cos \theta + j \sin \theta]^n &= \cos(n\theta) + j \sin(n\theta). \end{aligned} \quad (2.25)$$

From this we may recover the various multiple angle formulae from trigonometry. For example, when $n = 2$ we have

$$\begin{aligned} \cos 2\theta + j \sin 2\theta &= [\cos \theta + j \sin \theta]^2 \\ &= [\cos^2 \theta - \sin^2 \theta] + j[2 \sin \theta \cos \theta]. \end{aligned} \quad (2.26)$$

On equating real and imaginary parts, respectively, we get

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \quad \sin 2\theta = 2 \sin \theta \cos \theta.$$

When $n = 3$, we have

$$\begin{aligned} \cos 3\theta + j \sin 3\theta &= [\cos \theta + j \sin \theta]^3 \\ &= \cos^3 \theta + 3j \cos^2 \theta \sin \theta + 3j^2 \cos \theta \sin^2 \theta + j^3 \sin^3 \theta \\ &= [\cos^3 \theta - 3 \cos \theta \sin^2 \theta] + j[3 \cos^2 \theta \sin \theta - \sin^3 \theta]. \end{aligned} \quad (2.27)$$

This final line used the fact that $j^3 = j^2 j = -j$, and it may also be tidied up to give

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta, \quad \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta.$$

2.7 Roots of complex numbers

It is well-known that there are two square roots of positive numbers and, when that number is 9, then those roots are 3 and -3. But both negative numbers and complex numbers also have two square roots. So let us check out a few examples of square roots and other roots.

Example 2.2. Given that both $j^2 = -1$ and $(-j)^2 = -1$, it is clear that the two square roots of -1 are $\pm j$.

Example 2.3. Likewise, given that $(1 + j)^2 = 2j$ and $(-1 - j)^2 = 2j$, then the square roots of $2j$ are $\pm(1 + j)$.

Example 2.4. Once more, given that $(2 + j)^2 = 3 + 4j$ and $(-2 - j)^2 = 3 + 4j$, then the square roots of $3 + 4j$ are $\pm(2 + j)$.

The pattern so far is that each square root is precisely the negative of the other, which is exactly the same as for the square roots of positive numbers. No surprise there! Here's a different case:

Example 2.5. We have,

$$(1 + j)^4 = -4, \quad (-1 - j)^4 = -4, \quad (1 - j)^4 = -4 \quad \text{and} \quad (-1 + j)^4 = -4.$$

Therefore we have shown that -4 has four fourth roots:

$$(-4)^{1/4} = \pm 1 \pm j,$$

where all four possible combinations of plus/minus may be taken.

Example 2.6. One more case with a fourth power:

$$\begin{aligned} (1 + 2j)^4 &= -7 - 24j, & (-1 - 2j)^4 &= -7 - 24j, \\ (2 - j)^4 &= -7 - 24j & \text{and} & (-2 + j)^4 = -7 - 24j. \end{aligned}$$

Therefore the four fourth roots of $(-7 - 24j)$ are

$$\pm(1 + 2j) \quad \text{and} \quad \pm(2 - j).$$

Perhaps this still isn't sufficient to see the full pattern yet, although the fact that complex numbers have two square roots and four fourth roots does suggest something. So here is something a little different:

Example 2.7. Consider the following:

$$(-2)^3 = -8, \quad (1 + \sqrt{3}j)^3 = -8 \quad \text{and} \quad (1 - \sqrt{3}j)^3 = -8.$$

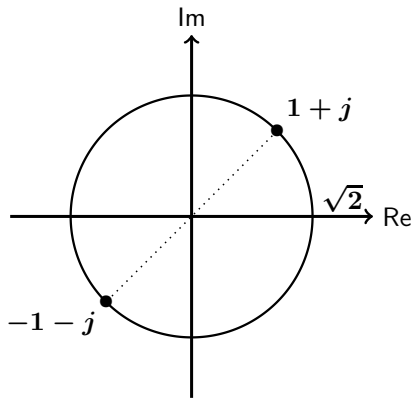
So we have three third roots of -8, namely

$$-2, \quad 1 + \sqrt{3}j \quad \text{and} \quad 1 - \sqrt{3}j$$

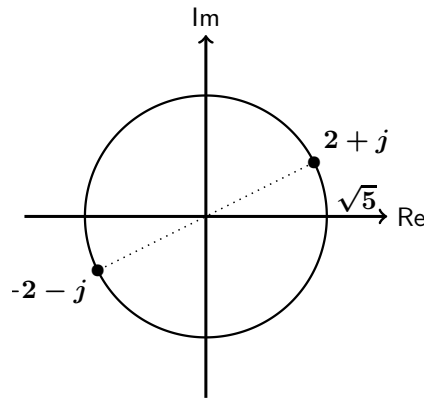
.

Example 2.8. If one were to find the sixth powers of 2, -2, $1 + \sqrt{3}j$, $1 - \sqrt{3}j$, $-1 + \sqrt{3}j$ and $-1 - \sqrt{3}j$, then each would yield 64. So 64 has six sixth roots.

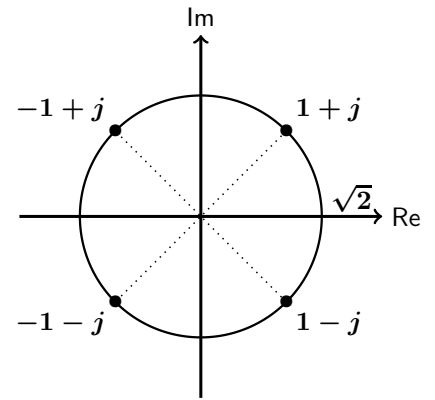
So that is a pattern, namely that there will be n n^{th} roots for a complex number. However, that is not the only pattern. The following Figure shows how the roots given within each example above are related to one another.



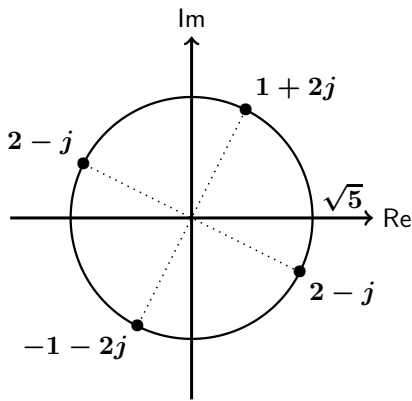
Example 2.3



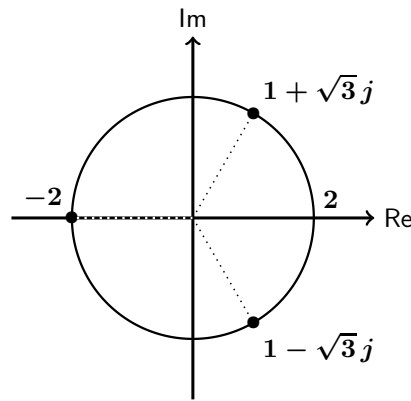
Example 2.4



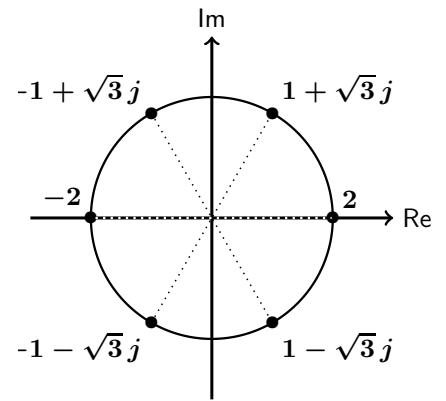
Example 2.5



Example 2.6



Example 2.7



Example 2.8

Figure 2.5. Displaying the roots corresponding to Examples 2.3 to 2.8.

The pattern that we need may now be seen in Fig. 2.5 and, for each case, the circle on which all the roots lie is split into equal segments.

The general method. We may motivate the general method using the data in Example 2.5. The four roots are given by

$$\sqrt{2}e^{j\pi/4}, \quad \sqrt{2}e^{3j\pi/4}, \quad \sqrt{2}e^{5j\pi/4} \quad \text{and} \quad \sqrt{2}e^{7j\pi/4}. \quad (2.28)$$

If we raise all of these to the 4th power, then we obtain,

$$4e^{j\pi}, \quad 4e^{3j\pi}, \quad 4e^{5j\pi} \quad \text{and} \quad 4e^{7j\pi}, \quad (2.29)$$

respectively. The arguments of these numbers are π , 3π , 5π and 7π , i.e. they are separated by 2π , and hence each one represents the same point in the Complex Plane. We may write these four values in the alternative form:

$$4e^{(1+2n)\pi j}, \quad n = 0, 1, 2, 3, \quad (2.30)$$

and therefore the roots listed in Eq. (2.28) may be written as,

$$z_n = \sqrt{2}e^{(1+2n)\pi j/4}, \quad n = 0, 1, 2, 3. \quad (2.31)$$

Note: In Eq. (2.31) n takes four successive values. Given the way we have motivated the method it is natural that the first value of n will be zero. However, any four successive values of n will be ok.

Note: The general method for finding roots uses the above approach but does it backwards! Three examples follow.

Example 2.9. Find the square roots of j .

We may write j in polar form as either $e^{(\pi/2)j}$ (the obvious one where the argument is $\pi/2$) or as $e^{(5\pi/2)j}$. We may combine these into the form:

$$e^{(\pi/2+2n\pi)j} \quad \text{where } n = 0 \text{ or } n = 1.$$

These two forms for j lie at the same point in the Complex Plane and must indeed do so! However, they will cease to do so when we take the square roots:

$$\sqrt{j} = e^{(\pi/4)j} \quad \text{or} \quad e^{(5\pi/4)j}. \quad (2.32)$$

For this particular example we can write the Cartesian form without the use of a calculator! We have,

$$\begin{aligned} \sqrt{j} &= \left(\cos \frac{1}{4}\pi + j \sin \frac{1}{4}\pi \right) \quad \text{or} \quad \left(\cos \frac{5}{4}\pi + j \sin \frac{5}{4}\pi \right) \\ &= \pm \left(\frac{1+j}{\sqrt{2}} \right). \end{aligned} \quad (2.33)$$

Example 2.10. Find the fifth roots of $(3 + 4j)$.

In polar form we have

$$z = 5e^{\theta j}, 5e^{(\theta+2\pi)j}, 5e^{(\theta+4\pi)j}, 5e^{(\theta+6\pi)j}, 5e^{(\theta+8\pi)j}$$

where we have written five different expressions for z , and where $\tan \theta = \frac{4}{3}$ with θ lying in the first quadrant (i.e. $\theta = 0.927295$). The five fifth roots are

$$z = 5^{1/5} e^{(\theta+2n\pi)j/5} \quad \text{for } n = 0, 1, 2, 3, 4.$$

If we were to plot these in the Complex Plane then we would see that the five points are equally distributed around a circle of radius $5^{1/5}$ centred on the origin.

Example 2.11. Find $z = (4 + 3j)^{2/3}$.

For such a power, it is probably better to find the square first and then take the cube root, although it could be done the other way around. Perhaps I ought to show that they give the same solutions. So we'll square first...

First way: Let

$$z = \left[(4 + 3j)^2 \right]^{1/3} = (7 + 24j)^{1/3}.$$

In polar form we have $7 + 24j = 25e^{(\theta+2n\pi)j}$ for $n = 0, 1, 2$, where $\tan \theta = \frac{24}{7}$ and θ lies in the first quadrant, we note that the modulus of $(7 + 24j)$ is 25. Therefore

$$z = 25^{1/3} e^{(\theta+2n\pi)j/3} \quad \text{for } n = 0, 1, 2 \text{ and where } \theta = 1.287002. \quad (2.34)$$

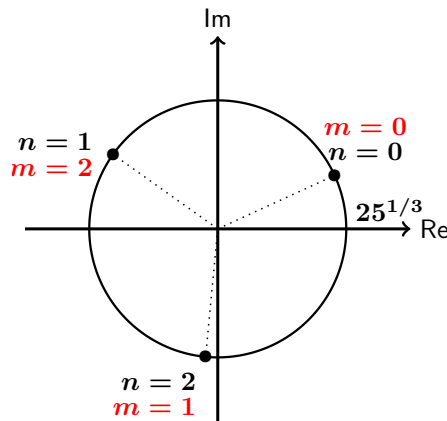
Second way: Let

$$z = (4 + 3j)^{2/3}.$$

In polar form we have $4 + 3j = 5e^{(\phi+2m\pi)j}$ for $m = 0, 1, 2$, where $\tan \phi = \frac{3}{4}$ and where ϕ also lies in the first quadrant. We have used m here instead of n , and ϕ instead of θ to distinguish the two analyses. The modulus of $(4 + 3j)$ is 5. Therefore

$$z = 5^{2/3} e^{2(\phi+2m\pi)j/3} \quad \text{for } m = 0, 1, 2 \text{ and where } \phi = 0.643501. \quad (2.35)$$

The following figure shows where the roots are as a consequence of these two analyses. The locations of the roots are identical, but manner in which they are counted is different.



Example 2.11

Note: Again I emphasize in the strongest terms that θ MUST be given in radians when using polar form.

2.8 Irrational exponents [For interest only]

Here we are thinking of powers such as $z^{\sqrt{2}}$.

We note that irrational numbers, such as $\sqrt{2}$, may not be expressed as the ratio of two whole numbers, whereas the method we have used above depends on this. Although $\sqrt{2}$ may be approximated by a rational fraction as closely as we might desire, then the more accurate the approximation is the larger is the denominator. For example, the absolute error in the fraction, $\frac{99}{70} = 1.41428571$ (8DP), is less than 10^{-4} and it is the first fraction (as the denominator increases) for which the error is smaller than 10^{-4} . For the errors, 10^{-6} and 10^{-8} , the corresponding fractions are,

$$\sqrt{2} \simeq \frac{1393}{985} = 1.41421320 \text{ (8DP)} \quad \text{and} \quad \sqrt{2} \simeq \frac{19601}{13860} = 1.41421356 \text{ (8DP)}.$$

Therefore increasingly accurate representations of $\sqrt{2}$ result in an increasing number of legitimate roots on a circle in the complex plane: 70, 985 and 13860 in our three examples. But typically the context will dictate if this is sensible, and it generally will not be. Therefore, in such situations we may restrict ourselves to what is called the **principal value** of the complex exponential. By this is meant that, if $z = re^{\theta j}$, then $z^{\sqrt{2}} = r^{\sqrt{2}} e^{\sqrt{2}\theta j}$. However, do be assured that such matters will not arise in engineering applications.

2.9 Relationship with the hyperbolic functions.

The functions $\cos \theta$ and $\sin \theta$ are known as either trigonometric or circular functions, while $\cosh \theta$ and $\sinh \theta$ are the hyperbolic functions. Given that

$$e^{\theta j} = \cos \theta + j \sin \theta \quad \text{and} \quad e^{-\theta j} = \cos \theta - j \sin \theta,$$

we may add and subtract these to obtain the relations,

$$\cos \theta = \frac{1}{2} [e^{\theta j} + e^{-\theta j}] \quad \text{and} \quad \sin \theta = \frac{1}{2j} [e^{\theta j} - e^{-\theta j}].$$

These are reminiscent of

$$\cosh \theta = \frac{1}{2} [e^{\theta} + e^{-\theta}] \quad \text{and} \quad \sinh \theta = \frac{1}{2} [e^{\theta} - e^{-\theta}],$$

and explains why there is so much similarity between the circular and hyperbolic functions, particularly when dealing with calculus and differential equations, even though the circular and the hyperbolic functions look so different from one another.

3 Differentiation

If we are given a function, $y = y(t)$, where the value of y is a function of t , then we say that y is the **dependent variable** and t is the **independent variable**. This is because the value that y takes *depends* on the value of t . If we interpret y to mean distance and t to mean time, then the dependence of y on t also makes physical sense. The following graph shows a typical function $y(t)$.

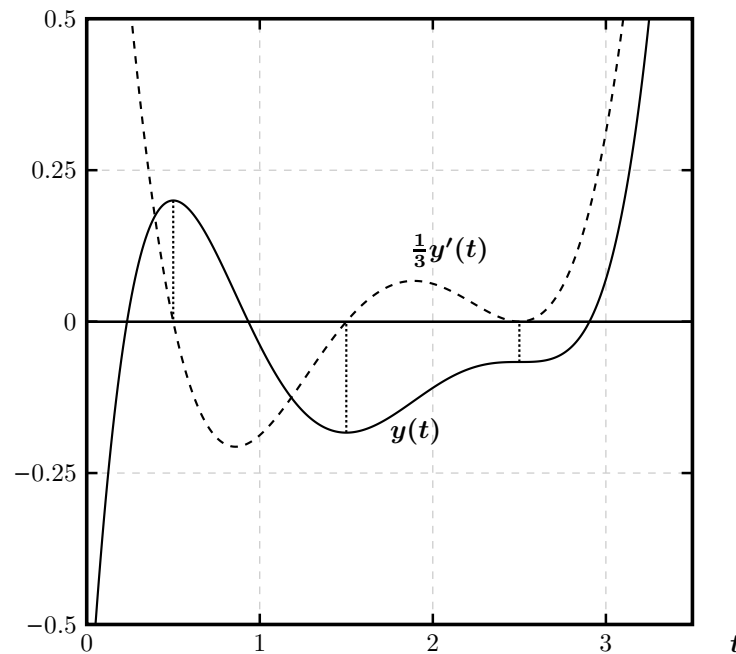


Figure 3.1. Depicting a typical function, $y(t)$ (continuous curve), and its derivative, $y'(t)$ (dashed curve). The dotted lines show where $y(t)$ has a zero slope.

The derivative of $y(t)$ at any point is defined as the slope or gradient of the tangent touching the curve at that point. Places where $y(t)$ has a horizontal tangent are places where the derivative is zero. The gradient of y is also shown in Figure 3.1; note the behaviour of the gradient as compared with the function, especially where the gradient is zero, i.e. where it achieves its maximum or minimum values. The point of inflexion at $t = 2.5$ corresponds to a double zero in $y'(t)$, and therefore $y''(2.5) = 0$ as well.

3.1 First derivative

This derivative is written in various ways:

$$\frac{dy}{dt}, \quad y'(t), \quad y', \quad Dy. \quad (3.1)$$

The first three of these are very common alternatives, while the last occurs only in certain specialised circumstances. The third one is often used as a short-hand form when the identity of the independent variable isn't in question. However, the first notation (that of Leibniz) is reminiscent of one way of defining the derivative mathematically using the notion of limits, as will be illustrated in Figure 3.2, later.

Although we have the common-sense definition that the gradient at a point is the slope of the tangent to the curve at that point, it requires calculus to find that tangent. So it makes sense that we can approximate the gradient of y at $t = t_0$ by the gradient of the straight line joining $y(t_0)$ and $y(t_0 + \delta t)$ since we can do that without calculus; this may be seen in Fig 3.2.

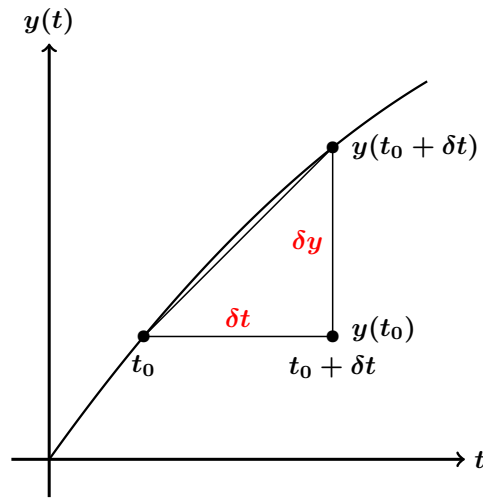


Figure 3.2. A sketch of how a derivative at a point may be approximated by the limit of a slope between two points on a curve.

If the change in y over this interval in time is denoted by δy , then we may define the derivative as

$$\left. \frac{dy}{dt} \right|_{t=t_0} = \lim_{\delta t \rightarrow 0} \frac{\delta y}{\delta t}. \quad (3.2)$$

Using the fact that $\delta y = y(t_0 + \delta t) - y(t_0)$ we may also write

$$\left. \frac{dy}{dt} \right|_{t_0} = \lim_{\delta t \rightarrow 0} \frac{y(t_0 + \delta t) - y(t_0)}{\delta t}. \quad (3.3)$$

If we now imagine what happens to the straight line in Fig. 3.2 as δt gets smaller, we should be able to accept that it approximates the curve increasingly well.

Note that the definition of dy/dt given in Eq. (3.3) is not unique — we may also define the derivative in the following way

$$\left. \frac{dy}{dt} \right|_{t_0} = \lim_{\delta t \rightarrow 0} \frac{y(t_0 + \delta t/2) - y(t_0 - \delta t/2)}{\delta t}, \quad (3.4)$$

where the target value of t is halfway between the points where y is evaluated, but the end result is exactly the same. To demonstrate this, we will apply both Eqs. (3.3) and (3.4) to the function $y = t^2$. In the case of Eq. (3.3) we have

$$\begin{aligned} \frac{dy}{dt} &= \lim_{\delta t \rightarrow 0} \left[\frac{(t + \delta t)^2 - t^2}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{(t^2 + 2t\delta t + \delta t^2) - t^2}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{2t\delta t + \delta t^2}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} [2t + \delta t] = 2t. \end{aligned} \quad (3.5)$$

Using Eq. (3.4) we have,

$$\begin{aligned}
\frac{dy}{dt} &= \lim_{\delta t \rightarrow 0} \left[\frac{(t + \delta t/2)^2 - (t - \delta t/2)^2}{\delta t} \right] \\
&= \lim_{\delta t \rightarrow 0} \left[\frac{(t^2 + t\delta t + \delta t^2/4) - (t^2 - t\delta t + \delta t^2/4)}{\delta t} \right] \\
&= \lim_{\delta t \rightarrow 0} \left[\frac{2t\delta t}{\delta t} \right] \\
&= \lim_{\delta t \rightarrow 0} [2t] = 2t.
\end{aligned} \tag{3.6}$$

In the same way we may also show that the derivative of t^n is nt^{n-1} when n is an integer. This will involve the use of the Binomial theorem.

This technique may also be used for other functions, although in the case of $\sin t$, for example, it becomes necessary to use the results,

$$\lim_{\delta t \rightarrow 0} \frac{\sin \delta t}{\delta t} = 1, \quad \lim_{\delta t \rightarrow 0} \frac{(1 - \cos \delta t)}{\delta t} = 0. \tag{3.7}$$

These results also require calculus for their proof but we do need to avoid circular arguments. However, they may at least be verified using a calculator (which must be set to radians); they will also be proved using l'Hôpital's rule, which we will meet later.

3.2 Higher derivatives

There is no reason why we should not take another derivative, which is called the second derivative. If $y(t)$ is again the position of an object at time t , then dy/dt is the **velocity** of that object and the second derivative, which is denoted by,

$$\frac{d^2y}{dt^2}, \quad y''(t), \quad y''(t) \quad \text{or} \quad D^2y, \tag{3.8}$$

is the **acceleration**. Third derivatives are denoted by

$$\frac{d^3y}{dt^3}, \quad y'''(t), \quad y''' \quad \text{and} \quad D^3y, \tag{3.9}$$

and fourth derivatives by

$$\frac{d^4y}{dt^4}, \quad y''''(t), \quad y'''' \quad \text{and} \quad D^4y. \tag{3.10}$$

For yet higher derivatives it is often the case that the 'primed' notation is replaced by a superscript. So 9th derivatives may be written as $y^{(9)}(t)$, as opposed to $y''''''''''''(t)$. And occasionally some textbooks use a Roman letter notation: $y^i, y^{ii}, y^{iii}, y^{iv}, y^v$ and so on.

Sometimes third and higher derivatives are of use. In the design of rides such as Nemesis at Alton Towers, the term **jerk** is used to denote y''' , the rate of change of acceleration. Further, the fourth derivative, y'''' , is called the **jounce**. For a successfully designed ride (i.e. one that is sufficiently exciting, but not too nauseous), the acceleration, jerk and jounce all have to be within certain limits in all three directions.

3.3 Examples of derivatives

It is useful to be able to memorise as many as possible of the following.

$y(t)$	$y'(t)$	$y(t)$	$y'(t)$
c	0	$\sin t$	$\cos t$
t	1	$\sin at$	$a \cos at$
t^n	nt^{n-1}	$\cos t$	$-\sin t$
ct^n	nct^{n-1}	$\cos at$	$-a \sin at$
e^t	e^t	$\tan t$	$\sec^2 t = 1 + \tan^2 t$
e^{at}	ae^{at}	$\tan at$	$a \sec^2 at$
$\ln t $	t^{-1}	$\cot at$	$-a \operatorname{cosec}^2 at$
$\sin^{-1} t$	$\frac{1}{\sqrt{1-t^2}}$	$\sinh at$	$a \cosh at$
$\tan^{-1} t$	$\frac{1}{t^2 + 1}$	$\cosh at$	$a \sinh at$

Some of these will be derived in the following sections.

3.4 Manipulation of derivatives

This is a very short subsection, but the message is extremely important.

It is important to be able to take the derivatives of various functions, even quite complicated functions, with speed and accuracy. In spoken English (or any other language, as far as I know) we may be understood perfectly clearly even when we have expressed ourselves extremely poorly in terms of grammar. But in mathematics it is essential to follow all the rules absolutely precisely in order to avoid errors — all sorts of strange things can happen otherwise. Four of the following five sections give the “rules of the game” for differentiation.

3.5 Linearity

If we are given the two functions $u(t)$ and $v(t)$, then the **first rule of linearity** states that

$$\frac{d(u + v)}{dt} = \frac{du}{dt} + \frac{dv}{dt}. \quad (3.11)$$

If you are already familiar with this result, then it is very likely that you apply it automatically. In many cases this is assumed without even thinking about it, but we need to bear in mind what it means. Essentially it says that you may interchange the order of application of (i) summation and (ii) the taking of derivatives. The left hand side is the derivative of a sum, while the right hand side is the sum of derivatives. This may seem to be a very obvious result, and it can be proved fairly quickly from the limiting definition given in Section 3.1, but similar interchanges of mathematical operations do not always work. For example, the act of walking 100 miles east followed by 100 miles north is not the same as walking 100 miles east followed by 100 north, although there are some isolated locations on Earth where they are the same. In the area of matrices (if you have covered it), if A and B are two matrices, then not only will $AB \neq BA$ in general, but it is also possible that one product exists while the other does not.

The **second rule of linearity** is that

$$\frac{d(ku)}{dt} = k \frac{du}{dt}, \quad (3.12)$$

where k is a constant. Again this may be proved using the limit definition of the derivative.

We may take the example of the function $4t^3 - 2t$ to demonstrate the use of the two linearity rules:

$$\begin{aligned}
 \frac{d}{dt}[4t^3 - 2t] &= \left[\frac{d}{dt}(4t^3) \right] + \left[\frac{d}{dt}(-2t) \right] && \text{using (3.11)} \\
 &= \left[4 \frac{d}{dt}(t^3) \right] + \left[(-2) \frac{d}{dt}(t) \right] && \text{using (3.12)} \\
 &= [4 \times 3t^2] + [(-2) \times 1] = 12t^2 - 2.
 \end{aligned} \tag{3.13}$$

This was very pedantic indeed and, in practice, we should strive to go immediately from $4t^3 - 2t$ to $12t^2 - 2$ without the intervening lines.

3.6 Product rule

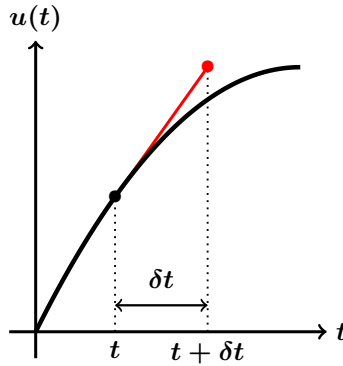
This is a rule which enables us to evaluate the derivative of a product of two functions. It is

$$\frac{d}{dt}(uv) = u \frac{dv}{dt} + v \frac{du}{dt} \quad \text{or} \quad (uv)' = uv' + u'v. \tag{3.14}$$

To prove this we first approximate the value of $u(t + \delta t)$ by

$$u(t + \delta t) \simeq u(t) + \delta t \frac{du}{dt} = u(t) + u'(t) \delta t; \tag{3.15}$$

this expression is equivalent to the equation of the straight line through $u(t)$ with the slope of the tangent to the curve at that point, as shown below



Here, the red disk corresponds to the right hand side of Eq. (3.5). We may also write down a similar expression for v . Now if we use the limiting definition of the derivative, see Eq. (3.3), we have the following analysis:

$$\begin{aligned}
 \frac{d}{dt}(uv) &= \lim_{\delta t \rightarrow 0} \left[\frac{u(t + \delta t)v(t + \delta t) - u(t)v(t)}{\delta t} \right] \\
 &= \lim_{\delta t \rightarrow 0} \left[\frac{(u + u'\delta t)(v + v'\delta t) - uv}{\delta t} \right] \\
 &= \lim_{\delta t \rightarrow 0} \left[\frac{\delta t(uv' + vu') + \delta t^2 u'v'}{\delta t} \right] \\
 &= \lim_{\delta t \rightarrow 0} [(uv' + vu') + \delta t u'v'] \\
 &= uv' + vu'.
 \end{aligned} \tag{3.16}$$

Note that this argument is not completely rigorous, but it is plausible. The lack of rigour takes place between lines 1 and 2 where the functions are replaced by the straight lines. However, the result is correct.

Example 3.1: Find the derivatives of $t^2 \sin t$ and $e^{-4t} \cos 5t$.

Given that the derivative of $\sin t$ is $\cos t$, we may use the product rule to show that

$$\frac{d}{dt}[t^2 \sin t] = t^2 \left[\frac{d}{dt}(\sin t) \right] + \sin t \left[\frac{d}{dt}(t^2) \right] = t^2 \cos t + 2t \sin t. \quad (3.17)$$

Similarly, the derivative of $e^{-4t} \cos 5t$ is

$$\frac{d}{dt}[e^{-4t} \cos 5t] = e^{-4t}(-5 \sin 5t) + (-4)e^{-4t} \cos 5t = -e^{-4t}(5 \sin 5t + 4 \cos 5t). \quad (3.18)$$

The product of three functions.

It is now fairly straightforward to write down the derivative of the product of three or more functions. If we replace v by vw in Eq. (3.14) where $w = w(t)$, then

$$\begin{aligned} \frac{d}{dt}(uvw) &= u \frac{d(vw)}{dt} + (vw) \frac{du}{dt} && \text{using the product rule on } u \text{ and } vw \\ &= u \left(v \frac{dw}{dt} + w \frac{dv}{dt} \right) + vw \frac{du}{dt} && \text{using (3.14) on } vw \\ &= uvw' + uv'w + u'vw \\ &= u'vw + uv'w + uvw'. \end{aligned} \quad (3.19)$$

Although an alternative way of writing this final answer is

$$\frac{d}{dt}(uvw) = \left[\frac{u'}{u} + \frac{v'}{v} + \frac{w'}{w} \right] uvw, \quad (3.20)$$

I prefer to remember the form given in Eq. (3.19).

Example 3.2: Find the derivative of $t^5 e^{4t} \sin 3t$.

We shall follow the formula given in Eq. (3.19):

$$\begin{aligned} \frac{d}{dt}[t^5 e^{4t} \sin 3t] &= (5t^4)e^{4t} \sin 3t + t^5(4e^{4t}) \sin 3t + t^5 e^{4t}(3 \cos 3t) \\ &= t^4 e^{4t} [(5 + 4t) \sin 3t + 3t \cos 3t]. \end{aligned} \quad (3.21)$$

The product of four functions.

In similar fashion the derivative of the product of four functions, u_1 , u_2 , u_3 and u_4 is

$$\begin{aligned} \frac{d}{dt}(u_1 u_2 u_3 u_4) &= u_1' u_2 u_3 u_4 + u_1 u_2' u_3 u_4 + u_1 u_2 u_3' u_4 + u_1 u_2 u_3 u_4', \\ &= \left[\frac{u_1'}{u_1} + \frac{u_2'}{u_2} + \frac{u_3'}{u_3} + \frac{u_4'}{u_4} \right] u_1 u_2 u_3 u_4. \end{aligned} \quad (3.22)$$

and so on for products of yet more functions. Again, though, I prefer to remember the first line of Eq. (3.22) rather than the second, but both have memorable patterns.

3.7 The chain rule

This rule applies for functions which are functions of other functions of t , two examples of which are $\sin(t^2)$ and $\sin^2 t$. These are the sine of the square of t and the square of the sine of t . In general, if we have $u(t) = u(v(t))$ then the derivative is,

$$\frac{d}{dt}u(v(t)) = \frac{du}{dv} \frac{dv}{dt}. \quad (3.23)$$

This rule may be remembered easily because it *looks* as though the dv may be cancelled on the right hand side of Eq. (3.23). Still, it looks unusual and possibly even implausible, so I have given an outline of the proof of this later in Section (3.8).

Example 3.3: Find the derivatives of $u = \sin t^2$ and of $u = \sin^2 t$.

First we need to decompose/decouple these functions from their basic nested states. For the first one we can say that $u = \sin v$ where $v = t^2$, and therefore

$$\frac{du}{dt} = \frac{du}{dv} \frac{dv}{dt} = [\cos v] [2t] = 2t \cos(t^2). \quad (3.24)$$

For the second one we can say that $u = v^2$ where $v = \sin t$. Therefore we have,

$$\frac{du}{dt} = \frac{du}{dv} \frac{dv}{dt} = [2v] [\cos t] = 2 \sin t \cos t. \quad (3.25)$$

And yes, I know that that last answer may be simplified further! Perhaps more important is that the fact that both answers need to be written as functions solely of t since we introduced v solely to help us to do the differentiation, and therefore that final result must not involve v .

Example 3.4: Find the derivative of $u = (t + a)^5$, where a is a constant.

For $u = (t + a)^5$ we let $u = v^5$ where $v = t + a$. The derivative is now

$$\frac{du}{dt} = \frac{du}{dv} \frac{dv}{dt} = [5v^4] [1] = 5(t + a)^4. \quad (3.26)$$

That final result is usually memorised.

Example 3.5: Find the derivative of $y = v^{-1}$ where v is a function of t .

To find this derivative we first note that $d(t^{-1})/dt = -t^{-2}$. Now we apply the chain rule to get,

$$\frac{dy}{dt} = \frac{dy}{dv} \frac{dv}{dt} = \left[-\frac{1}{v^2}\right] \times [v'] = -\frac{v'}{v^2}. \quad (3.27)$$

Here we are using primes to denote derivatives with respect to t . This result will be used a little later to prove the Quotient Rule.

A triply-nested function

The chain rule may be extended indefinitely. If y is a function of u which is a function of v which is a function of t , i.e.

$$y = y[u(v(t))],$$

then

$$\frac{dy}{dt} = \frac{dy}{du} \frac{du}{dv} \frac{dv}{dt}. \quad (3.28)$$

Again, this formula may be remembered because of the apparent cancellations hinted at by the colours.

Example 3.6: Find the derivative of $y = e^{\sin t^2}$.

We may decompose this function by setting, $y = e^u$ where $u = \sin v$ and where $v = t^2$. Therefore the derivative is given by

$$\frac{dy}{dt} = \frac{dy}{du} \frac{du}{dv} \frac{dv}{dt} = e^u \times \cos v \times 2t = 2t[\cos(t^2)] [e^{\sin(t^2)}]. \quad (3.29)$$

Example 3.7: The Chain Rule may also be used for finding the derivatives of the inverse trigonometrical functions. For example, if $y = \tan^{-1} t$, then $\tan y = t$. The left hand side is of the form of a function of a function, since $y = y(t)$. Using the chain rule, the derivative of $\tan y$ with respect to t is given by

$$\frac{d}{dt} \tan y = \frac{d \tan y}{dy} \frac{dy}{dt} = (1 + \tan^2 y) \frac{dy}{dt}, \quad (3.30)$$

where we have used the result which was stated in the above Table (and which will be proved below) for the derivative of $\tan y$. Hence the time derivative of the equation, $\tan y = t$, is

$$(1 + \tan^2 y) \frac{dy}{dt} = 1 \quad \Rightarrow \quad \frac{dy}{dt} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + t^2}. \quad (3.31)$$

Some say that the process used in Eq. (3.31) is called **Implicit Differentiation**, but it is no more than a subset of the chain rule.

Example 3.8: Find the derivative of $y = \ln |t|$.

The derivative of $y = \ln |t|$ may be obtained in like fashion by first taking exponentials of each side, but the detail is a little more complicated because of the presence of the modulus signs.

When $t > 0$ we have

$$y = \ln |t| = \ln t \quad \Rightarrow \quad e^y = t \quad \Rightarrow \quad e^y y' = 1 \quad \Rightarrow \quad y' = e^{-y} = t^{-1}, \quad (3.32)$$

but when $t < 0$ we have,

$$y = \ln |t| = \ln(-t) \quad \Rightarrow \quad e^y = -t \quad \Rightarrow \quad e^y y' = -1 \quad \Rightarrow \quad y' = -e^{-y} = t^{-1}. \quad (3.33)$$

Hence the derivative of $\ln |t|$ is t^{-1} .

3.8 Outline proof of the chain rule [For interest only]

We will let $u = u(v)$ where $v = v(t)$, so that u is ultimately a function of t . We'll write the function as $u = u[v(t)]$ where I've used the differently-shaped brackets here solely for the purposes of clarity/brevity. We will use the limit definition of a derivative as originally given in Eq. (3.3), above. Therefore we have

$$\begin{aligned}
\frac{du}{dt} &= \lim_{\delta t \rightarrow 0} \left[\frac{u[v(t + \delta t)] - u[v(t)]}{\delta t} \right] \\
&= \lim_{\delta t \rightarrow 0} \left[\frac{u\left[v(t) + \frac{dv}{dt}\delta t + \dots\right] - u[v(t)]}{\delta t} \right] && v(t + \delta t) \text{ approximated by its tangent} \\
&= \lim_{\delta t \rightarrow 0} \left[\frac{u[v(t) + \delta v] - u[v(t)]}{\delta t} \right] && \text{where } \delta v \equiv \frac{dv}{dt}\delta t \text{ for clarity} \\
&= \lim_{\delta t \rightarrow 0} \left[\frac{\left(u[v(t)] + \frac{du}{dv}\delta v + \dots\right) - u[v(t)]}{\delta t} \right] && u(v + \delta v) \text{ approximated by the tangent} \\
&= \lim_{\delta t \rightarrow 0} \left[\frac{\cancel{u[v(t)]} + \frac{du}{dv} \frac{dv}{dt} \delta t + \dots - \cancel{u[v(t)]}}{\delta t} \right] && \text{Using the definition of } \delta v. \text{ Red text cancels} \\
&= \frac{du}{dv} \frac{dv}{dt}.
\end{aligned}$$

(3.34)

3.9 Quotient rule

This is closely related to the product rule and makes use of the above result, Eq. (3.27), on the derivative of v^{-1} . The rule is derived as follows:

$$\begin{aligned}
\frac{d}{dt}\left(\frac{u}{v}\right) &= \frac{d}{dt}(uv^{-1}) = u \frac{d(v^{-1})}{dt} + v^{-1} \frac{du}{dt} && \text{using the product rule} \\
&= -\frac{uv'}{v^2} + \frac{u'}{v} && \text{using Eq. (3.27)} \\
&= \frac{vu' - uv'}{v^2}.
\end{aligned}$$

(3.35)

In the good old(?) days of cathode ray tubes I used to remember this result by stating the first term to be 'VDU' (v delta u or 'Visual Display Unit' — the old term for a computer terminal or monitor).

Example 3.9: Find the derivative of $y = \tan t$.

This takes the following route:

$$\begin{aligned}
\frac{d \tan t}{dt} &= \frac{d}{dt} \left(\frac{\sin t}{\cos t} \right) \\
&= \frac{\cos t (\sin t)' - \sin t (\cos t)'}{\cos^2 t} \\
&= \frac{\cos^2 t - \sin t(-\sin t)}{\cos^2 t} = \frac{\cos^2 t + \sin^2 t}{\cos^2 t} \\
&= \frac{1}{\cos^2 t} \quad \text{or} \quad 1 + \tan^2 t \quad \text{or} \quad \sec^2 t.
\end{aligned} \tag{3.36}$$

3.10 Critical points

The maximum and minimum values of a function are examples of **critical points** or **extrema** (singular: **extremum**). Points of inflexion* are also critical points. Generally the value of t at which this happens is found by setting the derivative of the function to zero, and solving the resulting equation for t . For example, if $y(t) = t^2 - t$, then $y'(t) = 2t - 1$. Therefore the function has an extremum when $y'(t) = 0$, which, in this case, is when $t = \frac{1}{2}$.

Sometimes it is not clear whether a particular critical point is a local maximum or a local minimum, but it is possible to determine this mathematically in almost all cases by considering the second derivative of the function. Returning to the above function, $y = t^2 - t$, a simple sketch is enough to tell us that this extremum is a minimum, but it is necessary also to have a set of mathematical criteria.

The second derivative at this point is $y''(\frac{1}{2}) = 2$ which is positive; a positive value of the second derivative always corresponds to a minimum (and a negative value corresponds to a maximum). That this is a general result may be seen by considering the typical situation: to the left of a minimum the slope of the function must be negative, whereas the slope is positive to the right of the minimum. Therefore the slope is increasing and hence the second derivative (which is the slope of the slope) is positive.

If the above paragraph has too much verbiage and too little imagery then it may be better to consider Fig. 3.3 on the next page which shows diagrammatically that a minimum in y corresponds to a zero value of y' and a positive value for y'' . On the other hand, while a maximum in y also corresponds to a zero value of y' , the value of y'' is negative. Hopefully this is clear in Fig. 3.3.

* Some authorities regard an inflexion point as being that value of t at which $y''(t) = 0$ with no condition on $y'(t)$. This means that the curve is S-shaped locally, and changes from being concave to convex or vice-versa. Here, my definition is more restrictive because I require both $y'(t) = 0$ and $y''(t) = 0$ simultaneously; this means that the inflexion point is also a critical point.

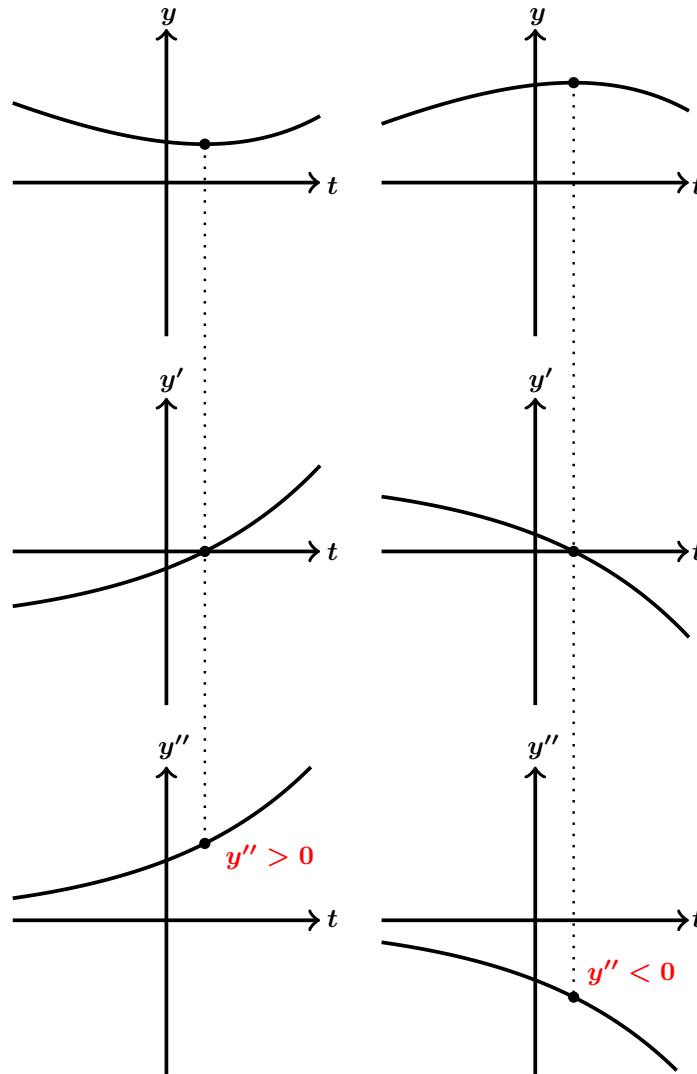


Figure 3.3. Depicting curves for y , y' and y'' (top to bottom) where y shows a minimum (left column) and a maximum (right column). The dotted lines and the black disks indicate the relationship between the behaviours of the y , y' and y'' curves in each case.

Example 3.10: Find the critical points of the function $y = x^3 - 3x$.

Note that we have now changed the identity of the independent variable from t to x . Maxima and minima also occur in space! We have

$$y' = 3x^2 - 3 \quad \text{which may be factorised: } y' = 3(x - 1)(x + 1).$$

Hence the critical points are at $x = -1, 1$. These may be classified by evaluating y'' at the critical points. Hence

$$y'' = 6x = \begin{cases} 6 & \text{at } x = 1 \Rightarrow \text{Minimum} \\ -6 & \text{at } x = -1 \Rightarrow \text{Maximum} \end{cases} \quad (3.37)$$

Figure 3.4 confirms this analysis.

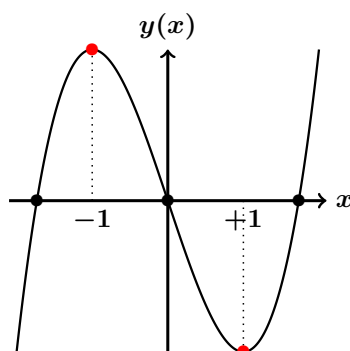


Figure 3.4. Showing $y = x^3 - 3x$ together with its roots (black disks) and the maximum and minimum values (red disks).

While the solution of $y'(x) = 0$ yields where the critical points are, their identification requires the use of secondary criteria. In this case a positive y'' corresponds to a minimum, while a negative value corresponds to a maximum. But what happens if $y'' = 0$ as well? The following example provides an illustration.

Example 3.11: Find the critical points of the function $y = x^3$.

This is a fairly trivial-looking function, but it works well and allows us to see the important details. I could have chosen to use $y = 2(x - 2)^3$ and while the analysis will work, it might prove more difficult to do should I have presented $y(x)$ as a power series, i.e. having multiplied out the cubic.

Clearly $y' = 3x^2$, and the setting of this to zero yields $x = 0$ as the location of the critical point.

If we now evaluate y'' then we obtain, $y''(x) = 6x$. When $x = 0$ then $y''(0) = 0$, and therefore this falls into an intermediate category. We know from applying our curve-sketching ideas (or also the idea of factorisation) that $y = x^3$ has a triple root at $x = 0$ and therefore $x = 0$ corresponds to a point of inflexion. The third derivative is $y'''(x) = 6$ which is positive, and therefore $y'''(0) = 6$ corresponds to a point of inflexion *which is rising*. Figure 3.5 shows this.

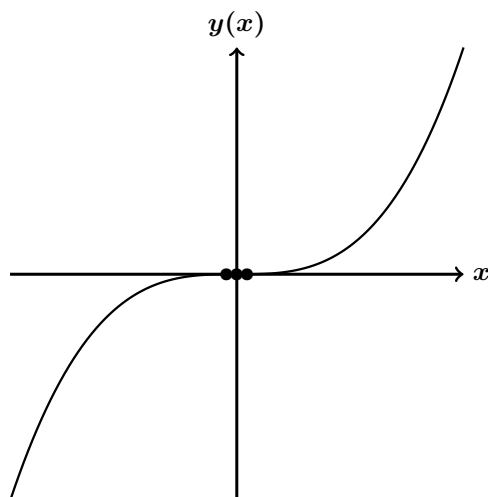


Figure 3.5. Showing $y = x^3$ which has a **rising point of inflexion** at $x = 0$.

If we had had a case where $y'' = 0$ but $y''' < 0$ at the critical point, then we could name it a **descending point of inflexion**. So a point of inflexion has $y' = 0$ and $y'' = 0$ as its primary criteria while the secondary criterion is that $y''' \neq 0$ at the critical point.

But what if $y''' = 0$ as well? Here's an example....

Example 3.12: Find the critical points of the function $y = ax^4$ where a is a **nonzero** constant. We already know from the curve-sketching section that this function has a quadruple root at $x = 0$ and has a quartic minimum when $a > 0$. Our analysis follows:

$$\begin{array}{llll}
 y' = 4ax^3 & \Rightarrow x = 0 & \text{is the sole critical point} \\
 y'' = 12ax^2 & \Rightarrow y''(0) = 0 & \Rightarrow \text{is not a max/min} \\
 y''' = 24ax & \Rightarrow y'''(0) = 0 & \Rightarrow \text{is not an inflexion point} \\
 y'''' = 24a & \Rightarrow y''''(0) = 24a & \Rightarrow \text{is a } \mathbf{\text{quartic minimum when } a > 0} \\
 & & \text{or is a } \mathbf{\text{quartic maximum when } a < 0}
 \end{array}$$

The case $a = 1$ is illustrated below.

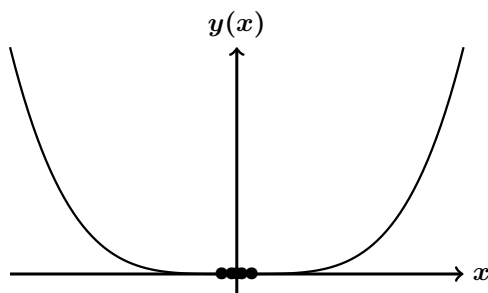


Figure 3.6. Showing $y = x^4$ which has a quartic minimum at $x = 0$.

Example 3.13: Find and classify the critical point/points for $y = x^4 - 8x^3 + 24x^2 - 32x + 16$.

This appears to be a nasty one, and it is, but it isn't a trivial one like Example 3.12 is. The analysis proceeds as follows.

$$\begin{array}{llll}
 y' = 4x^3 - 24x^2 + 48x - 32 & \Rightarrow x = 2 & \text{is the sole critical point — you'll need to trust me on this!} \\
 y'' = 12x^2 - 48x + 48 & \Rightarrow y''(2) = 0 & \Rightarrow \text{is not a max/min} \\
 y''' = 24x - 48 & \Rightarrow y'''(2) = 0 & \Rightarrow \text{is not an inflexion point} \\
 y'''' = 24 & \Rightarrow y''''(2) = 24 > 0 & \Rightarrow \text{is a quartic minimum}
 \end{array}$$

Clearly one could now ask what happens if we have $y'' = y''' = y'''' = 0$ at a critical point, but the fifth derivative is nonzero. Well, this will be a quintic inflexion point and it may be a rising or descending one depending on the sign of $y^{(5)}$ at the critical point.

On the next page I have compiled a Table which details how to detect critical points up to and including the septic (!!!) ones. At least you can be assured that the more exotic the critical point is the less likely it is to turn up on an exam paper because of the increased workload which is required — Exercise 3.13 was bad enough!

Table. Primary and secondary criteria for the different types of critical points.

y'	y''	y'''	y''''	$y^{(5)}$	$y^{(6)}$	$y^{(7)}$	
0	+						Minimum
0	−						Maximum
0	0	+					Rising inflexion
0	0	−					Descending inflexion
0	0	0	+				Quartic minimum
0	0	0	−				Quartic maximum
0	0	0	0	+			Rising quintic inflexion
0	0	0	0	−			Descending quintic inflexion
0	0	0	0	0	+		Sextic minimum
0	0	0	0	0	−		Sextic maximum
0	0	0	0	0	0	+	Rising septic inflexion
0	0	0	0	0	0	−	Descending septic inflexion

3.11 Notation

Throughout this part of the unit I have used either t or x as the independent variable. This has been motivated by the fact that very many physical quantities (e.g. speed, acceleration, electrical current, body temperature) are time-dependent or space dependent, and we are often interested in how quickly these quantities change.

However, there are many instances where the independent variable could be something else. At a given pressure, the density of water depends on temperature ($\rho = \rho(T)$). The mean atmospheric pressure depends on the height above the ground ($P = P(h)$). The intensity of an electric field around a point charge depends on the distance from the charge ($E = E(r)$). Therefore we need to be able to take derivatives with respect to any and every possible independent variable. Fortunately, all the rules described above apply, although it may be possible to become a little confused when variables play unusual roles, such as when t is a dependent variable (some people use this to mean temperature). Therefore all the following examples are, in effect, identical — it's just that the names of the variables have changed:

$$\begin{aligned}
 y = \sin x^2 &\Rightarrow \frac{dy}{dx} = 2x \cos x^2 \\
 x = \sin y^2 &\Rightarrow \frac{dx}{dy} = 2y \cos y^2 \\
 t = \sin v^2 &\Rightarrow \frac{dt}{dv} = 2v \cos v^2 \\
 \gamma = \sin \alpha^2 &\Rightarrow \frac{d\gamma}{d\alpha} = 2\alpha \cos \alpha^2 \\
 \heartsuit = \sin \heartsuit^2 &\Rightarrow \frac{d\heartsuit}{d\heartsuit} = 2\heartsuit \cos \heartsuit^2
 \end{aligned}$$

3.12 Partial differentiation

The height of a hill may be characterised mathematically by the function, $h = h(x, y)$. This statement means that h , the height above sea level, say, is a function of both x and y , which we may take to correspond to easterly and northerly directions, respectively, from a reference location. In this case h is the dependent variable, while x and y are the two independent variables.

At a typical location on the hill one may face east and then determine the gradient in that direction. Likewise, one may then turn 90° to face north and determine the gradient in that direction. Mathematically, these processes correspond to the taking of what are called **partial derivatives**, and these are denoted, respectively, by

$$\frac{\partial h}{\partial x} \quad \text{and} \quad \frac{\partial h}{\partial y}$$

(or, as h_x and h_y when used within text or for notational convenience). This slightly different notation from the ordinary derivative (i.e. the use of ∂ instead of d) merely reminds us that the function being differentiated is a function of more than one variable. Thus the process of differentiation with respect to x involves treating y as a constant — this is seen clearly by considering the hill example where one's northerly location does not change when finding the slope in the easterly direction. Indeed, this aspect is what causes the most difficulty with partial differentiation because x needs to be treated as a constant when taking the partial derivative with respect to y and that y needs to be treated as a constant when taking the partial derivative with respect to x . I would even go so far as to say that, when tired, avoid both driving and partial derivatives — each is likely to go badly wrong.

Example 3.14: Given the function, $f(x, y) = x^2 + 2xy^2 + 3y^5 + 1$, we may differentiate once with respect to each variable to obtain,

$$\frac{\partial f}{\partial x} = 2x + 2y^2 \quad \text{and} \quad \frac{\partial f}{\partial y} = 4xy + 15y^4.$$

Example 3.15: In suitable units, the height of Glastonbury Tor may be modelled by the the function,

$$h(x, y) = \frac{1}{x^2 + y^2 + 1}. \quad (3.38)$$

The partial derivative of this with respect to each variable is

$$\frac{\partial h}{\partial x} = \frac{-2x}{(x^2 + y^2 + 1)^2} \quad \text{and} \quad \frac{\partial h}{\partial y} = \frac{-2y}{(x^2 + y^2 + 1)^2}. \quad (3.39)$$

Note 1: that we have used the chain rule here, as described earlier for ordinary differentiation. Indeed all of our previous derivations and results for obtaining ordinary derivatives also apply for partial derivatives without change.

Note 2: that the expression for h is symmetric in the sense that, should you swap x and y around, then the same function is produced. The useful consequence of this is that $\partial h / \partial y$ may be found by swapping x and y around in $\partial h / \partial x$. This may be seen in Eq. (3.39).

Comment: Given that many physical quantities are functions of time and of all three spatial coordinates, it is quite possible to have quantities that are functions of many more independent variables than just two. The temperature, θ , of a solid may take the form,

$$\theta = \theta(x, y, z, t),$$

and therefore there are four possible first partial derivatives that could be found.

Example 3.16: Given the function, $\theta = y^2 e^{-ayz} \sin(\omega t - kx)$, where ω , k and a are constants, all four first partial derivatives are given by,

$$\begin{aligned}\frac{\partial \theta}{\partial t} &= \omega y^2 e^{-ayz} \cos(\omega t - kx), \\ \frac{\partial \theta}{\partial x} &= -k y^2 e^{-ayz} \cos(\omega t - kx), \\ \frac{\partial \theta}{\partial y} &= y(2 - azy) e^{-ayz} \sin(\omega t - kx), \\ \frac{\partial \theta}{\partial z} &= -a y^3 e^{-ayz} \sin(\omega t - kx).\end{aligned}\tag{3.40}$$

3.12.1 Higher partial derivatives

Just as in ordinary differentiation, it is possible to have higher partial derivatives, although there turns out to be quite a variety of these. Given the function, $f(x, y) = x^2 + 2xy^2 + 3y^5$, which was considered in Ex. 3.14, we may write down three different second partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = 4x + 60y^3 \quad \text{and} \quad \frac{\partial^2 f}{\partial x \partial y} = 4y.\tag{3.41}$$

The last of this trio is the simplest example of a **mixed partial derivative**. It may be interpreted as being the rate of change in the y -direction of the x -derivative or, alternatively, as being the rate of change in the x -direction of the y -derivative. Mathematically, this may be stated as

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = f_{yx}.\tag{3.42}$$

Yes! Honestly, it makes no difference in which order the partial derivatives are taken. This result is called **Clairaut's theorem**. One of the options can be easier/quicker than the other, so it better to choose that one.

Example 3.17: If a temperature within a solid satisfies the formula,

$$\theta = e^{-at} \sin(ax) e^{-by},$$

where the solid is contained within the region, $-\infty < x < \infty$ and $y \geq 0$, and where both a and b are constants, then the rate of heat transfer into the solid at any point on its boundary is given by, $q = -k \partial \theta / \partial y$ which is evaluated at $y = 0$, and where k is the thermal conductivity. Therefore the rate of change of the surface heat transfer in time is given by

$$\frac{\partial q}{\partial t} = \frac{\partial}{\partial t} \left[-k \frac{\partial \theta}{\partial y} \right]_{y=0} = -k \frac{\partial^2 \theta}{\partial t \partial y} \Big|_{y=0} = -k \left[ab e^{-at} \sin(ax) e^{-by} \right]_{y=0} = -abk e^{-at} \sin(ax).$$

Again, it does not matter whether the t -derivative is found before or after finding the y -derivative. Likewise, it does not matter whether we set $y = 0$ before or after taking the t -derivative, but clearly the y -derivative must be taken before $y = 0$ is set. [Note: The subscript, $y=0$, indicates that the quantity is evaluated at $y = 0$.]

One may also have more exotic partial derivatives such as

$$\frac{\partial^4 \theta}{\partial t \partial x^2 \partial y}, \quad \frac{\partial^4 \theta}{\partial x^2 \partial y^2}, \quad \frac{\partial^5 \theta}{\partial x \partial y^4},$$

which may also be written in the forms, θ_{txxy} , θ_{xxyy} and θ_{xyyy} . In all cases these are evaluated by taking each single partial derivative in turn in any order.

Example 3.18: Find $\frac{\partial^4 \theta}{\partial x^2 \partial y^2}$ where $\theta = x^2(x + y)^2$.

We have

$$\begin{aligned}
 \theta &= x^2(x + y)^2 \\
 \Rightarrow \quad \frac{\partial \theta}{\partial y} &= 2x^2(x + y) \\
 \Rightarrow \quad \frac{\partial^2 \theta}{\partial y^2} &= 2x^2 \\
 \Rightarrow \quad \frac{\partial^3 \theta}{\partial x \partial y^2} &= 4x \\
 \Rightarrow \quad \frac{\partial^4 \theta}{\partial x^2 \partial y^2} &= 4.
 \end{aligned} \tag{3.43}$$

Note: yet again these four derivatives may be taken in any order.

3.13 Some examples of surfaces.

In the next section we shall begin to find out how to categorise critical points on surfaces. The extra spatial dimension over and above what we considered earlier makes some difference in how this categorisation takes place, and it also increases the variety of critical points that are available.



Figure 3.7. Glastonbury Tor.

This looks like the function given in Eq. (3.38) but only from a distance! It gets a lot more complicated when you get closer to it, including a ridge which cannot be seen here. The source for the photograph is here [May be seen online].

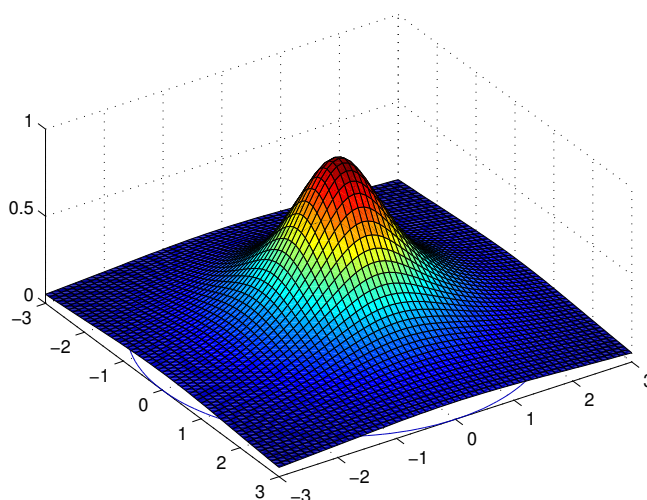


Figure 3.8. The function $h = \frac{1}{1 + x^2 + y^2}$.

This is the function given in Eq. (3.38) and is a stylized version of Glastonbury Tor. In this Figure we see that the surface has a single maximum point and every direction from that point takes us downwards. However, at that point the tangent surface is horizontal, i.e. the partial derivative in all possible directions is zero.

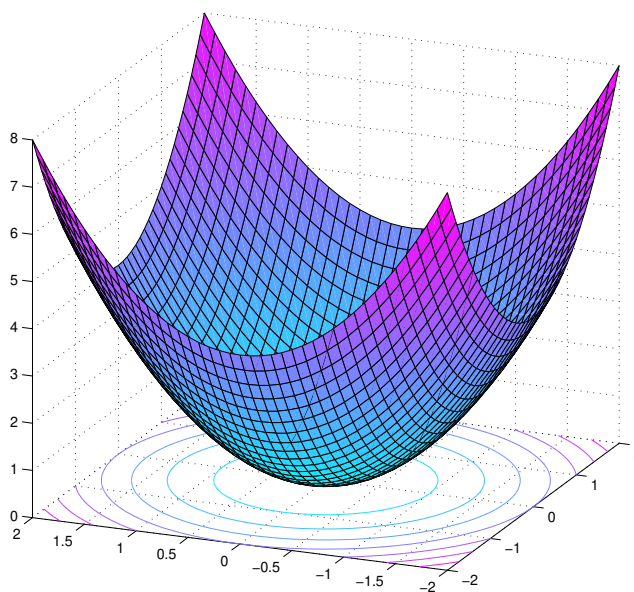
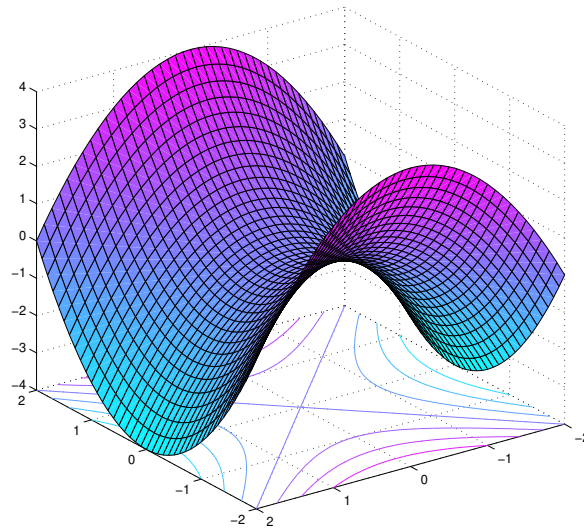


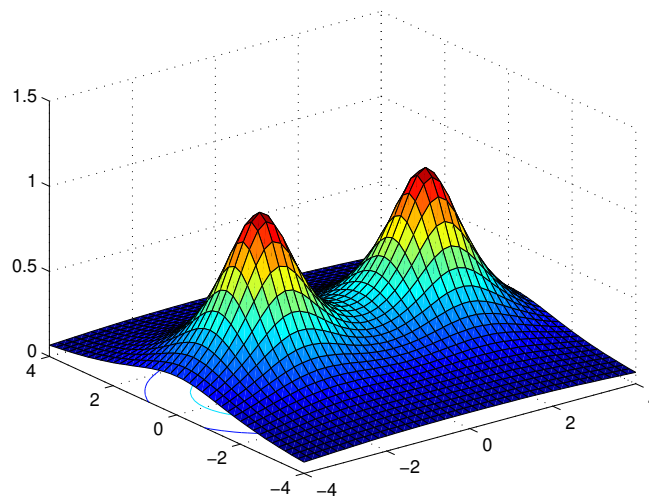
Figure 3.9. The function $h = x^2 + y^2$.

This function has a single minimum. Again, the partial derivative in any direction is zero at the minimum. Indeed, it will be sufficient to set both h_x and h_y to be zero **simultaneously** in order to find where the critical points are.

Figure 3.10. The function $h = x^2 - y^2$.

Ah, a Pringle! An alternative and more mathematical name is that it is a saddle point. This is the essential shape of a mountain pass or col. Under the surface is an indication of the contours of the surface. The two straight lines there both correspond to $h = 0$, and where they intersect is the critical point.

If we confine ourselves to a path on which $y = 0$ then $h = x^2$, and therefore this cross-section of the surface has a minimum at $(x, y) = (0, 0)$. If we confine ourselves to the path on which $x = 0$, then $h = -y^2$, and therefore the surface has a maximum at $(x, y) = (0, 0)$.

Figure 3.11. The function $h = \frac{1}{(x+2)^2 + y^2 + 1} + \frac{1}{(x-2)^2 + y^2 + 1}$.

This surface has two maxima, quite clearly, but we have a saddle point between them. For 2d graphs it is a minimum that tends to arise between two maxima.

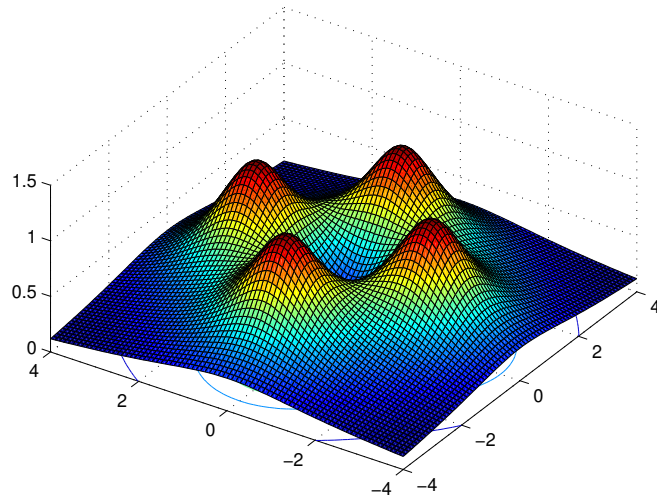


Figure 3.12. The function $h = \frac{1}{(x+1)^2 + y^2 + 1} + \frac{1}{(x-1)^2 + y^2 + 1} + \frac{1}{x^2 + (y+1)^2 + 1} + \frac{1}{x^2 + (y-1)^2 + 1}$.

So there are four maxima here with four saddle points and one minimum.

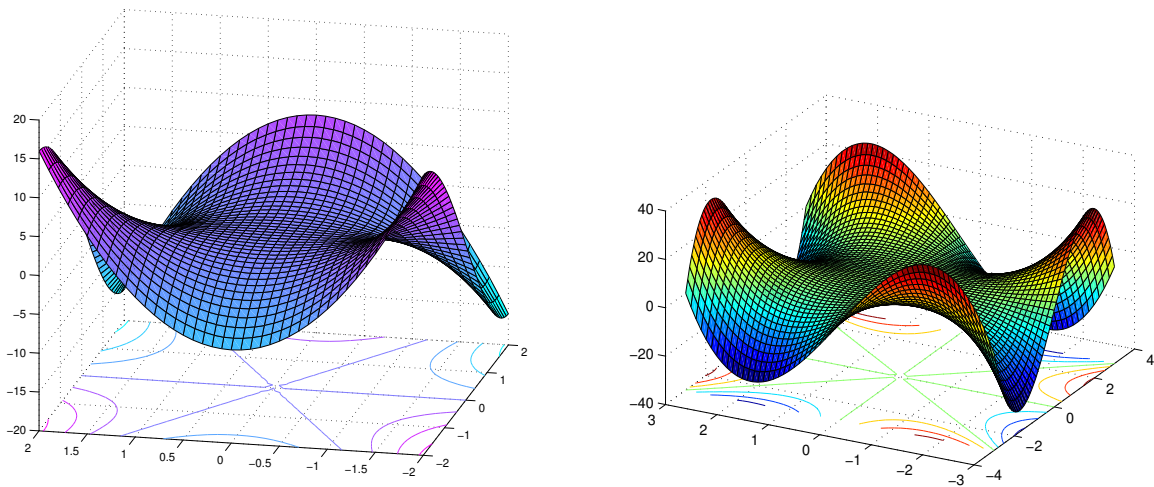


Figure 3.13. These functions may be called the Monkey Saddle (two legs and one tail!) and the Cat Saddle (four legs, no tail — presumably a Manx cat!).

The equations for these respective surfaces are,

$$h(x, y) = y^3 - 3x^2y = y(y - \sqrt{3}x)(y + \sqrt{3}x) \quad (3.44)$$

and

$$h(x, y) = xy(x^2 - y^2) = xy(x - y)(x + y). \quad (3.45)$$

Each has a solitary critical point at $(x, y) = (0, 0)$. Their structures are more complicated than the usual saddle point. Again, the contour lines beneath each surface help us to understand the surfaces a little more. For the Monkey saddle there are three directions in which the surface contains the line, $h = 0$, while it is four for the Cat saddle. For the Monkey saddle these are $y = 0$, $y = \sqrt{3}x$ and $y = -\sqrt{3}x$. For the Cat saddle they are $x = 0$, $y = 0$, $y = x$ and $y = -x$.

3.14 Critical points

As has already been mentioned, a critical point of a surface will satisfy both

$$\frac{\partial h}{\partial x} = 0 \quad \text{and} \quad \frac{\partial h}{\partial y} = 0 \quad (3.46)$$

simultaneously. These will yield a pair of equations to solve for the values of x and y . For example, the function,

$$h = x^2 + y^2,$$

has a well-defined minimum value at $x = y = 0$. If we set $h_x = 0$ and $h_y = 0$ then these yield $2x = 0$ and $2y = 0$, respectively, and hence the critical point is $(x, y) = (0, 0)$. Precisely the same conclusion is obtained when we analyse the Pringle function,

$$h = x^2 - y^2.$$

Note that this isn't its proper name!

Example 3.19: A more difficult example. Find the critical points for the function $h = xy(x + y - 1)$.

First, this function needs to be multiplied out: $h = x^2y + xy^2 - xy$. Now we set to zero the two first derivatives:

$$\frac{\partial h}{\partial x} = 2xy + y^2 - y = y(2x + y - 1) = 0 \quad (3.47)$$

and

$$\frac{\partial h}{\partial y} = x^2 + 2xy - x = x(x + 2y - 1) = 0. \quad (3.48)$$

Clearly there are two ways in which $h_x = 0$ and two in which $h_y = 0$. This gives us four combinations to look at:

$$\begin{array}{llll} y = 0 & \text{and} & x = 0 & \Rightarrow (x, y) = (0, 0), \\ y = 0 & \text{and} & x + 2y - 1 = 0 & \Rightarrow (x, y) = (1, 0), \\ 2x + y - 1 = 0 & \text{and} & x = 0 & \Rightarrow (x, y) = (0, 1), \\ 2x + y - 1 = 0 & \text{and} & x + 2y - 1 = 0 & \Rightarrow (x, y) = (\frac{1}{3}, \frac{1}{3}). \end{array} \quad (3.49)$$

therefore we get four possible critical points.

Example 3.20: Find the critical points for $h = (y - x^2)(1 - y)$.

The previous example had four possible combinations to satisfy and we obtained four critical points. However, the number of combinations isn't always a guide for how many critical points there are. This is an example of that. We have $h = y - y^2 - x^2 + yx^2$, hence

$$\frac{\partial h}{\partial x} = 2x(y - 1), \quad \text{and} \quad \frac{\partial h}{\partial y} = 1 - 2y + x^2. \quad (3.50)$$

Therefore we have only two combinations to try:

$$\begin{array}{ll} x = 0 & \text{and} \quad 1 - 2y + x^2 = 0 \Rightarrow (x, y) = (0, \frac{1}{2}), \\ y = 1 & \text{and} \quad 1 - 2y + x^2 = 0 \Rightarrow x^2 = 1 \Rightarrow (x, y) = (\pm 1, 1). \end{array} \quad (3.51)$$

So we have three critical points for this example.

3.15 Classification of the critical points.

Naïvely it may be thought that we could apply the obvious extension to the ordinary differential rules for determining whether an extremum is a maximum or a minimum, namely by checking the signs of the second partial derivatives with respect to x and y .

This strategy will work with the examples shown in Fig. 3.8 (Glastonbury Tor) and Fig. 3.9 (the paraboloid), although the evaluation of the second derivatives for Fig. 3.8 is quite lengthy. However, things begin to fall apart with the Pringle example. The equation is $h = x^2 - y^2$ and we get $h_{xx} = 2$ and $h_{yy} = -2$. The opposite signs here do reflect earlier comments about having a maximum in one direction and a minimum in the other, but things aren't so simple, unfortunately.

The simplest demonstration I can think of involves rotating the Pringle a little. The present one may be rewritten as $h = (x + y)(x - y)$ and therefore the lines on which $h = 0$ are $y = \pm x$. A different combination of lines is $h = xy$ where $h = 0$ on $x = 0$ and $y = 0$, and if one were to plot this out it would be similar to the first but rotated through 45° . Again the critical point is at the origin, but $h_{xx} = h_{yy} = 0$ at $(x, y) = (0, 0)$. Therefore our simple-minded approach involving the values of h_{xx} and h_{yy} is incorrect.

If we determine the value of

$$H = \frac{\partial^2 h}{\partial x^2} \frac{\partial^2 h}{\partial y^2} - \left(\frac{\partial^2 h}{\partial x \partial y} \right)^2, \quad (3.52)$$

then

$$\begin{array}{ll} H < 0 & \Rightarrow \text{saddle point,} \\ H > 0 \text{ and } \frac{\partial^2 h}{\partial x^2} > 0 & \Rightarrow \text{minimum,} \\ H > 0 \text{ and } \frac{\partial^2 h}{\partial x^2} < 0 & \Rightarrow \text{maximum,} \\ H = 0 & \Rightarrow \text{inconclusive.} \end{array} \quad (3.53)$$

Note: If $H > 0$ then h_{xx} and h_{yy} must have the same sign.

In Eqs. (3.52) and (3.53) all the derivatives are evaluated at the critical point which is being tested. I have called the test function, H , because its official name is the two-dimensional Hessian. The proof of this more complicated test is quite detailed and is omitted here. However, Eqs. (3.52) and (3.53) do not appear in the University Formula Book and will need to be memorised for a closed-book examination.

Here follows quite a few example cases, the first few of which have only one critical point. Some parts of the following are repetitions of some material above, but I have included the full analysis for the sake of having complete examples here.

Example 3.21: Find the critical points of the function, $h = x^2 + y^2$, and classify them.

We find that $h_x = 2x$ and $h_y = 2y$. This means that the sole critical point is $(x, y) = (0, 0)$.

The three second derivatives are $h_{xx} = 2$, $h_{yy} = 2$ and $h_{xy} = 0$. Therefore $H = 4$. Given that this is positive and that both h_{xx} and h_{yy} are positive, this means that the critical point is a minimum.

Example 3.22: Find the critical points of the function, $h = x^2 - y^2$, and classify them.

We find that $h_x = 2x$ and $h_y = -2y$. This means that the sole critical point is $(x, y) = (0, 0)$.

The three second derivatives are $h_{xx} = 2$, $h_{yy} = -2$ and $h_{xy} = 0$. Therefore $H = -4$. A negative value means that it is a saddle point.

Example 3.23: Find the critical points of the function, $h = xy$, and classify them. This is the alternative saddle point which was discussed a little earlier.

We find that $h_x = y$ and $h_y = x$. This means that the sole critical point is $(x, y) = (0, 0)$.

The three second derivatives are $h_{xx} = 0$, $h_{yy} = 0$ and $h_{xy} = 1$. Therefore $H = h_{xx}h_{yy} - (h_{xy})^2 = -1$. A negative value means that it is a saddle point.

Example 3.24: Find the critical points of the function, $h = 1/(x^2 + y^2 + 1)$, and classify them. This is the idealized Glastonbury Tor surface.

We find that $h_x = -2x/(x^2 + y^2 + 1)^2$ and $h_y = -2y/(x^2 + y^2 + 1)^2$. This means that the sole critical point is $(x, y) = (0, 0)$.

The three second derivatives are

$$h_{xx} = \frac{2(3x^2 - y^2 - 1)}{(x^2 + y^2 + 1)^3} = -2 \quad h_{yy} = \frac{2(3y^2 - x^2 - 1)}{(x^2 + y^2 + 1)^3} = -2, \quad h_{xy} = \frac{8xy}{(x^2 + y^2 + 1)^3} = 0.$$

Therefore $H = h_{xx}h_{yy} - (h_{xy})^2 = 4$. As H is positive and both h_{xx} and h_{yy} are negative, this implies that the critical point is a maximum.

Example 3.25: While the function, $h = x^2 + y^2$, has a minimum, what can we say about $h = x^2 + y^2 + \frac{5}{2}xy$? Does the addition of the xy term change the nature of the critical point?

We find that $h_x = 2x + \frac{5}{2}y$ and $h_y = 2y + \frac{5}{2}x$. This means yet again that the sole critical point is $(x, y) = (0, 0)$.

The three second derivatives are $h_{xx} = 2$, $h_{yy} = 2$ and $h_{xy} = \frac{5}{2}$, and therefore we get $H = 4 - \frac{25}{4} = -\frac{9}{4}$, which corresponds to a saddle point.

This example shows that one must not be guided by a quick look at the signs of the coefficients of x^2 and y^2 . In fact, this function may be factorised and rearranged to give $h = (y + 2x)(y + \frac{1}{2}x)$, and it is now quite clearly a saddle point example, being the product of linear factors.

Example 3.26: Consider the function $h = y^3 - 3x^2y$.

Given that $h_x = -6xy$ and $h_y = 3y^2 - 3x^2$, the only point at which both these derivatives are zero is $x = y = 0$. At this point we find that $h_{xx} = -6y = 0$, $h_{yy} = 6y = 0$ and $h_{xy} = -6x = 0$. Therefore $H = f_{xx}f_{yy} - (f_{xy})^2 = 0$, and we have a degenerate case which isn't covered above. This is the Monkey saddle example.

This is an unusual case for which some further analysis could be undertaken even if just for interest's sake. For example, one could see what happens to $h(x, y)$ along the line $y = \alpha x$. If this is substituted into the

expression for h , we get

$$h = \alpha(\alpha^2 - 3)x^3,$$

which shows that the origin is a point of inflexion when travelling in the direction of this line (the x^3 shows this). Just this information helps us to understand why we have a degenerate case. Further, $h = 0$ when $\alpha = 0, \pm\sqrt{3}$, and these corresponds to the directions of the straight lines in the contours of Fig. 3.13.

Example 3.27: Find the critical points of $h = x^2y + xy^2 - xy$ and classify them.

We find that,

$$h_x = 2xy + y^2 - y = y(2x + y - 1), \quad h_y = 2xy + x^2 - x = x(x + 2y - 1).$$

On setting both these derivatives to zero, we obtain four possible outcomes:

$$(x, y) = (0, 0), \quad (x, y) = (0, 1), \quad (x, y) = (1, 0), \quad (x, y) = \left(\frac{1}{3}, \frac{1}{3}\right).$$

Given that

$$h_{xx} = 2y, \quad h_{yy} = 2x, \quad h_{xy} = 2x + 2y - 1,$$

we obtain the following Table of data/information.

(x, y)	h_{xx}	h_{yy}	h_{xy}	H	Category
$(0, 0)$	0	0	1	-1	Saddle
$(0, 1)$	2	0	1	-1	Saddle
$(1, 0)$	0	2	1	-1	Saddle
$\left(\frac{1}{3}, \frac{1}{3}\right)$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	Minimum

(3.54)

The first three are clearly saddle points, while the fourth is a minimum due to the fact that $h_{xx} = \frac{2}{3} > 0$ at that point.

Note: Please note how I have laid out the above Table. This is perhaps the quickest and safest way of dealing with the data for these problems. It is the easiest way for you to check your work, and also the easiest for me to mark! This is a win/win situation where both you and I benefit!

Example 3.28: This is the last example, quite long but not too tricky, I hope. Find the critical points of the function $h = xy(x^2 + y^2 - 4)$ and classify them.

We find that,

$$h_x = 3x^2y + y^3 - 4y = y(3x^2 + y^2 - 4) \quad h_y = 3y^2x + x^3 - 4x = x(3y^2 + x^2 - 4).$$

On setting both these derivatives to zero, we obtain four possible combinations of outcomes:

$$\begin{aligned} y = 0 \quad \text{and} \quad x = 0 &\Rightarrow (x, y) = (0, 0), \\ y = 0 \quad \text{and} \quad 3y^2 + x^2 - 4 = 0 &\Rightarrow (x, y) = (\pm 2, 0), \\ 3x^2 + y^2 - 4 = 0 \quad \text{and} \quad x = 0 &\Rightarrow (x, y) = (0, \pm 2), \\ 3x^2 + y^2 - 4 = 0 \quad \text{and} \quad 3y^2 + x^2 - 4 = 0 &\Rightarrow (x, y) = (\pm 1 \pm 1) \text{ all four combinations.} \end{aligned}$$

Therefore we have nine critical points!

Given that

$$h_{xx} = 6xy, \quad h_{yy} = 6xy, \quad h_{xy} = 3x^2 + 3y^2 - 4,$$

we obtain the following Table of data/information.

(x, y)	h_{xx}	h_{yy}	h_{xy}	H	Category
$(0, 0)$	0	0	-4	-16	Saddle
$(2, 0)$	0	0	8	-64	Saddle
$(-2, 0)$	0	0	8	-64	Saddle
$(0, 2)$	0	0	8	-64	Saddle
$(0, -2)$	0	0	8	-64	Saddle
$(1, 1)$	6	6	2	32	Minimum ($h_{xx} > 0$)
$(-1, -1)$	6	6	2	32	Minimum ($h_{xx} > 0$)
$(1, -1)$	-6	-6	2	32	Maximum ($h_{xx} < 0$)
$(-1, 1)$	-6	-6	2	32	Maximum ($h_{xx} < 0$)

(3.55)

Perhaps there are questions being framed about why there should be quite so many critical points — a good question, one which I will answer so that one can gain an understanding of what is going on behind the scenes. The following is not examinable.

In the present example, $h = xy(x^2 + y^2 - 4)$, and the loci where $h = 0$ are the lines, $x = 0$ and $y = 0$ and the circle $x^2 + y^2 = 4$ (i.e. one of radius 2 centred at the origin). We may plot these in the (x, y) -plane.

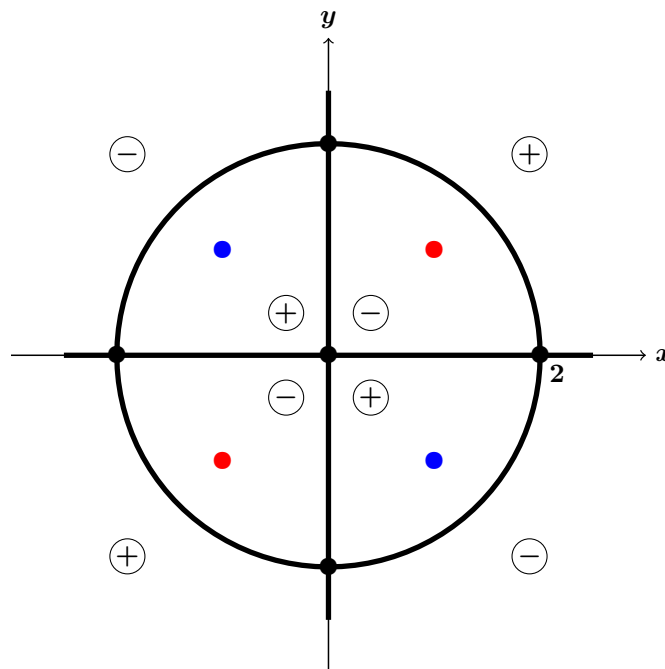


Figure 3.14. Showing where $h = xy(x^2 + y^2 - 4)$ takes the value zero, together with the locations of saddle points ($\bullet$), minima ($\bullet$) and maxima ($\bullet$). The encircled + and - signs show the sign of h within each subregion.

As we saw in Fig. 3.10, the saddle point appears when two lines corresponding to $h = 0$ cross. In Fig. 3.14 we see five places where two such lines/curves cross. If we now consider the interior of the circle which lies in the upper right quadrant, then h takes negative values. Given that these values are contained within a quadrant

whose boundaries correspond to $h = 0$, then there must be a minimum value in the interior. Therefore we have established the existence of that minimum, although it takes the full analysis above to determine where it is. This type of analysis provides a good qualitative indication of how many critical points exist and what type each one is. These ideas may also be used for Example 3.27, where it helps to factorise h first.

4 Integration

If we are given a function, $f(x)$, then the area, A , between the function and the x -axis and which lies between $x = a$ and $x = b$ is given by the yellow region below.

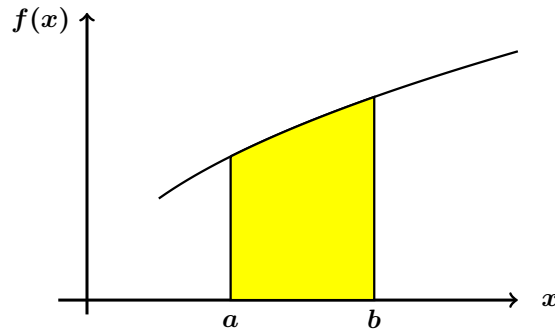


Figure 4.1. Showing the definition of the integral between $x = a$ and $x = b$ as being equivalent to the yellow area under the graph.

It is known as the “**area under graph between a and b** ” or as the “**integral of $f(x)$ between a and b** ”. This is denoted by

$$A = \int_a^b f(x) dx. \quad (4.1)$$

The notation, $\int$, represents a summation of the areas of small strips, as sketched below:

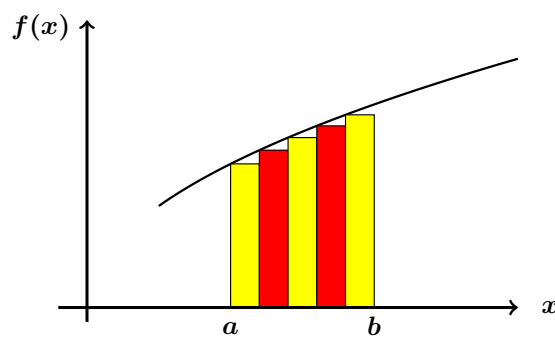


Figure 4.2. Showing how the integral between $x = a$ and $x = b$ may be approximated using 5 strips of equal width.

In this Figure each strip has width $\frac{1}{5}(b - a)$, but if we say more generally that each strip has width, δx , and that the height of each strip is $f(x_i)$, where $x_i = a + i \delta x$, then the area under the curve in this general version of Fig. 4.2 may be approximated by

$$A \simeq \sum_{i=0}^{N-1} f(x_i) \delta x. \quad (4.2)$$

As with the derivative, the integral may be written in terms of a limiting process where

$$A = \lim_{\delta x \rightarrow 0} \sum_{i=0}^{N-1} f(x_i) \delta x, \quad (4.3)$$

and where the number of strips is given by $N = (b - a)/\delta x$. So $N \rightarrow \infty$ as $\delta x \rightarrow 0$. In Fig. 4.3, which shows 20 strips, it is shown how much better Eq. (4.3) approximates the integral.

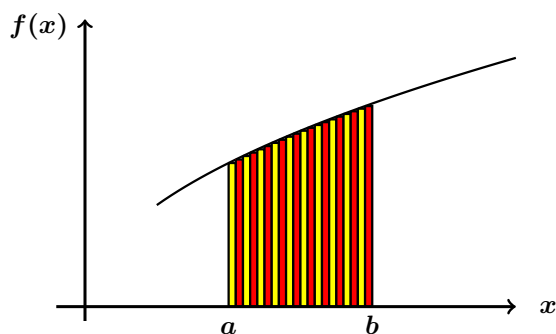


Figure 4.3. Showing the definition of the integral between $x = a$ and $x = b$ as being equivalent to the area under the graph.

The right hand side of Eq. (4.1) is said to be the **integral of $f(x)$ between $x = a$ and $x = b$** . The function $f(x)$ is called the **integrand**, while x is the **variable of integration**.

Note that, since the value of this integral is the constant value, A , then the symbol we use for the variable of integration is irrelevant. Hence

$$\int_a^b f(x) dx = \int_a^b f(y) dy. \quad (4.4)$$

For this reason we may call x a **dummy variable** because the final result is independent of its identity.

4.1 Relationship with the derivative [For interest and background]

Formally, integration and differentiation are mutually inverse operations. At first glance this seems like madness, for what does the area under a graph have to do with the slope of a graph?

This question may be answered by first defining $I(x)$ according to,

$$I(x) = \int_a^x f(y) dy. \quad (4.5)$$

Note the different viewpoint here: this is an integral, or area under the graph, between $y = a$ and $y = x$. We assume that a is a constant. The upper limit, x , is allowed to vary and therefore the value of the integral depends on the value of x . So $I(x)$ is a function of x .

We will now attempt to find the derivative of $I(x)$ using the limiting definition of the derivative which was given in Eq. (3) in the Differentiation notes. Therefore the derivative of $I(x)$ is defined according to,

$$\frac{dI}{dx} = \lim_{\delta x \rightarrow 0} \left[\frac{I(x + \delta x) - I(x)}{\delta x} \right], \quad (4.6)$$

or,

$$\frac{dI}{dx} = \lim_{\delta x \rightarrow 0} \frac{1}{\delta x} \left[\int_a^{x+\delta x} f(y) dy - \int_a^x f(y) dy \right]. \quad (4.7)$$

If we now consider our original definition of an integral as an area under a graph, then this expression reduces to the integral of a narrow strip (at least when δx is small):

$$\frac{dI}{dx} = \lim_{\delta x \rightarrow 0} \left[\frac{1}{\delta x} \int_x^{x+\delta x} f(y) dy \right]. \quad (4.8)$$

This transition from Eq. (4.7) to Eq. (4.8) is illustrated in Fig. 4.4:

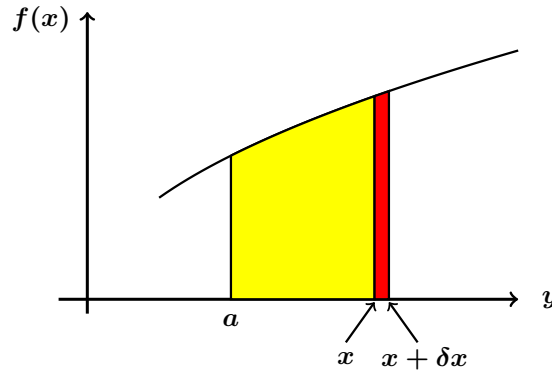


Figure 4.4. Illustrating the process behind finding the derivative of the area defined by Eq. (4.5).

Here $\int_a^x f(y) dy$ corresponds to the yellow region, $\int_a^{x+\delta x} f(y) dy$ to both the red and yellow regions, and $\int_x^{x+\delta x} f(y) dy$ to the red strip. When δx is very small, the thin red strip has an area which is well-approximated by a trapezium of width δx , and of sides $f(x)$ and $f(x + \delta x)$. Therefore the integral in Eq. (4.8) is, to a very good degree of approximation equal to,

$$\int_x^{x+\delta x} f(y) dy \simeq \frac{f(x + \delta x) + f(x)}{2} \delta x. \quad (4.9)$$

Therefore we have

$$\frac{dI}{dx} = \lim_{\delta x \rightarrow 0} \left[\frac{1}{\delta x} \left(\frac{f(x + \delta x) + f(x)}{2} \delta x \right) \right] = f(x). \quad (4.10)$$

Note 1: Should $I(x) = c + \int_a^x f(y) dy$ where c is an arbitrary constant, then an identical analysis also yields Eq. (4.10) simply because the derivative of c is zero.

It is this result that tells us that integration and the taking of derivatives are mutually inverse operations. Therefore the art of integration is the determination of a function which, when differentiated, yields the original function. So the integral of $3x^2$ is $x^3 + c$ because the derivative of $x^3 + c$ is $3x^2$. In mathematical terms we write,

$$\int 3x^2 dx = x^3 + c \quad (4.11)$$

where an integral which is written without limits is generally called an **indefinite integral** and is sometimes called the **antiderivative**. On the other hand, an integral with limits is called a **definite integral**.

Table of some common indefinite integrals

$$\begin{aligned}
\int \cos ax \, dx &= \frac{\sin ax}{a} + c, \\
\int \sin ax \, dx &= -\frac{\cos ax}{a} + c, \\
\int x^n \, dx &= \frac{x^{n+1}}{n+1} + c \quad (n \neq -1), \\
\int \frac{1}{x} \, dx &= \ln |x| + c, \\
\int e^{ax} \, dx &= \frac{e^{ax}}{a} + c.
\end{aligned}$$

Note 2: Protocol for the evaluation of a definite integral. If we take the expression,

$$I(x) = \int_a^x f(y) \, dy + c, \quad (4.12)$$

then the respective setting of $x = a$ and of $x = b$ yields the following,

$$\begin{aligned}
I(a) &= \int_a^a f(y) \, dy + c = c && \text{identical limits} \Rightarrow \text{zero area} \\
I(b) &= \int_a^b f(y) \, dy + c
\end{aligned} \quad (4.13)$$

By eliminating c we obtain,

$$\int_a^b f(y) \, dy = I(b) - I(a), \quad (4.14)$$

which will, no doubt, be familiar. And this result tells us immediately that swapping the limits is effectively the same as multiplication by -1 :

$$\int_b^a f(y) \, dy = I(a) - I(b) = -\int_a^b f(y) \, dy. \quad (4.15)$$

Example 4.1. Given that the indefinite integral of $3x^2$ is $x^3 + c$ (see Eq. (4.11)) find the definite integral between $x = 2$ and $x = 5$.

This is,

$$\int_2^5 3x^2 \, dx = \left[x^3 + c \right]_2^5 = (5^3 + c) - (2^3 + c) = 117. \quad (4.16)$$

This example also tells us that we don't need to use the arbitrary constant when evaluating definite integrals, i.e. we are, for once, allowed to be lazy and not bother with the arbitrary constant.

4.2 Linearity Rules

Integration obeys the same linearity rules as differentiation:

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx, \quad (4.17)$$

$$\int k f(x) dx = k \int f(x) dx \quad \text{where } k \text{ is a constant.} \quad (4.18)$$

Therefore we may interchange the order of summation and integration and the order of multiplication by a constant and integration. Here's an example.

Example 4.2.

$$\begin{aligned} \int (x^2 - 2x + 1) dx &= \int x^2 dx + \int (-2)x dx + \int 1 dx && \text{using (4.17)} \\ &= \int x^2 dx + (-2) \int x dx + \int 1 dx && \text{using (4.18)} \\ &= \frac{1}{3}x^3 - x^2 + x + c. \end{aligned}$$

This was long-winded and pedantic; in practice one should aim to go directly to the answer when the integrand consists of functions like those in the above Table.

4.3 Integration by substitution

Very few functions that we need to integrate take the precise form given in the above examples and exercises. Although differentiation may always be done even if it is lengthy, it is often the case that it can be quite a puzzle to find out how to do some integrations. This is what makes the topic more tricky than differentiation. Indeed, there are some integrals that cannot be done, an example being that of e^{-x^2} .

We will illustrate the "Integration by substitution" method by means of examples.

Example 4.3. Evaluate $I = \int \cos(\omega x) dx$.

We know how to integrate $\cos x$, but we will pretend not to know how deal with $\cos \omega x$, and therefore our aim will be to reduce the integrand to the form we know. This aim is achieved using the substitution, $\xi = \omega x$. We may now alter the integration by means of the following equivalent of the chain rule:

$$\begin{aligned} I = \int \cos(\omega x) dx &= \int \cos \xi \frac{dx}{d\xi} d\xi \\ &= \int \frac{\cos \xi}{\omega} d\xi \\ &= \frac{1}{\omega} \int \cos \xi d\xi && \text{taking the constant outside the integral} \quad (4.19) \\ &= \frac{\sin \xi}{\omega} + c && \text{on integration} \\ &= \frac{\sin \omega x}{\omega} + c && \text{returning to the original variable.} \end{aligned}$$

The equivalent of the chain rule is involved in the step with the red font. Generally, though, most people do not follow the above method even though it looks more rigorous and reasonable than the method which is usually taught and which I will describe now.

Given that $\xi = \omega x$ we may deduce that $d\xi = \omega dx$ by differentiation of both sides — essentially this says if we change x to $x + \delta x$, where δx is small, then we expect ξ to change too. Hence, $\xi + \delta\xi = \omega(x + \delta x)$, and hence $\delta\xi = \omega \delta x$.

We now have all the information required to change variable from x to ξ . Therefore the above integration-by-substitution analysis becomes the following.

$$\begin{aligned}
 I &= \int \cos \omega x \, dx \\
 &= \int \cos \xi \, \frac{d\xi}{\omega} \quad \text{the substitution, all in one go. } \left(dx = \frac{d\xi}{\omega} \right) \\
 &= \frac{1}{\omega} \int \cos \xi \, d\xi \\
 &= \frac{\sin \xi}{\omega} + c \\
 &= \frac{\sin(\omega x)}{\omega} + c. \quad \text{returning from } \xi \text{ to } x
 \end{aligned} \tag{4.20}$$

Example 4.4. Evaluate $I = \int 2x \sin(x^2) \, dx$.

We know how to evaluate the integral of $\sin x$, but $\sin(x^2)$ is not one of the standard integrals. However, we may make this part of the integrand integrable by using the substitution $y = x^2$. From this we have $dy = 2x \, dx$ where the $2x$ is the derivative of x^2 ; this also follows from the small-increment analysis: $y + \delta y = (x + \delta x)^2 = x^2 + 2x\delta x + \dots$

Hence the integrand immediately transforms to

$$I = \int \underbrace{\sin(x^2)}_{\sin y} \underbrace{2x \, dx}_{dy} = \int \sin(y) \, dy = -\cos(y) + c = -\cos(x^2) + c. \tag{4.21}$$

Example 4.5. Find $I = \int \sin^2(t) \cos(t) \, dt$.

As $\cos t$ is the differential of $\sin t$, and there is only one $\cos t$, we shall let $z = \sin t$ and hence $dz = \cos t \, dt$. The integral becomes,

$$I = \int \underbrace{\sin^2 t}_{z^2} \underbrace{\cos t \, dt}_{dz} = \int z^2 \, dz = \frac{1}{3} z^3 + c = \frac{1}{3} \sin^3 t + c. \tag{4.22}$$

Of course, in all of these cases one may check the final answer by differentiating it.

Example 4.5 (reprise). Find $I = \int_0^{\pi/2} \sin^2(t) \cos(t) \, dt$.

This rerun of Example 4.5 will show us how important it is to consider carefully the limits of a definite integral when using the substitution method. I'll pretend that Example 4.5 hasn't happened...

We have the following pieces of information which will need to be used:

$$z = \sin t \Rightarrow dz = \cos t \, dt.$$

$$t = 0 \Rightarrow z = 0.$$

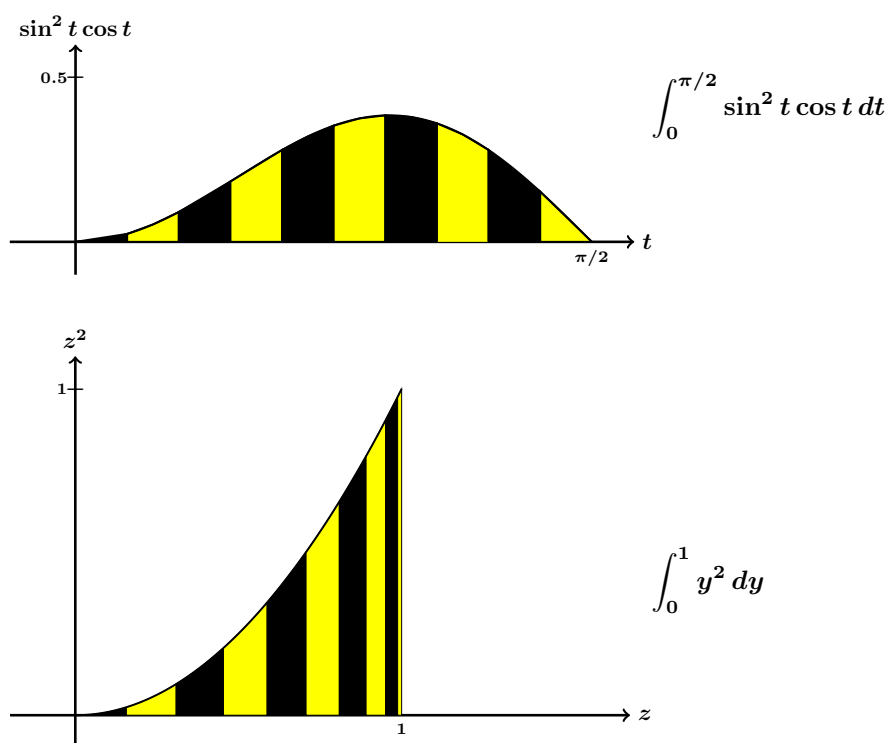
$$t = \frac{1}{2}\pi \Rightarrow z = 1.$$

Now we may transform the given integral immediately from being a t -integral to being a z -integral:

$$I = \int_0^{\pi/2} \sin^2 t \cos t \, dt = \int_0^1 z^2 \, dz = \left[\frac{1}{3}z^3 + c \right]_0^1 = \frac{1}{3}. \quad (4.23)$$

Of course, this final numerical value could have been obtained easily by using the final answer in Eq. (4.22) and evaluating that between $t = 0$ and $t = \frac{1}{2}\pi$.

Although the integrands and the limits in two integrals in Eq. (4.22) are so very different, the fact is that the overall value of the integral is conserved. We may illustrate this fairly well in the following figure.



Showing the manner in which areas are conserved when using Integration-by-Substitution, as given by the reprise of Example 4.5.

Each of the integrals shown above is split into ten intervals. In the t -integral the strips have equal widths and the edges of these strips are located at $t = (n/10) \times (\pi/2) = n\pi/20$ for $n = 0, 1, \dots, 9, 10$. This yields ten separate regions that are illustrated as alternating black and yellow strips. On the other hand the edges of the strips in the z -integral correspond to where $z = \sin(n\pi/20)$, which is why they are of unequal widths. However, each of the ten strips for the z -integral has an equal area to the corresponding strip in the t -integral. This looks very feasible for the first two or three strips (working from the left) but it remains true even for the very last strips on the right where the two tenth strips have very different aspect ratios.

Example 4.6. Find the value of $I = \int_0^{\sqrt{\pi}} 2x \sin(x^2) dx$.

The integrand is another reprise, but of Example 4.4, above. Using the notation of Example 4.4, when $x = 0$ then $y = 0$, and when $x = \sqrt{\pi}$ then $y = \pi$. Therefore the present integral is

$$\begin{aligned}
 I &= \int_0^{\sqrt{\pi}} 2x \sin(x^2) dx \\
 &= \int_0^{\pi} \sin y dy && \text{the } x \text{ to } y \text{ substitution, including the limits} \\
 &= [-\cos y]_0^{\pi} = 2.
 \end{aligned} \tag{4.24}$$

Note: It is essential that such integrals use only one dummy variable at any time — there must never be a “half-way house” containing both x and y simultaneously — if that were to happen, then which are the correct limits, those belonging to x or those for y ? The transformation from one dummy variable to the other must take place instantaneously and then the maths is completely unambiguous.

An alternative route which avoids transforming the limits of integration is as follows. First find the indefinite integral using substitution, then write the answer in terms of the original variable, and finally apply the limits. So here we would have the following:

$$\begin{aligned}
 \text{Let } I &= \int 2x \sin(x^2) dx \\
 &= \int \sin y dy && \text{the } x \text{ to } y \text{ substitution} \\
 &= [-\cos y] \\
 &= -\cos(x^2) && \text{back to } x. \text{ No need for the arbitrary constant}
 \end{aligned} \tag{4.25}$$

$$\begin{aligned}
 \text{Hence } \int_0^{\sqrt{\pi}} 2x \sin(x^2) dx \\
 &= [-\cos(x^2)]_0^{\sqrt{\pi}} = 2.
 \end{aligned}$$

Example 4.7. Find $I = \int \frac{x}{\sqrt{1-x^2}} dx$.

This integral may be obtained using either the substitution, $y = x^2$ or $y = 1 - x^2$, but I will use a third substitution, $x = \sin \theta$. This one is motivated by the fact that the denominator of the integrand will transform to $\cos \theta$, while the dx part will also involve a $\cos \theta d\theta$, and there will be the potential for cancellation. Given our chosen transformation, we have

$$x = \sin \theta \quad \Rightarrow \quad dx = \cos \theta d\theta,$$

and hence

$$\begin{aligned}
 I &= \int \frac{x}{\sqrt{1-x^2}} dx = \int \frac{\sin \theta}{\sqrt{1-\sin^2 \theta}} \cos \theta d\theta && \text{the substitution} \\
 &= \int \frac{\sin \theta \cos \theta}{\cos \theta} d\theta && \text{tidying up} \\
 &= \int \sin \theta d\theta && \text{some cancellation} \\
 &= -\cos \theta + c = -\sqrt{1-x^2} + c.
 \end{aligned} \tag{4.26}$$

In the last line of this evaluation of I we chose $\cos \theta = +\sqrt{1-\sin^2 \theta}$ rather than $-\sqrt{1-\sin^2 \theta}$. The correctness of this choice may (and perhaps should) be tested by differentiating the final answer.

If we were to consider the very similar integral, $I = \int \frac{x}{\sqrt{a^2-x^2}} dx$, then we could use $x = a \sin \theta$.

Example 4.8. Find the integral of $\int \frac{1}{x^2+a^2} dx$.

This is a classic integral. The secret here is try find a substitution which, like the previous one, employs a trigonometric identity to simplify the integral. This one will be based on the result that $\sec^2 \theta = 1 + \tan^2 \theta$. Given the a^2 in the integral, we shall use the substitution,

$$x = a \tan \theta \quad \Rightarrow \quad dx = a \sec^2 \theta d\theta. \tag{4.27}$$

Hence,

$$\int \frac{1}{x^2+a^2} dx = \int \frac{a \sec^2 \theta}{a^2(1+\tan^2 \theta)} d\theta = \int \frac{a \sec^2 \theta}{a^2 \sec^2 \theta} d\theta = \int \frac{1}{a} d\theta = \frac{1}{a} \theta + c = \frac{1}{a} \tan^{-1} \frac{x}{a}. \tag{4.28}$$

Example 4.9. $I = \int \frac{1}{x^2+2x+2} dx$.

Given that the denominator may be rewritten as $(x+1)^2+1$ by the process of completing the square, we see that the integral is in the same category as the previous one. So we set,

$$x+1 = \tan \theta \quad \Rightarrow \quad dx = \sec^2 \theta d\theta,$$

and hence,

$$\begin{aligned}
 I &= \int \frac{1}{x^2+2x+2} dx \\
 &= \int \frac{1}{(x+1)^2+1} dx && \text{using the completing of the square} \\
 &= \int \frac{\sec^2 \theta}{\sec^2 \theta} d\theta && \text{using } x+1 = \tan \theta \\
 &= \theta + c \\
 &= \tan^{-1}(x+1) + c. && \text{returning to } x.
 \end{aligned} \tag{4.29}$$

Example 4.10. $I = \int \frac{1}{x^2 + 4x + 13} dx$.

Another on the same theme. The denominator may be rewritten as $(x + 2)^2 + 9$ and therefore we need the substitution,

$$x + 2 = 3 \tan \theta \quad \Rightarrow \quad dx = 3 \sec^2 \theta d\theta,$$

and hence,

$$\begin{aligned} I &= \int \frac{1}{x^2 + 4x + 13} dx \\ &= \int \frac{1}{(x + 2)^2 + 3^2} dx && \text{using the completing of the square} \\ &= \int \frac{3 \sec^2 \theta}{9 \sec^2 \theta} d\theta && \text{using } x + 2 = 3 \tan \theta \\ &= \frac{1}{3} \theta + c \\ &= \frac{1}{3} \tan^{-1} \left(\frac{1}{3}(x + 2) \right) + c && \text{returning to } x. \end{aligned} \tag{4.30}$$

Example 4.11. Evaluate $I = \int \frac{f'(x)}{f(x)} dx$.

Here $f(x)$ is an unknown function of x . This particular example of an integral is very important indeed and the final result is frequently used as a *method* in and of itself. The evaluation of I follows from the substitution, $y = f(x)$, and hence $dy = f'(x) dx$. The integral transforms to

$$I = \int \frac{1}{y} dy = \ln |y| + c = \ln |f(x)| + c. \tag{4.31}$$

This general integral illustrates what is often called the *f dashed over f* form. So if it is recognised that the numerator is the derivative of the denominator, then this result may be applied immediately. The following example uses this.

Example 4.12. Determine $I = \int \frac{2x + 2}{x^2 + 2x + 2} dx$.

This has the same denominator as the one in Example 4.9, but we note first that the integrand has the precise form, f'/f where $f = x^2 + 2x + 2$. Hence,

$$I = \ln |x^2 + 2x + 2| + c. \tag{4.32}$$

Example 4.13. It is rare that an integrand has the precise f'/f form but it may be close. This example illustrates such a case: Determine $I = \int \frac{x}{x^2 + 2x + 2} dx$.

The numerator, x , may be coerced to yield an integral which involves cases that we have already seen in Examples 4.9 and 4.11. Here we go....

$$\begin{aligned}
I &= \int \frac{x}{x^2 + 2x + 2} dx \\
&= \frac{1}{2} \int \frac{\textcolor{red}{2}x + \textcolor{red}{2} - 2}{x^2 + 2x + 2} dx && \text{red is the } f'/f \text{ case, black is Ex. 4.9} \\
&= \frac{1}{2} \int \frac{2x + 2}{x^2 + 2x + 2} dx - \int \frac{1}{x^2 + 2x + 2} dx && \text{to get the two known forms} \\
&= \frac{1}{2} \ln |x^2 + 2x + 2| - \tan^{-1}(x + 1) + c.
\end{aligned} \tag{4.33}$$

Example 4.14. Find $\int \cot x \, dx$.

Given that $\cot x = \cos x / \sin x$ then the integrand is already in f'/f form. Hence the integral is $I = \ln |\sin x| + c$.

4.4 Integration using Partial Fractions

The technique of partial fractions may be used to integrate functions which are the ratio of two polynomials such as $1/(1 - x^2)$ and $(x - 1)/(x^3 + 1)$. It begins with the factorisation of the denominator of the fraction, and the original function then gets split into smaller and more easily integrable parts.

Example 4.15. Determine $I = \int \frac{1}{1 - x^2} dx$.

We proceed by first reducing the integrand to more manageable bits (i.e. the more easily integrable bits).

$$\text{Let } \frac{1}{1 - x^2} = \frac{1}{(1 - x)(1 + x)} = \frac{A}{1 - x} + \frac{B}{1 + x}. \tag{4.34}$$

The expression involving the constants A and B is the essential core idea of the method of Partial Fractions. We now need to find A and B , and I prefer to start it by multiplying by the original denominator to get

$$1 = (1 + x)A + (1 - x)B. \tag{4.35}$$

We may now follow either of two ways, although the first way I present becomes very cumbersome when applied to cases with three or more unknown constants.

Method 1: We equate coefficients of 1 and of x separately. . .

$$\text{Constant terms: } 1 = A + B$$

$$\text{Coefficients of } x: 0 = A - B$$

This pair of simultaneous equations is solved easily to give $A = B = \frac{1}{2}$.

Method 2: choose appropriate values of x . . .

$$x = 1 \Rightarrow 1 = 2A$$

$$x = -1 \Rightarrow 1 = 2B.$$

Clearly the two methods yield the same value for A and B , but when we choose values of x which are roots of the original denominator $((x - 1)(x + 1))$, in this case, then the equations for the unknown constants decouple, and the answer is obtained much more quickly.

To complete this problem we have

$$\begin{aligned}
 \int \frac{1}{1-x^2} dx &= \int \left[\frac{\frac{1}{2}}{1-x} + \frac{\frac{1}{2}}{1+x} \right] dx \\
 &= \int \left[-\frac{\frac{1}{2}}{x-1} + \frac{\frac{1}{2}}{x+1} \right] dx && \text{I prefer the first integral this way — it's safer} \\
 &= \frac{1}{2} \left[\ln|x+1| - \ln|x-1| \right] + c \\
 &= \frac{1}{2} \ln \left| \frac{x+1}{x-1} \right| + c.
 \end{aligned}
 \tag{4.36}$$

Example 4.16. $I = \int \frac{2}{x^3 + 3x^2 + 2x} dx.$

The denominator of the integrand factorises to

$$x^3 + 3x^2 + 2x = x(x+1)(x+2),$$

and hence we set

$$\frac{2}{x^3 + 3x^2 + 2x} = \frac{2}{x(x+1)(x+2)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2},$$

where A , B and C are to be found. On multiplying by the original denominator we obtain,

$$2 = A(x+1)(x+2) + Bx(x+2) + Cx(x+1). \tag{4.37}$$

We will use the three values, $x = 0$, -1 and -2 , to evaluate the constants since these are the roots of the linear factors of the original denominator. Substituting these in turn into Eq. (4.37) yields,

$$2 = 2A \quad 2 = -B \quad 2 = 2C,$$

in turn and hence $A = C = 1$ and $B = -2$. The integral now becomes,

$$\begin{aligned}
 I &= \int \left[\frac{1}{x} - \frac{2}{x+1} + \frac{1}{x+2} \right] dx \\
 &= \ln|x| - 2\ln|x+1| + \ln|x+2| + c \\
 &= \ln \left| \frac{x(x+2)}{(x+1)^2} \right| + c \\
 &= \ln \left| \frac{x^2 + 2x}{x^2 + 2x + 1} \right| + c.
 \end{aligned}
 \tag{4.38}$$

Example 4.17. Evaluate $I = \int \frac{t-1}{t^3+t^2+t+1} dt$.

This example illustrates what happens when one cannot find real linear factors of the denominator. First note that the denominator may be factorised to get

$$t^3 + t^2 + t + 1 = (t+1)(t^2+1),$$

where (t^2+1) does not have real linear factors and is said to be an **irreducible quadratic**. The modification to the standard method of using partial fractions is to use a linear numerator (rather than a constant) when dealing with a quadratic denominator which has complex roots. Mathematically we set

$$\frac{t-1}{t^3+t^2+t+1} = \frac{t-1}{(t+1)(t^2+1)} = \frac{A}{t+1} + \frac{Bt+C}{t^2+1}. \quad (4.39)$$

Multiplying by (t^3+t^2+t+1) we obtain,

$$t-1 = A(t^2+1) + (Bt+C)(t+1).$$

We may find A , B and C using a mixture of the methods used in Example 4.15. It's a bit of an art spotting the quickest/safest way through.

$$\text{Let } t = -1: \quad 2A = -2 \Rightarrow A = -1$$

$$\text{Coefficient of } t^2: \quad A + B = 0 \Rightarrow B = 1 \quad (4.40)$$

$$\text{Let } t = 0: \quad A + C = -1 \Rightarrow C = 0.$$

Using this information we are now in a position to find the original integral, which again uses the $\int f'/f$ result in the middle, somewhere — I'll leave you to spot it!

$$\begin{aligned} I &= \int \frac{t-1}{t^3+t^2+t+1} dt \\ &= \int \left[\frac{t}{t^2+1} - \frac{1}{t+1} \right] dt \\ &= \frac{1}{2} \int \frac{2t}{t^2+1} dt - \int \frac{1}{t+1} dt \\ &= \frac{1}{2} \ln|t^2+1| - \ln|t+1| + c \\ &= \ln \left| \frac{(t^2+1)^{1/2}}{t+1} \right| + c. \end{aligned} \quad (4.41)$$

Note: The final page of this Chapter gives a brief checklist of the “rules of the game” when applying the method of partial fractions.

Example 4.17 (reprise). Evaluate $I = \int \frac{t-1}{t^3+t^2+t+1} dt$.

I'll repeat Example 4.17 solely to demonstrate why one needs to use a linear function, $Bt + C$, in the numerator of the quotient, $(Bt + C)/(t^2 + 1)$, in Eq. (4.39). The quadratic, $t^2 + 1$, is irreducible but it is so only when one is seeking real linear factors. Given that $t^2 + 1 = (t - j)(t + j)$, we could proceed with the analysis using these instead. So we have,

$$\frac{t-1}{t^3+t^2+t+1} = \frac{t-1}{(t+1)(t^2+1)} = \frac{t-1}{(t+1)(t-j)(t+j)} = \frac{A}{t+1} + \frac{B}{t-j} + \frac{C}{t+j}. \quad (4.42)$$

Multiplying up in the usual way we have,

$$t-1 = A(t-j)(t+j) + B(t+1)(t+j) + C(t+1)(t-j). \quad (4.43)$$

Now we evaluate the unknown coefficients:

$$\begin{aligned} \text{Let } t = -1: \quad -2 &= A(-1-j)(-1+j) = 2A \Rightarrow A = -1 \\ \text{Let } t = j: \quad -1+j &= 2j(1+j)B = 2B(j-1) \Rightarrow B = \frac{1}{2} \\ \text{Let } t = -j: \quad -1-j &= -2j(1-j)C = -2(j+1) \Rightarrow C = \frac{1}{2}. \end{aligned} \quad (4.44)$$

Hence we obtain,

$$\begin{aligned} \frac{t-1}{t^3+t^2+t+1} &= \frac{-1}{t+1} + \frac{1/2}{t-j} + \frac{1/2}{t+j} \\ &= \frac{-1}{t+1} + \frac{\frac{1}{2}(t+j) + \frac{1}{2}(t-j)}{t^2+1} \\ &= \frac{-1}{t+1} + \frac{t}{t^2+1}, \end{aligned} \quad (4.45)$$

as in Example 4.17.

So clearly the strategy of using a linear function as the numerator when the denominator is an irreducible quadratic works for this particular case, but does it work in general? Yes, it does, and while I won't supply a general proof of this, the following analysis comes close.

Suppose we have the following two complex linear factors of an irreducible quadratic: $t = a \pm bj$. Hence this general case may be written down as,

$$\frac{\text{polynomial}}{(\text{polynomial}) \times (t-a-bj)(t-a+bj)} = \frac{A}{t-a-bj} + \frac{B}{t-a+bj} + \text{other terms}, \quad (4.46)$$

where neither of the two polynomials contain either of these complex factors. Given that the left hand side of Eq. (4.46) is real then the right hand side must also be real. On the right hand side the linear factors are complex conjugates of one another, and this implies that A and B must also be complex conjugates of one another. So we may therefore let $A = C + Dj$ and $B = C - Dj$.

Hence,

$$\begin{aligned}
\frac{A}{t-a-bj} + \frac{B}{t-a+bj} &= \frac{C+Dj}{t-a-bj} + \frac{C-Dj}{t-a+bj} \\
&= \frac{(C+Dj)(t-a+bj) + (C-Dj)(t-a-bj)}{(t-a-bj)(t-a+bj)} \\
&= \frac{2C(t-a) - 2Db}{(t+a)^2 + b^2} \\
&= \frac{Ct + \mathcal{D}}{(t+a)^2 + b^2},
\end{aligned} \tag{4.47}$$

where the as-yet unknown **real** constants, $\mathcal{C}$ and $\mathcal{D}$, are given in terms of the earlier unknown constants, C and D , and these may be traced back to the original A and B . So the use of the form,

$$\frac{A}{t-a-bj} + \frac{B}{t-a+bj},$$

for the two complex linear factors is equivalent to the linear/quadratic form,

$$\frac{Ct + \mathcal{D}}{(t+a)^2 + b^2}, \tag{4.48}$$

which was used without explanation in Eq. (4.39). In practice it is usually much quicker to use the linear/quadratic form given in (4.48).

4.5 Integration of products

In many important applications, such as Fourier series and integral transforms both of which you will meet in ME10305 Mathematics 2 next semester, it is often necessary to integrate a product of two functions. I find that the classic method for doing this, which is called **integration by parts**, is very time-consuming and susceptible to more errors than one would expect of a technique which, after all, is often taught in school and subsequently used frequently in undergraduate work. The fault, I believe, lies not in the ability of those who teach but in the formula itself which, as far as I am aware, is used almost universally.

So there we are, outrageously I have trashed something that almost everyone uses! Maybe I had better demonstrate a better way in the next few pages.

The standard derivation uses the well-known result for the derivative of the product of two functions, $u(x)$ and $v(x)$:

$$\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}. \quad (4.49)$$

This is rearranged slightly and integrated to obtain the result,

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx, \quad (4.50)$$

or the other way around,

$$\int v \frac{du}{dx} dx = uv - \int u \frac{dv}{dx} dx. \quad (4.51)$$

As already mentioned, the application of this expression should be straightforward enough — in Eq. (4.50) one chooses which function is u , which is dv/dx , then find both v and du/dx , substitute into Eq. (4.50) and then see what happens next. Here's an example.

Example 4.18. Integrate $\int x e^{ax} dx$ which involves the product of x and e^{ax} .

The overall aim of the formula in Eq. (4.50) is to replace the integral on the left with a right hand one which may be solved or at least is easier to solve. In the light of this, if we were to identify dv/dx with x then the resulting integral would involve $\frac{1}{2}x^2$ which yields a more difficult product to integrate. On the other hand, the identification of v with x means that the resulting integral involves 1 instead.

Therefore a typical way in which this is written down is like this:

$$\text{Let } \begin{cases} \frac{du}{dx} = e^{ax} \\ v = x \end{cases} \implies \begin{cases} u = e^{ax}/a \\ \frac{dv}{dx} = 1. \end{cases} \quad (4.52)$$

Substitution into Eq. (4.50) gives,

$$\int x e^{ax} dx = \frac{x e^{ax}}{a} - \int \frac{1 \times e^{ax}}{a} dx. \quad (4.53)$$

This right hand side integral becomes e^{ax}/a^2 , and hence the final result is,

$$\int x e^{ax} dx = \frac{x e^{ax}}{a} - \frac{e^{ax}}{a^2} = \frac{1}{a^2}(ax - 1)e^{ax}, \quad (4.54)$$

after a little tidying-up. Of course there should be an arbitrary constant in there too.

So what is the problem with this? I reckon that this example is fine, but if we were to be asked to integrate $x^2 e^{ax}$ then the selection and writing out of information corresponding to Eq. (4.52) would need to be done twice as well the equivalent of Eq. (4.53) twice. This will be getting on for half a page of writing.

If, in the other hand, I were to ask for the integral of $x^5 e^{ax}$, then this represents five successive integrations by parts, and correct choices would need to be made for each one, together with more than a page of writing which would be difficult to check. I regard that as unsatisfactory.

There has to be a better way....

...and there is. For integrals such as $x^5 e^{ax}$ there is a way to write the answer down in one line, but we'll need to do a little derivation first. Sorry about that...

Referring to Equation (4.50), it seems to me rather unnatural to start off with the product of two functions, one of which is a *derivative*. Since we are faced with a product of two functions, it would be better for both functions to be represented explicitly in the integrand on the left hand side of the equation. Therefore let us set

$$u(x) = f(x) \quad \text{and} \quad \frac{dv}{dx} = g(x)$$

in Eq. (4.50). We will need to use the superscript notation mentioned in the Differentiation chapter as a shorthand, for convenience. Therefore we let,

$$f^{(1)} = \frac{df}{dx}, \quad f^{(2)} = \frac{d^2 f}{dx^2},$$

and

$$g^{(-1)} = \int g \, dx, \quad g^{(-2)} = \int g^{(-1)} \, dx,$$

and so on. So positive superscripts denote the number of differentiations, while negative superscripts denote how many times an indefinite integral has been taken.

Equation (4.50) now becomes,

$$\int f g \, dx = f g^{(-1)} - \int f^{(1)} g^{(-1)} \, dx. \quad (4.55)$$

As before, we choose one function to be differentiated (called f here) and one to be integrated (called g here).

Thus far we have only achieved a change in notation. The reason for that change becomes clearer when we apply this latest formula recursively. In other words, we will apply the way the formula works to the right hand side integral. This will give,

$$\int f^{(1)} g^{(-1)} \, dx = f^{(1)} g^{(-2)} - \int f^{(2)} g^{(-2)} \, dx. \quad (4.56)$$

Nice! And now we may substitute Eq. (4.56) into Eq. (4.55) to obtain,

$$\int f g \, dx = f g^{(-1)} - f^{(1)} g^{(-2)} + \int f^{(2)} g^{(-2)} \, dx, \quad (4.57)$$

and this represents two integrations by parts. One may continue this process as many times as one wishes. So the formulae for five and six integrations by parts are,

$$\int f g \, dx = f g^{(-1)} - f^{(1)} g^{(-2)} + f^{(2)} g^{(-3)} - f^{(3)} g^{(-4)} + f^{(4)} g^{(-5)} - \int f^{(5)} g^{(-5)} \, dx. \quad (4.58)$$

$$\int f g \, dx = f g^{(-1)} - f^{(1)} g^{(-2)} + f^{(2)} g^{(-3)} - f^{(3)} g^{(-4)} + f^{(4)} g^{(-5)} - f^{(5)} g^{(-6)} + \int f^{(6)} g^{(-6)} \, dx. \quad (4.59)$$

So do I wish you to remember all of these new formulae? No, not at all. Definitely not. Rather, I would prefer everyone to follow a set of rules that will work every time and for however many integrations-by-parts that are needed. Here they are:

1. The first product on the right hand side, $f g^{(-1)}$, has the g -term integrated but the f -term has been left alone. So we start by integrating, which feels natural given that the overall aim is to find an integral!

- Each successive product which appears is related to the previous product by having the term called f being differentiated and the term called g being integrated.
- The signs alternate between the terms on the right hand side of (4.59) including the one for the final integral.
- The final integral (should it be required) uses a final differentiation of the preceding f term.

Do check all of these statements against the formula given in Eq. (4.59). Use of these rules will make Integration by Parts quite astonishingly fast and easily checkable afterwards.

The following two examples won't need to use the final integral because the power of x will eventually become zero after a sufficient number of differentiations.

Example 4.19. Find the integral, $\int x^2 e^{-ax} dx$.

We will choose to differentiate the x^2 terms and to integrate the exponential. We obtain the following,

$$\int \underbrace{x^2}_D \underbrace{e^{-ax}}_I dx = \underbrace{\left[x^2\right]}_{D_0} \underbrace{\left[\frac{e^{-ax}}{-a}\right]}_{I_1} - \underbrace{\left[2x\right]}_{D_1} \underbrace{\left[\frac{e^{-ax}}{a^2}\right]}_{I_2} + \underbrace{\left[2\right]}_{D_2} \underbrace{\left[\frac{e^{-ax}}{-a^3}\right]}_{I_3} - \underbrace{\left[0\right]}_{D_3} \underbrace{\left[\frac{e^{-ax}}{a^4}\right]}_{I_4} + c. \quad (4.60)$$

Here I have placed red annotations under the different terms to indicate how many differentiations or integrations have taken place — if needed, these may be checked against Eq. (4.59). The final product, $D_3 I_4$, is zero, and therefore there is no need to continue because every further term will also be zero. And I have included the arbitrary constant.

The beauty of the solution in Eq. (4.60) is that one may hop through all the D terms to check that all the differentiations are correct, and through all the I terms to check the integrations.

The final point to make is that all of the D and I terms are sitting cosily inside their own pair of brackets. This means that there is no need to try to tidy up any minus signs during the writing down of a solution like Eq. (4.60) because that will be done on a second line. So the first line, Eq. (4.60), is the integration while the second line, which I am just about to do, is the tidying up. So here's the tidied-up version:

$$\int x^2 e^{-ax} dx = -\frac{(a^2 x^2 + 2ax + 2)}{a^3} e^{-ax} + c. \quad (4.61)$$

Example 4.20. Find the integral, $\int x^5 e^{ax} dx$.

This one would normally be the stuff of nightmares, but now we can take it in our stride.... So it's the same as the last Example where we differentiate the power.

$$\begin{aligned} \int \underbrace{x^5}_D \underbrace{e^{ax}}_I dx &= \underbrace{\left[x^5\right]}_{D_0} \underbrace{\left[\frac{e^{ax}}{a}\right]}_{I_1} - \underbrace{\left[5x^4\right]}_{D_1} \underbrace{\left[\frac{e^{ax}}{a^2}\right]}_{I_2} + \underbrace{\left[20x^3\right]}_{D_2} \underbrace{\left[\frac{e^{ax}}{a^3}\right]}_{I_3} - \underbrace{\left[60x^2\right]}_{D_3} \underbrace{\left[\frac{e^{ax}}{a^4}\right]}_{I_4} \\ &\quad + \underbrace{\left[120x\right]}_{D_4} \underbrace{\left[\frac{e^{ax}}{a^5}\right]}_{I_5} - \underbrace{\left[120\right]}_{D_5} \underbrace{\left[\frac{e^{ax}}{a^6}\right]}_{I_6} + c \\ &= \frac{(a^5 x^5 - 5a^4 x^4 + 20a^3 x^3 - 60a^2 x^2 + 120ax - 120)}{a^6} e^{ax} + c. \end{aligned} \quad (4.62)$$

Example 4.21. Find the integral, $\int x^2 \cos ax \, dx$.

We have,

$$\int \underbrace{x^2}_D \underbrace{\cos ax}_I \, dx = \underbrace{\left[x^2\right]}_{D_0} \underbrace{\left[\frac{\sin ax}{a}\right]}_{I_1} - \underbrace{\left[2x\right]}_{D_1} \underbrace{\left[\frac{-\cos ax}{a^2}\right]}_{I_2} + \underbrace{\left[2\right]}_{D_2} \underbrace{\left[\frac{-\sin ax}{a^3}\right]}_{I_3}, \quad (4.63)$$

which may, of course, be tidied up a bit and the arbitrary constant included.

Example 4.22. Find $\int e^{ax} \sin bx \, dx$.

This one is a little different because the integration by parts process doesn't terminate. Given that exponentials yield exponentials when differentiated or integrated, and the sines are reproduced every two differentiations or integrations, then (i) it doesn't matter which of the two functions is differentiated and (ii) after two integrations one should obtain an integral which is proportional to the original one.

For this I will choose to integrate the sine, but it worth you attempting the other way to see if a correct solution will be obtained. Hence,

$$\begin{aligned} \text{Let } I &= \int \underbrace{e^{ax}}_D \underbrace{\sin bx}_I \, dx = \underbrace{\left[e^{ax}\right]}_{D_0} \underbrace{\left[\frac{-\cos bx}{b}\right]}_{I_1} - \underbrace{\left[ae^{ax}\right]}_{D_1} \underbrace{\left[\frac{-\sin bx}{b^2}\right]}_{I_2} + \int \underbrace{\left[a^2 e^{ax}\right]}_{D_2} \underbrace{\left[\frac{-\sin bx}{b^2}\right]}_{I_2} \, dx \\ &= \frac{e^{ax}}{b^2} \left[-b \cos bx + a \sin bx\right] - \frac{a^2}{b^2} I. \end{aligned} \quad (4.64)$$

The terms involving I may be brought together on the left to obtain,

$$\left[1 + \frac{a^2}{b^2}\right] I = \frac{e^{ax}}{b^2} [a \sin bx - b \cos bx] \quad (4.65)$$

and therefore we obtain,

$$I = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]. \quad (4.66)$$

Example 4.23. Find $\int \ln |x| \, dx$.

Another classic, and although it doesn't look like a product we can make it so: $\int 1 \times \ln x \, dx$. Once you know the trick, you know the trick. We will differentiate the $\ln x$ because we can! We will only integrate by parts once because it isn't entirely obvious what will happen after that, so then we'll take stock at that point.

$$\begin{aligned} \int \underbrace{1}_I \times \underbrace{\ln |x|}_D \, dx &= \underbrace{\left[x\right]}_{I_1} \underbrace{\left[\ln |x|\right]}_{D_0} - \int \underbrace{\left[x\right]}_{I_1} \underbrace{\left[\frac{1}{x}\right]}_{D_1} \, dx && \text{one integration by parts} \\ &= x \ln x - \int 1 \, dx && \text{tidied up the integral} \\ &= x \ln x - x + c. \end{aligned} \quad (4.67)$$

Note: An additional rule — if in doubt, do it once!

4.6 Mean and Root Mean Square

It is frequently necessary to evaluate the mean value of a function and its root mean square (or RMS). The mean of a function, $f(x)$, over the interval $a \leq x \leq b$, is defined as

$$\mu = \text{mean of } f(x) = \frac{1}{b-a} \int_a^b f(x) dx. \quad (4.68)$$

This definition may be derived easily by insisting that a rectangle of length $b - a$ and with a height which is the mean of $f(x)$ has the same area as the graph of $f(x)$ in the same range. This is illustrated in Fig. 4.5.

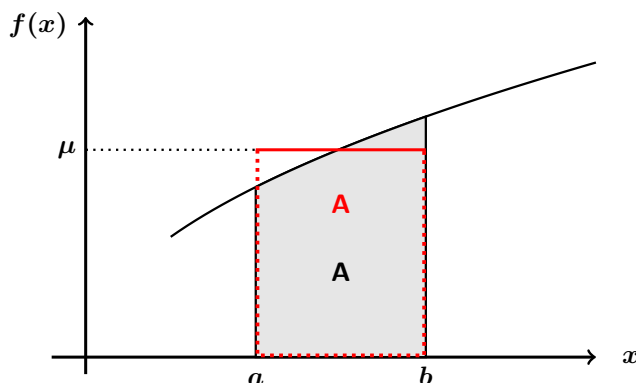


Figure 4.5. Demonstrating the mean value of the function, $f(x)$, in the range $a \leq x \leq b$. The areas of the grey region and of the red rectangle need to be the same.

The RMS is precisely what it says it is, the square root of the mean of the square of the function. Therefore we have the following definition,

$$\text{RMS of } f(x) = \left[\frac{1}{b-a} \int_a^b [f(x)]^2 dx \right]^{1/2}. \quad (4.69)$$

Example 4.24. What are the mean and RMS of the function $f(x) = x$ in the interval $1 \leq x \leq 3$?

They are,

$$\mu = \frac{1}{3-1} \int_1^3 x dx = 2, \quad \text{RMS} = \left[\frac{1}{3-1} \int_1^3 x^2 dx \right]^{1/2} = \sqrt{\frac{13}{3}}. \quad (4.70)$$

4.7 Recurrence relations

The following integral may be found easily:

$$\int_0^{\infty} e^{-x} dx = 1,$$

but it will be quite time-consuming to evaluate the integral,

$$\int_0^{\infty} x^{20} e^{-x} dx,$$

even with a fast integration by parts method. The aim of this section is to find a way of evaluating such an integral without resorting to 20 integrations by parts. This will be called a **recurrence relation**.

Perhaps it is best to illustrate what it is by just doing one.

Example 4.25. If we let,

$$I_n = \int_0^{\infty} x^n e^{-x} dx,$$

then if we undertake one integration by parts, we should obtain a way of linking I_n with I_{n-1} . So one integration by parts yields,

$$\begin{aligned} I_n &= \int_0^{\infty} x^n e^{-x} dx, \\ &= \left[x^n \right] \left[-e^{-x} \right]_0^{\infty} - \int_0^{\infty} \left[nx^{n-1} \right] \left[-e^{-x} \right] dx, \\ &= 0 + n \int_0^{\infty} x^{n-1} e^{-x} dx, \\ &= n I_{n-1}. \end{aligned} \tag{4.71}$$

Therefore we have derived the recurrence relation:

$$I_n = n I_{n-1}. \tag{4.72}$$

It is straightforward to show that the $n = 0$ case is $I_0 = \int_0^{\infty} e^{-x} dx = 1$. We may now use this recurrence relation to find I_1 , I_2 , I_3 and so on, and hopefully we'll be able to spot the pattern. Hence,

$$\begin{aligned} n = 1 &\Rightarrow I_1 = 1 \times I_0 = 1 \\ n = 2 &\Rightarrow I_2 = 2 \times I_1 = 2 \\ n = 3 &\Rightarrow I_3 = 3 \times I_2 = 2 \times 3 \\ n = 4 &\Rightarrow I_4 = 4 \times I_3 = 2 \times 3 \times 4 = 4! \\ n = 5 &\Rightarrow I_5 = 5 \times I_4 = 2 \times 3 \times 4 \times 5 = 5! \end{aligned} \tag{4.73}$$

So it has become clear that $I_n = n!$. Hence $I_{20} = 20! = 2\,432\,902\,008\,176\,640\,000$.

The remaining part of this chapter is on applications of integration, the finding of volumes under surfaces, lengths of lines and areas and volumes of revolution.

4.8 Volumes under surfaces

The simple question is: given the function, $h(x, y)$, which describes the height of a surface above the (x, y) -plane, what is the volume under that surface? The manner in which this volume is determined is motivated by the following Figure.

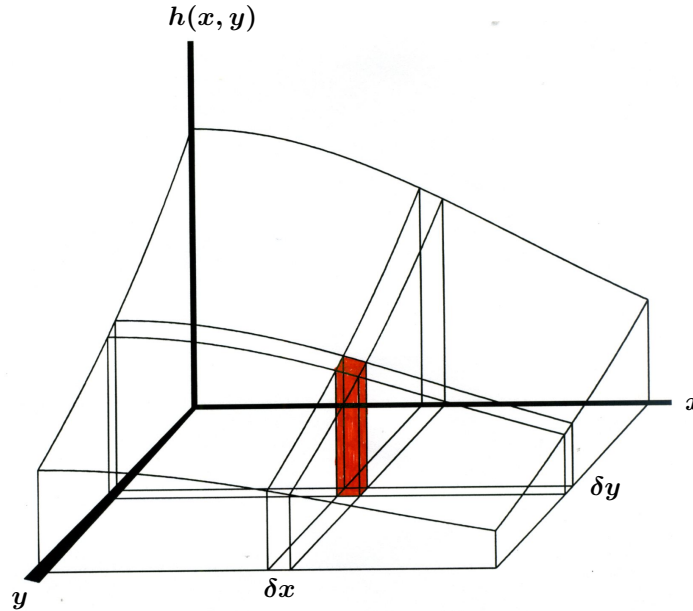


Figure 4.6. Depicting a typical surface, $h(x, y)$, and the manner in which the volume integral may be determined. The shaded volume is $\delta V = h(x, y) \delta x \delta y$.

In this figure there is a narrow strip of width δy in the x -direction and a narrow strip of width δx in the y -direction. These intersect and yield a small rectangle of area equal to $\delta x \delta y$. The volume above this rectangle will then be just the area of the rectangle multiplied by the height, $h(x, y)$, of the surface. [This height could be considered to be the height directly above the centre of the rectangle, or else we could use a formula which is akin to the Trapezium rule, but the final result will not differ.]

Therefore the volume of the shaded rectangular cylinder is

$$\delta V = h(x, y) \delta x \delta y. \quad (4.74)$$

If we need to determine the volume in the ranges, $0 \leq x \leq a$ and $0 \leq y \leq b$, where a and b are constants, then we may sum (integrate) all these small volumes to give,

$$V = \int_0^b \int_0^a h(x, y) dx dy. \quad (4.75)$$

It is important to note that the manner in which this double integral is to be evaluated is to begin with the inner integral (i.e. the integration with respect to x , here), and then to proceed to integrate with respect to y . Therefore Eq. (4.75) should be interpreted as being equivalent to,

$$V = \int_0^b \left[\int_0^a h(x, y) dx \right] dy, \quad (4.76)$$

where the computation in the square brackets must be completed first *including* the substitution of the limits. When the inner integral is being performed, y is treated as a parameter, i.e. as if it were a constant.

It is also important to note that it is quite possible to integrate with respect to y first, and then to undertake the x -integration. In this case the above double integral must be written in the form,

$$V = \int_0^a \int_0^b h(x, y) dy dx \equiv \int_0^a \left[\int_0^b h(x, y) dy \right] dx. \quad (4.77)$$

Given how we derived Eqs. (4.76) and (4.77), clearly they should give exactly the same values!

Example 4.26. Determine the value of $I = \int_0^\infty \int_0^1 (2x + y)e^{-y} dx dy$.

This surface lies over a strip of unit width in the x -direction and of infinite length in the y -direction. On following the standard procedure the inner integral is evaluated first, particularly since the y -dependent part of the integrand suggests some integration by parts later. Therefore we get,

$$\begin{aligned} \int_0^\infty \int_0^1 (2x + y)e^{-y} dx dy &= \int_0^\infty \left[(x^2 + xy)e^{-y} \right]_{x=0}^{x=1} dy \\ &= \int_0^\infty (1 + y)e^{-y} dy \\ &= \left[(1 + y)(-e^{-y}) - (1)(e^{-y}) \right]_{y=0}^{y=\infty} \quad \text{Using integration by parts} \\ &= 2. \end{aligned} \quad (4.78)$$

Further examples could be given, but no further difficulty over and above what has been covered earlier in this chapter will arise. However, if one needs to integrate over nonrectangular regions, then the determination of the correct limits is not so straightforward, especially the swapping around of the order of integration, but this aspect is not covered in the present syllabus.

4.9 Triple integrals

In like manner one may perform triple integrals. A simple example of this would be the determination of the mass of a cuboid if we are given a function describing its density. Thus, if $\rho(x, y, z)$ is the density of a solid at a given point, then the mass of a very small cuboid of dimensions, $\delta x \times \delta y \times \delta z$, is simply

$$\delta m = \rho(x, y, z) \delta x \delta y \delta z, \quad (4.79)$$

i.e. mass = (mass per unit volume) $\times$ volume.

If the cuboidal solid has dimensions, $a \times b \times c$, then its mass is

$$m = \int_0^c \int_0^b \int_0^a \rho(x, y, z) dx dy dz. \quad (4.80)$$

Example 4.27. What is the mass of the unit cube with density distribution, $\rho = xy^2e^{-z}$ which occupies the region, $0 \leq x \leq 1$, $0 \leq y \leq 1$ and $0 \leq z \leq 1$.

We get

$$\begin{aligned}
 m &= \int_0^1 \int_0^1 \int_0^1 xy^2e^{-z} dx dy dz \\
 &= \int_0^1 \int_0^1 \left[\frac{1}{2}x^2y^2e^{-z} \right]_{x=0}^{x=1} dy dz \\
 &= \int_0^1 \int_0^1 \frac{1}{2}y^2e^{-z} dy dz \\
 &= \int_0^1 \left[\frac{1}{6}y^3e^{-z} \right]_{y=0}^{y=1} dz \tag{4.81} \\
 &= \int_0^1 \frac{1}{6}e^{-z} dz \\
 &= \left[-\frac{1}{6}e^{-z} \right]_{z=0}^{z=1} \\
 &= \frac{1}{6}(1 - e^{-1}).
 \end{aligned}$$

Note: If a double integral has an integrand of the form, $f(x)g(y)$, then it may be simplified into the product of two single integrals, as follows,

$$\int_{y_{\min}}^{y_{\max}} \int_{x_{\min}}^{x_{\max}} f(x)g(y) dx dy = \left[\int_{x_{\min}}^{x_{\max}} f(x) dx \right] \left[\int_{y_{\min}}^{y_{\max}} g(y) dy \right]. \tag{4.82}$$

A similar result holds for triple integrals with integrand, $f(x)g(y)h(z)$:

$$\int_{z_{\min}}^{z_{\max}} \int_{y_{\min}}^{y_{\max}} \int_{x_{\min}}^{x_{\max}} f(x)g(y)h(z) dx dy dz = \left[\int_{x_{\min}}^{x_{\max}} f(x) dx \right] \left[\int_{y_{\min}}^{y_{\max}} g(y) dy \right] \left[\int_{z_{\min}}^{z_{\max}} h(z) dz \right]. \tag{4.83}$$

4.10 Volumes under surfaces in polar coordinates

We turn now to functions which are expressed in polar coordinates. Here the radial variable is r and this measures the distance from the origin; we may think of this as being within the (x, y) -plane, as in Fig. 4.6. Then θ is the angular direction compared with the direction of the positive x -axis. The height of the surface above the (x, y) -plane is now written as $h(r, \theta)$.

For this we may again use the approach of adding up small volumes, and this process is illustrated in Fig. 4.7,

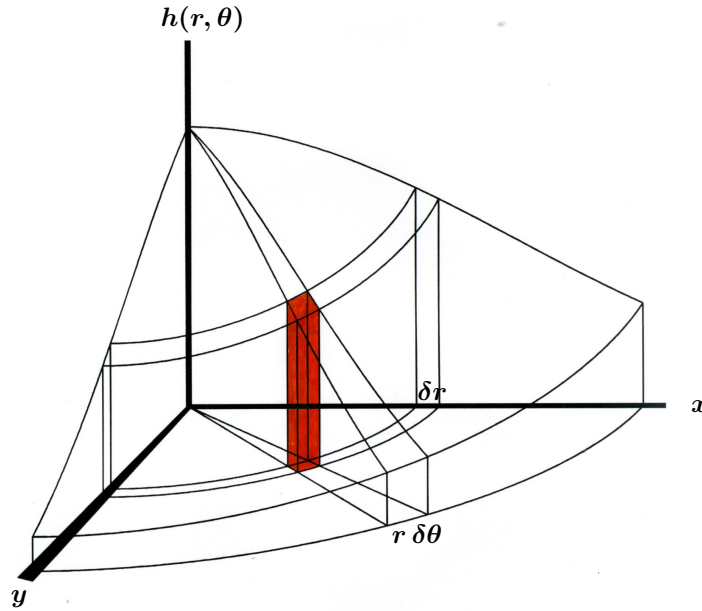


Figure 4.7. Depicting a typical surface, $h(r, \theta)$, and the manner in which the volume integral may be determined. The small element on the (x, y) -plane has area, $\delta r \times (r \delta \theta)$. The shaded volume is given by $\delta V = h(r, \theta) r \delta r \delta \theta$.

The small element of area on the (x, y) or (r, θ) -plane corresponds to a small change in the radius of size, δr , and a small change in the polar angle, $\delta \theta$. However, this small angle is equivalent to a length of $r \delta \theta$ (on noting that the circumference of a circle is $2\pi r$, where 2π is the angle subtended by a full circle). Therefore we may take the volume of the cylinder to be

$$\delta V = h(r, \theta) \times r \delta \theta \times \delta r = h(r, \theta) r \delta r \delta \theta. \quad (4.84)$$

Therefore the full volume is found by integration of this quantity. Thus,

$$V = \int_{\theta_{\min}}^{\theta_{\max}} \int_{r_{\min}}^{r_{\max}} h(r, \theta) r dr d\theta. \quad (4.85)$$

Once again, one may change the order of integration should it prove convenient to do so:

$$V = \int_{r_{\min}}^{r_{\max}} \left[\int_{\theta_{\min}}^{\theta_{\max}} h(r, \theta) d\theta \right] r dr. \quad (4.86)$$

Note: that there is an extra r in the integrand. We are, in effect, integrating $r h(r, \theta)$.

Example 4.28. Find the volume under the function, $h(r, \theta) = r(1 - r) \sin \theta$, in a unit semicircle lying in the range $0 \leq \theta \leq \pi$.

This shape ought to be familiar, although the mathematical definition obscures it a fair bit. Here she be:

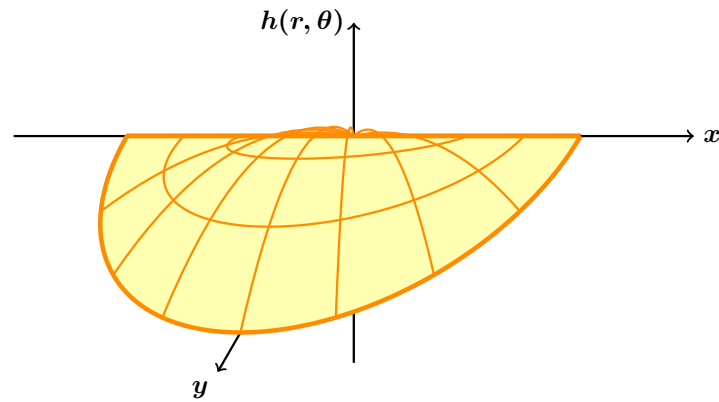


Figure 4.8. Showing the Cornish pasty whose volume is being sought. There is no crimping on the circular edge, though. Sorry about that.

We have,

$$\begin{aligned}
 V &= \int_0^\pi \int_0^1 (r - r^2) \sin \theta \, r \, dr \, d\theta && \text{by definition} \\
 &= \int_0^\pi \int_0^1 (r^2 - r^3) \sin \theta \, dr \, d\theta && \text{tidying up} \\
 &= \int_0^\pi \left[\left(\frac{1}{3} r^3 - \frac{1}{4} r^4 \right) \sin \theta \right]_{r=0}^{r=1} d\theta && (4.87) \\
 &= \int_0^\pi \frac{1}{12} \sin \theta \, d\theta \\
 &= \left[-\frac{1}{12} \cos \theta \right]_{\theta=0}^{\theta=\pi} \\
 &= 1/6.
 \end{aligned}$$

I haven't told you what the units are, but if they were metres, then that would be one for sharing.

4.11 The length of a curve

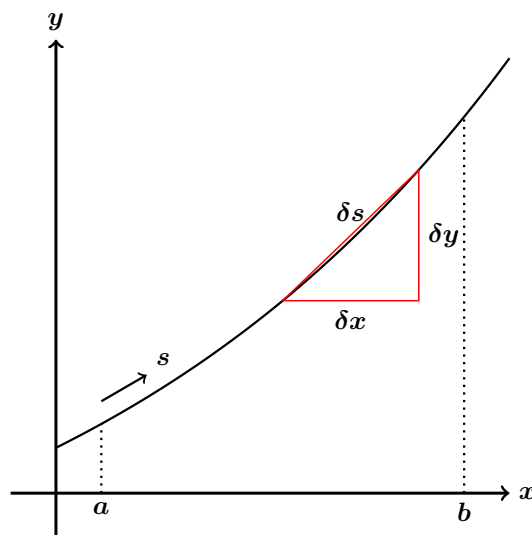


Figure 4.9. Showing the basic configuration for calculating the length of a curve.

What is the length of the curve, $y(x)$, between $x = a$ and $x = b$?

We will use $s(x)$ to measure the distance along the curve as measured from $x = a$. We could denote by δs a small element of length; Fig. 4.9 shows a rather large version of this, but our hope is that we can add together all the small elements which make up the curve. Therefore we could write the total length in the form,

$$L = \int_{x=a}^{x=b} ds,$$

although this doesn't yet appear to be useful! Now, our limits are in terms of x but we're integrating with respect to s , and therefore we could draw on an idea from integration by substitution:

$$L = \int_a^b \frac{ds}{dx} dx, \quad (4.88)$$

and therefore we need to find an expression for $\frac{ds}{dx}$.

In Fig. 4.9 we may use Pythagoras's theorem to find δs in terms of δx and δy :

$$(\delta s)^2 = (\delta x)^2 + (\delta y)^2,$$

and hence $\delta s = \sqrt{(\delta x)^2 + (\delta y)^2}$. If we now divide by δx , then

$$\frac{\delta s}{\delta x} = \sqrt{1 + \left(\frac{\delta y}{\delta x}\right)^2}. \quad (4.89)$$

This expression will be increasingly accurate as δx gets smaller and therefore we can take the limit as $\delta x \rightarrow 0$; hence,

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}. \quad (4.90)$$

Therefore we have the following expression for the length of the curve,

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx. \quad (4.91)$$

Note 1: When the line is horizontal then $dy/dx = 0$, and the integral reduces to $\int_a^b 1 dx = b - a$.

Note 2: When the line is at 45° then $dy/dx = 1$, and the integral reduces to $\int_a^b \sqrt{2} dx = \sqrt{2}(b - a)$.

Note 3: When the curve displays any deviation from being horizontal, then the presence of the square root term in the integral serves to increase its length from when it is a horizontal line. Intuitively that makes sense.

Example 4.29. Find the length of a quarter of the circumference of a circle of radius, R . We will use $0 \leq x \leq R$ and hence take the first quadrant of the circle.

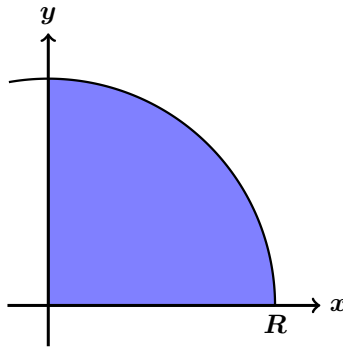


Figure 4.10. A quadrant of a circle with radius, R .

The equation is $y = \sqrt{R^2 - x^2}$ and so $\frac{dy}{dx} = \frac{-x}{\sqrt{R^2 - x^2}}$. Hence,

$$\begin{aligned} L &= \int_0^R \sqrt{1 + \left(\frac{-x}{\sqrt{R^2 - x^2}}\right)^2} dx && \text{using Eq. (4.91)} \\ &= \int_0^R \sqrt{1 + \frac{x^2}{R^2 - x^2}} dx \\ &= \int_0^R \sqrt{\frac{R^2}{R^2 - x^2}} dx \\ &= R \int_0^R \frac{1}{\sqrt{R^2 - x^2}} dx. \end{aligned} \quad (4.92)$$

This integral may be simplified by substituting $x = R \sin \theta$ where $dx = R \cos \theta d\theta$. With regard to the limits: $x = 0 \Rightarrow \theta = 0$ and $x = R \Rightarrow \theta = \frac{1}{2}\pi$. Hence L is now given by,

$$L = R \int_0^{\pi/2} \frac{R \cancel{\cos \theta}}{R \cancel{\cos \theta}} d\theta = \frac{1}{2}\pi R. \quad (4.93)$$

This means that the circumference of the full circle is $2\pi R$.

4.12 Volumes of revolution

If we are given a function, $y = y(x)$, and this function is rotated about the x -axis as depicted below, then there follows two natural questions, namely (i) what is the volume contained within this object, and (ii) what is its external surface area?

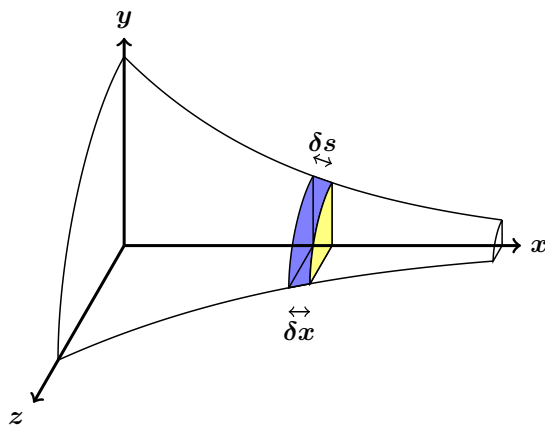


Figure 4.11. A Volume of revolution about the x -axis.

This figure displays a small element of the volume of revolution of the function, $y(x)$, about the x -axis, or rather it shows a quarter of it. This thin coin-like disk has a volume which is the product of the surface area of the disk (yellow shading) and its width, δx . Therefore the volume of the full disk is,

$$\delta V = \pi y^2 \delta x, \quad (4.94)$$

where πy^2 is the usual formula for the area of a circle with radius, y . This may be integrated to give the total volume:

$$V = \int_{x_{\min}}^{x_{\max}} \pi y^2 dx. \quad (4.95)$$

Note: This formula needs to be memorised because it doesn't appear in the University's formula book.

Example 4.30. Find the volume of a sphere of radius, R .

The appropriate function to use is that of a circle, $y = \sqrt{R^2 - x^2}$, in the range $-R \leq x \leq R$. Therefore we have,

$$\begin{aligned} V &= \int_{-R}^R \pi \left[\sqrt{R^2 - x^2} \right]^2 dx && \text{by definition} \\ &= \int_{-R}^R \pi (R^2 - x^2) dx && \text{tidying up} \\ &= 2 \int_0^R \pi (R^2 - x^2) dx && \text{integrand is even!} \\ &= 2\pi \left[R^2 x - \frac{1}{3} x^3 \right]_0^R && \text{note the limits and the 2} \\ &= \frac{4}{3} \pi R^3. \end{aligned} \quad (4.96)$$

4.13 Surfaces of revolution.

Figure 4.11 may also be used to begin the analysis of the surface area obtained by rotating a function about the x -axis. We are now interested in the surface area of the thin disk which is depicted in blue. Its area may be written in the form,

$$\delta A = (2\pi y) \times \delta s, \quad (4.97)$$

where $2\pi y$ is the circumference of the disk, and δs is the distance between the two sides of the disk while travelling along the surface. NOTE that $\delta s \neq \delta x$ unless the function $y(x)$ is a constant; see Fig. 4.11. We have already determined how δs is calculated in Section 4.11, where Pythagoras made an appearance. Therefore Eq. (4.97) becomes,

$$\delta A = (2\pi y) \times \sqrt{1 + \left(\frac{\delta y}{\delta x}\right)^2} \delta x. \quad (4.98)$$

On taking the limits as $\delta x \rightarrow 0$, and integrating between suitable limits, we get,

$$A = 2\pi \int_{x_{\min}}^{x_{\max}} y \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{1/2} dx. \quad (4.99)$$

This formula also needs to be memorised.

Example 4.31. Find the surface area of a sphere of radius, R .

Again we will use $y = \sqrt{R^2 - x^2}$ in the range, $-R \leq x \leq R$. First we need to evaluate the derivative of y ; this is

$$\frac{dy}{dx} = -\frac{x}{\sqrt{R^2 - x^2}}.$$

Therefore we have,

$$\begin{aligned} A &= 2\pi \int_{-R}^R \sqrt{R^2 - x^2} \left[1 + \frac{x^2}{R^2 - x^2}\right]^{1/2} dx && \text{by definition} \\ &= 2\pi \int_{-R}^R \sqrt{R^2 - x^2} \left[\frac{R^2}{R^2 - x^2}\right]^{1/2} dx && \text{simplification} \\ &= 2\pi \int_{-R}^R \sqrt{R^2 - x^2} \left[\frac{R}{\sqrt{R^2 - x^2}}\right] dx && (4.100) \\ &= 2\pi \int_{-R}^R R dx \\ &= 2\pi \left[Rx\right]_{-R}^R \\ &= 4\pi R^2. \end{aligned}$$

A nice simple integral to end the Chapter....

Appendix — a checklist for Partial Fractions

This is a checklist of the various complications which can arise when using partial fractions to simplify the ratio of two polynomials, and it shows what procedure to use in each case.

I will assume that the degree of the polynomial in the denominator is less than that of the numerator, i.e. that the ratio is “bottom heavy”. An example would be a quadratic divided by a cubic. If the ratio is not bottom heavy (e.g. a quintic divided by a cubic), then it is always possible to make it so by subtracting out suitable polynomials.

If all the factors of the denominator are different, then let

$$\frac{?}{(x+a)(x+b)(x+c)\cdots} = \frac{A}{x+a} + \frac{B}{x+b} + \frac{C}{x+c} + \cdots.$$

If one factor is repeated, then let

$$\frac{?}{(x+a)^2(x+b)(x+c)\cdots} = \frac{A_1}{x+a} + \frac{A_2}{(x+a)^2} + \frac{B}{x+b} + \frac{C}{x+c} + \cdots.$$

A more complicated example: let

$$\frac{?}{(x+a)^3(x+b)^2(x+c)(x+d)\cdots} = \frac{A_1}{x+a} + \frac{A_2}{(x+a)^2} + \frac{A_3}{(x+a)^3} + \frac{B_1}{x+b} + \frac{B_2}{(x+b)^2} + \frac{C}{x+c} + \frac{D}{x+d} + \cdots.$$

If one of the factors is an irreducible quadratic then let

$$\frac{?}{(x^2+ax+b)(x+c)(x+d)\cdots} = \frac{Ax+B}{x^2+ax+b} + \frac{C}{x+c} + \frac{D}{x+d} + \cdots.$$

A repeated quadratic factor:

$$\frac{?}{(x^2+ax+b)^2(x+c)(x+d)\cdots} = \frac{Ax+B}{x^2+ax+b} + \frac{Cx+D}{(x^2+ax+b)^2} + \frac{E}{x+c} + \frac{F}{x+d} + \cdots.$$

Note that it is always possible to reduce (what is generally known as) an irreducible quadratic factor to two complex linear factors, and then one may proceed in the usual way for linear factors. It might take a little while!

An example of a top-heavy case:

$$\begin{aligned} \frac{x^3}{(x+a)(x+b)} &= \frac{x(x+a)(x+b) + A(x+a)(x+b) + Bx + C}{(x+a)(x+b)} \\ &= x + A + \frac{Bx+C}{(x+a)(x+b)} = x + A + \frac{D}{x+a} + \frac{E}{x+b}, \end{aligned}$$

where $a \neq b$ here, and where A , D and E are to be found.

5 Series

5.1 Definition of a sequence

A sequence is an ordered list of numbers which may be either random or generated by experimental readings or else generated according to a given rule. It may be finite or infinite in length. Examples include the following,

$$1, \quad 2, \quad 4, \quad 8, \quad 16, \quad 32, \quad 64, \quad 128, \quad \dots \quad (i)$$

$$1, \quad \frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}, \quad \frac{1}{16}, \quad \frac{1}{32}, \quad \frac{1}{64}, \quad \frac{1}{128}, \quad \dots \quad (ii)$$

$$2, \quad 4, \quad 6, \quad 8, \quad 10, \quad 12, \quad 14, \quad 16, \quad \dots \quad (iii)$$

$$1, \quad -1, \quad 1, \quad -1, \quad 1, \quad -1, \quad 1, \quad -1, \quad \dots \quad (iv)$$

$$1, \quad -\frac{1}{2}, \quad \frac{1}{4}, \quad -\frac{1}{8}, \quad \frac{1}{16}, \quad -\frac{1}{32}, \quad \frac{1}{64}, \quad -\frac{1}{128}, \quad \dots \quad (v)$$

$$1, \quad 4, \quad 1, \quad 4, \quad 2, \quad 1, \quad 3, \quad 5, \quad \dots \quad (vi)$$

$$19, \quad 13, \quad 15, \quad 6, \quad 19, \quad 17, \quad 1, \quad 3, \quad \dots \quad (vii)$$

Can you identify the rule which governs each member of the different sequences?

Generally we use a subscript notation for sequences and series: x_1, x_2, x_3 and so on. Or, should there be a good reason to do, we could begin with a zero subscript: x_0, x_1, x_2 and so on. Therefore we write the general term (or the n^{th} term) simply as x_n .

Sequence (i) is formed by doubling the preceding term. We could write it in the form,

$$x_n = 2^{n-1} \quad \text{for } n = 1, 2, 3, \dots, \quad (5.1)$$

or in the alternative way,

$$x_n = 2^n \quad \text{for } n = 0, 1, 2, \dots. \quad (5.2)$$

or even,

$$x_n = 2^{n-163} \quad \text{for } n = 163, 164, 165, \dots, \dots. \quad (5.3)$$

Tastes differ, but I prefer the second one because it is a little tidier. The third is correct but just plain silly.

Sequence (ii). The terms here are the reciprocals of the respective terms in Sequence (i). Therefore,

$$x_n = 2^{-(n-1)} \quad \text{for } n = 1, 2, 3, \dots \quad \text{or} \quad x_n = 2^{-n} \quad \text{for } n = 0, 1, 2, \dots. \quad (5.4)$$

Sequence (iii). Clearly these are twice the value of the positive integers. Hence

$$x_n = 2n, \quad \text{for } n = 1, 2, 3, \dots. \quad (5.5)$$

Sequence (iv). Alternating signs. This feature happens very frequently in sequences and series and it is worth knowing about them now, right at the outset.

$$x_n = (-1)^n, \quad \text{for } n = 0, 1, 2, \dots \quad \text{or} \quad x_n = (-1)^{n+1}, \quad \text{for } n = 1, 2, 3, \dots. \quad (5.6)$$

Note that an alternative form for $(-1)^n$ is $\cos \pi n$. Strange but true. This cosine version will eventually crop up naturally in Fourier Series in ME10305 Mathematics 2 and we usually convert it back into the $(-1)^n$ form.

Sequence (v) is a combination of (ii) and (iv). Hence,

$$x_n = \frac{(-1)^n}{2^n} \quad \text{for } n = 0, 1, 2, \dots, \quad \text{or} \quad x_n = \frac{(-1)^{n-1}}{2^{n-1}} \quad \text{for } n = 1, 2, 3, \dots \quad (5.7)$$

In this case I prefer the first version because it looks tidier.

Sequence (vi) is a bit of a trick question. These are the successive digits of $\sqrt{2}$, and although this knowledge is sufficient to determine, say, the 100th significant figure, there isn't a simple formula to tell us what that is without finding all the previous ones first. The following analysis is for interest only, but it is possible to write down some sort of expression; it's a bit of a rocky ride to get there and has no relevance to the unit but I hope that some may enjoy the experience.

First I will introduce the floor function. This is the greatest integer less than or equal to a real number. For positive numbers it is more easily defined as what you get when the decimal places are ignored! So $\text{floor}(2.4) = 2$ and $\text{floor}(\pi) = 3$. Given that $\sqrt{2} = 1.414\,213\,562$ to 9SFs we can see that $\text{floor}(\sqrt{2}) = 1$ and $\text{floor}(100\sqrt{2}) = 141$. To be honest I hadn't come across this analysis or even the idea that it is possible before typing these notes, so what follows marks out my voyage of discovery. Therefore a little bit of playing first:

$$\begin{aligned} x_1 = 1 &= \text{floor}[\sqrt{2}] \\ x_2 = 4 &= \text{floor}[14.14213562\dots - 10] = \text{floor}[10\sqrt{2} - 10x_1] \\ x_3 = 1 &= \text{floor}[141.4214562\dots - 140] = \text{floor}[100\sqrt{2} - 100x_1 - 10x_2] \\ x_4 = 4 &= \text{floor}[1414.214562\dots - 1410] = \text{floor}[1000\sqrt{2} - 1000x_1 - 100x_2 - 10x_3] \\ x_5 = 2 &= \text{floor}[14142.14562\dots - 14140] = \text{floor}[10000\sqrt{2} - 10000x_1 - 1000x_2 - 100x_3 - 10x_4] \end{aligned} \quad (5.8)$$

This gives the idea of the pattern. The general case, then, is

$$x_n = \text{floor}[10^{n-1}\sqrt{2} - 10^{n-1}x_1 - 10^{n-2}x_2 - \dots - 10x_{n-1}]. \quad (5.9)$$

and this may be written more compactly using the summation notation:

$$x_n = \text{floor}\left[10^{n-1}\sqrt{2} - \sum_{i=1}^{n-1} 10^{n-i}x_i\right]. \quad (5.10)$$

Worthy, perhaps interesting, but not at all useful! Just to say that I don't expect this of you in the exams (!!!) but these ideas may prove useful at some point. Possibly.

Sequence (vii). This one is random. The numbers came from the first eight rolls of an icosahedral die where the faces are numbered from 1 up to 20, as opposed to from 1 up to 6 for a standard cubical die.

5.2 Definition of a series

However, the present topic is about series. A series (singular) may be defined as the sum of the terms in a sequence, and series (plural) may also be either finite or infinite in length. The following contrasts sequences and series.

Sequence: $x_1, x_2, x_3, x_4, \dots, x_n,$

$$\text{Series: } x_1 + x_2 + x_3 + x_4 + \dots + x_n = \sum_{i=1}^n x_i = S_n. \quad (5.11)$$

So we may create a series by adding the terms in a sequence, but it is possible too to make a sequence from a series. The value, S_n , which is the sum of the first n terms of the series and which is called the n^{th} **partial sum**), can form a sequence: S_1, S_2, S_3 and so on. In more detail, this sequence is,

$$S_1 = x_1, \quad S_2 = x_1 + x_2, \quad S_3 = x_1 + x_2 + x_3, \quad S_4 = x_1 + x_2 + x_3 + x_4, \dots \quad (5.12)$$

In many cases we are usually interested in determining the sum of the series and, when it is of infinite length, whether that sum converges to a finite limit or not. The convergence of an infinite series is not a straightforward concept but there are two main ways of judging whether or not a series converges and these will be covered below. As a taster, it is quite clear that the series formed by adding the terms in sequence (i), above, will diverge and become infinite. On the other hand, the series corresponding to sequence (ii) is a geometric series and converges to 2.

In some cases it is possible to find ways of summing series analytically, or else of proving divergence, but these methods are often quite specific to the type of series being considered. A bit of a pain, really.

We will be interested here in creating series which represent functions; this will involve both Binomial expansions and Taylor's Series. When these power series exist we also need to determine their convergence properties, by which we mean, for what range of values of x do the series converge?

Ideas drawn from all of this will be used to answer the age-old question of determining the values of $0/0$ and ∞/∞ . The general non-committal (but completely correct) answer is that it all depends....

5.3 The binomial expansion, theorem and series

First it is essential to state that $(x + y)$ is a **binomial** because there are two quantities involved. Although much less well-known, the following is a **trinomial**: $(x + y + z)$. No doubt one can guess what $(a + b + c + d)$ should be called!

We may find an alternative form for $(x + y)^2$ easily by multiplying out the brackets in $(x + y)(x + y)$; it is $x^2 + 2xy + y^2$. This fact may be visualised by means of the area of the following square.

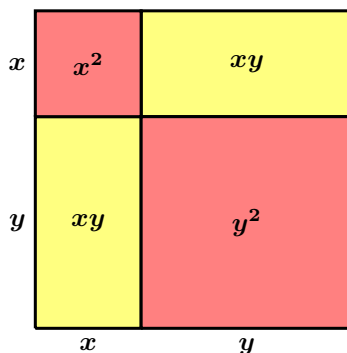


Figure 5.1. Illustrating the binomial expansion of $(x + y)^2$ using areas within a square of side, $(x + y)$.

$$\begin{aligned}
 (1.01)^5 &= 1^5 + [5 \times 1^4 \times 0.01] + [10 \times 1^3 \times 0.01^2] + \dots \\
 &= 1 + 0.05 + 0.001 + \dots \\
 &\simeq 1.051.
 \end{aligned}
 \tag{5.15}$$

Note firstly that one always has to be careful about how large the first neglected term is. In this case it will be 10×0.01^3 , i.e. it is roughly 10^{-5} in magnitude and therefore negligible if we intend to keep to 3DPs. Secondly, this method works well only if $x \gg y$ (or the other way around, obviously). It would not be quite as efficient to evaluate 1.1^5 by this method since all the terms would be required.

Example 5.3. A second example is to find a 4DP approximation to 2.004^5 .

$$\begin{aligned}
 2.004^5 &= 2^5(1.002)^5 && \text{to get a leading "1"} \\
 &= 2^5(1 + 0.002)^5 && \text{now in Binomial form} \\
 &= 2^5 \left[1 + [5 \times 0.002] + [10 \times 0.002^2] + \dots \right] && \text{three terms should be enough?} \\
 &= 2^5 \left[1 + 0.01 + 0.000004 + \dots \right] \\
 &= 2^5 \times 1.01004 = 32.3214 && \text{4DPs.}
 \end{aligned}
 \tag{5.16}$$

Here we factored out the 2 in order to make life a little easier — such expansions are easier and safer with a leading 1. Those who have used a calculator to check the above answer will note that the last decimal place is incorrect and that the number should be **32.3213** to four decimal places. This indicates how much care should be taken with such expansions, and I should have taken one more term in the above calculation.

5.4 Explicit formula for the Binomial coefficients

The binomial coefficients are of particular use when the binomial is raised to a large power, but it is clearly impractical to use Pascal's triangle for, say, a 100th power! Fortunately there is an alternative way.

$$(x + y)^n = \binom{n}{0} x^n + \binom{n}{1} x^{n-1}y + \binom{n}{2} x^{n-2}y^2 + \dots + \binom{n}{n-1} xy^{n-1} + \binom{n}{n} y^n, \tag{5.17}$$

where the Binomial coefficients are defined as,

$$\binom{n}{i} = \frac{n!}{i!(n-i)!} = {}^nC_i, \tag{5.18}$$

and where $0!$ is defined as being equal to 1.

If you think that that is a bit of a cheat, then if we were to define $n!$ using the following formula,

$$n! = \int_0^\infty x^n e^{-x} dx,$$

for integer values of n , it is straightforward to show that $0! = 1$ by setting $n = 0$. This definition also provides for strange quantities such as $(\frac{1}{2})!$.

Note: the symmetry about the vertical midline of Pascal's triangle is reflected in its coefficients:

$$\binom{n}{i} = \frac{n!}{i!(n-i)!} = \binom{n}{n-i}. \quad (5.19)$$

So let us check if Eqs. (5.17) and (5.18) work. We'll use $n = 5$:

$$\begin{aligned} \binom{5}{0} &= \frac{5!}{0!5!} = 1, \\ \binom{5}{1} &= \frac{5!}{1!4!} = 5, \\ \binom{5}{2} &= \frac{5!}{2!3!} = 10, \\ \binom{5}{3} &= \frac{5!}{3!2!} = 10, \\ \binom{5}{4} &= \frac{5!}{4!1!} = 5, \\ \binom{5}{5} &= \frac{5!}{5!0!} = 1. \end{aligned} \quad (5.20)$$

That matches up nicely with Eq. (5.13).

There is a second check that we can do (which is beyond the examinable part of this unit, but it is quick). Referring to the red digits in Fig. 5.3 and the summation property of the coefficients which was demonstrated in Eq. (5.14), we may try to derive the general case. So,

$$\begin{aligned} &\binom{n}{m} + \binom{n}{m+1} && \text{neighbouring coefficients on row } n \\ = &\frac{n!}{m!(n-m)!} + \frac{n!}{(m+1)!(n-m-1)!} && \text{by definition} \\ = &\frac{n!}{(m+1)!(n-m)!} \times [(m+1) + (n-m)] && \text{need your wits on this one!} \\ = &\frac{n! \times (n+1)}{(m+1)!(n-m)!} \\ = &\frac{(n+1)!}{(m+1)![(n+1)-(m+1)]!} && \text{sneaky treatment of the bottom-right term} \\ = &\binom{n+1}{m+1}. \end{aligned} \quad (5.21)$$

That was a bit of a tour-de-force, but that very last step could have been predicted from the values of n and m corresponding to the red values in Fig. 5.3.

5.5 From the Binomial expansion to the Binomial series.

We may now manipulate this new explicit definition of the binomial coefficients, Eq. (5.18), in order to rewrite them in a different form. On taking the general case we have,

$$\begin{aligned}
 \binom{n}{0} &= \binom{n}{n} = \frac{n!}{0! n!} = 1, \\
 \binom{n}{1} &= \binom{n}{n-1} = \frac{n!}{1! (n-1)!} = n, \\
 \binom{n}{2} &= \binom{n}{n-2} = \frac{n!}{2! (n-2)!} = \frac{n(n-1)}{2}, \\
 \binom{n}{3} &= \binom{n}{n-3} = \frac{n!}{3! (n-3)!} = \frac{n(n-1)(n-2)}{3!},
 \end{aligned} \tag{5.22}$$

and so on following that pattern. Therefore Eq. (5.17) with $y = 1$ begins as

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4 + \dots \tag{5.23}$$

Again this has a pattern that is useful to memorise.

Surprisingly, this latest formula, Eq. (5.23), is also valid for both positive fractional values of n and for negative values — this will be proved using Taylor's series which will follow later.

Example 5.4. Find the Binomial Series for $(1+x)^{-1}$.

We simply apply Eq. (5.23) with $n = -1$. Hence

$$\begin{aligned}
 (1+x)^{-1} &= 1 + (-1)x + \frac{(-1)(-2)}{2}x^2 + \frac{(-1)(-2)(-3)}{3!}x^3 + \dots \\
 &= 1 - x + x^2 - x^3 + \dots = \sum_{n=0}^{\infty} (-1)^n x^n.
 \end{aligned} \tag{5.24}$$

We went as far as x^3 on the first line because this was far enough to observe the pattern which was then simplified (though cancellations) on the next line, and then eventually expressed in summation form. The acquisition of a general term is the usual final objective of problems like this.

Note that it is possible to use the series in Eq. (5.24) to write down others. For example, if we were to replace all instances of “ x ” by “ $-x$ ” then we would obtain,

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n. \tag{5.25}$$

We could also replace all instances of “ x ” in Eq. (5.24) by “ x^2 ” to obtain,

$$(1+x^2)^{-1} = 1 - x^2 + x^4 - x^6 + \dots = \sum_{n=0}^{\infty} (-1)^n x^{2n}. \tag{5.26}$$

Honestly, that isn't cheating!

Example 5.5. Find the Binomial Series for $(1 - x)^{-2}$.

Again we apply Eq. (5.23) with $n = -2$ where x is replaced by $-x$. Hence

$$\begin{aligned}
 (1 - x)^{-2} &= 1 + (-2)(-x) + \frac{(-2)(-3)}{2}(-x)^2 + \frac{(-2)(-3)(-4)}{3!}(-x)^3 + \dots \\
 &= 1 + 2x + 3x^2 + 4x^3 + \dots \\
 &= \sum_{n=0}^{\infty} (n+1)x^n.
 \end{aligned} \tag{5.27}$$

Example 5.6 Find the Binomial series for $(1 + x)^{-1/2}$. This one is much trickier.

We apply Eq. (5.23) with $n = -1/2$. Hence

$$(1 + x)^{-1/2} = 1 + \left(-\frac{1}{2}\right)x + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}x^2 + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}x^3 + \dots \tag{5.28}$$

This looks like a mess, and while it is indeed true that it is a mess it can be tidied up after some work. So we'll have a look at the coefficient of x^4 as an exemplar. This is the Binomial series equivalent of white water rafting; we have,

$$\begin{aligned}
 \text{Coefficient of } x^4 &= \frac{(-1)^4 1 \times 3 \times 5 \times 7}{2^4 4!} && \text{leave the } (-1)^4 \text{ as it is} \\
 &= \frac{(-1)^4 1 \times \color{red}{2} \times 3 \times \color{red}{4} \times 5 \times \color{red}{6} \times 7 \times \color{red}{8}}{2^4 4! \times \color{red}{2} \times \color{red}{4} \times \color{red}{6} \times \color{red}{8}} && \text{now we have a factorial in the numerator} \\
 &= \frac{(-1)^4 8!}{2^4 4! \times \color{red}{2^4 4!}} && \text{note carefully what happens with the red terms.}
 \end{aligned} \tag{5.29}$$

The form taken by this coefficient tells us that the general coefficient is $\frac{(-1)^n (2n)!}{2^{2n} n! n!}$, and hence

$$(1 + x)^{-1/2} = \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{2^{2n} n! n!} x^n. \tag{5.30}$$

Note that it is possible to integrate or differentiate these power series in order to derive others although the constant of integration needs to be calculated when integrating. Two very useful examples of such an integration follow.

Example 5.7. Find a power series representation of $y = \ln(1 + x)$ using Binomial series.

In its present form we cannot apply Eq. (5.23), but we will be able to after differentiation:

$$\begin{aligned}
 y &= \ln(1 + x) \\
 \Rightarrow \frac{dy}{dx} &= (1 + x)^{-1} \\
 &= 1 - x + x^2 - x^3 + x^4 \dots && \text{using the previous result in Eq. (5.24)} \\
 \Rightarrow y &= c + x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 \dots && c \text{ is an arbitrary constant} \\
 &= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 \dots && y(0) = 0, \text{ hence } c = 0 \\
 &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}.
 \end{aligned} \tag{5.31}$$

Example 5.8. Find the power series representation of $\sin^{-1}(x)$.

The following is how one may write the inverse sine function in integral form.

$$\sin^{-1} x = \int_0^x \frac{dt}{\sqrt{1-t^2}}. \tag{5.32}$$

Give our experience with Example 5.7, we shall begin by finding the Binomial series of $(1 - t^2)^{-1/2}$. Rather than undertaking that expansion from scratch, we may use Eq. (5.30) and replace all instances of ' x ' by ' $-t^2$ '. This gives us,

$$(1 - t^2)^{-1/2} = \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n} n! n!} t^{2n}. \tag{5.33}$$

We may now integrate each term in turn and evaluate these between $t = 0$ and $t = x$. Hence,

$$\sin^{-1} x = \int_0^x (1 - t^2)^{-1/2} dt = \sum_{n=0}^{\infty} \frac{(2n)!}{(2n+1) 2^{2n} n! n!} x^{2n+1}. \tag{5.34}$$

Hence,

$$\begin{aligned}
 \sin^{-1} x &= \sum_{n=0}^{\infty} \frac{(2n)!}{(2n+1) 2^{2n} n! n!} x^{2n+1} \\
 &= x + \frac{2!}{3 \times 2^2 \times 1! \times 1!} x^3 + \frac{4!}{5 \times 2^4 \times 2! \times 2!} x^5 + \frac{6!}{7 \times 2^6 \times 3! \times 3!} x^7 + \dots,
 \end{aligned} \tag{5.35}$$

where the first terms have been written out.

Note: One has to be a little careful with $(2n)!$ — this is **not** the same as $2 \times (n!)$. So if $n = 5$, then $(2n)! = 10! = 3628800$ while $2(n!) = 2 \times 5! = 240$. This is another example where it is possible to write something down ambiguously, or to make an incorrect assumption about what a formula means. Consider the following three expressions:

$$(2n)! \qquad 2n! \qquad 2(n!).$$

Of these the first and the third are written correctly, and the brackets have been used (and hopefully will be interpreted) correctly. The problem with the second one is that it could be interpreted as either the first of the third, and therefore my recommendation is that it should never be used. One could fit the act of taking factorials into the BODMAS/PEMDAS/BEDMAS/BIDMAS rules by saying that it has a higher priority than multiplication and division. In that case $2n!$ ought to be the double of $n!$, but I still think that it is open to misinterpretation and therefore it should be avoided.

5.6 Power series, Maclaurin's series and Taylor's series

Any series of the type

$$a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots \quad (5.36)$$

is called a **power series** simply because it consists of powers of x . An example of such a series is

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad (5.37)$$

but how is such an expression obtained, particularly since we cannot use the Binomial expansion?

Let us first assume that it is possible to express an arbitrary function, $f(x)$, in a power series form with positive integer powers, and therefore we set

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots \quad (5.38)$$

We may find the value of a_0 by substituting $x = 0$ into Eq. (5.38). This yields $a_0 = f(0)$. Now we shall differentiate Eq. (5.38) and substitute $x = 0$ again. We get,

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots, \quad \Rightarrow \quad a_1 = f'(0). \quad (5.39)$$

Now, two terms is rarely a good basis for extrapolation — yes, a_2 will not be equal to $f''(0)$! (By the way, that was an exclamation mark, not a factorial!). So we'll differentiate Eq. (5.39),

$$f''(x) = 2a_2 + (3 \times 2)a_3x + (4 \times 3)a_4x^2 + (5 \times 4)a_5x^3 + \dots, \quad \Rightarrow \quad a_2 = \frac{1}{2}f''(0). \quad (5.40)$$

Actually, three terms are also not a good basis for extrapolation. The denominators of the numerical coefficients for a_0 , a_1 and a_2 are, 1, 1, 2, which could represent various things including the Fibonacci sequence. So we need to go a little further. Actually, we'll do it twice more, and we get, in turn:

$$f'''(x) = (3 \times 2)a_3 + (4 \times 3 \times 2)a_4x + (5 \times 4 \times 3)a_5x^2 + \dots \quad \Rightarrow \quad a_3 = f'''(0)/(3 \times 2), \quad (5.41)$$

and

$$f''''(x) = (4 \times 3 \times 2)a_4 + (5 \times 4 \times 3 \times 2)a_5x + \dots \quad \Rightarrow \quad a_4 = f''''(0)/(4 \times 3 \times 2). \quad (5.42)$$

So these denominators are factorials. We have just derived what is usually called **Maclaurin's series**:

$$\begin{aligned} f(x) &= f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \frac{f''''(0)x^4}{4!} + \dots \\ &= \sum_{n=0}^{\infty} \frac{d^n f}{dx^n} \bigg|_{x=0} \frac{x^n}{n!}. \end{aligned} \quad (5.43)$$

The Maclaurin expansion focuses on $x = 0$ as the point ‘about which’ the function is expanded. More generally we may expand about any other point, and therefore we could also have

$$\begin{aligned} f(x) &= f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \frac{f^{(4)}(a)(x-a)^4}{4!} + \dots \\ &= \sum_{n=0}^{\infty} \frac{d^n f}{dx^n} \bigg|_{x=a} \frac{(x-a)^n}{n!}. \end{aligned} \quad (5.44)$$

Equation (5.44) may also be derived in exactly the same way as for Eq. (5.43), namely by successive differentiation followed by the substitution of $x = a$. This more general expression is called **Taylor's series about $x = a$** . Thus Maclaurin's series is simply Taylor's series about $x = 0$.

Note: that while Eq. (5.44) is the one which we shall use for this unit, the University's Formula Book has it written down slightly differently; there it is

$$f(x) = f(x_0) + \delta f'(x_0) + \frac{\delta^2}{2!} f''(x_0) + \frac{\delta^3}{3!} f'''(x_0) + \dots + \frac{\delta^n}{n!} f^{(n)}(x_0) + R_{n+1}. \quad (5.45)$$

In this expression $f^{(n)}$ is the n^{th} derivative, $\delta = x - x_0$ and the value, R_{n+1} is the error due to using only a finite series. Clearly x_0 plays the same role that a does in the above analysis, but we will not be considering the error term.

Example 5.9. Find the Taylor's series of e^{bx} about $x = 0$.

An alternative way of phrasing this example is: Expand e^{bx} about $x = 0$.

Essentially all we need to do is find successive derivatives of the function, let $x = 0$ in those derivatives and then assemble the Taylor's series using Eq. (5.43). I prefer to use a Table to do this because it is tidy and may be checked easily. So we have,

n	$f^{(n)}$	$f^{(n)}(0)$
0	e^{bx}	1
1	$b e^{bx}$	b
2	$b^2 e^{bx}$	b^2
3	$b^3 e^{bx}$	b^3

Equation (5.43) yields

$$e^{bx} = 1 + bx + \frac{b^2 x^2}{2!} + \frac{b^3 x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{b^n x^n}{n!}. \quad (5.46)$$

Once more this type of result may be manipulated. By the simple act of replacing b by $-b$ we obtain,

$$e^{-bx} = 1 - bx + \frac{b^2 x^2}{2!} - \frac{b^3 x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n b^n x^n}{n!}. \quad (5.47)$$

This may look like a sleazy manoeuvre but it is correct and rigorous. A more acceptable way might be to replace b by c in (5.46) and afterwards to use the substitution, $c = -b$.

Example 5.10. Find the Taylor's series of $\cos bx$ about $x = 0$.

Again using a Table:

n	$f^{(n)}$	$f^{(n)}(0)$
0	$\cos bx$	1
1	$-b \sin bx$	0
2	$-b^2 \cos bx$	$-b^2$
3	$b^3 \sin bx$	0
4	$b^4 \cos bx$	b^4
5	$-b^5 \sin bx$	0

I took a few more terms to be assured of the pattern of coefficients. The Taylor's series is

$$\cos bx = 1 - \frac{b^2 x^2}{2!} + \frac{b^4 x^4}{4!} \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n b^{2n} x^{2n}}{(2n)!}. \quad (5.48)$$

Note carefully the uses of n and $2n$ in the final summation and how this enables us to get the alternating signs. A similar analysis yields,

$$\sin bx = bx - \frac{b^3 x^3}{3!} + \frac{b^5 x^5}{5!} \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n b^{2n+1} x^{2n+1}}{(2n+1)!}. \quad (5.49)$$

Sorry, but I want to play briefly with these last two results. If all the instances of b in Eq. (5.46) were to be replaced by bj then we have,

$$e^{jbx} = 1 + jbx + \frac{j^2 b^2 x^2}{2!} + \frac{j^3 b^3 x^3}{3!} + \frac{j^4 b^4 x^4}{4!} + \frac{j^5 b^5 x^5}{5!} + \cdots = \sum_{n=0}^{\infty} \frac{j^n b^n x^n}{n!}. \quad (5.50)$$

This may be rearranged to give,

$$\begin{aligned} e^{jbx} &= \left[1 - \frac{b^2}{2!} x^2 + \frac{b^4}{4!} x^4 - \frac{b^6}{6!} x^6 + \cdots \right] + j \left[bx - \frac{b^3}{3!} x^3 + \frac{b^5}{5!} x^5 - \frac{b^7}{7!} x^7 + \cdots \right] \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n b^{2n} x^{2n}}{(2n)!} + j \sum_{n=0}^{\infty} \frac{(-1)^n b^{2n+1} x^{2n+1}}{(2n+1)!} \\ &= \cos bx + j \sin bx \quad \text{using Eqs. (5.48) and (5.49).} \end{aligned} \quad (5.51)$$

So this is the proof of $e^{jbx} = \cos bx + j \sin bx$ which was promised earlier in the Complex Numbers chapter.

Example 5.11. Find the Taylor's series of $(1+x)^{-1}$ about $x=0$ and about $x=1$.

Here's the Table of data for the two cases:

n	$f^{(n)}$	$f^{(n)}(0)$	$f^{(n)}(1)$
0	$(1+x)^{-1}$	1	$1/2$
1	$-(1+x)^{-2}$	-1	$-1/2^2$
2	$2(1+x)^{-3}$	2	$2/2^3$
3	$-3!(1+x)^{-4}$	-3!	$-3!/2^4$
4	$4!(1+x)^{-5}$	4!	$4!/2^5$
5	$-5!(1+x)^{-6}$	-5!	$-5!/2^6$

From these data and Eq. (5.44) we obtain the desired Taylor's series:

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots = \sum_{n=0}^{\infty} (-1)^n x^n, \quad (5.52)$$

and

$$\frac{1}{1+x} = \frac{1}{2} \left[1 - \frac{x-1}{2} + \frac{(x-1)^2}{2^2} - \frac{(x-1)^3}{2^3} + \frac{(x-1)^4}{2^4} - \frac{(x-1)^5}{2^5} + \dots \right] = \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^n}{2^n}. \quad (5.53)$$

Example 5.12 Expand the integral $y(x) = \int_0^x e^{-t^2} dt$ about $x=0$ and hence evaluate $y(0.2)$.

This integral is very closely related to the Normal Distribution but the indefinite integral of e^{-x^2} cannot be written in terms of familiar functions, which is a bit awkward. Bizarrely, it is possible to show that $y(x) \rightarrow \sqrt{\pi}/2$ as $x \rightarrow \infty$, as we found in the final integration problem sheet.

We begin by differentiating the integral; we get

$$y'(x) = e^{-x^2},$$

an exponential function. Now, I would not advise trying to apply the Taylor's series formula directly. Rather it is better to note that we already have a Taylor's series representation of e^{-x} in Eq. (5.46) and therefore we may simply replace all occurrences of x by x^2 to find that,

$$y'(x) = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} \dots \quad (5.54)$$

Integration of this (noting that $y(0) = 0$ so that we can evaluate the constant of integration) yields,

$$y(x) = x - \frac{x^3}{3} + \frac{x^5}{5 \cdot 2!} - \frac{x^7}{7 \cdot 3!} + \frac{x^9}{9 \cdot 4!} \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1) \times n!}. \quad (5.55)$$

Three terms of this series gives $y(0.2) = 0.197365$. The fourth term has magnitude 3×10^{-7} and therefore just these three terms are enough for six decimal places of accuracy since four terms yields no change.

5.7 The convergence of power series

Following on from Example 5.12 a natural question is whether we can use any value of x in Eq. (5.55) or any other series. It is this which forms the topic of the present subsection, but we will study a couple of series from earlier in this chapter in order to illustrate that different power series may have different convergence properties. For convenience Eqs. (5.52), (5.49) (with $b = 1$) and (5.46) (also with $b = 1$) are quoted here:

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots, \quad (5.56)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad (5.57)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad (5.58)$$

5.7.1 Numerical demonstration of convergence/divergence

The series in Eq. (5.56) is a geometric series and therefore we could have found its sum, the left hand side, using those ideas. When we choose $x = 0.1$ then it feels certain that the right hand side will converge because the successive terms are decreasing rapidly. Even more obvious is the choice, $x = -0.1$, for then the partial sums become, 1, 1.1, 1.11, 1.111 and on to the value 10/9. On the other hand, if we were to choose $x = 2$ then the series looks like,

$$1 - 2 + 2^2 - 2^3 + 2^4 - 2^5 + \dots$$

and there is no hope of this converging to a finite value. Although the signs alternate, the magnitudes of the **partial sums** increase: 1, -1, 3, -5, 11, -21, 43.... After a bit of work I determined that this sequence satisfies $[(-1)^2 2^{n+1} + 1]/3$ — big deal! Our conclusion is that convergence for the series in Eq. (5.56) is conditional: if x is small enough then it converges, but it will diverge otherwise. The boundary between those behaviours is likely to be $x = 1$ for then the series is

$$1 - 1 + 1 - 1 + 1 \dots,$$

and the partials sums alternate: 1, 0, 1, 0, and so on.

A similar playing around with numbers for Eqs. (5.57) and (5.58) using a calculator takes time. The fact that the denominators are factorials gives a hint that these series might well be convergent. If one were to choose $x = 0.1$ in Eq. (5.58), then only a few terms are required to obtain, say, 6DP accuracy. This may be seen in the following Table (where u_n represents the successive terms, and S_n the partial sums):

Showing successive terms and partial sums for Eq. (5.58) when $x = 0.1$

n	u_n	S_n
0	1.000000	1.000000
1	0.100000	1.100000
2	0.005000	1.105000
3	0.000167	1.105167
4	0.000004	1.105171
5	4×10^{-7}	1.105171

That final answer is correct to 6DPs.

On the other hand, if we choose $x = 10$ then we will have the following.

Showing successive terms and partial sums for Eq. (5.58) when $x = 10$

n	u_n	S_n
0	1.000000	1.000000
1	10.000000	11.000000
2	50.000000	61.000000
3	166.666667	227.666667
4	416.666667	644.333333
5	833.333333	1477.666667
6	1388.888889	2866.555556
7	1984.126984	4850.682540
8	2480.158730	7330.841270
9	2755.731922	10086.573192
10	2755.731922	12842.305115
11	2505.210839	15347.515953
12	2087.675699	17435.191652
13	1605.904384	19041.096036
14	1147.074560	20188.170595
15	764.716373	20952.886969
16	477.947733	21430.834702
17	281.145725	21711.980427
18	156.192070	21868.172497
19	82.206352	21950.378849
20	41.103176	21991.482026
21	19.572941	22011.054967
22	8.896791	22019.951758
23	3.868170	22023.819928
24	1.611738	22025.431666
25	0.644695	22026.076361
26	0.247960	22026.324321
27	0.091837	22026.416157
28	0.032799	22026.448956
29	0.011310	22026.460266
30	0.003770	22026.464036
31	0.001216	22026.465252
32	0.000380	22026.465632
33	0.000115	22026.465748
34	0.000034	22026.465781
35	0.000010	22026.465791
36	0.000003	22026.465794
37	0.000001	22026.465795
38	0.000000	22026.465795
39	0.000000	22026.465795
40	0.000000	22026.465795

My apologies for the length of this Table but there are some important points which arise. The first is that, even for a value of x that is as large as 10, the series converges. The second point is that the magnitude of successive terms in the series decrease once $n > 10$. Given the form of the series, successive terms will decrease when $x = 100$ is taken only when $n > 100$, and therefore this suggests that the series will converge for any value of x . The third point is illustrated by compiling a similar Table for $x = -10$:

Showing successive terms and partial sums when $x = -10$

n	u_n	S_n
0	1.000000	1.000000
1	-10.000000	-9.000000
2	50.000000	41.000000
3	-166.666667	-125.666667
4	416.666667	291.000000
5	-833.333333	-542.333333
6	1388.888889	846.555556
7	-1984.126984	-1137.571429
8	2480.158730	1342.587302
9	-2755.731922	-1413.144621
10	2755.731922	1342.587302
11	-2505.210839	-1162.623537
12	2087.675699	925.052162
13	-1605.904384	-680.852222
14	1147.074560	466.222338
15	-764.716373	-298.494035
16	477.947733	179.453698
17	-281.145725	-101.692027
18	156.192070	54.500042
19	-82.206352	-27.706310
20	41.103176	13.396866
21	-19.572941	-6.176075
22	8.896791	2.720716
23	-3.868170	-1.147454
24	1.611738	0.464284
25	-0.644695	-0.180411
26	0.247960	0.067548
27	-0.091837	-0.024289
28	0.032799	0.008510
29	-0.011310	-0.002800
30	0.003770	0.000970
31	-0.001216	-0.000246
32	0.000380	0.000134
33	-0.000115	0.000019
34	0.000034	0.000053
35	-0.000010	0.000043
36	0.000003	0.000046
37	-0.000001	0.000045
38	0.000000	0.000045
39	-0.000000	0.000045
40	0.000000	0.000045

The terms in the u_n column here have the same magnitude as their counterparts when $x = 10$ but they alternate in sign. Despite the cancellations due to having both positive and negative values with a large magnitude, the final converged value of this series is $e^{-10} \simeq 0.00004540$, which is the same value that I get on my calculator.

For a more extreme case, $x = -40$, the ‘converged’ value of the series turns out to be about -3.166 which is clearly wrong — exponentials can’t be negative. So what has happened? This has arisen because the 13

significant figures of accuracy in my computer calculations are no longer sufficient to cope with the very great amount of cancellation that arises when adding the terms in the series. For $x = -40$ my calculator gives 4.248×10^{-18} which I have checked to be correct via other means, and therefore I conclude that my calculator doesn't use Taylor's series! Even here, one must always be aware of significant figures and whether accuracy has been compromised because of the necessary use of finite precision arithmetic.

5.7.2 Graphical representation of convergence/divergence

Figure 5.3 shows the variation of $\sin x$ and of the first 11 partial sums of the Taylor's series (the third partial sum, for example, being the first three terms, $x - x^3/3! + x^5/5!$):

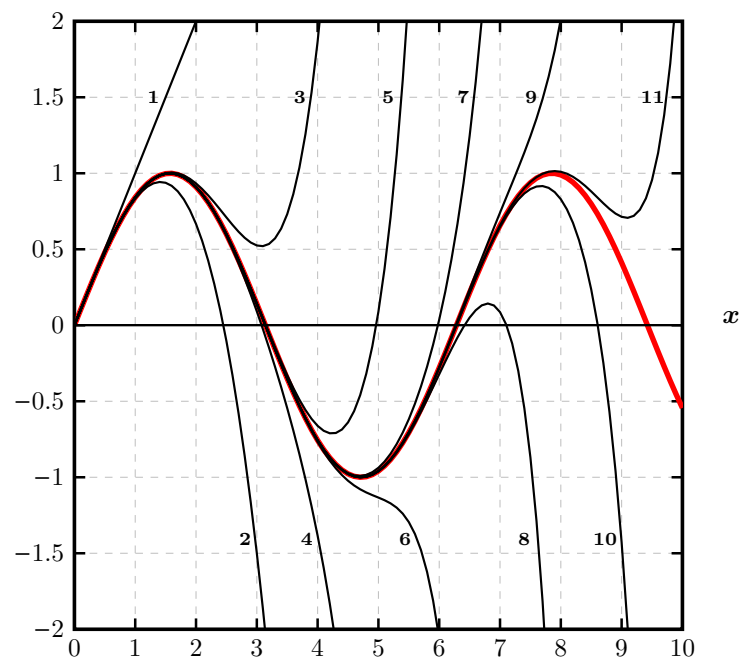


Figure 5.3. Demonstrating the convergence properties of the Taylor's series for $\sin x$. Showing the function itself (red) and the first 11 partial sums.

The first partial sum, x , forms the tangent to $\sin x$ at $x = 0$, and this figure demonstrates that just this one term is quite a good approximation up to, perhaps $x = 0.5$.

The second partial sum, $x - x^3/3!$, matches the sine wave over a larger range of values of x , and the graph suggests that that series does well when $x < 1$.

As further terms in the Taylor's series are included, the sequence of curves gradually continues to envelop the sine wave, and it appears that the ninth partial sum provides quite an accurate estimate for its first period. There is no suggestion or hint here that the Taylor's series will diverge for some larger value of x , although we have already seen that there will be a limitation on the usefulness of the series for sufficiently large values of x due to the use of fixed precision arithmetic.

So our tentative conclusion is that convergence happens irrespective of the value of x .

Figure 5.4 shows the variation of $(1+x)^{-1}$ and of the first 10 partial sums of the Taylor's series

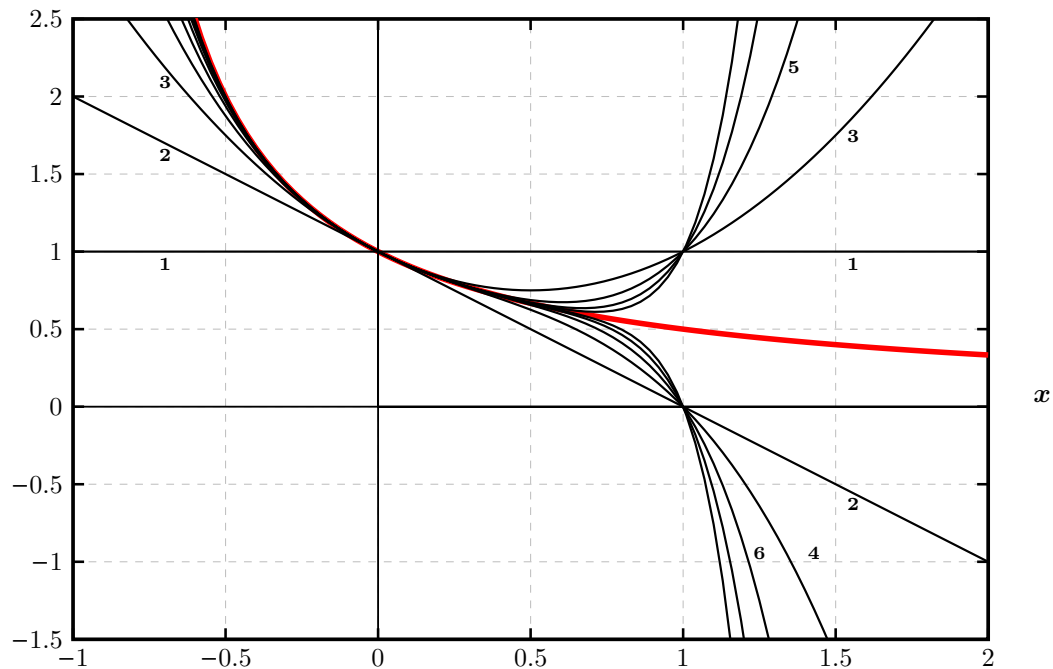


Figure 5.4. Demonstrating the convergence properties of the Taylor's series for $(1+x)^{-1}$. Showing the function itself (red) and the first 10 partial sums.

Once more, when x lies within a certain range, the partial sums approach the original curve. For positive values of x this certainly happens when $x < 1$ (strictly less than 1). Once $x > 1$ be it ever so slightly, then successive partial sums diverge away from the original function.

When $x < 0$ the convergence towards $(1+x)^{-1}$ is monotonic. It is more difficult to see what happens when x is large and negative, but $(1+x)^{-1}$ tends to infinity as $x \rightarrow -1$ and this presents a difficulty. When $x < -1$ the successive terms in the Taylor's series get larger in magnitude and therefore convergence fails to happen.

Our tentative conclusion from this numerical experiment is that we have convergence when $-1 < x < 1$.

5.8 d'Alembert's ratio test

If the series under consideration has the form

$$S = \sum_{n=1}^{\infty} u_n = u_1 + u_2 + u_3 + \cdots, \quad (5.59)$$

then **d'Alembert's ratio test** states the following:

$\lim_{n \rightarrow \infty} \left \frac{u_{n+1}}{u_n} \right < 1$	$\Rightarrow$	absolute convergence	(5.60)
$\lim_{n \rightarrow \infty} \left \frac{u_{n+1}}{u_n} \right > 1$	$\Rightarrow$	divergence	
$\lim_{n \rightarrow \infty} \left \frac{u_{n+1}}{u_n} \right = 1$	$\Rightarrow$	inconclusive.	

Note: that this test relies on the long-term behaviour of the coefficients, and therefore the starting point (viz. $n = 1$ in Eq. (5.59)) is of no consequence.

d'Alembert's test is an excellent method for finding the range of values of x for power series, but when the series consists solely of numerical data inconclusive conclusions apply to a depressingly wide range of useful series. You'll see what I mean....

5.8.1 Convergence of numerical series

Example 5.13. The Maclaurin's series for e^1 is $e^1 = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$. Determine if it converges to a value.

The first step always is to determine an expression for the general term. In this case the general term is the reciprocal of successive factorials. Hence $u_n = 1/n!$ and $u_{n+1} = 1/(n+1)!$. The second step is to evaluate $|u_{n+1}/u_n|$:

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{n!}{(n+1)!} \right| = \left| \frac{n!}{n! \times (n+1)} \right| = \frac{1}{n+1}. \quad (5.61)$$

Hence,

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0. \quad (5.62)$$

Given that $0 < 1$, this series converges.

Example 5.14. Use d'Alembert's test to check for the convergence of the geometric series, $r + r^2 + r^3 + \dots$.

Here u_n may be taken to be r^n . Hence $u_{n+1} = r^{n+1}$ and so $u_{n+1}/u_n = r$. Therefore we have,

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = |r|. \quad (5.63)$$

For convergence we require $|r| < 1$.

Example 5.15. Consider the sum of the reciprocals of the positive integers,

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

Here $u_n = 1/n$ and therefore

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{n}{n+1} \right| = \frac{1}{1 + \frac{1}{n}} \rightarrow 1 \text{ as } n \rightarrow \infty. \quad (5.64)$$

Therefore the test is inconclusive because the limit is precisely 1.

However, it is possible to show that the series is divergent by considering the series in the following way.

$$\begin{aligned} & 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4} \right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \right) + \left(\frac{1}{9} + \dots + \frac{1}{16} \right) + \dots \\ & > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4} \right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} \right) + \left(\frac{1}{16} + \dots + \frac{1}{16} \right) + \dots \\ & = 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots \end{aligned} \quad (5.65)$$

Clearly this process continues indefinitely and if we continue to N such sets of brackets then we will accumulate N instances of $\frac{1}{2}$, and therefore

$$S_{2^N} = \sum_{n=1}^{2^N} \frac{1}{n} > 1 + \frac{1}{2}N,$$

and hence the series diverges.

This is a trick that can only very rarely be deployed, but it is interesting.

Having settled the convergence properties of this example, we may return to Example 5.7 which yields the following when $x = 1$ is substituted,

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

This differs from the present example by having alternating signs, yet Example 5.7 shows that this series adds to $\ln 2$ and hence it is convergent.

These two series demonstrate the idea of **conditional convergence**. The respective magnitudes of the terms in the two series are identical but the pattern of signs differ, so the convergence properties then depend on that pattern of signs.

Example 5.16. Consider the sum of the reciprocals of the squares of the integers,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots.$$

Here $u_n = 1/n^2$ and hence

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{n^2}{(n+1)^2} \right| = \frac{1}{(1 + \frac{1}{n})^2} \rightarrow 1 \text{ as } n \rightarrow \infty. \quad (5.66)$$

Again we have an inconclusive conclusion from d'Alembert! This is a little depressing for these coefficients decay more rapidly than those in Example 5.15. This series does in fact converge, but it requires us to use Fourier Series methods (to be done in ME10305 Mathematics 2) to show that the sum is $\pi^2/6$. The equivalent sum of the reciprocals of the fourth powers of the positive integers is $\pi^4/90$, although d'Alembert's test remains inconclusive even for this series.

5.9 Application of d'Alembert's test to power series

Now we take the power series,

$$a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \cdots, \quad (5.67)$$

and apply d'Alembert's ratio test to it using $u_n = a_nx^n$ as the general term. Therefore we obtain convergence when

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}x^{n+1}}{a_nx^n} \right| < 1. \quad (5.68)$$

Although this may be simplified and rearranged to get the convergence criterion,

$$|x| < \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|. \quad (5.69)$$

I really don't like to teach this form because the subscripts here (n on top, $n+1$ below) are the opposite way around from those in the original ratio test given in Eq. (5.60) and this could be confusing. My preference is to

employ the formula in Eq. (5.60) as it is and then to follow whatever the maths tells us. I will do this in all of the examples below.

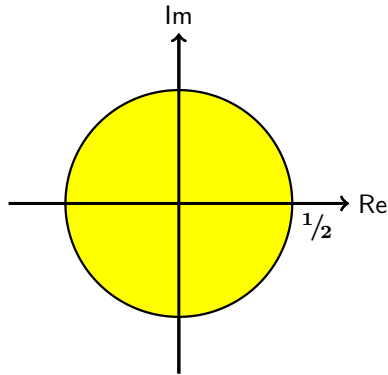
Example 5.17. Find the values of x for which the following power series converges.

$$1 + 2x + 4x^2 + 8x^3 + 16x^4 + \dots$$

The general term may be written as $u_n = 2^n x^n$ and therefore we have

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{2^{n+1} x^{n+1}}{2^n x^n} \right| = 2|x|. \quad (5.70)$$

Therefore we have convergence when $2|x| < 1$, i.e. when $|x| < \frac{1}{2}$. Perhaps this may be a surprise, but this expression is also valid for complex values of x and therefore we may sketch the region of convergence in the complex plane as follows:



The yellow region is the region of convergence while the radius of the black perimeter is called the **radius of convergence** of the series; in this case we say that the radius of convergence is $\frac{1}{2}$. Thus values of x which are larger than $\frac{1}{2}$ in magnitude will cause the series to diverge.

Example 5.18. Find the values of x for which the Taylor series given in Eq. (5.53) converges.

The series is

$$\frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^n}{2^n}$$

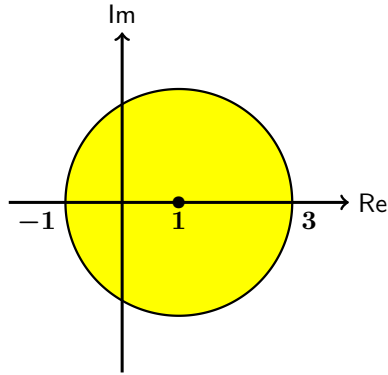
and therefore the general term is

$$u_n = \frac{(-1)^n (x-1)^n}{2^n}.$$

Hence,

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (x-1)^{n+1}}{2^{n+2}} \times \frac{2^{n+1}}{(-1)^n (x-1)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x-1)}{2} \right| = \frac{1}{2} |x-1|. \quad (5.71)$$

This needs to be less than 1 for convergence, which means that $|x-1| < 2$. So the radius of convergence is 2, as illustrated in the following Figure.



Should we wish to confine ourselves to real values of x , this means that convergence will then take place only when $-1 < x < 3$, noting the strict inequalities in this expression.

Example 5.19. Find the radius of convergence of the Taylor's series for e^x given in Eq. (5.58).

The series is $e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \dots$. Hence the general term is $u_n = x^n/n!$, and therefore d'Alembert's test gives,

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| = \lim_{n \rightarrow \infty} \left| \frac{x}{n+1} \right| = 0. \quad (5.72)$$

While we saw this sort of result in Example 5.13, in the present context the limit is zero for all possible values of x , and therefore this function has **an infinite radius of convergence**. This confirms our deduction from numerical evidence in Section 5.7.1.

Example 5.20. Find the radius of convergence of the Taylor's series for $\sin x$ which is given in Eq. (5.57).

The series is $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ and therefore the general term is $u_n = \frac{(-1)^n x^{2n+1}}{(2n+1)!}$.

The correct expression for u_{n+1} is $\frac{(-1)^{n+1} x^{2n+3}}{(2n+3)!}$. **Note** that the replacement of n by $n+1$ in $2n+1$ yields $2n+3$ rather than $2n+2$, a favourite error of so many undergraduates in an exam!

Hence

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{2n+3}}{(2n+3)!} \times \frac{(2n+1)!}{(-1)^n x^{2n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^2}{(2n+3)(2n+2)} \right| = 0. \quad (5.73)$$

So again we have an infinite radius of convergence, just as we suspected from the graphical evidence in Fig. 5.3.

Finally, I'll leave it to you to show that the series, $1 + (1!)x + (2!)x^2 + (3!)x^3 + \dots$, has a zero radius of convergence. By this is meant that the series always diverges except for when $x = 0$.

5.10 l'Hôpital's rule

5.10.1 A graphical motivation

This rule allows us to evaluate the ratio of two functions, $f(x)$ and $g(x)$, as $x \rightarrow a$ when both $f(a) = 0$ and $g(a) = 0$. The situation is well-illustrated by Figure 5.5.

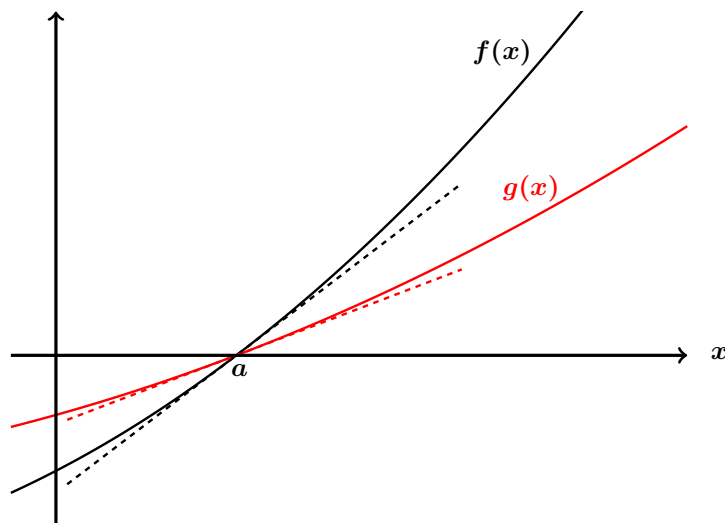


Figure 5.5. Displaying $f(x)$ and $g(x)$ both of which are zero at $x = a$, together with their tangents (dotted lines) at $x = a$.

What is zero-divided-by-zero? More precisely in this context, we wish to find the value of $f(x)/g(x)$ when $x = a$ and this diagram tells us the basic answer: it depends on how quickly the two functions move towards zero as x moves towards a . Figure 5.5 shows two generic functions, $f(x)$ and $g(x)$, and both of them are zero at $x = a$. Also shown as dashed lines are the tangents to each curve at $x = a$. One could say that as $x = a$ is approached then each curve more closely resembles its tangent and when $x = a$ they are identical. So this produces what I call the graphical motivation for finding the value of the ratio of $f(x)$ and $g(x)$ at $x = a$.

The tangents are given, respectively, by

$$f(x) \simeq (x - a)f'(a) \quad g(x) \simeq (x - a)g'(a). \quad (5.74)$$

These may be written down immediately when it is noted (i) tangents are linear functions, (ii) these specific tangents must pass through $x = a$ and hence $(x - a)$ must be a factor, and (iii) their slopes are the slopes of the functions at $x = a$.

Therefore the ratio of $f(x)$ and $g(x)$ may be written as,

$$\frac{f(x)}{g(x)} = \frac{(x - a)f'(a)}{(x - a)g'(a)} = \frac{f'(a)}{g'(a)}. \quad (5.75)$$

Given that $f(x)$ isn't quite equal to $(x - a)f'(a)$ unless $x = a$, it is better to write this down as

$$\boxed{\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}}. \quad (5.76)$$

OK, if you are someone like me then your reaction is going to be that this seems plausible, but you would rather have a more rigorous proof than this. Well, we can bring Taylor's series into the fray and let Taylor provide a better proof.

5.10.2 Proof of l'Hôpital's rule using Taylor's series

Equations (5.74) use what is, in reality, the first nonzero term of the Taylor's series of the two functions. We need to use a few more just to check that all is well. So let us write out this ratio again and work on it.

$$\begin{aligned}
 \frac{f(x)}{g(x)} &= \frac{\cancel{f(a)} + (x-a)f'(a) + \frac{1}{2}(x-a)^2 f''(a) + \frac{1}{3!}(x-a)^3 f'''(a) + \dots}{\cancel{g(a)} + (x-a)g'(a) + \frac{1}{2}(x-a)^2 g''(a) + \frac{1}{3!}(x-a)^3 g'''(a) + \dots} && \text{by definition} \\
 &= \frac{f'(a) + \frac{1}{2}(x-a)f''(a) + \frac{1}{3!}(x-a)^2 f'''(a) + \dots}{g'(a) + \frac{1}{2}(x-a)g''(a) + \frac{1}{3!}(x-a)^2 g'''(a) + \dots} && \text{cancelling } (x-a) \\
 &\longrightarrow \frac{f'(a)}{g'(a)} \text{ as } x \rightarrow a.
 \end{aligned}
 \tag{5.77}$$

The conclusion here is that the terms we neglected in Eq. (5.74) do not contribute to the final answer, but it was necessary to check this for our peace of mind. So Eq. (5.76) has been retrieved using a better analysis.

Example 5.21 Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$.

Both the numerator and the denominator are zero at $x = 0$ and therefore we apply l'Hôpital's rule.

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = \left. \frac{2 \cos 2x}{1} \right|_{x=0} = 2.$$

Example 5.22 Find $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$.

The numerator and the denominator are zero at $x = \pi$ and therefore,

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi} = \left. \frac{\cos x}{1} \right|_{x=\pi} = -1.$$

Example 5.23 Find $\lim_{x \rightarrow 0} \frac{\sin^2 x}{x}$.

We have

$$\lim_{x \rightarrow 0} \frac{\sin^2 x}{x} = \left. \frac{2 \sin x \cos x}{1} \right|_{x=0} = 0.$$

This example shows that zero is a legitimate final answer.

Example 5.24 Find $\lim_{x \rightarrow 0} \frac{\sin x}{x^2}$.

We have

$$\lim_{x \rightarrow 0} \frac{\sin x}{x^2} = \left. \frac{\cos x}{2x} \right|_{x=0} = \infty.$$

So this example shows that it is legitimate for an infinite solution to be obtained. One can say that x^2 tends to zero much faster than $\sin x$ does.

5.10.3 Nested applications of l'Hôpital's rule

This follows from the question: what happens to the derivation of l'Hôpital's rule if, in addition to $f(a) = g(a) = 0$, we also have $f'(a) = g'(a) = 0$? These conditions tell us that both $f(x)$ and $g(x)$ have a double zero at $x = a$ and therefore we are comparing two **parabola**e.

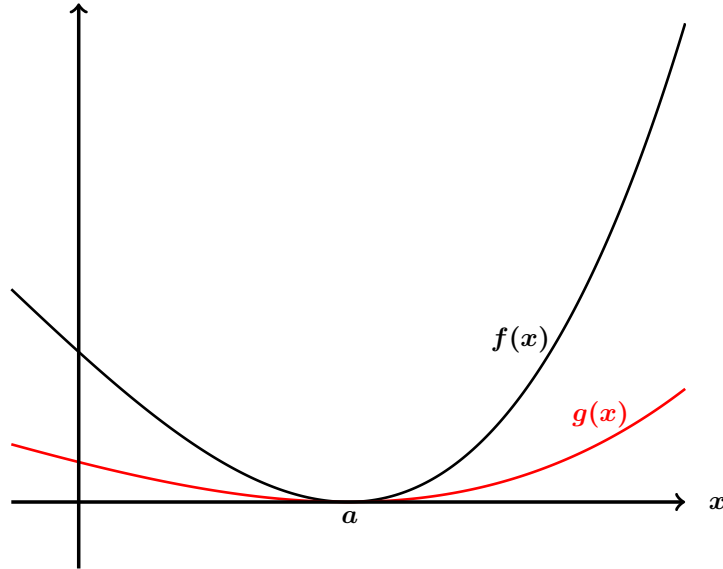


Figure 5.6. Displaying $f(x)$ and $g(x)$ both of which are zero and have a zero first derivative at the point $x = a$.

In this case Eq. (5.77) becomes,

$$\begin{aligned}
 \frac{f(x)}{g(x)} &= \frac{\cancel{f(a)} + \cancel{(x-a)}f'(a) + \frac{1}{2}(x-a)^2 f''(a) + \frac{1}{3!}(x-a)^3 f'''(a) + \dots}{\cancel{g(a)} + \cancel{(x-a)}g'(a) + \frac{1}{2}(x-a)^2 g''(a) + \frac{1}{3!}(x-a)^3 g'''(a) + \dots} && \text{by definition} \\
 &= \frac{\frac{1}{2}f''(a) + \frac{1}{3!}(x-a)f'''(a) + \dots}{\frac{1}{2}g''(a) + \frac{1}{3!}(x-a)g'''(a) + \dots} && \text{cancelling } (x-a)^2 \\
 &\longrightarrow \frac{f''(a)}{g''(a)} \text{ as } x \rightarrow a.
 \end{aligned}
 \tag{5.78}$$

So when the functions and their derivatives are all zero at $x = a$, l'Hôpital's rule becomes,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f''(a)}{g''(a)}.
 \tag{5.79}$$

Although we won't prove this, if these second derivatives are also zero then we need to compute the ratio of the third derivatives, and so on. So it is my view that the best way to write down l'Hôpital's rule is the following way:

$$\boxed{\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}}.
 \tag{5.80}$$

There is a subtle difference between this and Eq. (5.76) in that the right hand side also uses the limiting process. Thus if we have applied the rule to a zero-divide-zero ratio and have obtained another one, then we merely apply the same formula to the new case. In essence this is a recursive formula which can continue to be applied to successive ratios until the ratio has a definitive limit (i.e. zero, nonzero or infinite, as opposed to being indeterminate).

Example 5.25 Find $\lim_{x \rightarrow 0} \frac{1 - \cos ax}{x^2}$.

$$\lim_{x \rightarrow 0} \frac{1 - \cos ax}{x^2} \stackrel{\textcolor{red}{l'H}}{=} \lim_{x \rightarrow 0} \frac{a \sin ax}{2x}$$

This too is a $0/0$ case, hence l'H again

$$\stackrel{\textcolor{red}{l'H}}{=} \lim_{x \rightarrow 0} \frac{a^2 \cos ax}{2}$$

This isn't a $0/0$ case

$$= \frac{1}{2}a^2.$$

Note that I have introduced my own notation here: $\stackrel{\textcolor{red}{l'H}}{=}$. You won't find it in textbooks, but I am using it simply to indicate that I have used l'Hôpital's rule as opposed to merely simplifying what is inside the limit.

Example 5.26 Find $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$.

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} \stackrel{\textcolor{red}{l'H}}{=} \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2}$$

This too is a $0/0$ case, hence l'H again

$$\stackrel{\textcolor{red}{l'H}}{=} \lim_{x \rightarrow 0} \frac{-\sin x}{6x}$$

This too is a $0/0$ case, hence l'H again

$$\stackrel{\textcolor{red}{l'H}}{=} \lim_{x \rightarrow 0} \frac{-\cos x}{6} = \frac{1}{6}$$

This isn't a $0/0$ case

5.10.4 A few more exotic examples

Example 5.27 Find $\lim_{x \rightarrow \infty} \frac{e^{ax}}{x}$.

This example demonstrates two different variations on the standard l'Hôpital's rule theme. First we are considering $x \rightarrow \infty$ and second we are considering two functions that become infinite in that limit. In this case l'Hôpital's rule compares the speed at which the two functions grow in magnitude. We have

$$\lim_{x \rightarrow \infty} \frac{e^{ax}}{x} \stackrel{\textcolor{red}{l'H}}{=} \frac{ae^{ax}}{1} \Big|_{x \rightarrow \infty} = \infty.$$

Example 5.28 [A more difficult example.] Find $\lim_{x \rightarrow 0} x^x$.

This doesn't look like a l'Hôpital's rule question because there is no ratio! However, if we let $y = x^x$, then

$$\ln y = x \ln x = \frac{\ln x}{1/x}.$$

This is an $x \rightarrow 0$ problem where the ratio is an infinity-over-infinity case. Now we may apply l'Hôpital's rule:

$$\lim_{x \rightarrow 0} \ln y \stackrel{\textcolor{red}{l'H}}{=} \lim_{x \rightarrow 0} \frac{\ln x}{1/x}$$

This is an ∞/∞ case, hence use l'H

$$\stackrel{\textcolor{red}{l'H}}{=} \lim_{x \rightarrow 0} \frac{1/x}{-1/x^2}$$

Also ∞/∞ but we can simplify the quotient

$$= \lim_{x \rightarrow 0} (-x)$$

I haven't used l'Hôpital's rule here, just normal algebra

$$= 0.$$

Since $\ln y \rightarrow 0$ then $y \rightarrow 1$. Therefore the $x \rightarrow 0$ limit of x^x is 1.

Example 5.29 [Also a more difficult example.] Find $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

This problem is associated with compound interest. When $n = 1$ the factor $\left(1 + \frac{1}{n}\right)^n$ may be said to be equivalent to the adding of 100% interest to an investment, I , after a year. The yield is that the investment doubles to $2I$.

When $n = 2$, then 50% interest is applied twice per year, and the return is now $\left(1 + \frac{1}{2}\right)^2 I = 2.25I$.

When $n = 4$, then 25% interest is added quarterly, and the return is $\left(1 + \frac{1}{4}\right)^4 I = 2.441406I$.

If monthly, then the return will be $\left(1 + \frac{1}{12}\right)^{12} I = 2.613035I$, and if it is daily then the return will be $\left(1 + \frac{1}{365}\right)^{365} I = 2.714567I$. The ultimate question is: what is your return if interest is added continuously at the appropriate infinitesimal rate?

Again this is not a ratio, but again we may take logs. So if $y = \left(1 + \frac{1}{n}\right)^n$ then

$$\ln y = n \ln \left(1 + \frac{1}{n}\right) = \frac{\ln \left(1 + \frac{1}{n}\right)}{\frac{1}{n}}.$$

This looks very messy, but with a little sleight of hand we may transform a large- n problem into a small- x problem by setting $x = 1/n$. Therefore the problem becomes,

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} \stackrel{\textcolor{red}{l'H}}{=} \lim_{x \rightarrow 0} \frac{1/(1+x)}{1} = 1.$$

This, then, is the value of $\ln y$ and therefore the $n \rightarrow \infty$ limit of y is e . The return on the investment will be eI or $2.718282I$ (6DPs). You won't get any more than that!

6 Vector methods

6.1 Introduction

In this Chapter we will introduce what a vector is together with concepts such as the length of a vector, unit vectors and addition. Thereafter we meet both the scalar and vector products of two vectors, so-named because the results of these products are a scalar and a vector, respectively. Then the scalar triple product appears. Using these concepts the idea of orthogonality may be defined, and we then apply vector theory to physical problems like (i) what is the angle between two directions in three-dimensional space, (ii) how far is a given point from a given line, (iii) how close do two lines approach each other, and (iv) how far is a point from a plane? Some of these geometrical ideas would be considerably more difficult to work out without vectors. That lot would normally take up four lectures' worth of time.

6.2 Elementary definitions

There is a simple difference between scalars and vectors:

Scalar: This has a magnitude. Examples include my mass, and the fact that London is 100 miles away.

Vector: This has a magnitude and a direction. Examples include the size and direction of the force induced on the ground due to my weight, and the fact that London is 100 miles east of the University.

Other scalar quantities include temperature, pressure, voltage, how much cash I have in the bank and the number of children I have. Other vector quantities include position relative to a fixed point (a so-called position vector), velocities, accelerations and forces. It is worth noting that there is a scientific distinction between speed and velocity: the former is a scalar while the latter is a vector. The same is true formally, at least, between mass and weight (though perhaps not in your local grocer shop).

6.3 Notation

In the following figure we introduce the notations which are used in the context of vectors.

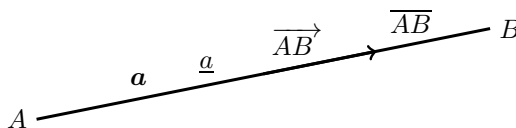


Figure 6.1. Showing a vector in the direction from point A to point B, together with some of the notations which are often used.

Some more detail on the notations:

- $\underline{a}$ is a very common notation where the underbar denotes the fact that it is a vector.
- $\mathbf{a}$ is what often appears in textbooks; it's a bold italic font. Strange, because it is quite difficult to write bold characters on a piece of paper or on a blackboard/whiteboard!
- $\vec{AB}$ and $\overline{AB}$ tell us that the vector corresponds to the position of B relative to A , or to the direction and distance that one has to travel to get from A to B .

The length of a vector also has a variety of notations: $|\underline{a}|$, $|\mathbf{a}|$, $|\vec{AB}|$ and AB . Given all of these, it is clear that $|\underline{a}| = |-\underline{a}|$ and $AB = BA$ and so on.

6.4 Addition of two vectors

This is most easily seen in the following figure:

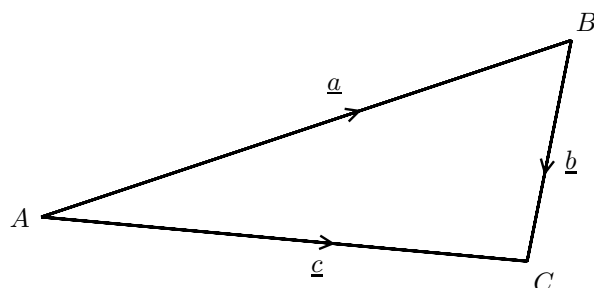


Figure 6.2. Showing vector addition diagrammatically. Here $\underline{a} + \underline{b} = \underline{c}$.

We have

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC} = \underline{a} + \underline{b} = \underline{c}. \quad (6.1)$$

If we now create a point D which has a position vector $\underline{b}$ relative to A , as in the following Figure, then what is the vector denoted by $\underline{v}$?

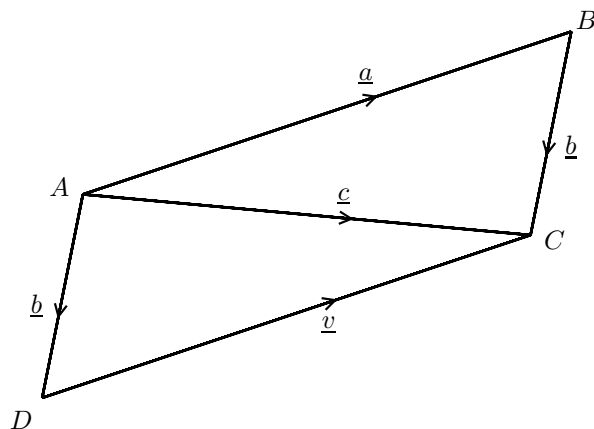


Figure 6.3. Showing that $\underline{v} = \underline{a}$.

From this Figure we see that,

$$\underline{v} = \overrightarrow{DA} + \overrightarrow{AC} = -\underline{b} + \underline{c} = -\underline{b} + \underline{a} + \underline{b} = \underline{a}. \quad (6.2)$$

6.5 Cartesian Coordinates and Unit Vectors

While it is possible to derive many results involving vectors without having a coordinate system in mind, it is sometimes helpful think of vectors in terms of their Cartesian components. The following Figure shows how this occurs.

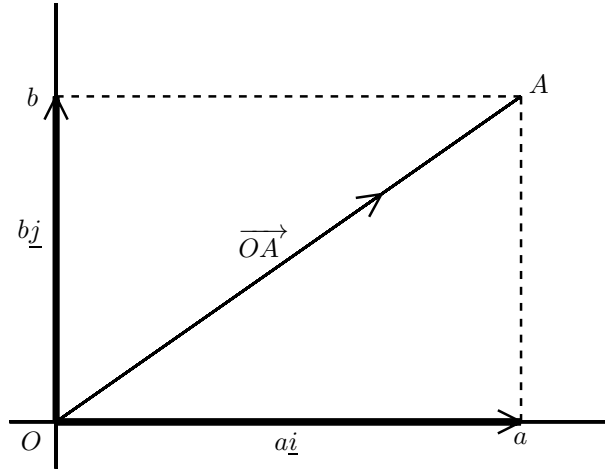


Figure 6.4. Showing how a 2D vector may be decomposed into its components in the x and y -directions, and how these components are proportional to the respective unit vectors, $\underline{i}$ and $\underline{j}$.

From this Figure we see that the x -component of the vector $\overrightarrow{OA}$ is a , while the y -component is b . As vectors which are equivalent to travelling unit distances in the x and y directions may be written in the form,

$$\underline{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \underline{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (6.3)$$

respectively, we may write $\overrightarrow{OA}$ in the equivalent forms,

$$\overrightarrow{OA} = \begin{pmatrix} a \\ b \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix} = a\underline{i} + b\underline{j}. \quad (6.4)$$

Here, it is to be noted that this notation, $\begin{pmatrix} a \\ b \end{pmatrix}$, is distressingly similar that for the Binomial coefficients, but this present context is very different and therefore confusion should not arise.

Given this representation we have

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix} = \begin{pmatrix} c \\ d \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}, \quad (6.5)$$

and therefore vector addition is **commutative** in Cartesian coordinates, i.e. the same answer is obtained independently of the order in which two vectors are added, just as with $2 + 3 = 3 + 2 = 5$.

From this diagram we see that the magnitude of the vector $\overrightarrow{OA}$ may be obtained using Pythagoras's theorem:

$$|\overrightarrow{OA}| = \sqrt{a^2 + b^2}. \quad (6.6)$$

In three dimensions we have the analogous result:

$$|\overrightarrow{OA}| = \sqrt{a^2 + b^2 + c^2}, \quad (6.7)$$

where

$$\overrightarrow{OA} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}, \quad (6.8)$$

as shown in the following Figure.

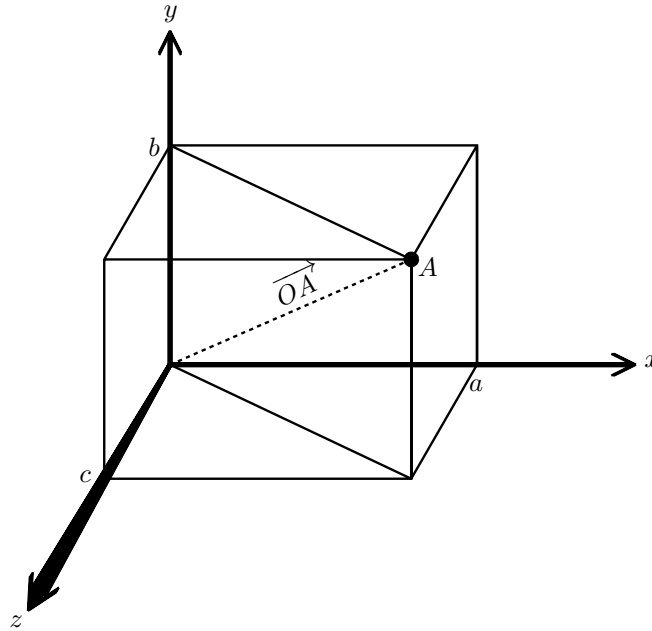


Figure 6.5 Showing how a 3D vector, $\overrightarrow{OA}$, may be understood in terms of its components in the x , y and z -directions.

6.6 The scalar product of two vectors

There are two ways which have been defined for the multiplication of two vectors, namely the scalar (or dot) product, and the vector (or cross) product.

The principal formula for the scalar product is

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta, \quad (6.9)$$

where θ is the angle between the directions of the two vectors, as illustrated in Figure 6.6.

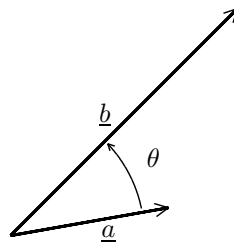


Figure 6.6 Showing two different vectors, and the angle, θ between their directions.

6.6.1 The angle between two vectors

It is possible to use the formula in Eq. (6.9) to calculate the scalar product without having to evaluate θ , itself. Then, we may use Eq. (6.9) directly so that $\cos \theta$, and hence θ itself, may be found.

Restricting ourselves to 2D for now, we need to note first that

$$\underline{i} \cdot \underline{i} = \underline{j} \cdot \underline{j} = 1 \quad \text{and} \quad \underline{i} \cdot \underline{j} = 0, \quad (6.10)$$

since the angles between the various vectors are either zero or $\frac{1}{2}\pi$. This means that any two perpendicular vectors have a zero scalar product. Now if we define

$$\underline{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = a_1 \underline{i} + a_2 \underline{j} \quad \text{and} \quad \underline{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = b_1 \underline{i} + b_2 \underline{j}, \quad (6.11)$$

then

$$\begin{aligned} \underline{a} \cdot \underline{b} &= \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \\ &= (a_1 \underline{i} + a_2 \underline{j}) \cdot (b_1 \underline{i} + b_2 \underline{j}) \\ &= a_1 b_1 \underline{i} \cdot \underline{i} + a_2 b_2 \underline{j} \cdot \underline{j} + (a_1 b_2 + a_2 b_1) \underline{i} \cdot \underline{j} \\ &= a_1 b_1 + a_2 b_2. \end{aligned} \quad (6.12)$$

Thus the scalar product may be found simply by multiplying corresponding components of the two vectors and adding the resulting products. Given that we now have two different formulae for the scalar product, they may be equated:

$$a_1 b_1 + a_2 b_2 = |\underline{a}| |\underline{b}| \cos \theta, \quad (6.13)$$

and therefore θ may be found.

A similar result applies in three dimensions, for if

$$\underline{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = a_1 \underline{i} + a_2 \underline{j} + a_3 \underline{k} \quad \text{and} \quad \underline{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = b_1 \underline{i} + b_2 \underline{j} + b_3 \underline{k}, \quad (6.14)$$

then

$$\begin{aligned} \underline{a} \cdot \underline{b} &= \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \\ &= (a_1 \underline{i} + a_2 \underline{j} + a_3 \underline{k}) \cdot (b_1 \underline{i} + b_2 \underline{j} + b_3 \underline{k}) \\ &= a_1 b_1 + a_2 b_2 + a_3 b_3. \end{aligned} \quad (6.15)$$

So the angle between general 3D vectors is given by,

$$a_1 b_1 + a_2 b_2 + a_3 b_3 = |\underline{a}| |\underline{b}| \cos \theta. \quad (6.16)$$

Example 6.1. What is the angle between the vectors $\underline{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$?

Using the formula Eq. (6.13) the left hand side gives,

$$\underline{a} \cdot \underline{b} = (2 \times 1) + (1 \times 2) = 4.$$

while the right hand side give,

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta = \sqrt{5} \sqrt{5} \cos \theta = 5 \cos \theta.$$

As these quantities are equal, then

$$5 \cos \theta = 4 \quad \Rightarrow \quad \cos \theta = \frac{4}{5} \quad \Rightarrow \quad \theta = 0.643501 \text{ (36.8699}^\circ\text{)}.$$

Example 6.2. What is the angle between the vectors $\underline{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$?

These two look as though they are quite similar in direction so my initial guess is that the angle is fairly small. The left hand side of Eq. (6.16) gives,

$$\underline{a} \cdot \underline{b} = (1 \times 4) + (2 \times 5) + (3 \times 6) = 32.$$

while the right hand side is

$$\underline{a} \cdot \underline{b} = \sqrt{1^2 + 2^2 + 3^2} \sqrt{4^2 + 5^2 + 6^2} \cos \theta = 32.8329 \cos \theta.$$

Therefore

$$\cos \theta = \frac{32}{32.8329} = 0.97463 \quad \Rightarrow \quad \theta = 0.22573 \text{ (12.933}^\circ\text{)}.$$

Yes, quite a small angle. Smug mode....

Example 6.3. Two vectors are called **orthogonal** if the angle between them is 90° . If $\underline{a} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 2 \\ \alpha \end{pmatrix}$ are orthogonal, what is the value of α ?

We are told that the vectors are orthogonal, which means that $\underline{a} \cdot \underline{b} = 0$ because $\cos(\pi/2) = 0$. We find that,

$$\underline{a} \cdot \underline{b} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ \alpha \end{pmatrix} = (4 \times 2) + (1 \times \alpha) = 8 + \alpha \quad \Rightarrow \quad \alpha = -8.$$

6.7 The unit vector

If we have a vector such as $\underline{a} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$ then its corresponding unit vector $\hat{\underline{a}}$ has a unit length but is pointed in the same direction as $\underline{a}$. Here $|\underline{a}| = \sqrt{2^2 + 3^2 + 6^2} = 7$ (woo, a Pythagorean quadruple...) and therefore,

$$\hat{\underline{a}} = \frac{1}{7} \underline{a} = \begin{pmatrix} 2/7 \\ 3/7 \\ 6/7 \end{pmatrix}. \quad (6.17)$$

More generally we can say that $\underline{b} = |\underline{b}| \hat{\underline{b}}$, where $|\underline{b}|$ takes care of the magnitude, and $\hat{\underline{b}}$, the direction of $\underline{b}$.

6.8 The vector product of two vectors

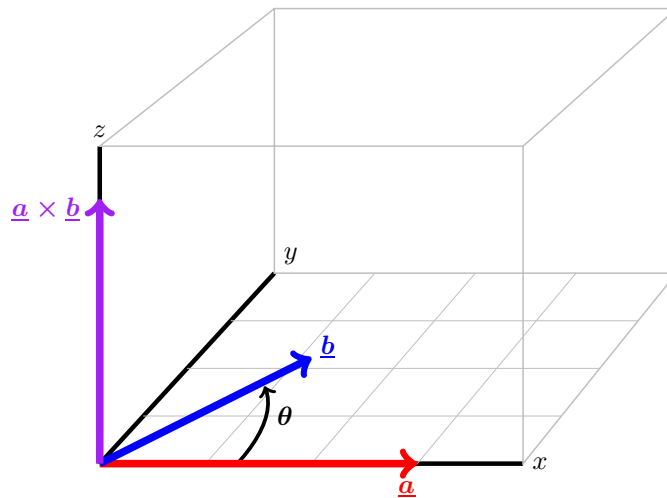
For the *scalar* product of two vectors we had

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta, \quad (6.18)$$

a scalar. The standard formula for the *vector* product is

$$\underline{a} \times \underline{b} = |\underline{a}| |\underline{b}| \sin \theta \hat{p}, \quad (6.19)$$

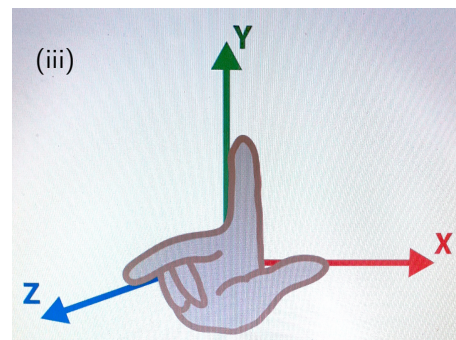
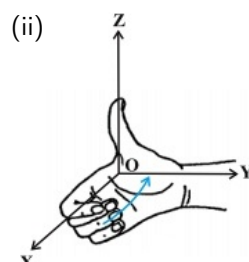
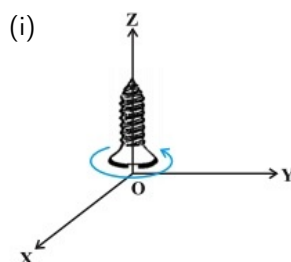
where $\hat{p}$ is the unit vector which is perpendicular to both $\underline{a}$ and $\underline{b}$, and which points in the direction given by the right hand rule. Here's a diagram which illustrates this product:



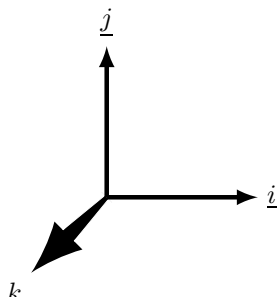
There are various ways of illustrating this; one of these is to imagine the direction of motion of a screw (i.e. inwards or outwards) when the screwdriver is rotated from the direction given by $\underline{a}$ to that given by $\underline{b}$. Personally, I find that difficult to visualise easily, so a good alternative is that given by considering the thumb and fingers of the right hand when hitchhiking. Here the rotational direction given by the curved fingers correspond to the rotation required to move from the direction of $\underline{a}$ to that of $\underline{b}$, and then the direction in which the thumb is pointing gives the direction of $\hat{p}$.

A third alternative is given by setting one's thumb and first two fingers of the right hand at right angles to one another, taking $\underline{a}$ to correspond to the thumb, $\underline{b}$ to the first finger, and $\hat{p}$ to the second finger. I find that one the easiest to remember.

After all of those words, it is probably better if I were to illustrate these ways of recalling the right hand rule (these are for $\underline{z} = \underline{x} \times \underline{y}$):



If we take $\underline{i} \times \underline{j} = \underline{k}$ as the definition of a right handed system of orthogonal unit vectors, $\underline{i}$, $\underline{j}$ and $\underline{k}$, then the application of the above right hand rule to the various unit vector combinations yields the results,



$\underline{i} \times \underline{j} = \underline{k},$	$\underline{j} \times \underline{i} = -\underline{k},$
$\underline{i} \times \underline{k} = -\underline{j},$	$\underline{k} \times \underline{i} = \underline{j},$
$\underline{j} \times \underline{k} = \underline{i},$	$\underline{k} \times \underline{j} = -\underline{i}.$

(6.20)

Notice here that the vector product is not commutative since, for example,

$$\underline{i} \times \underline{j} \neq \underline{j} \times \underline{i}.$$

This may be verified using one of three illustrations above for what constitutes a right-handed system. In fact, the following is true in general:

$$\underline{a} \times \underline{b} = -\underline{b} \times \underline{a}. \quad (6.21)$$

6.8.1 Lexicographical ordering

Note that it is easy to remember cross products of the unit vectors if we recall the order in which they are usually stated: $\underline{i}, \underline{j}, \underline{k}, \underline{i}, \underline{j}$ etc.. If we evaluate the cross product of any two neighbouring vectors in the order in which they appear in this sequence, then the result is *plus* the next vector in the sequence:

$$\underline{i} \times \underline{j} = \underline{k} \quad \underline{j} \times \underline{k} = \underline{i} \quad \underline{k} \times \underline{i} = \underline{j}, \quad (6.22)$$

and it is this feature which is called **Lexicographical ordering**. If the two vectors being multiplied are in sequence from right to left, then the product is *minus* the next vector in the sequence. Hence

$$\underline{j} \times \underline{i} = -\underline{k} \quad \underline{k} \times \underline{j} = -\underline{i} \quad \underline{i} \times \underline{k} = -\underline{j}. \quad (6.23)$$

We also note that the cross product of a vector with itself is zero, i.e. that

$$\underline{a} \times \underline{a} = 0, \quad (6.24)$$

since $\sin \theta = 0$ here.

6.8.2 The determination of vector products

We have

$$\begin{aligned}
 \underline{a} \times \underline{b} &= (a_1 \underline{i} + a_2 \underline{j} + a_3 \underline{k}) \times (b_1 \underline{i} + b_2 \underline{j} + b_3 \underline{k}) \\
 &= (a_2 b_3 - a_3 b_2) \underline{i} + (a_3 b_1 - a_1 b_3) \underline{j} + (a_1 b_2 - a_2 b_1) \underline{k} \quad \text{using (6.22) and (6.23)} \\
 &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad (6.25)
 \end{aligned}$$

This construction is a determinant. It is likely that a substantial proportion of the class won't have come across this before — it is part of next's semester content on matrices — but for now I offer the following way to evaluate it.

The secret to remembering the formulae is to note how the coefficients of the unit vectors in the penultimate line of Eq. (6.25) are obtained from the determinant notation on the last line. For the $\underline{i}$ coefficient we have,

$$(a_2 b_3 - a_3 b_2) \underline{i} + (a_3 b_1 - a_1 b_3) \underline{j} + (a_1 b_2 - a_2 b_1) \underline{k} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}. \quad (6.26)$$

Note how the terms typeset in red are multiplied out. This is called cross-multiplication. For the $\underline{j}$ coefficient we have,

$$(a_2 b_3 - a_3 b_2) \underline{i} + (a_3 b_1 - a_1 b_3) \underline{j} + (a_1 b_2 - a_2 b_1) \underline{k} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \begin{matrix} a_1 & a_2 \\ b_1 & b_2 \end{matrix} \quad (6.27)$$

where I have written 'ghostly' terms to the right of the determinant purely in order to preserve the multiplying-out pattern used in Eq. (6.26). And finally, for the $\underline{k}$ coefficient, we have,

$$(a_2 b_3 - a_3 b_2) \underline{i} + (a_3 b_1 - a_1 b_3) \underline{j} + (a_1 b_2 - a_2 b_1) \underline{k} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \begin{matrix} a_1 & a_2 \\ b_1 & b_2 \end{matrix} \quad (6.28)$$

I hope that this is clear. There are other ways of doing this, but I prefer this as a particularly safe way to proceed.

Example 6.4. If $\underline{a} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$, then find a unit vector which is perpendicular to them both.

This is easily done by applying the determinantal formula for the vector product. We have

$$\underline{p} = \underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 1 & 1 \\ 1 & 0 & 3 \end{vmatrix} = \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix}.$$

We'll test out the answer by taking the dot products of both $\underline{a}$ and $\underline{b}$ with $\underline{p}$:

$$\underline{a} \cdot \underline{p} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} = 6 - 5 - 1 = 0.$$

$$\underline{b} \cdot \underline{p} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} = 3 + 0 - 3 = 0.$$

This confirms that $\underline{p}$ is perpendicular to both $\underline{a}$ and $\underline{b}$.

The length of the vector is $\sqrt{3^2 + 5^2 + 1^2} = \sqrt{35}$. Hence,

$$\underline{\hat{p}} = \frac{1}{\sqrt{35}} \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix}.$$

Given the phrasing of the Exercise (Find $\underline{\hat{a}}$ unit vector), then one could also legitimately write down,

$$\underline{\hat{p}} = \frac{1}{\sqrt{35}} \begin{pmatrix} -3 \\ 5 \\ 1 \end{pmatrix}.$$

Therefore there are always two unit vectors which are perpendicular to two given vectors and one is the negative of the other.

Example 6.5. Prove that the vectors $\underline{a} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix}$ and $\underline{c} = \begin{pmatrix} 1 \\ -3 \\ 5 \end{pmatrix}$ all lie in the same plane.

Note that this question doesn't ask about three points lying in a plane because three points always lie in a plane, hence the widespread use of tripods! When three vectors lie in a plane it is assumed that they all come from a common origin, and therefore we are really considering four points. It is illustrated below.



Figure 6.8. Showing three vectors lying in a plane.

In general four points don't lie in a plane; this could be illustrated by raising, say, the end of the red pen above the piece of paper the pens are on. In such a case, three points lie on a plane and the fourth doesn't.

So the question is: is there a general condition that will be satisfied when three vectors lie in a plane?

If $\underline{a}$, $\underline{b}$ and $\underline{c}$ all lie in the same plane, then one approach could be to find values of α and β which are such that

$$\underline{c} = \alpha \underline{a} + \beta \underline{b}. \quad (6.29)$$

This approach corresponds to using $\underline{a}$ and $\underline{b}$ like Cartesian coordinates, albeit ones that are not necessarily at right angles. If one can find the values of α and β , then the three vectors lie in the same plane, but if there is no solution (always likely because Eq. (6.29) represents three equations for two unknowns), then they don't lie

in the same plane. However, this approach can be fairly time consuming. A better approach is to realise that, if the three vectors lie in the same plane, then $\underline{a} \times \underline{b}$ must be perpendicular to $\underline{c}$. In mathematical terms this **co-planarity condition** is

$$\underline{a} \times \underline{b} \cdot \underline{c} = 0. \quad (6.30)$$

This is called a **scalar triple product**. For the specific vectors in this exercise, we can show that,

$$\underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & -2 & 1 \\ 2 & 1 & -4 \end{vmatrix} = \begin{pmatrix} 7 \\ 14 \\ 7 \end{pmatrix},$$

and therefore

$$\underline{a} \times \underline{b} \cdot \underline{c} = \begin{pmatrix} 7 \\ 14 \\ 7 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -3 \\ 5 \end{pmatrix} = 7 - 42 + 35 = 0.$$

So these three vectors lie in the same plane.

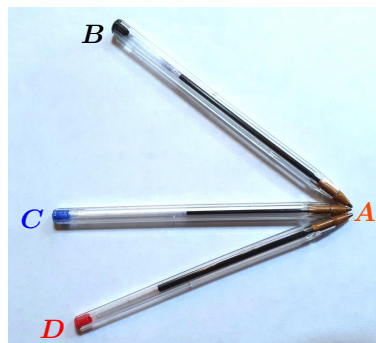
Note: The condition $\underline{a} \times \underline{b} \cdot \underline{c} = 0$ is the condition for coplanarity. When this is satisfied then the vectors are also called **linearly dependent**, and in such a case it is possible to find the values of α and β in Eq. (6.29). By linearly dependent we mean that any one of the three vectors may be written in terms of the other two. If the three vectors do not satisfy co-planarity condition then they are **linearly independent** and may then be used as a 3D coordinate system — this means that any point in space may be reached using a unique combination of these vectors. Of course the unit vectors, $\underline{i}$, $\underline{j}$ and $\underline{k}$, are linearly independent vectors.

Example 6.6. Determine if four points are co-planar.

This is a variation on the previous example. These are points, not vectors. If the four points are labelled, A , B , C and D , then we could choose A , say, as a reference point and create three vectors:

$$\overrightarrow{AB}, \quad \overrightarrow{AC} \quad \text{and} \quad \overrightarrow{AD},$$

as shown below.



Then we may apply the analysis of Example 6.5 to these three vectors.

6.9 The area of a triangle

This is the first section of what might be called applications of vectors.

The finding of the area of a triangle is also a rather unusual application. Here is the diagram.

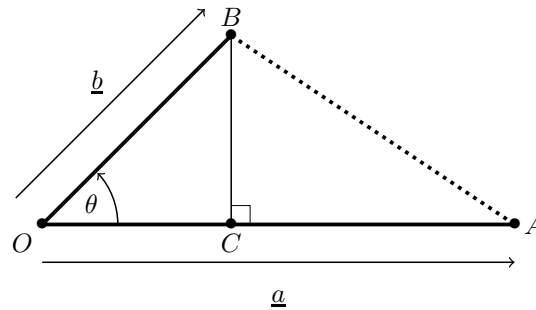


Figure 6.9. Showing how to find the area of the triangle OAB using vector methods.

The aim is find the area of the triangle, OAB , and all the quantities that we need are shown in Fig. 6.9. For simplicity we shall assume that the triangle lies in the (x, y) -plane. We start with the standard verbal description of the area of a triangle and gradually coerce that into an overtly mathematical form.

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} \text{ base} \times \text{the perpendicular height} \\
 &= \frac{1}{2} OA \times CB && \times = \text{scalar multiplication} \\
 &= \frac{1}{2} |\underline{a}| \times |\underline{b}| \sin \theta && \text{resolving the angle} \\
 &= \frac{1}{2} (\underline{a} \times \underline{b}) \cdot \underline{k}. && \times = \text{vector multiplication}
 \end{aligned} \tag{6.31}$$

This form of the solution takes account of the fact that $\underline{a} \times \underline{b}$ is in the positive $\underline{k}$ -direction. Then the dot product with $\underline{k}$ merely reveals the coefficient of $\underline{k}$ in $\underline{a} \times \underline{b}$. Given the way that Fig. 6.9 has been drawn with the relative placement of $\underline{a}$ and $\underline{b}$ as shown, then $\sin \theta > 0$ and this answer will always be positive.

There could well be scenarios when a negative value of the area might be obtained. This will happen when the rotation from the direction of $\underline{a}$ to that of $\underline{b}$ is in the clockwise direction. Sometimes a negative area is a legitimate answer because of what it represents — there's a question on a problem sheet where it is essential to use a negative answer — but if one wishes to avoid negative values then the absolute value of the cross product,

$$\text{Area} = \frac{1}{2} |\underline{a} \times \underline{b}|,$$

is always positive.

6.10 The distance of points from lines

We'll off with a general line. The theory isn't too tricky, he says somewhat optimistically, and relies on translating the words into a diagram. After that it is a case of using what we know that vectors can do. Let's see how we do....

Example 6.7. A straight line passes through a point with position vector, $\underline{a}$, and in the direction, $\underline{b}$. How close does the line get to the point? Here's the diagram which will help us.

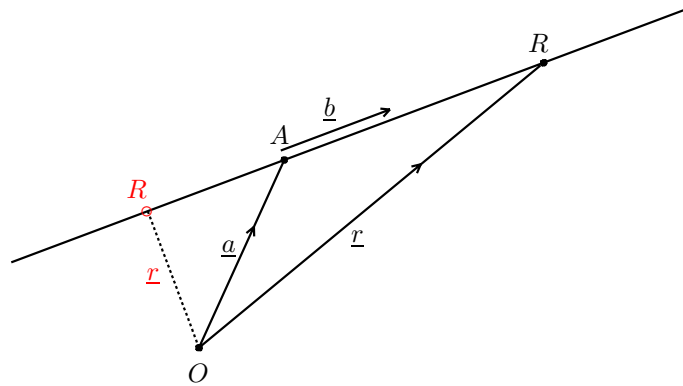


Figure 6.10. Showing the line which passes through the point with position vector, $\underline{a}$, and in the direction, $\underline{b}$. The general point on the line is depicted as $\underline{r}$. The dotted line shows where the line is closest to the origin.

Starting with the origin, O , the vector, $\underline{a}$, has been drawn with the end of the vector defining the point, A . The line from A to what I have labelled as R is in the direction of $\underline{b}$. Therefore, if we wish to travel from O to the arbitrary point, R , on this line, then we start from O , travel to A and then on to R . The vector taking us from A to R is an arbitrary multiple of $\underline{b}$. Mathematically that gives us,

$$\overrightarrow{OR} = \overrightarrow{OA} + \overrightarrow{AR} \quad \text{or} \quad \underline{r} = \underline{a} + \lambda \underline{b}, \quad (6.32)$$

where λ is the parameter which tells us how far along the line we are from A .

The red labels on Figure 6.10 represent the closest approach of the line and the origin. The essential criterion is that $\overrightarrow{OR}$ and $\underline{b}$ must be perpendicular. As perpendicular vectors have a zero dot product, this means that,

$$\underline{r} \cdot \underline{b} = 0. \quad (6.33)$$

Given the expression for $\underline{r}$ in Eq. (6.32), Eq. (6.33) gives,

$$(\underline{a} + \lambda \underline{b}) \cdot \underline{b} = 0 \quad \Rightarrow \quad \underline{a} \cdot \underline{b} + \lambda \underline{b} \cdot \underline{b} = 0. \quad (6.34)$$

We may now complete this general case by writing, $\lambda = -(\underline{a} \cdot \underline{b})/(\underline{b} \cdot \underline{b})$, then finding a final expression for r using this value for λ , and ultimately declaring that the required distance we seek is $|\underline{r}|$. However, when one has a numerical exercise, the dot products have already been done and the final answer obtained more easily. So we had better do one.

Example 6.8. A straight line passes through a point with position vector, $(2, 1, 1)^T$, and in the direction, $(1, 0, 1)^T$. How close does this get to the origin?

Ah, I have used the *Transpose* notation here. Formally, in matrix theory (that'll happen next semester), the rows and columns are swapped around when the transpose is taken. So here a normal 3-vector, which is written as a column of numbers, is transposed into a form where the three numbers aligned horizontally as a row. There's nothing special about this apart from the fact that the typesetting is much easier with evenly-spaced lines. So

please don't get freaked out by this — I am simply trying to make the typesetting look nicer, and the presence of the T shows the rest of the world that I am not being careless but I truly know what I doing! I generally do use the transpose version when within a body of text, but not in the displayed equations.

OK, we'll power through this. I would normally draw roughly the same diagram as above but would populate it with the numerical data. Hence

$$\underline{r} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

At nearest approach the line joining the origin to the given line is perpendicular to the direction of the line. In other words, we have

$$\begin{aligned} \underline{r} \cdot \underline{b} &= 0 \\ \Rightarrow \left[\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} &= 0 \\ \Rightarrow 3 + 2\lambda &= 0 \\ \Rightarrow \lambda &= -\frac{3}{2}. \end{aligned} \tag{6.35}$$

Hence $\underline{r} = \begin{pmatrix} \frac{1}{2} \\ 1 \\ -\frac{1}{2} \end{pmatrix}$, and therefore the distance of the point from the line is $|\underline{r}| = \sqrt{\frac{3}{2}}$.

Example 6.9. We have $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and $\overrightarrow{OC} = \underline{c}$. A straight line passes through points A and B . How close does the line get to C ? Here,

$$\underline{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \underline{b} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \quad \text{and} \quad \underline{c} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Here is the diagram:

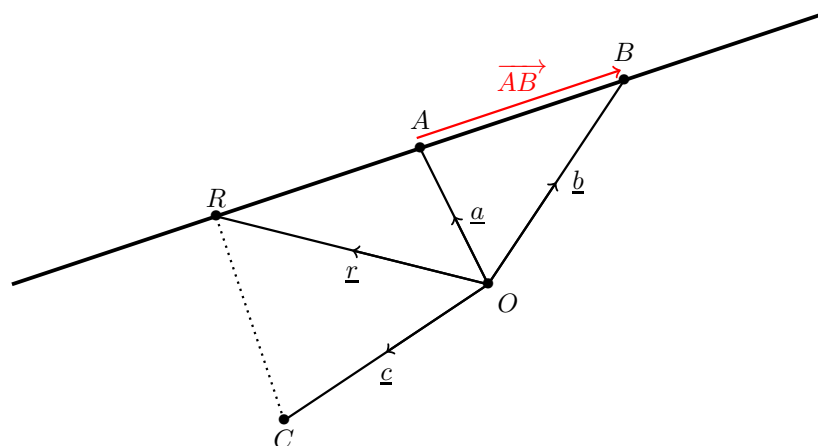


Figure 6.10. The shortest distance from C to a line which passes through both A and B .

This example is a bit of a variation on the previous example. This time we don't have the direction of the line but we'll have to work it out. And we are looking for the closest approach to C rather than to the origin, O .

First we need to write down a vector which takes us from O to a general point on the line. This will be,

$$\overrightarrow{OR} = \overrightarrow{OA} + \lambda \overrightarrow{AB} \quad \text{or} \quad \underline{r} = \underline{a} + \lambda(\underline{b} - \underline{a}) \quad \text{or} \quad \underline{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix}.$$

We now need $\overrightarrow{CR}$ to be perpendicular to $\overrightarrow{AB}$, and so

$$(\underline{r} - \underline{c}) \cdot (\underline{b} - \underline{a}) = 0. \quad (6.36)$$

Hence,

$$\begin{aligned} 0 &= \left[\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix} \right] \cdot \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix} \\ &= \left[\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix} \right] \cdot \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix} \\ &= -9 + 18\lambda \end{aligned}$$

$$\Rightarrow \lambda = \frac{1}{2}.$$

Hence,

$$\underline{r} - \underline{c} = \begin{pmatrix} 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \Rightarrow |\underline{r} - \underline{c}| = \sqrt{1/2}$$

is the distance of closest approach.

Example 6.10. One line passes through a point with position vector $\underline{a}_1$ in the direction of $\underline{b}_1$, while a second passes through a point with position vector $\underline{a}_2$ in the direction of $\underline{b}_2$. How close do they pass? We have

$$\underline{a}_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad \underline{b}_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad \underline{a}_2 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad \text{and} \quad \underline{b}_2 = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}.$$

A general diagram is given by,

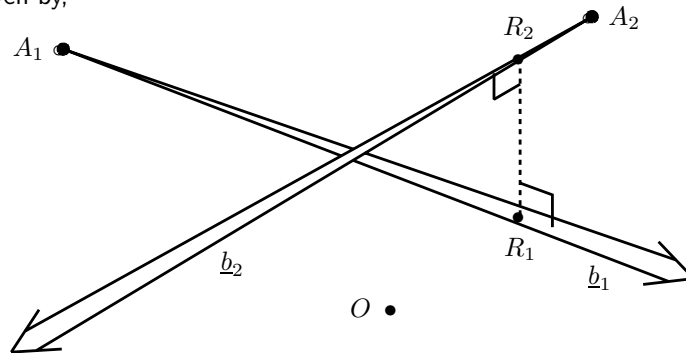


Figure 6.12. Showing two lines travelling in different directions but which do not cross. The dotted line indicates where the lines have their closest approach to one another.

Arbitrary points on the two lines are given by,

$$\underline{r}_1 = \underline{a}_1 + \lambda \underline{b}_1 \quad \text{and} \quad \underline{r}_2 = \underline{a}_2 + \sigma \underline{b}_2, \quad (6.37)$$

where λ and σ are parameters. The following is then a vector which takes us from an arbitrary point on one of the lines to an arbitrary point on the other:

$$\underline{r}_2 - \underline{r}_1 = \underline{a}_2 - \underline{a}_1 + \sigma \underline{b}_2 - \lambda \underline{b}_1. \quad (6.38)$$

At closest approach $\underline{r}_2 - \underline{r}_1$ will be the dotted line in Fig. 6.12 and it must be perpendicular to the directions of both the lines, $\underline{b}_1$ and $\underline{b}_2$. If this is difficult to visualise, then consider the following Figure, which is a real-life example of Fig. 6.12 and was a real-life balancing act!

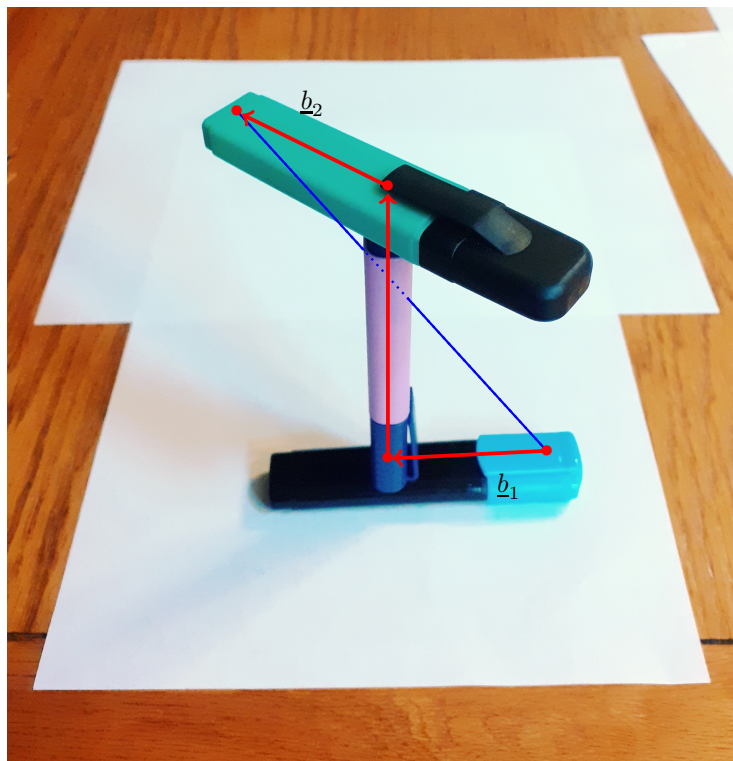


Figure 6.12b. A real-life version of Fig. 6.12.

The marker pens are meant to represent lines/vectors in 3D space. There is no need to specify where the points corresponding to $\underline{a}_1$ and $\underline{a}_2$ are, but each one lies somewhere within the lines corresponding to the respective marker pen. The directions of the vectors, $\underline{b}_1$ and $\underline{b}_2$ are as shown. Although the blue and green marker pens are horizontal, which seems like a special case, you will need to convince yourself that it is indeed fully representative of a general case: all fixed configurations of two lines may be rotated in space without change in their relative positions so that both become horizontal. If this concept is not immediately apparent, then pretend that the three marker pens have been glued together, and then one may move and rotate the full configuration in any way you like! Conversely, a random configuration may then be moved and rotated to yield two horizontal lines.

The shortest distance we seek is then represented by the length of the vertical line. The uppermost red disk on that line is given by the position vector, $\underline{r}_2$, and the lower one is given by $\underline{r}_1$. Thus the vector corresponding to the vertical line is $(\underline{r}_2 - \underline{r}_1)$ and this must be perpendicular to both $\underline{b}_1$ and $\underline{b}_2$ simultaneously since both are horizontal.

There are two ways of achieving our aim and both will be described.

Method 1. Given the last paragraph on the previous page, its mathematical translation is that we must have,

$$(\underline{r}_2 - \underline{r}_1) \cdot \underline{b}_1 = 0 \quad \text{and} \quad (\underline{r}_2 - \underline{r}_1) \cdot \underline{b}_2 = 0, \quad (6.39)$$

simultaneously, where

$$\underline{r}_2 - \underline{r}_1 = \underline{a}_2 - \underline{a}_1 + \sigma \underline{b}_2 - \lambda \underline{b}_1.$$

Hence,

$$0 = \underline{b}_1 \cdot (\underline{r}_2 - \underline{r}_1) = \underline{b}_1 \cdot (\underline{a}_2 - \underline{a}_1) + \sigma \underline{b}_1 \cdot \underline{b}_2 - \lambda \underline{b}_1 \cdot \underline{b}_1,$$

$$0 = \underline{b}_2 \cdot (\underline{r}_2 - \underline{r}_1) = \underline{b}_2 \cdot (\underline{a}_2 - \underline{a}_1) + \sigma \underline{b}_2 \cdot \underline{b}_2 - \lambda \underline{b}_1 \cdot \underline{b}_2.$$

Given that $\underline{a}_1$, $\underline{a}_2$, $\underline{b}_1$ and $\underline{b}_2$ are given, all the dot products will be known values, and this leaves us with a pair of simultaneous equations to solve for λ and σ . Once these are found we can determine $\underline{r}_2 - \underline{r}_1$, and then the modulus of that is the required distance, $|\underline{r}_2 - \underline{r}_1|$.

Method 2. This begins with Eq. (6.38). If $\underline{r}_2 - \underline{r}_1$ is to be perpendicular to the directions of both the lines, then it also means that it will be parallel to the vector, $\underline{p} = \underline{b}_1 \times \underline{b}_2$, because that, definition, is perpendicular to both $\underline{b}_1$ and $\underline{b}_2$. The required distance will then be a multiple of the corresponding unit vector, $\hat{\underline{p}}$, and so $(\underline{r}_2 - \underline{r}_1) \cdot \hat{\underline{p}}$ will be the distance between the two lines.

If you are like me, then your first reaction to that last paragraph may well have been a “what?” followed by a shaking of the head! Apologies. But read it again with Fig. 6.12b in mind.

Here’s a slightly different argument. The dotted line in Fig. 6.12, which illustrates the closest approach of the lines, will correspond to a unique pair of values of λ and σ . So if we have precisely the right values, then the dot product of $(\underline{r}_2 - \underline{r}_1)$ with $\hat{\underline{p}}$ must give the distance because these two vectors are in precisely the same direction.

A third possible argument, this time a little more mathematical. If instead, we take the dot product of Eq. (6.38) with $\hat{\underline{p}}$, then we get,

$$\begin{aligned} (\underline{r}_2 - \underline{r}_1) \cdot \hat{\underline{p}} &= (\underline{a}_2 - \underline{a}_1) \cdot \hat{\underline{p}} + \sigma \underline{b}_2 \cdot \hat{\underline{p}} - \lambda \underline{b}_1 \cdot \hat{\underline{p}} \\ &= (\underline{a}_2 - \underline{a}_1) \cdot \hat{\underline{p}} \end{aligned} \quad (6.40)$$

given that $\underline{b}_1 \cdot \hat{\underline{p}} = 0$ and $\underline{b}_2 \cdot \hat{\underline{p}} = 0$ by definition of $\hat{\underline{p}}$. This means that the value of $(\underline{r}_2 - \underline{r}_1) \cdot \hat{\underline{p}}$ is independent of the values of λ and σ , and so

$$(\underline{a}_2 - \underline{a}_1) \cdot \hat{\underline{p}} \quad (6.41)$$

is the shortest distance between the two lines. The reasonableness of Eq. (6.41) as the final answer may be understood if we again consider the two lines to be horizontal, and so $\hat{\underline{p}}$ is now vertical. Hence Eq. (6.41) merely tells us the vertical distance between these two points.

Now we’ll use the given numbers.

The equations for the lines are,

$$\underline{r}_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \underline{r}_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix},$$

and so

$$\underline{r}_2 - \underline{r}_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \sigma \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} - \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}. \quad (6.42)$$

Method 1. We need $\underline{r}_2 - \underline{r}_1$ to be perpendicular to both $(0, 2, -1)^T$ and $(2, 1, 0)^T$. Therefore we need to take the appropriate dot products:

$$0 = (\underline{r}_2 - \underline{r}_1) \cdot \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = -2 + 5\sigma - 2\lambda,$$

$$0 = (\underline{r}_2 - \underline{r}_1) \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = -2 + 2\sigma - 5\lambda.$$

This yields $\lambda = \sigma = \frac{2}{3}$. After some arithmetic we obtain,

$$\underline{r}_2 - \underline{r}_1 = \begin{pmatrix} -1/3 \\ 2/3 \\ 4/3 \end{pmatrix} \Rightarrow |\underline{r}_2 - \underline{r}_1| = \sqrt{7/3}.$$

Method 2. The vector which is perpendicular to the direction of the lines is,

$$\begin{aligned} \underline{p} = \underline{b}_1 \times \underline{b}_2 &= \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \\ &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 1 & 0 \\ 0 & 2 & -1 \end{vmatrix} \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} \\ &= \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}. \end{aligned}$$

Hence $\hat{\underline{p}} = \frac{1}{\sqrt{21}} \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$. And finally we obtain,

$$(\underline{r}_2 - \underline{r}_1) \cdot \hat{\underline{p}} = (\underline{a}_2 - \underline{a}_1) \cdot \hat{\underline{p}} = \frac{1}{\sqrt{21}} \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \frac{7}{\sqrt{21}} = \sqrt{\frac{7}{3}}.$$

6.11 The distance of a plane from the origin

The final topic is to determine the shortest distance from a plane to the origin. Figure 6.13 gives the general set-up where three points, A , B and C , define the plane. Use

$$\overrightarrow{OA} = \underline{a} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \overrightarrow{OB} = \underline{b} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \overrightarrow{OC} = \underline{c} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

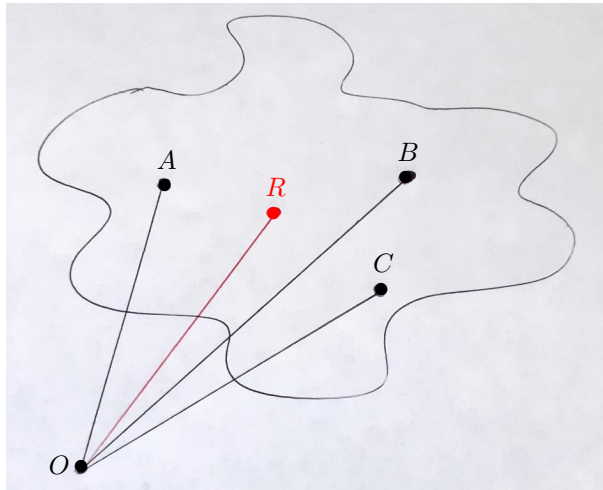


Figure 6.13. Showing a plane defined by three points.

Any point in the plane may be found by first travelling from O to the plane, and then by travelling different distances in two different directions. We will choose the following route:

$$\overrightarrow{OR} = \overrightarrow{OA} + \lambda \overrightarrow{AB} + \sigma \overrightarrow{AC},$$

where I have used A as a proxy origin for within the plane, and the vectors which are multiplied by λ and σ are vectors relative to that point. One could use any of the other possible choices of proxy origin. Therefore we have,

$$\underline{r} = \underline{a} + \lambda(\underline{b} - \underline{a}) + \sigma(\underline{c} - \underline{a}). \quad (6.43)$$

Once more there are two possible methods to follow from here which may be motivated by the following Figure:

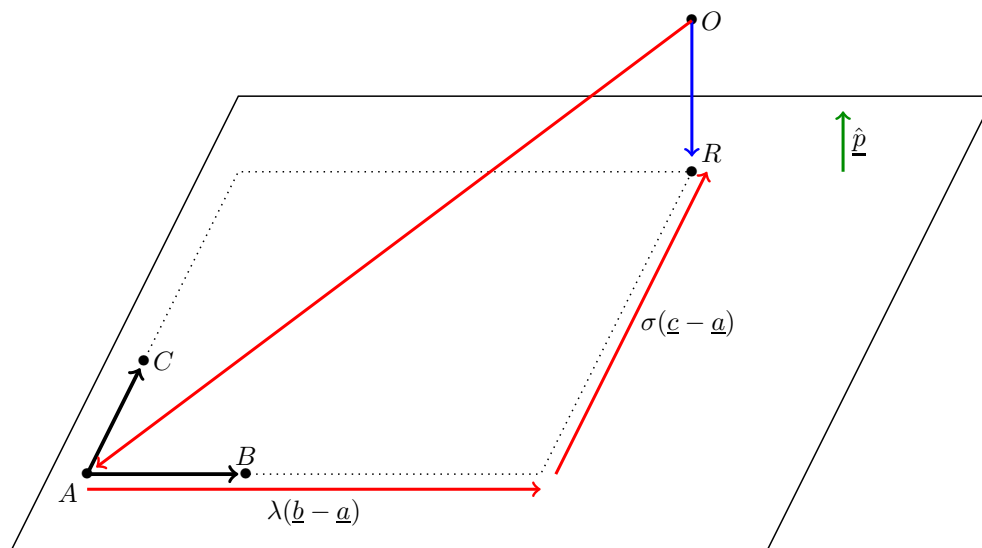


Fig. 6.14. Showing the shortest distance from a plane to the origin.

Method 1. In the above Figure, the point, R , may be anywhere on the plane but here it corresponds to that point on the plane which is closest to the origin. The shortest distance to the plane arises when $\underline{r}$ is perpendicular to the plane. Therefore $\underline{r}$ needs to be perpendicular to both of our vectors in the plane, i.e. $(\underline{b} - \underline{a})$ and $(\underline{c} - \underline{a})$.

Mathematically, this means that both of the following need to be satisfied:

$$\begin{aligned} 0 &= (\underline{b} - \underline{a}) \cdot \underline{r} = (\underline{b} - \underline{a}) \cdot \left[\underline{a} + \lambda(\underline{b} - \underline{a}) + \sigma(\underline{c} - \underline{a}) \right] \\ 0 &= (\underline{c} - \underline{a}) \cdot \underline{r} = (\underline{c} - \underline{a}) \cdot \left[\underline{a} + \lambda(\underline{b} - \underline{a}) + \sigma(\underline{c} - \underline{a}) \right] \end{aligned}$$

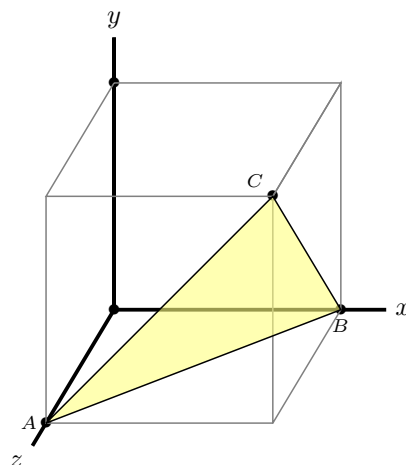
When the numerical data is substituted into this we obtain,

$$\begin{aligned} 0 &= \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdot \left[\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \sigma \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right] = -1 + 2\lambda + \sigma, \\ 0 &= \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \cdot \left[\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \sigma \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right] = 0 + \lambda + 2\sigma. \end{aligned}$$

Hence $\lambda = \frac{2}{3}$ and $\sigma = -\frac{1}{3}$. Therefore,

$$\underline{r} = \begin{pmatrix} 1/3 \\ -1/3 \\ 1/3 \end{pmatrix} \Rightarrow |\underline{r}| = \sqrt{1/3}.$$

The following Figure shows that this value is reasonable, given that the points A and B lie a unit distance from the origin..



Method 2. Again, we need to bear in mind that the shortest distance corresponds to a vector which is perpendicular to the plane. Given that $(\underline{b} - \underline{a})$ and $(\underline{c} - \underline{a})$ are in the plane, then

$$\underline{p} = (\underline{b} - \underline{a}) \times (\underline{c} - \underline{a})$$

is perpendicular to it. Thus $\underline{r} \cdot \hat{\underline{p}}$ will yield the required distance by isolating the component of a general $\underline{r}$ in that direction. This value could turn out naturally to be negative, and so we would normally take its modulus.

Therefore,

$$\begin{aligned} \underline{p} = (\underline{b} - \underline{a}) \times (\underline{c} - \underline{a}) &= \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \\ &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{vmatrix} \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} \\ &= \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}. \end{aligned}$$

$$\text{Hence } \hat{\underline{p}} = \begin{pmatrix} 1/3 \\ -1/3 \\ 1/3 \end{pmatrix}.$$

If we take the dot product of Eq. (6.43) with $\hat{\underline{p}}$ then we obtain,

$$\begin{aligned} \underline{r} \cdot \hat{\underline{p}} &= \underline{a} \cdot \hat{\underline{p}} + \lambda(\underline{b} - \underline{a}) \cdot \hat{\underline{p}} + \sigma(\underline{c} - \underline{a}) \cdot \hat{\underline{p}} \\ &= \underline{a} \cdot \hat{\underline{p}} \\ &= \frac{1}{\sqrt{3}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{3}}. \end{aligned}$$

So this matches with Method 1. This second method shows explicitly that the final result is independent of the values of λ and σ , and therefore we have found the component of $\underline{a}$ which is perpendicular to the plane. Given

that we chose $\underline{a}$ arbitrarily at the start of our analysis to represent a point on the plane, then we should obtain the same distance for the other two main choices. So we'll check this:

$$\underline{b} \cdot \underline{\hat{p}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{3}},$$

$$\underline{c} \cdot \underline{\hat{p}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{3}}.$$

If this all feels rather dodgy, then consider a horizontal plane (a sheet of paper, for example) which is above a table and parallel with it. There is an origin somewhere on the table. Then, whatever point is chosen on the plane has the same vertical component.

Note: The above analyses have told us that the two-parameter equation for the plane is Eq. (6.43). If we multiply this by the unit perpendicular vector, then it gives

$$\underline{r} \cdot \underline{\hat{p}} = \underline{a} \cdot \underline{\hat{p}},$$

where $\underline{a}$ represents the position vector of any point within the plane. If we set $\underline{r} = (x, y, z)$, then the above numerical case this gives us the following version for the equation for the plane:

$$\frac{x}{\sqrt{3}} - \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

By its construction the coefficients of x , y and z are the coefficients of the unit perpendicular vector and the right hand side is the nearest distance. Therefore if we were to have the following as the equation of the plane,

$$ax + by + cz = d,$$

then we may use this to find the perpendicular vector to the plane and its closest distance to the origin. The perpendicular vector is,

$$\underline{\hat{p}} = \frac{1}{\sqrt{a^2 + b^2 + c^2}} \begin{pmatrix} a \\ b \\ c \end{pmatrix},$$

and the closest approach is $d/\sqrt{a^2 + b^2 + c^2}$.

Example 6.11. To fix ideas, if we were given $x + 2y + 2z = 6$ as an equation for the plane, then $\underline{p} = (1, 2, 2)$. Given that $|\underline{p}| = 3$, we may divide the equation for the plane by 3 to yield,

$$\frac{1}{3}x + \frac{2}{3}y + \frac{2}{3}z = 2.$$

Hence $\underline{\hat{p}} = \begin{pmatrix} 1/3 \\ 2/3 \\ 2/3 \end{pmatrix}$ and the closest distance from the origin is 2.

Example 6.12 The above type of analysis also works when we are confined to 2D. Consider the equation, $3x - 4y = 60$. The vector, $(3, 4)$, has magnitude, 5, and therefore the equation may also be written in the form,

$$\frac{3}{5}x - \frac{4}{5}y = 12 \quad \text{or} \quad \begin{pmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 12,$$

after division by 5. This tells us immediately that the line is a distance of 12 from the origin.

If that doesn't feel quite right but you were convinced by the earlier fully 3D analysis, then we may rerun that analysis using the equation, $3x - 4y + 0z = 60$. Once more the vector formed from the left-hand-side coefficients has length, 5, and therefore division by 5 yields,

$$\frac{3}{5}x - \frac{4}{5}y + 0z = 12 \quad \text{or} \quad \begin{pmatrix} \frac{3}{5} \\ -\frac{4}{5} \\ 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 12.$$

Although this is an essentially 2D example with a line confined to the $z = 0$ plane, when we are in full glorious 3D we have a plane which is independent of z and therefore the nearest point to the origin will be when $z = 0$.

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Mathematics 1 ME10304

Problem Sheet 1 — Curve Sketching

Sketch all the following curves showing either y against x , or y against t , as appropriate. Use your knowledge of what the basic functions look like, together with the locations of poles and zeros, and the asymptotic behaviours for large values of x or t . The examples are not in any particular order of difficulty.

1. Sketch the following functions. *[Do this by factorisation and by identifying the zeros of the function. Do not try to evaluate the polynomials at a number of points. Don't use calculus to identify where maxima and minima are — that's for later. The aim here is just to get a rough idea of the shapes of the curves.]*

(a) $y = 1 - x^2$, (b) $y = x^3 - x^2$, (c) $y = x^3 + 1$, (d) $y = x^3 + x^2 - x - 1$,
 (e) $y = 8x^3 + 4x^2 - 2x - 1$, (f) $y = x^5 - x^4$, (g) $y = x^3 - 6x^2 + 11x - 6$.

2. Sketch the following functions.

(a) $y = |x|$, (b) $y = |x^2|$, (c) $y = \sec x$, (d) $y = |\sin x|$, (e) $y = \sin^3 x$, (f) $y = \cos^3 x$,
 (g) $y = \tan x$, (h) $y = \tan^2 x$, (i) $y = \cot x$, (j) $y = \sin^{-1} x$, (k) $y = (\sin x)^{-1}$.

3. Sketch the following functions. [Some of these are quite tricky, but one could sketch $e^{g(x)}$ by first sketching $g(x)$ as a first step.]

(a) $y = e^{-t}$, (b) $y = e^{-25t}$, (c) $y = e^{-t^2}$, (d) $y = e^{-|t|}$, (e) $y = e^{-1/t^2}$, (f) $y = e^{-1/t}$,
 (g) $y = e^{\sin t}$, (h) $y = e^{1/(t^2-t)}$.

4. Sketch the following functions.

(a) $y = t^2 \cos t$, (b) $y = (\sin t)/t$, (c) $y = te^{-t}$, (d) $y = te^{-t^2}$, (e) $y = (t^2 - 1)e^{-t^2}$,
 (f) $y = (\sin 100t) \tanh t$, (g) $y = \sin(t^2)$, (h) $y = \sin(t^{-1})$, (i) $y = t \cos(t^{-1})$.

5. Sketch the following functions. [Note that some of these functions do not exist for some ranges of values of x ; you will need to specify where the function exists.]

(a) $y = x \ln x$, (b) $y = \ln \cos x$, (c) $y = \ln(e^x)$, (d) $y = e^{\ln x}$, (e) $y = \ln |\cos x|$.

6. Sketch the following functions.

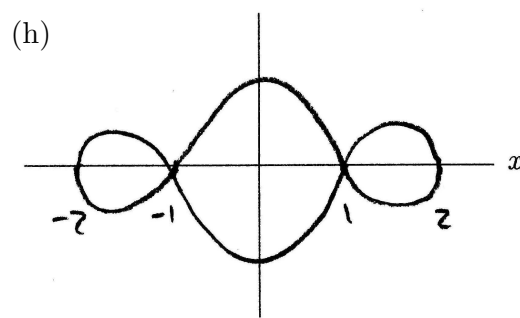
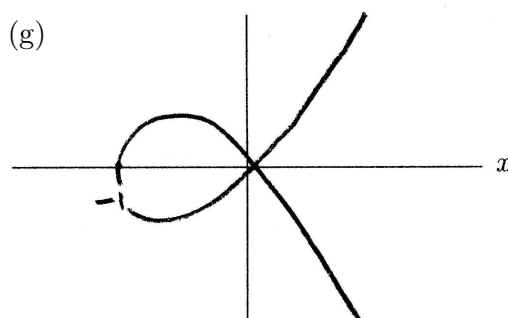
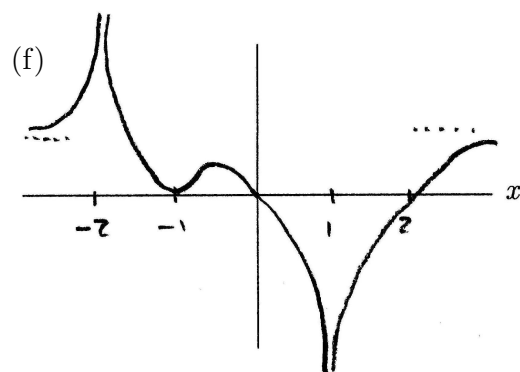
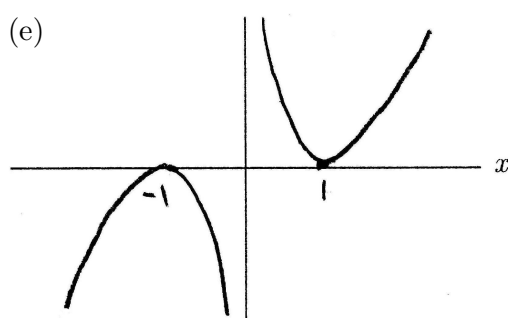
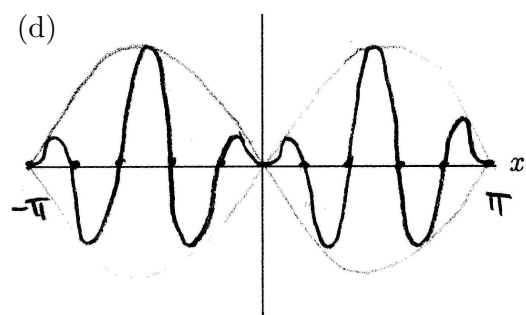
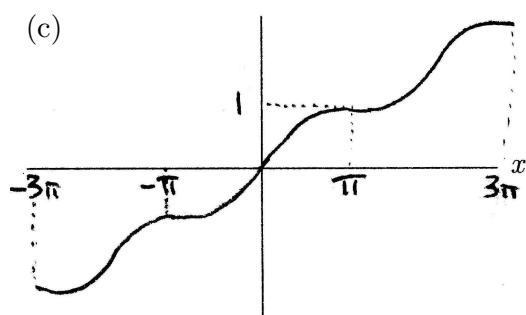
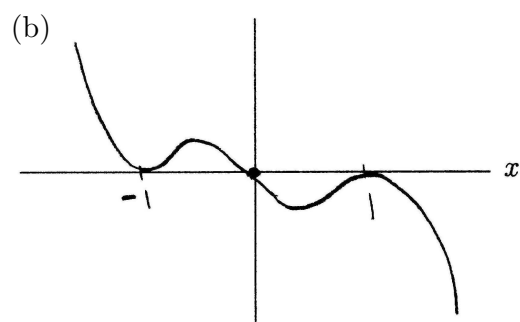
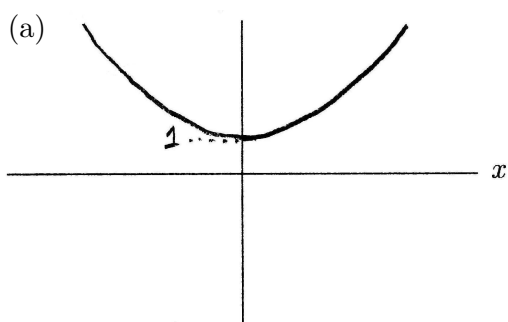
(a) $y = \frac{1}{x^2 + x + 1}$, (b) $y = \frac{x^2 - 1}{x + 2}$, (c) $y = \frac{x + 1}{(x - 2)(x - 4)}$, (d) $y = \frac{x^2 + 1}{x + 1}$,
 (e) $y = \frac{(x - 1)^2(x - 4)}{(x - 2)(x - 3)^2}$, (f) $y = \left| \frac{x}{x - 1} \right|$, (g) $y = \frac{x^2 + 2x + 1}{x^2 - 2x + 1}$, (h) $y = \frac{(x^2 - 1)^2}{x^2}$.

7. Sketch the curves given by the following equations.

(a) $y^2 = 4x^2 - x^4$, (b) $x^2 + y^2 = x^2 y^2$, (c) $x^2 + 4x + y^2 = -3$, (d) $x^2 y^2 = 1$,
 (e) $y^4 + x^4 = 1$.

8. The functions, $y = 1/x$ and $y = 1/x^3$, have poles of different kinds at $x = 0$; one is of first order and the other is of third order. In the range, $0 < x \leq 1$, which one is closer to the vertical axis, and what is the consequence of this answer on how one would sketch the two functions on the same axes?

9. Determine the equations which correspond to the following eight sketches. Note that some of the answers are not necessarily unique.



Department of Mechanical Engineering, University of Bath**Mathematics 1 ME10304****Problem Sheet 1b — Curve Sketching — additional examples**

Sketch all the following curves showing y against x . Use your knowledge of what the basic functions look like, together with the locations of poles and zeros, and the asymptotic behaviours for large values of x . The examples are not in any particular order of difficulty and are selected randomly. As in sheet 1, do not try to evaluate the polynomials at a number of points. Don't use calculus to identify where maxima and minima are. The aim here is just to get a rough idea of the shapes of the curves.

1. $y = x^3 + 3x^2 + 3x + 2$

2. $y = x \sin^2 x$

3. $y = \frac{(x-1)^2}{x^2(x-2)}$

4. $y = \frac{\cos x}{x}$

5. $y = \frac{x^3 - 4x}{(x-1)^2}$

6. $y = \sin(\pi e^{-x})$

7. $y = \sin(2\pi e^{-x})$

8. $y = |x^3 - x|$

9. $y = \operatorname{sech}^2(x-1) + \operatorname{sech}^2(x+1)$

10. $y = x e^{-x^2}$

11. $y = e^{-x^2} \cos(100\pi x)$

12. $y = \left(\frac{x+1}{x-1}\right)^3 x$

13. $y^2 = \sin x$

14. $y^2 = x \sin x$

15. $x^2 + y^2 + 4x + 2y + 4 = 0$

16. $y = \frac{(x-3)^2(x^2-4)x}{(x+3)(x^2-1)}$

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Problem Sheet 2 — Complex Numbers

1. Simplify the following:

- (a) j^3 , (b) j^4 , (c) j^5 , (d) j^{10} , (e) j^{2022} , (f) $(1+j)(2+j)$, (g) $(1+j)(1-j)$,
 (h) $(2+j)(1+3j) + (2-j)(1-3j)$, (i) $(2+j)(2+3j) - (2-j)(2-3j)$, (j) $2/(1-j)$,
 (k) $(3+j)/(4+3j)$, (l) $(1+j)^2$, (m) $(1+j)^{100}$.

2. Find $(1 + \sqrt{3}j)^3$, and hence find $(1 + \sqrt{3}j)^{60}$.

3. Find the modulus and argument of each of the following complex numbers, and hence write them in complex exponential form.

- (a) $2 + 3j$, (b) $-j$, (c) $40 + 9j$, (d) -5 , (e) $1 - 100j$, (f) $1 + \sqrt{3}j$.

4. Convert the following numbers from complex exponential form to Cartesian form and write in as simple form as possible.

- (a) $e^{j\pi/3}$, (b) $4e^{2\pi j/3}$, (c) $\sqrt{2}e^{3\pi j/4}$.

5. Use de Moivre's theorem to find $\cos 4\theta$ and $\sin 4\theta$ in terms of $\cos \theta$ and $\sin \theta$. [Note that $\cos 4\theta$ may be written in terms of cosines only or sines only; find both forms.] Also find $\cos 5\theta$ and $\sin 5\theta$ in terms of $\cos \theta$ and $\sin \theta$.

6. Evaluate the following roots of complex numbers:

- (a) $(5 + 12j)^{1/2}$, (b) $(-16)^{1/4}$, (c) $1^{1/5}$, (d) $(-1)^{1/100}$,
 (e) $(1 + 2j)^{2/7}$, (f) $(336 + 527j)^{1/4}$, (g) $(2j)^{1/2}$, (h) $(-15 + 8j)^{3/5}$.

7. Solve the following quadratic equations and plot the roots in the Argand diagram (or the complex plane). Find the modulus and argument of each root, and hence write them in complex exponential form.

- (a) $x^2 + 2 = 0$ (b) $y^2 + 2jy - 2 = 0$ (c) $z^2 + 2\sqrt{2}z + 2 - 4j = 0$.

The following questions are for interest only and are definitely not examinable. If you are bored, then do have a go at them!

8. de Moivre's theorem is used to express $\cos n\theta$ and $\sin n\theta$ in terms of $\cos \theta$ and $\sin \theta$ when n is an integer. The equivalent for the hyperbolic sine and cosine is not quite as easy to write down. However, write $\cosh \theta$ in terms of e^θ and $e^{-\theta}$ and square both sides of the equation. Hence find $\cosh 2\theta$ in terms of $\cosh \theta$.

Further, can you think of a quick way of determining a simple expression for $\sinh 2\theta$ in terms of $\sinh \theta$ and $\cosh \theta$? Also, find an expression for $\sinh 3\theta$ in terms of powers of $\sinh \theta$, and an expression for $\cosh 3\theta$ in terms of powers of $\cosh \theta$.

9. Leibniz, he of the invention-of-calculus fame, came late to complex numbers. He found that

$$\sqrt{1 + \sqrt{-3}} + \sqrt{1 - \sqrt{-3}} = \sqrt{6},$$

but no-one was surprised! How does one prove this one?

10. Express $z = \frac{1}{2}(\sqrt{3} + 1) + \frac{1}{2}(\sqrt{3} - 1)j$ in complex exponential form. Now find the argument in terms of degrees, and hence find the first integer power of z for which it is a real value.

11. Some very weird ones. Not examinable. You may need to resort to some lateral thinking...

(a) $\ln(-1)$, (b) $\ln j$, (c) j^j , (d) $y = \cos^{-1} 2$.

12. Definitely a maths department type of question, but let's fill up space. A Pythagorean triple is a set of three integers which satisfy, $n^2 + m^2 = q^2$. Now if $a^2 + b^2$ is such a square, as is $c^2 + d^2$, then the product, $(a^2 + b^2)(c^2 + d^2)$ may be written as the sum of the two squares, $u^2 + v^2$, in two different ways. I've only just recently discovered this result and my initial reaction was disbelief! Your task is to prove it but I'll give a hint or two.

One may factorise $(a^2 + b^2)$ into complex factors (think complex conjugates). This means that $(a^2 + b^2)(c^2 + d^2)$ may be split into four complex factors. Now there are two different ways of pairing up these factors before they are multiplied. The rest is up to you.... Test out your result using

$$(3^2 + 4^2)(5^2 + 12^2).$$

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Problem Sheet 3 — Differentiation

1. Use the 'small increment' method (i.e. the approach using limits) described in the lecture notes to find the derivative of the following functions: [NOTE: **(i)** this type of question will *not* appear in the exam; **(ii)** part (d) begs the question – you'll see what I mean; **(iii)** there's a sneaky trick that you'll need to find for part (e) which is related in some way to finding z^{-1} in complex numbers.]

(a) x^3 , (b) x^4 , (c) x^{-1} , (d) $\sin x$, (e) $x^{1/2}$.

2. Find the derivatives of the following functions with respect to x :

(a) $4 \sin x + 2x$, (b) $4e^{2x} + 5x^{-1}$, (c) $(bx)^{-1}$, (d) $-4 - 5x^{-2}$, (e) e^{3x-4} , (f) $\ln |2x^3|$,
(g) $|x|$, (h) $\sin |x|$.

3. Find the derivatives of the following functions with respect to t :

(a) $t \sin t$, (b) $t^{-2}e^{3t}$, (c) $t \ln t - t$, (d) $te^{-t} \cos 2t$, (e) $\sin 2t \sinh 3t$, (f) $|t| \sin |t|$.

4. Differentiate the following problems with respect to t :

(a) e^{t^2} , (b) $\sqrt{1+t^2}$, (c) $(1+\sqrt{t})^2$, (d) $\sin[\sin(\sin t)]$, (e) $\tan(t^{1/2})$, (f) $e^{-\sin t^2}$,
(g) $(\sin t)^{1/2}$, (h) $|t|^b$.

5. Using the quotient rule, differentiate the following with respect to x :

(a) $\tan(ax)$, (b) $\tanh(ax)$, (c) $\operatorname{cosec}(ax)$, (d) $e^x/(1+x)$, (e) e^{3x}/x^2 , (f) $x/(1+x^2)$.

Use $\tan = \sin / \cos$, $\tanh = \sinh / \cosh$ and $\operatorname{cosec} = 1 / \sin$.

6. Find an expression for dy/dx in the following cases:

(a) $y^2 + y = x$, (b) $\sin(xy) = x$, (c) $\ln |y| = y - \cos x$.

7. Find the derivatives with respect to x of the following functions. You will need to use more than one of the above rules in some cases. Part (j) is rather lengthy.

(a) $\sin^{-1}(ax+b)$ (hint: let $y = \sin^{-1}(ax+b)$, find x in terms of y and then differentiate),

(b) $\sin^{-1}(\sin 2x)$,

(c) $e^{x \sin x}$,

(d) $(\sin x)e^{x^2}$,

(e) 2^x (hint: first show that $2^x = e^{x \ln 2}$),

(f) x^x ,

(g) $\log_{10}|x|$,

(h) $\sinh^{-1}(ax+b)$, [N.b. this is an inverse sinh, not a reciprocal]

(i) $xe^{(x/\sqrt{1+x^2})}$,

(j) $\sin(x^2)e^{x \sin x}/(1+x^2)$.

8. **[Lengthy and challenging.]** Find the first, second and third derivatives of $x^n e^{ax}$, where we can assume that $n > 3$. Can you write down a compact expression for the m^{th} derivative of this function?
9. **[Challenging.]** This one is the strangest one.... By considering a simple sketch it is easy to be convinced of the fact that $dy/dx = 1/(dx/dy)$. Use this result and the chain rule to find the appropriate formula for d^2y/dx^2 in terms of d^2x/dy^2 . Check that your final formula is correct by applying it to $y = \ln x$ (for $x > 0$) and to $y = x^2$.
10. **[To be answered after Lecture 7.]** Find the critical points of the following functions. Which are maxima and which are minima? Find the values of the functions at these points and sketch the functions.
- (a) $f(x) = 2x^2 - 3$, (b) $g(t) = t^2(t^2 - 1)$, (c) $h(y) = ye^{-y}$,
(d) $F(x) = x^2 \sin^2 x$, (e) $G(x) = x^4 + 2x^3 - 2x - 1$.
11. **[To be answered after Lecture 7.]** Find and classify any critical points of the function,

$$y = 10x^6 - 36x^5 + 45x^4 - 20x^3 + 1.$$

12. **To be answered after Lecture 7.]** Find and classify any critical points of the function, $y = e^x/(x^2 + 1)$.
[Hint, this becomes very very messy after the first derivative. Therefore I would suggest that one should set $y' = (x - 1)^2 f(x)$ at that point before obtaining any higher derivatives.]

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Problem Sheet 4 — Partial Differentiation

1. In the following, find both $\partial f/\partial x$ and $\partial f/\partial y$ for all cases.

(a) $f(x, y) = \sin(x + y)$, (b) $f(x, y) = x/y$, (c) $f(x, y) = \ln|y/x|$, (d) $f(x, y) = \frac{1}{x^2 + y^2}$.

2. Find all the possible 1st and 2nd partial derivatives of the following functions. Hence determine how many of them satisfy Laplace's equation,

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0.$$

(a) $h = x^2 - y^2$, (b) $h = (x^2 - y^2)^2$, (c) $h = \frac{y}{x^2 + y^2}$, (d) $h = \ln(x^2 + y^2)$,

(e) $h = \frac{xy}{(x^2 + y^2)^2}$, (f) $h = e^{-ax} \cos ay$, (g) $h = \tan^{-1}(x/y)$, (h) $h = f(x - cy)$,

(i) $h = x^3y - xy^3$, (j) $h = x^4 - 6x^2y^2 + y^4$, (k) $h = x^5 - 10x^3y^2 + 5xy^4$.

[Note: In the above, please note that f is a given function (although I don't have to specify one!), and that both a and c are constants.]

3. If a function $f(x, y)$ has a local maximum or minimum at a particular point, then at that point both $\partial f/\partial x$ and $\partial f/\partial y$ are zero. Let V be the volume of a cuboid whose sides have lengths x , y and z , and suppose that its total surface area is the constant, A . First show that

$$V(x, y) = \frac{xy(\frac{1}{2}A - xy)}{x + y}.$$

Hence show that the largest cuboid with a given surface area is a cube.

4. Find all the critical points for the following functions, and classify them if possible.

(a) $z = x(x^2 + y^2 - 1)$, (b) $z = y^2 - (x^2 + 1)y + x^2$, (c) $z = xye^{-x-y}$,

(d) $z = \frac{xy}{1 + x^2 + y^2}$, (e) $z = xy(x^2 + y^2 - 1)$, (f) $z = (y - x^2)(y - x)$.

(g) $z = \cos \pi x \cos \pi y$. (h) $z = x^3y + xy^2 - x^3 - xy^3$.

[Note: Part (h) is quite long and a little awkward.]

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Problem Sheet 5: Integration

1. Find the following integrals:

$$(a) \int x^3 dx, \quad (b) \int x^{-5} dx, \quad (c) \int e^{-4t} dt, \quad (d) \int \cosh 2y dy,$$

$$(e) \int_0^1 (2x + 3) dx, \quad (f) \int_{-1}^2 (4x^2 - 3x) dx, \quad (g) \int_0^{\pi/4} (2 \cos \alpha + \sin \alpha) d\alpha.$$

2. Find the following integrals:

$$(a) \int (x + 3)^{-2} dx, \quad (b) \int_0^1 2xe^{x^2} dx \text{ (hint: substitute } z = x^2),$$

$$(c) \int \frac{1}{x^2 + a^2} dx \text{ (hint: substitute } x = a \tan \theta),$$

$$(d) \int_0^{\pi/2} \frac{\sin x}{1 + \cos^2 x} dx \text{ (hint: substitute } z = \cos x),$$

$$(e) \int \frac{x}{1 + x^2} dx, \quad (f) \int \cos x \cos(\sin x) dx, \quad (g) \int \theta^2 \sin(\theta^3) d\theta, \quad (h) \int_0^{\pi/2} \sqrt{\sin t} \cos t dt,$$

$$(i) \int [f(x)]^n f'(x) dx, \quad (j) \int (ff'' + f'f') dx \text{ (Try to find the integral by inspection),}$$

$$(k) \int_0^1 \sqrt{1 - x^2} dx, \quad (l) \int_0^1 \frac{1}{\sqrt{1 - x^2}} dx,$$

$$(m) \int_0^1 (1 + \ln x)x^x dx \text{ [you will need to rewrite } x^x \text{ in an alternative form (hint: take logs), and to assume that } \lim_{x \rightarrow 0} x^x = 1].$$

3. Use partial fractions to simplify the integrand in the following integrals, and hence find those integrals:

$$(a) \int \frac{2t + 3}{t^2 + 3t + 2} dt, \quad (b) \int \frac{1}{t^2 + 3t + 2} dt, \quad (c) \int \frac{1}{t^3 + t} dt, \quad (d) \int \frac{t + 3}{t^3 + 3t^2 + 2t} dt,$$

$$(e) \int \frac{2t + 1}{t^2(t + 1)} dt, \quad (f) \int \frac{t^2 - 2}{(t + 1)(t^2 + 2t + 2)} dt, \quad (g) \int \frac{1}{t(t^2 + 1)^2} dt.$$

Please note that parts (f) and (g) are quite lengthy, and for (g) you may need to look at the partial fractions handout.

4. Find the following integrals which involve top-heavy quotients. Do this by first performing a kind of long division.

$$(a) \int_0^1 \frac{t^2 + 4t + 5}{t + 1} dt, \quad (b) \int_0^1 \frac{t^4(1 - t)^4}{1 + t^2} dt.$$

Although part (b) is a little long, you will know if the answer is correct because it should give you an "Oh, that's interesting" moment.

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Problem Sheet 6: Integration by Parts

1 Obtain the following integrals.

$$(a) \int x^5 \cos ax \, dx, \quad (b) \int x^5 \sin ax \, dx, \quad (c) \int x^5 e^{ax} \, dx, \quad (d) \int x^5 e^{ajx} \, dx.$$

(e) Can you find the integral, $\int x^5 e^{-ax} \, dx$, directly from the answer to part (c)?

(f) Have you seen the real part of the answer to Q1d before? And the imaginary part?

2. Evaluate the following integrals by integrating by parts twice.

$$(a) \int \sin ax \sinh bx \, dx, \quad (b) \int \sin ax \cosh bx \, dx, \quad (c) \int_0^\infty e^{-ax} \cos \omega x \, dx, \quad (d) \int_0^\infty e^{-ax} \sin \omega x \, dx.$$

Can you think of an easy way of doing the last two integrals simultaneously which doesn't involve integration by parts?

3. Evaluate

$$(a) \int [\ln |x|]^2 \, dx, \quad (b) \int [\ln |x|]^3 \, dx, \quad (c) \int [\ln |x|]^n \, dx, \quad (d) \int_0^1 \frac{\ln |x|}{x^{1/2}} \, dx \quad (\text{use: } x^{1/2} \ln x \rightarrow 0 \text{ as } x \rightarrow 0^+),$$

$$(e) \int_0^2 x^3 \ln |x| \, dx, \quad (f) \text{ Evaluate Q3a by first using the substitution } x = e^y.$$

4. The mandatory silly integral: $\int \sin(x) \ln \left[\tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right] \, dx$.

5. Use integration by parts once to obtain a formula for $I_n = \int_0^\infty x^n e^{-ax} \, dx$ in terms of I_{n-1} — such a formula is called a **Reduction Formula**. Find I_0 directly, and use the reduction formula to evaluate I_6 . Check your answer by evaluating I_6 using integration by parts.

6. **Something weird.** Evaluate the following indefinite integral using one integration by parts where you'll choose f' to integrate first:

$$I = \int \frac{1}{f} f' \, dx.$$

Can you explain why the answer is incorrect? What happens if you choose to integrate between $x = a$ and $x = b$?

7. **Also weird.** Define $I(a)$ according to $I(a) = \int_0^\infty e^{-ax} \, dx$, and evaluate this integral. Clearly the value taken by $I(a)$ depends on a , and therefore we can differentiate it with respect to a . Do this and find $I'(a)$ both as an integral and as a function of a . Continue to differentiate in this way with the aim of eventually finding $\int_0^\infty x^4 e^{-ax} \, dx$.

Using this idea and the solution to Q2c, find $\int_0^\infty x e^{-ax} \cos \omega x \, dx$, the integral of the product of three functions.

8. **This is a set of miscellaneous questions and you may wish to wait until the revision period to tackle them.**

I won't give any hints at all about how to do them! Sorry. Most are substitutions but (e) should be done both as an integration by parts and by using a suitable multiple angle formula. Some will require the use of more than one method. Questions (g) and (h) are trick questions — it all depends on how quickly you can see the trick. As a guide, when I saw (h) for the very first time I managed 5 seconds, and so I thought that I would use it here!

$$(a) \int_0^{\pi^2} \sin \sqrt{x} \, dx, \quad (b) \int_0^{\infty} e^{-\sqrt{x}} \, dx,$$

$$(c) \int_0^{\infty} e^{-x^{1/4}} \, dx, \quad (d) \int_1^e \sin(\ln x) \, dx,$$

$$(e) \int_0^{\pi/2} \sin x \cos 5x \, dx, \quad (f) \int \frac{x+3}{\sqrt{x^2+6x+10}} \, dx,$$

$$(g) \int_{-\infty}^{\infty} e^{-x^4} \sin 3x \, dx, \quad (h) \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} \, dx.$$

$$(i) \int \frac{\sqrt{1-x}}{\sqrt{x}} \, dx, \quad (j) \int \frac{1}{\sqrt{x+1} + \sqrt{x}} \, dx, \quad (k) \int_0^{\infty} \frac{1}{1+e^x} \, dx.$$

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Problem Sheet 7: Mean, RMS, Recurrence, Volumes, Lines, Surfaces and Volumes of Revolution.

Note: the notation, $0 \leq x, y < \infty$, is equivalent to writing both $0 \leq x < \infty$ and $0 \leq y < \infty$.

Note: Questions 8f, 13, 14 and 15 are either too long, too tricky or too unusual to form a realistic examination question. However, they are definitely worth doing.

1. Evaluate the mean and RMS of the following functions

(a) $f(t) = t^2 \quad (0 \leq t \leq 1),$

(b) $f(t) = \sin t \quad (0 \leq t \leq 2\pi),$

(c) $f(t) = |\sin t| \quad (0 \leq t \leq 2\pi),$

(d) $f(t) = e^{-t} \quad (0 \leq t \leq 1).$

2. Obtain a reduction formula (a.k.a. as a recurrence relation) for $I_n = \int_0^\pi x^n \sin x \, dx$ in terms of I_{n-2} . Find I_0 directly, and use the reduction formula to evaluate I_6 . Check your answer by evaluating I_6 using integration by parts. Also, find I_1 directly using integration by parts and then use the reduction formula to evaluate I_7 .

3. **[Lengthy]** The aim is to determine a reduction formula for $I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$. This may be done by rewriting the integral in the form,

$$I_n = \int_0^{\pi/2} [\cos \theta] [\cos^{n-1} \theta] \, d\theta,$$

and by integrating by parts once. You will need to evaluate I_0 and I_1 explicitly in order to write down final expressions for I_n .

4. Find the volumes under the following surfaces:

(a) $h(x, y) = xy^2 \quad (0 \leq x \leq 1, \quad 1 \leq y \leq 2)$

(b) $h(x, y) = e^{-x-y} \quad (0 \leq x, y < \infty)$

(c) $h(x, y) = \sin(x + y) \quad (0 \leq x, y \leq \pi)$

(d) $h(x, y) = xye^{-x-y} \quad (0 \leq x, y < \infty)$

(e) $h(x, y) = (x + 2y)^2 \quad (0 \leq x, y \leq 1)$

(f) $h(x, y) = xy \sin(xy) - \cos(xy) \quad (0 \leq x, y \leq \pi) \quad [\text{Be careful with the integration by parts.}]$

5. Find the volumes under the following surfaces:

(a) $h(r, \theta) = r \cos^2 \theta \quad (0 \leq r \leq 1, \quad 0 \leq \theta \leq \frac{1}{2}\pi)$

(b) $h(r, \theta) = r^5 e^{-r} \cos(\theta) \quad (0 \leq r < \infty, \quad 0 \leq \theta \leq \pi)$

(c) $h(r, \theta) = \sin^2 \theta \quad (1 \leq r \leq 2, \quad 0 \leq \theta \leq 2\pi)$

[Note: One of the above is a trick question, and it may be evaluated immediately by inspection.]

6. The indefinite integral, $\int e^{-x^2} dx$, cannot be written down using elementary functions such as polynomials, or the trigonometric or hyperbolic functions. However, the definite integral,

$$I = \int_0^\infty e^{-x^2} dx,$$

which is related to the Normal Distribution in probability and statistics, may be evaluated as follows. Let,

$$I^2 = \left[\int_0^\infty e^{-x^2} dx \right] \left[\int_0^\infty e^{-y^2} dy \right] = \int_0^\infty \int_0^\infty e^{-x^2-y^2} dx dy.$$

This may be interpreted as an expression for the volume under a surface. On noting the ranges of integration, rewrite I^2 in terms of a double integral in polar coordinates ($x = r \cos \theta$, $y = r \sin \theta$), ensuring that the limits of integration are correct, and then carry out the integrations. Hence find I .

7. An object takes the form of a cube with the density distribution, $\rho(x, y, z) = (x + y + z)^2$, where $0 \leq x, y, z \leq 1$. What is the total mass of the object?
8. This question is about finding the length of six different lines, and the difficulty of doing this varies widely between these cases.

Despite looking innocuous, part (f) is by far the most difficult. Once you have the integral written down, there are two possible substitutions that could be used, one trigonometric, the other hyperbolic. Try each of them in turn, and hope for the answers being the same in the end! So, prior to attempting part (f) please find and write down the respective formulae for $\cosh^2 x + \sinh^2 x$ and $\cosh^2 x - \sinh^2 x$; these will be useful. Likewise, I would like you to determine the value of $\sinh^{-1} 2$. This may be done by first setting, $z = \sinh^{-1} 2$, taking the $\sinh$ of both sides, rewriting $\sinh$ in terms of e^z and e^{-z} , obtaining a quadratic equation for e^z , and then finally obtaining a simple numerical expression for z . Also, use integration by parts (or a substitution) to find the indefinite integrals of both $\sec x$ and $\sec^3 x$ — nasty stuff!

Find the lengths of the following curves in the given ranges.

- (a) $y = x$ in the range $0 \leq x \leq 1$; (b) $y = \frac{1}{3}x^{3/2}$ in the range $0 \leq x \leq 5$;
 (c) $y = \ln \cos x$ in the range $0 \leq x \leq \frac{1}{4}\pi$; (d) $y = \cosh x$ in the range $0 \leq x \leq X$;
 (e) $y = \sqrt{1-x^2}$ in the range $0 \leq x \leq 1$; (f) $y = \frac{1}{2}x^2$ in the range $0 \leq x \leq 1$.
9. Find both the surface area and the volume of the shape obtained by rotating the function, $y = \sqrt{x}$ about the x -axis. Consider the range, $0 \leq x \leq 1$.
10. A cone has a base of radius, r , and a height, h . Its surface may be modelled by the equation, $y = r[1 - (x/h)]$, in the range $0 \leq x \leq h$. Sketch this function. Determine the surface area and volume of revolution about the x -axis.
11. Find the volume of the ellipsoid which is formed by rotating the ellipse,

$$\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1,$$

about the x -axis. What is the volume if the ellipsoid is rotated about the y -axis? [N.b. there are two ways of doing this.]

12. This question is concerned with the famous paradox associated with what is known as either Gabriel's Horn or Torricelli's Trumpet. The function $y = 1/x$ is rotated about the x -axis. (a) Find the volume contained by this shape in the range, $1 \leq x < \infty$. (b) It isn't possible to evaluate the indefinite integral which is associated with the surface area, but it is possible to show that the surface area is infinite — go on, have a go!

13. [Lengthy] A torus is defined by rotating the circle,

$$(y - a)^2 + x^2 = b^2,$$

about the x -axis. It is necessary that $a > b > 0$ so that the torus has a hole in the middle. Sketch the function given above. Find the surface area and the volume of the torus.

[Note: you will need to split the problem into 'upper' and 'lower' problems corresponding to the upper and lower parts of the above circle. I recommend strongly to begin with a sketch.]

14. [Lengthy] The function, $y(x) = \sin x$, is rotated about the x -axis. Assuming that $0 \leq x \leq \pi$, find both the volume and the surface area of this object.

[Note: While the integral for the volume is relatively straightforward, you will need two successive substitutions for the surface area integral. For the second substitution, use an identity based on hyperbolic functions. Further, bear in mind that it is possible to find the value of θ when $\sinh \theta = 1$ by solving a quadratic equation for e^θ .]

15. Find both the surface area and the volume of revolution of $y = \cosh x$ about the x -axis in the range $0 \leq x \leq X$, where X is any positive value. What is strange about your answer?

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Problem sheet 8 — Series – Binomial Series

- Find the binomial expansions for **(a)** $(1+x)^{-2}$ and **(b)** $(1+x)^{-3}$, and express the final solution in summation form. Check your result for part (b) by differentiating the result for part (a). [You may need to alter the summation variable to get a perfect match.]
- Following on from the previous question, differentiate the Binomial Series form of $(1+x)^{-3}$ to find the series for $(1+x)^{-4}$. Confirm this by using the general expression for the Binomial Series.
- Find the binomial expansions for

$$\textbf{(a)} (1+x)^{-1/2}, \quad \textbf{(b)} (1+x)^{1/2} \quad \text{and} \quad \textbf{(c)} (1+x)^{-3/2}.$$

Note that part (a) is in the lecture notes, but do resist the temptation to look before trying this out! Check your answers to (b) and (c) by the respective integration and differentiation of the result of (a).

- Express $(1+x)^{-1/3}$ as a Binomial Series. Note that this one cannot be expressed in summation form....unless you decide to define your own notation! [Hint: do a google search for what are called double factorials.]
- Find the Binomial Series representations for **(a)** $(1+2x)^{-1}$ and **(b)** $(3+2x)^{-1}$.
- Obtain the binomial series for $(1+x^2)^{-1}$ and hence find a power series representation for $\tan^{-1} x$. Use this result to show that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$.
- (a)** Use the chain rule (i.e. implicit differentiation) to find the derivative of $y = \tanh^{-1} x$. Use this result and the Binomial Series to find a power series representation for y .
(b) At some point in part (a) you will have an expression for x as an explicit function of y ; replace the sinh and cosh terms by the correct exponentials, solve for e^y and hence find a different expression for y from the one given initially.
(c) Use the results of parts (a) and (b) to show that,

$$\frac{1}{2} + \frac{(\frac{1}{2})^3}{3} + \frac{(\frac{1}{2})^5}{5} + \frac{(\frac{1}{2})^7}{7} + \dots = \frac{1}{2} \ln 3.$$

- [For fun only!]** Use the binomial expansion of $(1+x)^n$, where n is a positive integer, to show that

$$\sum_{i=0}^n \binom{n}{i} = 2^n \quad \text{and} \quad \sum_{i=0}^n (-1)^i \binom{n}{i} = 0.$$

- [For fun only!]** Write out the Binomial Expansions for both $(1+x)^n$ and $(1+x^{-1})^n$ where n is an integer. Use these results to show that

$$\sum_{r=0}^n \left[\binom{n}{r} \right]^2 = \binom{2n}{n}.$$

[Hint: where might the Binomial coefficient on the right hand side have come from?]

- [For fun only!]** Try to determine a general formula for the coefficient of $x^i y^j z^k$ in the expansion of $(x+y+z)^n$, where $i+j+k=n$. This expansion is called a Trinomial expansion.

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Problem sheet 9 — Series – Taylor's series, convergence, l'Hôpital's rule

1. Find a cubic polynomial, $f(x)$, which is such that

(a) $f(1) = 2$, $f'(1) = -1$, $f''(1) = 0$ and $f'''(1) = 6$;

(b) $f(0) = 3$, $f'(0) = 0$, $f''(0) = -6$ and $f'''(0) = 12$;

(c) $f(2) = 7$, $f'(2) = 12$, $f''(2) = 18$ and $f'''(2) = 12$.

In parts (a) and (c) rewrite the polynomials as finite power series in x .

2. **(A strange one!)** Find a way to use Taylor's series to rewrite $y = 1 + x + x^2 + x^3$ as a power series in $(x - 1)$. By this I mean that $y = \sum_{n=0}^3 a_n(x - 1)^n$.

3. Find the Taylor's series of $y = (1 + x)^{-2}$ about $x = 0$ and the one about $x = 1$. What are their radii of convergence?

4. Find the Taylor's series of $(1 + x)^{-1}$ about $x = 2$. Find its radius of convergence. It is possible to use the Binomial series to find this power series, but it may need a little bit of thinking...

5. Find the Maclaurin series for $\sin ax$ and use this to find the following limits:

$$(a) \lim_{x \rightarrow 0} \frac{\sin(ax)}{x}, \quad (b) \lim_{x \rightarrow 0} \frac{\sin(ax) - ax}{x^3}.$$

6. Find the Maclaurin's series for both (a) $\cos(ax)$ and (b) $\cosh(ax)$. Use these results (not l'Hôpital's rule) to determine,

$$\lim_{x \rightarrow 0} \frac{\cos(ax) + \cosh(ax) - 2}{x^4}.$$

You may, of course, use l'Hôpital's rule to confirm your solutions.

7. Find the Maclaurin series representation of $\ln(2 + x)$. What is its radius of convergence?

8. In the lecture notes we found a power series representation of e^{-x^2} by first finding a Maclaurin expansion of e^{-x} followed by the replacing of x by x^2 . It was also mentioned that the direct determination of the Maclaurin series of e^{-x^2} is much more challenging. So here's your chance! Use the definition of the Maclaurin series to determine the power series of e^{-x^2} up to and including the x^6 term. If you have the stamina, go as far as the x^8 term!

9. Determine the convergence properties of the following numerical series.

$$(a) \frac{2}{1^3} + \frac{3}{2^3} + \frac{4}{3^3} + \frac{5}{4^3} + \cdots \quad (b) \frac{1}{10} - \frac{2}{11} + \frac{3}{12} - \frac{4}{13} + \cdots \quad (c) \sum_{k=1}^{\infty} \frac{k^p}{k!} \quad (p > 0)$$

$$(d) \sum_{k=1}^{\infty} \frac{k!}{(2k)!} \quad (e) \sum_{k=1}^{\infty} \frac{k! k!}{(2k)!} \quad (f) \sum_{n=1}^{\infty} \frac{n!}{n^n}$$

In part (f) you will need to use the result that $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$, which is to be proven in a later question.

10. Determine the radii of convergence of the following power series.

$$(a) \sum_{n=1}^{\infty} x^n$$

$$(b) \sum_{n=1}^{\infty} nx^n$$

$$(c) \sum_{n=1}^{\infty} n^{100} x^n$$

$$(d) \cosh \sqrt{x} = 1 + \frac{x}{2!} + \frac{x^2}{4!} + \frac{x^3}{6!} + \dots$$

$$(e) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} \dots$$

$$(f) \sum_{n=1}^{\infty} (-1)^n 2^n x^{2n}$$

$$(g) \sum_{n=1}^{\infty} n! x^n$$

$$(h) \sum_{n=1}^{\infty} \frac{n! x^n}{n^n}$$

$$(i) \sum_{n=0}^{\infty} \frac{(1+2n)}{(1+2^n)} x^{n/3}$$

11. Use l'Hôpital's rule to find the following limits.

$$(a) \lim_{x \rightarrow \frac{1}{2}} \frac{\cos \pi x}{2x - 1}$$

$$(b) \lim_{x \rightarrow 1} \frac{\sin \pi x}{\sqrt{x^2 - 1}}$$

$$(c) \lim_{x \rightarrow 0} \frac{\cosh ax - 1}{x}$$

$$(d) \lim_{x \rightarrow \infty} \frac{x^2 + 2x + 3}{3x^2 + 2x + 1}$$

$$(e) \lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$$

$$(f) \lim_{x \rightarrow 0} \frac{\sin^2 x - x^2}{x^3}$$

$$(g) \lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x^2 + 9} - 5}$$

$$(h) \lim_{x \rightarrow 0} \frac{\sin(x^2)}{x \tan x}$$

$$(i) \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$$

$$(j) \lim_{x \rightarrow 0} \frac{\cos(ax) + \cosh(ax) - 2}{x^4}$$

$$(k) \lim_{x \rightarrow 0} \frac{\sinh x + \sin x - 2x}{x^5}$$

12. This is not the type of question which I will give in the exam, but I would like to think that it is of interest. Your aim is to use l'Hôpital's rule to find the following three limits:

$$y = \lim_{x \rightarrow 0} x^x, \quad y = \lim_{x \rightarrow \infty} x^{1/x}, \quad \text{and} \quad y = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

None look like candidates for the use of l'Hôpital's rule, but if one takes the natural logarithm and then rearrange the resulting expressions appropriately, then you'll find that $\ln y$ does indeed take a suitable form. Note that, after you take the logarithm, then (i) the first two cases are of the form of ∞/∞ , and (ii) the third case will eventually need the substitution $x = 1/n$ where the limit $x \rightarrow 0$ is taken. This third one arises when considering compound interest....more information in the solutions.

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Problem Sheet 10 — Vectors

I will use the row vector notation (a, b, c) to represent a three-dimensional vector, rather than $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ (which is suitable for use with matrices) in order to make the typesetting look tidier and to save some space. An alternative notation is $(a, b, c)^T$ where the “ T ” superscript denotes Transpose. I have been lazy here and have omitted the “ T ”.

1. Given the vectors, $\underline{a} = (1, 0, 1)$, $\underline{b} = (2, 1, 2)$ and $\underline{c} = (0, 1, 1)$, evaluate the following:

- (a) $\underline{a} + \underline{b}$ (b) $\underline{a} + \frac{1}{2}\underline{b} + 2\underline{c}$ (c) $\underline{b} - 2\underline{a}$ (d) $|\underline{a}|$ (e) $|\underline{b}|$
 (f) $|\underline{a} - \underline{b}|$ (g) $\hat{\underline{a}}$ (h) $\hat{\underline{b}}$ (i) $\underline{a} \times \underline{b}$
 (j) The angle between $\underline{a}$ and $\underline{b}$ (k) The angle between $\underline{b}$ and $\underline{c}$.

2. Use vectorial methods to find the internal angles of the triangle ABC where

(i) $\overrightarrow{OA} = (1, 3), \quad \overrightarrow{OB} = (4, 6), \quad \overrightarrow{OC} = (-3, 7),$

(ii) $\overrightarrow{OA} = (-1, 3, 1) \quad \overrightarrow{OB} = (2, 6, 2) \quad \overrightarrow{OC} = (3, 3, 8).$

[Hint: Draw a schematic of the triangle and the origin; you'll need to find vectors such as $\overrightarrow{AB}$.]

3. Find the unit vector which is perpendicular to both $(1, 4, 1)$ and $(2, 1, -1)$.
4. Find the value of α for which the following two vectors are perpendicular: $(1, 2, 7)$ and $(4, 5, \alpha)$. Determine the vector product of these vectors, and check that your answer is correct by taking the appropriate scalar products.
5. Prove that the four points with position vectors $(2, 1, 0)$, $(2, -2, -2)$, $(7, -3, -1)$ and $(12, 2, 4)$ are coplanar.
6. The vectors $(2, -4, -1)$, $(3, 2, -2)$ and $(5, -2, -3)$ correspond to the directions and lengths of the three sides of a triangle. Confirm that they are co-planar. What is the length of each side? Show that the triangle is right angled. There are three different cross products that may be found upon taking different pairs of the above vectors; find them and make a comment about the result.
7. My house has a south-facing aspect. From my front door I walk 1 mile south, followed by 1 mile east and then 1 mile north, whereupon I have returned to my house. A bear chooses that moment to walk past. What colour is the bear?

8. Lagrange's identity is

$$|\underline{a} \times \underline{b}|^2 + |\underline{a} \cdot \underline{b}|^2 = |\underline{a}|^2 |\underline{b}|^2.$$

Can you prove it? [Hint: this is ridiculously easy.]

9. This is a lengthy question which is of some interest but the content isn't examinable. In this we derive quite an interesting formula.

(a) The triangle OAB lies in the (x, y) -plane and has one vertex at the origin. Regard $\underline{k}$ as being the unit vector in the z -direction. The other two vertices are at the points with the position vectors, $\underline{a}$ and $\underline{b}$. Use one of the definitions of the vector cross product to show that the area of this triangle is given by

$$\mathcal{A} = \frac{1}{2} (\underline{a} \times \underline{b}) \cdot \underline{k}.$$

(b) One may now find the area of a regular pentagon with unit 'radius' by saying that it is 5 times the area of the triangle with vertices, $(0, 0)$, $(1, 0)$ and $(\cos \frac{2}{5}\pi, \sin \frac{2}{5}\pi)$. What is the area of a regular pentagon? What is the area of a regular dodecagon?

(c) Extend the result of (b) and find a formula for the area of a regular polygon with n sides. By setting $m = 1/n$, use l'Hôpital's rule on this formula to show that the area tends towards π as $n \rightarrow \infty$.

(d) It is now possible to find the area of irregular polygons by splitting them into triangles with the origin as one vertex and two neighbouring points as the other two vertices. Suppose we have a polygon with n vertices, where vertex i is located at (x_i, y_i) . Using the formula given in part (a), show that the area of the polygon may be written as,

$$\mathcal{A} = \frac{1}{2} \sum_{i=1}^n [x_i y_{i+1} - x_{i+1} y_i],$$

where $(x_{n+1}, y_{n+1}) = (x_1, y_1)$, for notational convenience.

(e) Use the formula given in part (d) to find the areas corresponding to the following polygons:

- (i) The triangle, $(1, 1)$, $(1, 3)$, $(-1, 2)$, $(1, 1)$;
- (ii) The triangle, $(-1, 2)$, $(1, 3)$, $(1, 1)$, $(-1, 2)$;
- (iii) The bowtie, $(1, 1)$, $(1, 2)$, $(-1, 1)$, $(-1, 2)$, $(1, 1)$.

Some aspects of the results will need some discussion.

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Problem Sheet 11 — Vectors

Question 1 is very lengthy but it contains everything (and a little more besides) that is of relevance for the equations of lines.

1. [Quite a long question, but it involves almost everything concerning lines and distance.]

(a) Find the vector equation of the line through the points with position vectors, $\underline{a} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

(b) What is the equivalent Cartesian form of this equation?

[Note that there are two versions: either the parametric form, $x = a_1 + b_1\lambda$, $y = a_2 + b_2\lambda$, $z = a_3 + b_3\lambda$, or the parameter-free version, $y = m_1x + c_1$, $z = m_2x + c_2$. Attempt both. Note also that $\underline{r} = (x, y, z)^T$.]

(c) Does this line intersect with the line through the points with position vectors, $\underline{c} = (0, 0, 1)^T$ and $\underline{d} = (1, 0, 1)^T$?

(d) Do either of these lines intersect with the line through the points with position vectors, $\underline{e} = (0, 4, 5)^T$ and $\underline{f} = (3, -2, -1)^T$?

(e) We now have three lines defined in the question. How close to the origin do these lines pass? How close do they pass to the point with position vector, $(2, 0, 1)^T$?

2. Find the shortest distance between two lines where the first passes through $\underline{b}_1$ in the direction, $\underline{c}_1$ and the second passes through $\underline{b}_2$ in the direction, $\underline{c}_2$. Take the following data,

$$\underline{b}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \underline{c}_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \underline{b}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \underline{c}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

3. A plane passes through three given points. For each of the following four cases, (a) to (d), below:

- determine two vectors which lie in the plane;
- write the parametric form of the equation for the plane (i.e. involving two arbitrary parameters);
- find the values of the parameters and thereby determine $\underline{r}$ and hence the distance from the origin;
- find a vector, $\underline{p}$, which is perpendicular to the plane using the results of part (i);
- using the scalar product of $\underline{p}$, find the Cartesian form of the equation of the plane and hence verify the result of part (iii).

[You may wish to do parts (a) and (b) right now, and delay the others until the revision period.]

(a) $\underline{a} = (1, 1, 0)^T \quad \underline{b} = (3, 1, -1)^T \quad \underline{c} = (-1, 0, 2)^T.$

(b) $\underline{a} = (0, 2, 4)^T \quad \underline{b} = (2, 3, 2)^T \quad \underline{c} = (4, 3, 0)^T.$

(c) $\underline{a} = (1, 0, 1)^T \quad \underline{b} = (0, 1, 2)^T \quad \underline{c} = (-1, 1, 1)^T.$

(d) $\underline{a} = (1, 1, 1)^T \quad \underline{b} = (0, 1, 2)^T \quad \underline{c} = (-1, 1, -1)^T.$

4. A plane passes through the points, A , B and C , with position vectors,

$$\overrightarrow{OA} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

How close does the plane get to the point, D , with position vector, $\overrightarrow{OD} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$?

5. Find both the unit normal to the following planes, and their minimum distances to the origin.

(i) $x + 12y + 12z = 34$; (ii) $20x - 12y + 9z = 50$;

(iii) $12x + 16y - 15z = 50$; (iv) $x + 2y + 3z = 14$.

6. This question involves vectors which are confined to the 2D Cartesian plane, (x, y) .

- (a) The equation for a straight line is given by:

$$7x + 24y = 50.$$

Although this is a 2D question, you can think of it as 3D with an extra “+0z” added to the left hand side. What is the shortest distance to the origin?

- (b) [Extra to the lecture notes.] Find the equation for the line which passes through the origin and which is orthogonal to the line given in part (a).

- (c) The results of parts (a) and (b) should give the hint for you to be able to write down the corresponding results immediately when $ax + by = c$ is the line.

7. [Over and above exam level.] A plane contains the points given by the position vectors, $\underline{a}_1$, $\underline{a}_2$ and $\underline{a}_3$. A second plane contains the points given by the position vectors, $\underline{b}_1$, $\underline{b}_2$ and $\underline{b}_3$. Find the direction of the line of intersection of the two planes.

Department of Mechanical Engineering, University of Bath**Mathematics ME10304****Problem sheet 12 — Some example Q10 exam questions**

Four years ago was the first time that I had introduced into the Mathematics exam a question 10 which is composed of a variety of techniques and ideas drawn from many different parts of the syllabus. Therefore the following questions are an indication of the type of question which I might possibly set. I have to say that my enthusiasm has overridden my sense of what constitutes a fair length for a question in some cases here (especially Q4), and although Q5 was on the the 2017/18 paper, some of the others are longer than you'll meet in January. The aim here is simply to give you an indication of how I might construct question 10 in the exam, and the last few years' papers show how much I intend to set on an exam paper.

1. The question concerns the function $y = x/\sqrt{1-x^2}$.
 - (a) Find the derivative of $y(x)$ and express it in as simple a form as possible.
 - (b) Sketch $y(x)$.
 - (c) Find the integral of $y(x)$ between $x = 0$ and $x = 1$.
 - (d) Find the volume of the surface of revolution of $y(x)$ about the x -axis between $x = 0$ and $x = \frac{1}{2}$.
 - (e) Find the volume of the surface of revolution of $y(x)$ about the y -axis between $y = 0$ and $y = 1$.
2.
 - (a) Find a power series representation of $y = (1-x)^{-1/2}$ using either a Taylor's series or a Binomial series approach.
 - (b) Hence write down a power series representation of $(1-x^2)^{-1/2}$.
 - (c) Apply d'Alembert's test to the series obtained in part (a) to determine its radius of convergence.
 - (d) Given that,

$$\sin^{-1} y = \int_0^y \frac{dx}{(1-x^2)^{1/2}},$$

find a power series representation of $\sin^{-1} y$.

3. In this question we consider various aspects of the function,

$$y = \frac{(x-1)^2}{x^2+2x}.$$

- (a) Sketch this function making sure that all salient features are described clearly.
 - (b) Find all of the critical points and add these to the sketch. (There is no need to categorise them.)
 - (c) Determine the definite integral of y between $x = 1$ and $x = 2$.
 - (d) Sketch the reciprocal of y .

4. The so-called sinc function is defined as $\text{sinc } x = \frac{\sin x}{x}$.
- (a) Use a Taylor/Maclaurin series approach to find a power series representation of $\sin x$ in terms of powers of x .
 - (b) Use the result of part (a) to determine $\lim_{x \rightarrow 0} \text{sinc } x$.
 - (c) Confirm the result of part (b) by using l'Hôpital's rule to find $\lim_{x \rightarrow 0} \text{sinc } x$.
 - (d) Sketch $\text{sinc } x$ making sure that all salient features are described clearly.
 - (e) After writing the power series form for $\text{sinc } x$, integrate it to find a power series representation for $\int_0^X \text{sinc } x \, dx$. Write this series in summation form. Determine its radius of convergence using d'Alembert's test.
 - (f) Find the x -derivative of $\text{sinc } x$ and hence show that all the critical points occur when $\tan x = x$.
 - (g) Find the second x -derivative of $\text{sinc } x$, and hence show that the sinc function satisfies the differential equation,

$$xy'' + 2y' + xy = 0.$$

5. In this question we consider various aspects of the function,

$$y = \frac{x^2}{x^2 + 8x + 12}.$$

- (a) Sketch this function making sure that all salient features are shown and described clearly.
- (b) Find all of the critical points and add these to the sketch. (There is no need to categorise them.)
- (c) Determine the definite integral of y between $x = 0$ and $x = 1$.
- (d) Sketch the reciprocal of y .